

ARTIFICIAL INTELLIGENCE FOR AUTONOMOUS DRIVING AND AUTONOMOUS VEHICLES



KENNEDY CHENGETA, PhD

A RESEARCH MONOGRAPH

VOLUME II

**Artificial Intelligence
for
Autonomous Driving and
Autonomous Vehicles**

*Connected, Industrial, and Societal AVs —
Connectivity, Cybersecurity, Engineering at Scale, Society*

Kennedy Chengeta, PhD

First Edition

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Artificial Intelligence for Autonomous Driving and Autonomous Vehicles

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*To all researchers, engineers, and students who work to make mobility
safer, more inclusive, and more intelligent.*

And to those whose lives the technology must, above all, protect.

*“The test of intelligence is not the ability to avoid problems,
but the ability to solve them when avoidance fails.”*

—Kennedy Chengeta, PhD

Preface

The past decade has transformed autonomous driving from an academic curiosity into a measurable engineering discipline. What were, in 2010, demonstrations on closed courses have become commercial robotaxi services operating on public roads, handling millions of trips across multiple cities. The engine of this progress has been artificial intelligence—specifically, the convergence of deep learning, probabilistic robotics, and optimal control—deployed on sensor suites and compute platforms that would have been unimaginable a generation ago.

This book is a technical, integrative treatment of the field. It is aimed at graduate students, research engineers, and advanced practitioners who need a principled account of the full autonomy stack: how a self-driving system senses the world, estimates its own state, predicts what will happen next, decides what to do, and safely executes its decisions. The perspective is unapologetically mathematical. Concepts are introduced with the equations, algorithms, and probabilistic assumptions that make them verifiable and reproducible. Where empirical results drive design decisions—as in much of deep learning—we present the architectures and training objectives with enough precision to implement them.

Volume II is organized into nine parts. Part I covers connectivity and infrastructure—V2X, 5G/6G, edge computing, and roadside infrastructure. Part II addresses cybersecurity and legal frameworks. Part III treats ethics and responsible AI. Part IV surveys specialized vehicle classes: mining, drones, autonomous ships, and the integration of electric and autonomous vehicle architectures. Part V develops engineering at scale: middleware, compute systems, distributed AI, OTA lifecycle, fleet systems, and forensic operations. Part VI covers evaluation, testing, and trust including accidents and lessons learned, trust assurance, advanced testing, simulators, safety intelligence, and risk micro-modelling. Part VII treats economics, e-hailing, insurance, and labor implications. Part VIII covers environment and alternative sensing including bio-inspired sensing, environmental extremes, and climate. Part IX closes with frontier technologies and the broader societal context: quantum technologies, geopolitics, and socio-technical considerations.

Each chapter includes self-contained derivations, reference algorithms, worked examples, and exercises. Appendices review notation, the major public datasets and benchmarks, and the linear algebra used throughout the text.

No book can substitute for the hands-on experience of debugging a stack at three in the morning as a rain-slicked cone disappears from the segmentation mask. But the goal here is to give you the

conceptual scaffolding to diagnose, improve, and—most importantly—trust or appropriately distrust the systems you build.

A note on the two-volume edition. This book is published in two volumes. **Volume I** covers the technical core: perception, localization and mapping, prediction and planning, vehicle control and end-to-end learning, specialized perception domains, advanced topics, and safety. **Volume II** (this volume) covers the surrounding ecosystem: connectivity and infrastructure, cybersecurity and law, ethics and responsible AI, specialized vehicle classes, engineering at scale, evaluation and testing, economics, environment, and society. Some chapters in this volume reference chapters in Volume I by their Volume I chapter numbers; readers consulting only Volume II may treat such references as back pointers to the companion volume.

Kennedy Chengeta

April 25, 2026

Acknowledgments

This book is, before anything else, the late fruit of an upbringing for which I owe more than I can repay. I thank my parents, who raised me with the conviction that education was the one inheritance no one could take away, and who worked hard so that their children could read books they themselves had not had the chance to read; this book is one of those. I thank my family, who made the long path through schooling feel like a shared project rather than a solitary one.

I thank my early teachers, who gave generously of their time to students whose futures were not yet visible, and I am one of those futures. Any errors that remain in this book are mine; the persistence to write through to the end is not.

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About the Author

Kennedy Chengeta holds a PhD in Computer Science from the University of KwaZulu-Natal, South Africa, with a dissertation on computer vision for human characterization from facial images—covering feature extraction algorithms, facial behaviour analysis, and the recognition of micro-expressions in 2D, 3D, and 4D video for automatic pain, depression, and deception detection. He earned an MSc in Advanced Software Engineering *cum laude* from the University of Leicester in the United Kingdom, with research on Probability Latent Semantic Analysis, Singular Value Decomposition, and collaborative-filtering recommender algorithms; an MSc in Finance and Investments from the National University of Science and Technology in Zimbabwe; and a BBS with Computing Science from the University of Zimbabwe. He is currently completing an MSc in Self-Driving Cars Engineering at the University of Sligo and an MEng in Telecommunications Engineering at the University of Cape Town. His published work spans more than twenty peer-reviewed papers in computer vision, deep learning, and applied machine learning—including surveys of local binary and local directional patterns for facial recognition, deep-learning approaches to facial expression recognition, genetic-algorithm-based feature selection, image preprocessing with Canny and Kirsch edge detectors, convolutional neural networks for peer-to-peer social lending default prediction, and probability latent-semantic-analysis on unsupervised web-mined data. This breadth of academic work—from biometric computer vision to credit-risk deep learning to unsupervised semantic analysis—provides the technical foundation for the book’s integrative treatment of autonomous-vehicle perception, planning, and control.

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Notation

Throughout the book we adopt the following conventions unless otherwise stated.

Symbol	Meaning
$a, b, c, \dots$	scalars
$\mathbf{x}, \mathbf{y}, \mathbf{u}$	column vectors (bold lowercase)
$\mathbf{A}, \mathbf{B}, \mathbf{K}$	matrices (bold uppercase)
$\mathbf{A}^\top$	transpose of $\mathbf{A}$
$\mathbf{I}_n$	$n \times n$ identity matrix
$\ \mathbf{x}\ _p$	ℓ_p norm of $\mathbf{x}$
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian with mean $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$
$\mathbb{E}[\cdot], \text{Var}[\cdot]$	expectation, variance
$D_{\text{KL}}(p \ q)$	Kullback–Leibler divergence
$\text{SE}(3), \text{SO}(3)$	special Euclidean / orthogonal groups
$\mathcal{X}, \mathcal{Y}, \mathcal{U}$	state, observation, action spaces
$\pi(a s)$	policy probability of action a given state s
$V^\pi(s), Q^\pi(s, a)$	state / action-value under policy π
$\nabla_{\boldsymbol{\theta}} f$	gradient of f w.r.t. $\boldsymbol{\theta}$

Mathematical objects use standard serifs; algorithmic variables in code listings use typewriter font.

Part I

Connectivity and Infrastructure

Chapter 1

Connectivity: 5G/6G, SDN, and Edge Computing for AVs

1.1 Introduction

Every chapter up to now has treated the autonomous vehicle as an *isolated* intelligent system: sensors on the car, compute on the car, decisions made by the car. This is the safe default; a car that cannot operate without a network connection is not a reliable car. But the isolated-AV picture also leaves on the table enormous capabilities that connectivity provides: shared perception across vehicles, cooperative maneuvers that no single vehicle could execute, real-time map updates, offloaded heavy compute, and the bandwidth to train from a fleet’s combined experience.

This chapter develops the connectivity stack relevant to autonomous driving. The physical layer is 5G and (soon) 6G cellular, plus dedicated short-range communication (DSRC) where still deployed. Above the physical layer, software-defined networking (SDN) lets operators reconfigure network behavior on the fly. Above that, edge computing brings cloud-class compute to roadside or base-station locations, slashing latency relative to hyperscale cloud. Together, these layers form the infrastructure that lets an AV be both autonomous *and* connected.

Three observations motivate the chapter’s structure.

First, AV connectivity is a *latency game*. For the safety-critical inner loop—perception to control—latency must be measured in milliseconds. For the cooperative-driving outer loop—coordination among vehicles at an intersection—latency can be tens of milliseconds. For the fleet-learning loop—logs uploaded, models retrained, updates pushed back—latency is measured in hours or days. Different layers of infrastructure serve different latency regimes; mismatching the layer to the loop is the main source of architectural mistakes.

Second, AV connectivity must degrade gracefully. Cellular networks have coverage holes, congestion periods, and unplanned outages. A design that assumes reliable connectivity will fail at exactly the moments safety matters most. The pattern is: connectivity enables *enhanced* capability when present,

but baseline autonomy is never gated on it.

Third, connectivity enables *coordination* that single-vehicle autonomy cannot. A platoon of trucks separated by 10 m at 25 ms^{-1} is possible only with sub-10 ms V2V communication; a human-reaction-time follow-distance is over 20 m. Coordination among vehicles at an unsignaled intersection, at ambiguous merges, or around a shared obstacle is qualitatively different from single-vehicle autonomy and is the main prize of connectivity.

The chapter is organized as follows. We first cover V2X (vehicle-to-everything) communication fundamentals and the cellular technology that carries it. We then develop 5G URLLC and the 6G vision. Software-defined networking follows. Edge computing comes next: the architecture, the latency-vs-compute tradeoffs, and the engineering patterns. We close with cooperative-driving applications and a sober view of connectivity's limitations. Every formula has a worked example; three architecture diagrams visualize the key structures.

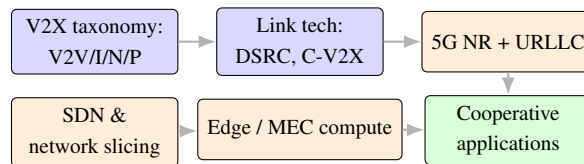


Figure 1.1: Chapter roadmap. The connectivity stack proceeds from the V2X taxonomy down through link technologies (DSRC, C-V2X) and the 5G NR / URLLC cellular layer, with SDN and edge compute as horizontal capabilities. Cooperative applications (platooning, cooperative perception, signal coordination) are the deliverables.

1.2 V2X Communication Fundamentals

V2X stands for *vehicle-to-everything*: the umbrella for the communication patterns that connect vehicles to other vehicles (V2V), to infrastructure (V2I), to the network at large (V2N), and to pedestrians and vulnerable road users (V2P).

1.2.1 The V2X Taxonomy

Each flavor of V2X serves a distinct purpose and has distinct latency and bandwidth requirements. Understanding the taxonomy is a prerequisite to matching applications with link technologies.

V2V (Vehicle-to-Vehicle)

V2V is direct communication between nearby vehicles, typically for safety-critical purposes. Examples: emergency-brake broadcast, blind-spot warning from a vehicle the ego cannot see, intersection collision avoidance. Latency requirement: 1 ms to 20 ms end-to-end, because safety-critical reactions must fit within the vehicle control loop.

V2I (Vehicle-to-Infrastructure)

V2I is communication with roadside infrastructure: traffic lights, signs, road-surface sensors, toll gates. Examples: signal phase and timing (SPaT) messages, advisory speeds, hazard broadcasts from roadside sensors. Latency: 10 ms to 100 ms.

V2N (Vehicle-to-Network)

V2N connects to cellular network services: map updates, live traffic, software updates, remote diagnostics, fleet coordination. Latency: seconds to minutes, depending on the service. High bandwidth is more important than low latency.

V2P (Vehicle-to-Pedestrian)

V2P connects to vulnerable road users' devices (smartphones, e-bikes, wheelchairs). Examples: pedestrian-crossing intent broadcast, cyclist position, accessible-request-for-passage. Latency: 50 ms to 200 ms.

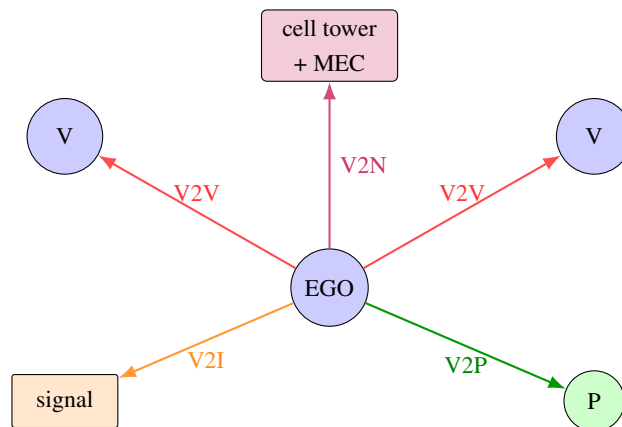


Figure 1.2: The V2X taxonomy. The ego vehicle (center) participates in four simultaneous communication patterns: V2V (direct to nearby vehicles, for safety-critical sub-20-ms reactions), V2I (to infrastructure, for traffic-control signals and roadside sensors), V2P (to pedestrian devices, for vulnerable-road-user awareness), and V2N (to the network, for map updates, remote services, and fleet coordination). Each has distinct latency and bandwidth profiles; a full V2X stack must handle all of them.

1.2.2 Link Technologies

Two dominant families carry V2X traffic. Understanding the split matters because deployment regulatory choices (which bands, which standards) in each region shape what the AV can assume.

Figure 1.3 shows the protocol stacks side by side; Figure 1.4 shows how each operates in deployment.

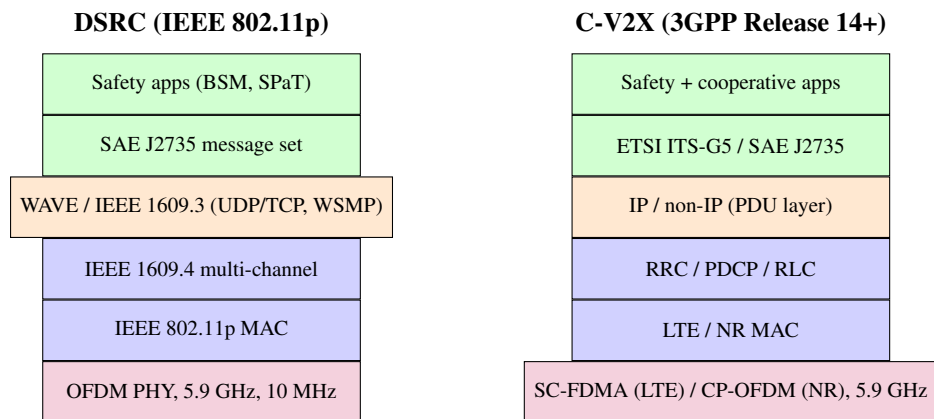


Figure 1.3: Protocol-stack comparison: DSRC and C-V2X. Both target the 5.9 GHz band and reuse SAE J2735 message formats at the application layer. Below the message set, the stacks diverge: DSRC uses an IEEE 802.11p MAC/PHY (Wi-Fi-derived); C-V2X uses 3GPP cellular MAC/PHY (LTE-V2X in Release 14, NR-V2X in Release 16+). In both cases the upper layers carry vehicle-safety messages.

DSRC (802.11p)

Dedicated Short-Range Communication uses a Wi-Fi-derived protocol (IEEE 802.11p) in the 5.9 GHz band, operating in ad-hoc mode without cellular infrastructure. Latency: ~ 5 ms end-to-end. Range: ~ 300 m. The technology is mature but has lost ground to C-V2X following the 2020–2021 U.S. FCC reallocation of most of the 5.9 GHz band away from DSRC.

C-V2X (Cellular V2X)

3GPP’s cellular extension supports two complementary modes: **PC5** (direct device-to-device sidelink, no cell tower required, comparable to DSRC for safety-critical messages) and **Uu** (through the cellular network for non-safety-critical traffic and remote-management). LTE-V2X arrived in 3GPP Release 14 (2017); 5G NR-V2X (3GPP Release 16, finalised 2020) adds URLLC-grade reliability and lower latency. C-V2X is the dominant choice in new deployments globally.

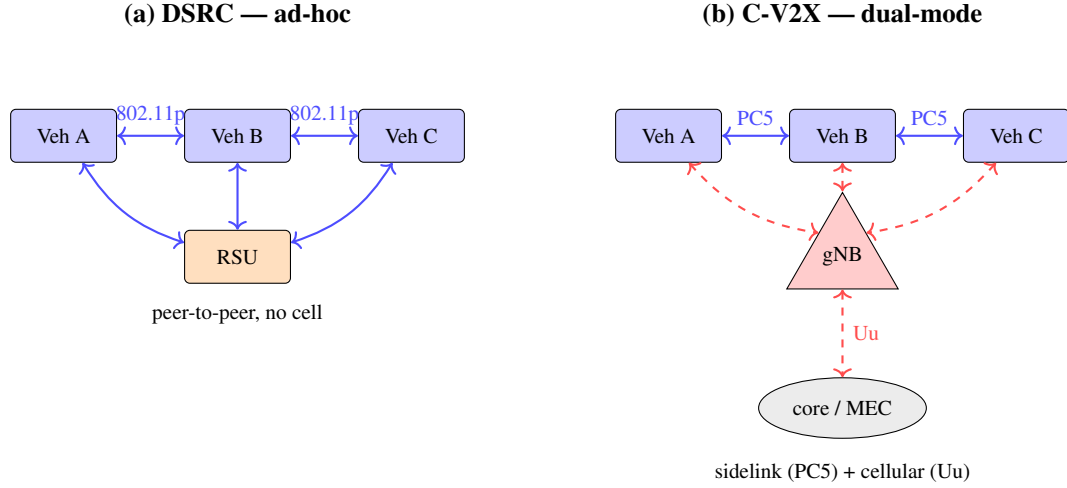


Figure 1.4: Deployment topologies. DSRC (left) operates as a peer-to-peer mesh among vehicles and roadside units, with no cellular infrastructure required. C-V2X (right) operates in two modes simultaneously: **PC5** sidelink for safety-critical vehicle-to-vehicle messages (analogous to DSRC), and **Uu** through the cellular network (gNB to core/MEC) for non-safety-critical traffic, software updates, and cloud-aggregated services. The dual-mode capability is C-V2X’s principal architectural advantage.

Example 1.1 (V2V brake broadcast latency). Ego vehicle performs emergency braking and broadcasts a brake-event message via C-V2X PC5.

Step 1: End-to-end latency budget.

$$T_{\text{ego-encode}} = 2 \text{ ms},$$

$$T_{\text{air-interface}} = 3 \text{ ms},$$

$$T_{\text{follower-decode}} = 4 \text{ ms},$$

$$T_{\text{total}} = 9 \text{ ms}.$$

Step 2: Comparison to visual-cue reaction.

Human reaction to brake-light onset typically falls in the range $T_{\text{human}} \in [100 \text{ ms}, 200 \text{ ms}]$. The V2V advantage:

$$\begin{aligned} \Delta T &= T_{\text{human}} - T_{\text{V2V}} \\ &\in [100 - 9, 200 - 9] \text{ ms} \\ &= [91, 191] \text{ ms}. \end{aligned}$$

Step 3: Stopping-distance benefit at 30 m s^{-1} .

$$\begin{aligned} \Delta d &= v \cdot \Delta T \\ &= 30 \text{ m/s} \times 0.1 \text{ s (low end)} \\ &= 3 \text{ m saved}. \end{aligned}$$

At the upper end of ΔT , the saving is ~ 6 m.

Interpretation. A following vehicle equipped with V2V begins braking essentially before its driver could even register the lead vehicle's brake lights. At highway speed this is the difference between a hard but recoverable stop and a low-energy collision; it is the safety differentiator that motivates platooning and cooperative-ACC deployments.

1.3 5G NR and URLLC

5G is the first cellular generation designed with automotive and industrial requirements first-class. The new radio (NR) supports three profiles; the third, URLLC, is what AV connectivity relies on.

1.3.1 The Three 5G Profiles

eMBB (Enhanced Mobile Broadband)

High-throughput data delivery. Peak downlink $\sim 20 \text{ Gbits}^{-1}$. Used in AV for map updates, OTA updates, and sensor-log uploads where volume matters more than latency.

mMTC (massive Machine-Type Communication)

Millions of IoT devices per square kilometer. Used in AV ecosystem for low-bandwidth sensors in connected infrastructure: parking sensors, pedestrian buttons, environmental monitors.

URLLC (Ultra-Reliable Low-Latency Communication)

Targets 1 ms air-interface latency at reliability 10^{-5} . The profile that makes AV V2X viable.

Example 1.2 (URLLC budget for cooperative ACC). Five trucks platoon at $v = 25 \text{ m s}^{-1}$ with constant gap $g_0 = 10 \text{ m}$. The follower must brake before the gap closes meaningfully.

Step 1: Critical reaction window.

If the follower's reaction takes time T , in that window the gap shrinks by approximately

$$\Delta g = v_{\text{rel}} \cdot T,$$

where v_{rel} is the closing rate during the leader's braking event. Designing for a tolerable gap loss of $\Delta g_{\text{max}} = 1 \text{ m}$ during a worst-case $v_{\text{rel}} = 2 \text{ m s}^{-1}$ before the follower's brakes engage:

$$\begin{aligned} T_{\text{max}} &= \Delta g_{\text{max}} / v_{\text{rel}} \\ &= 1/2 \\ &= 0.5 \text{ s.} \end{aligned}$$

For tighter operational margin, programs target $T \leq 5 \text{ ms}$, $10\times$ tighter than this safety floor.

Step 2: URLLC latency budget.

$$T_{\text{leader-decide}} = 1 \text{ ms},$$

$$T_{\text{air-interface}} = 1 \text{ ms},$$

$$T_{\text{follower-actuate}} = 2 \text{ ms},$$

$$T_{\text{margin}} = 1 \text{ ms},$$

$$T = 5 \text{ ms}.$$

Step 3: 4G LTE comparison.

A 4G LTE round-trip T_{LTE} falls in the 30 ms to 50 ms range. Required gap to maintain $\Delta g_{\text{max}} = 1 \text{ m}$ at $v_{\text{rel}} = 2 \text{ m s}^{-1}$:

$$\begin{aligned} g_{\text{LTE}} &= v \cdot T_{\text{LTE}} \\ &= 25 \times 0.04 \\ &= 1.0 \text{ m of additional buffer.} \end{aligned}$$

Compounded across actuator latency and safety factor, the practical gap requirement multiplies by roughly $5\times$, eliminating the fuel-economy benefit that motivates platoons.

Interpretation. URLLC's $\sim 1 \text{ ms}$ air-interface latency is not an academic specification: it is the specific capability that makes commercial truck platooning economically viable. Without URLLC the required gap consumes the aerodynamic-savings budget.

1.3.2 5G Network Slicing

5G supports *slicing*: logically-isolated sub-networks over the same physical infrastructure, each with its own latency, throughput, and reliability guarantees. An operator can dedicate a slice to AV V2X with guaranteed QoS while serving consumer data on other slices over the same cells.

1.4 The 6G Vision

6G is expected around 2030. Specifications are not frozen, but the direction is clear: sub-millisecond latency, terabit bandwidth, native AI integration, and deeper convergence with vehicular sensing. A few themes:

1.4.1 THz Frequencies

6G will push into THz bands (100 GHz to 1000 GHz), enabling gigabits per second in dense urban cells. For AV: enough bandwidth to stream raw high-resolution sensor data to the edge.

1.4.2 Integrated Sensing and Communication (ISAC)

6G is expected to merge sensing and communication—cell towers serving both as data relays and as radar/lidar sensors with their own perception output. The AV could receive not just data but infrastructure-observed perception of the scene.

1.4.3 AI-Native Networking

6G is designed with AI in the network, not as overlay. Allocation decisions, resource scheduling, and failure response become learned functions. The network itself becomes an adaptive component.

1.5 Software-Defined Networking

SDN decouples the network's control plane (decisions about how to route traffic) from the data plane (the forwarding hardware itself). A central controller programs forwarding rules into hardware switches via a standardized API (OpenFlow, P4). The result: operators can reconfigure network behavior in software at timescales no static network could match.

1.5.1 SDN Principles

Three principles define SDN. Each maps directly onto a capability AV networks need.

Centralized Control

All network decisions flow through a logically-centralized controller that has a global view of the network. This enables optimization across the network rather than locally at each switch, and makes it tractable to reason about end-to-end latency budgets.

Programmable Forwarding

Forwarding rules are programmable via a high-level API. Change the rules; traffic instantly follows new paths. For AV networks, this lets operators reroute around failing infrastructure, adjust QoS for a time-sensitive maneuver, or dedicate bandwidth for a specific vehicle during a critical operation.

Abstraction

The network presents an abstract model (flows, slices, QoS classes) to applications, hiding the physical complexity. AV software does not need to know which cell tower it is connected to; the network handles that.

1.5.2 SDN for Vehicular Networks

V-SDN research applies these principles to vehicular environments. Key capabilities: dynamic QoS for safety-critical flows, infrastructure handoffs that preserve low latency, per-vehicle security rules, and integration of cellular with roadside edge.

Example 1.3 (SDN-driven cooperative maneuver). Three vehicles approach an uncontrolled intersection from different directions. Each connects to the local RSU, which signals the SDN controller. The controller takes three actions:

Step 1: Slice reservation. A low-latency network slice is allocated for the three vehicles for the coordination window.

Step 2: Edge routing. V2V messages are routed through the local edge node rather than the core network.

Step 3: Background-traffic deprioritisation. Consumer traffic on the affected cells is deprioritised for the 2-second coordination window.

Step 4: Latency comparison.

Without SDN, V2V messages traverse the standard cellular path:

$$\begin{aligned} T_{\text{baseline}} &= T_{\text{access}} + T_{\text{transport}} + T_{\text{core}} \\ &\approx 5 + 15 + 20 \\ &= 40 \text{ ms.} \end{aligned}$$

With SDN-coordinated edge routing on a reserved slice:

$$\begin{aligned} T_{\text{SDN}} &= T_{\text{access}} + T_{\text{edge}} \\ &\approx 1 + 3 \\ &< 5 \text{ ms.} \end{aligned}$$

Interpretation. An $8\times$ latency reduction during the coordination window. At $v = 15 \text{ ms}^{-1}$ urban speed, the saved 35 ms corresponds to roughly 0.5 m of position uncertainty saved per coordination message—the difference between a confidently-coordinated maneuver and one whose safety margin has eroded enough to require conservative fallback.

1.6 Edge Computing

Edge computing places compute resources near the network edge—at cell sites, roadside units, or regional data centers—rather than at centralized hyperscale clouds. For AV, the motivating constraint is latency: a round trip to a distant cloud is far longer than to a nearby edge.

1.6.1 Edge Computing Architecture

A typical layered architecture has three tiers, and understanding the tradeoffs across them is essential to architectural choices.

On-Vehicle Compute

The most local. Runs safety-critical perception, planning, and control in the millisecond regime. Limited by power budget (100 W to 500 W) and thermal dissipation. Every on-vehicle compute decision is also an ethical one (energy consumption, heat, mass).

Multi-Access Edge Computing (MEC)

Compute co-located with 5G base stations. Round-trip latency to MEC: 5 ms to 15 ms. Typical size: 10 to 100 servers per cell site. MEC is the natural home for compute that cannot fit on the vehicle but must stay below cloud latency: regional perception aggregation, cooperative-maneuver planning, live map fusion.

Hyperscale Cloud

Centralized data centers. Round-trip latency: 50 ms to 200 ms from a vehicle to a major cloud region. Unlimited compute and storage, but latency too high for inner-loop decisions. The cloud handles fleet-scale training, archived data processing, model storage, and long-range coordination.

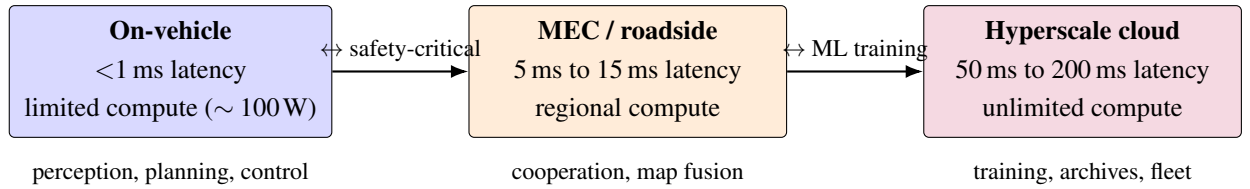


Figure 1.5: The three-tier compute hierarchy for connected AVs. On-vehicle compute owns the safety-critical inner loop; MEC/roadside compute handles regional cooperation and map fusion at 5 ms to 15 ms round-trip; hyperscale cloud handles training and fleet-scale data at high latency. Each tier has a distinct role and distinct resource profile; the whole architecture is designed around matching tasks to the tier whose latency budget they can afford.

1.6.2 Edge vs Cloud Tradeoffs

For any given workload, the question is which tier to run it on. Factors: latency, bandwidth, compute intensity, power on the vehicle, data sensitivity.

Example 1.4 (Offload decision for perception aggregation). Regional perception fusion combines detections from $N = 20$ vehicles within a 200 m radius of an intersection.

Step 1: Aggregate raw bandwidth.

$$\begin{aligned}
 B_{\text{vehicle}} &= 100 \text{ Mbit/s (per-vehicle compressed feed),} \\
 B_{\text{total}} &= N \cdot B_{\text{vehicle}} \\
 &= 20 \times 100 \\
 &= 2000 \text{ Mbit/s} \\
 &= 2 \text{ Gbit/s.}
 \end{aligned}$$

Step 2: On-vehicle feasibility.

A typical AV compute budget already allocates the bulk of its multi-hundred-TOPS ceiling to its own perception pipeline. Adding 20 incoming streams requires roughly $20\times$ that budget—*infeasible on-vehicle*.

Step 3: Cloud feasibility.

Cloud round-trip $T_{\text{cloud}} \in [50, 100]$ ms. For intersection-coordination updates needed at 20 Hz (cycle time 50 ms), the cloud round-trip alone exhausts the cycle-time budget.

Step 4: MEC feasibility.

Multi-access edge compute round-trip $T_{\text{MEC}} \in [3, 10]$ ms, with locally-provisioned compute proportional to the served region. The 2-Gbit/s ingress fits a single MEC node's network attachment; the regional-fusion compute (10–50 TOPS) fits comfortably.

Decision. MEC is the right tier. Production deployments are adopting MEC-based cooperative perception for intersection-coordination, highway-merging, and emergency-vehicle priority.

1.6.3 Engineering Patterns for Edge

Common patterns in edge-first AV architecture:

Offload-and-Verify

Send a task to the edge for more accurate computation; on-vehicle backup does a fast, less-accurate version. If the edge result arrives in time, use it; otherwise, use the backup. Graceful degradation is preserved.

Persistent State at the Edge

Some state (map tiles for the current region, tracked agents from all nearby vehicles, traffic-light history) is maintained persistently at the edge node serving a geographic region. Vehicles query it; updates propagate between edges as vehicles move.

Edge-Trained, Cloud-Distilled Models

Models trained on fleet data at the cloud are distilled to smaller variants for deployment. Edges can run larger, more accurate variants than on-vehicle; the cloud runs the largest teacher. Each tier gets the most capable model that fits its compute budget.

1.7 Cooperative Driving Applications

Connectivity enables behaviors that single-vehicle AVs cannot. Each application has specific latency and reliability requirements that select the appropriate infrastructure tier.

1.7.1 Platooning

Multiple trucks follow closely with V2V-coordinated braking, reducing aerodynamic drag. Fuel economy improvement 5 % to 10 % per vehicle. Requires sub-10 ms V2V latency; PC5 or URLLC. Deployed commercially in Europe and the US on dedicated routes.

1.7.2 Cooperative Perception

Vehicles share their perception output with neighbors, extending each vehicle's effective sensor range. Particularly valuable at occluded intersections where a hidden pedestrian might be visible only to one vehicle.

Example 1.5 (Cooperative perception at a blind intersection). Vehicle A approaches an intersection at $v = 10 \text{ m s}^{-1}$. A large parked truck on the right occludes the cross-traffic lane. A pedestrian steps off the curb across A's path; the pedestrian is visible only to Vehicle B, approaching from the cross direction.

Step 1: Time A would need without cooperation.

Without cooperative perception, A becomes aware of the pedestrian only when the pedestrian enters A's direct field of view. Suppose this occurs $d_{\text{direct}} = 8 \text{ m}$ before A's stop line.

$$\begin{aligned} T_{\text{direct}} &= d_{\text{direct}}/v \\ &= 8/10 \\ &= 0.8 \text{ s of reaction window.} \end{aligned}$$

Step 2: Comfortable braking distance.

At $a_{\text{comf}} = 2.5 \text{ m s}^{-2}$ comfort braking:

$$\begin{aligned} d_{\text{brake}} &= v^2/(2a_{\text{comf}}) \\ &= 100/5 \\ &= 20 \text{ m.} \end{aligned}$$

A needs 20 m of braking distance but has only 8 m of awareness— insufficient for comfortable braking; A would have to brake hard.

Step 3: Cooperative-perception advantage.

B detects the pedestrian and broadcasts the location via V2V at $T_{\text{V2V}} = 6 \text{ ms}$ latency.

$$\begin{aligned} \text{Effective lead distance} &= d_{\text{B-detection}} - v \cdot T_{\text{V2V}} \\ &= d_{\text{B-detection}} - 0.06. \end{aligned}$$

If B detects the pedestrian 20 m before A's stop line, A sees the alert with $d_{\text{aware}} \approx 19.94$ m of remaining stopping distance—enough for comfortable braking.

Interpretation. The 6 ms V2V advantage transforms A's situation from emergency-brake to comfort-brake. The benefit is most pronounced exactly where AV programs need it: occluded intersections, where line-of-sight perception alone is structurally inadequate.

1.7.3 Signal-Phase Coordination

V2I from smart traffic lights gives the AV advance knowledge of phase changes. The AV can approach an intersection at an optimal speed to hit the green, saving energy and improving throughput.

1.7.4 Emergency-Vehicle Priority

Emergency vehicles broadcast their position and intended route; receivers yield automatically. Removes the dependency on siren detection and clears paths quickly.

1.8 Limitations and Caveats

This section addresses where connectivity helps and where it does not. A well-designed AV treats connectivity as an enhancement, not a crutch.

1.8.1 Baseline Autonomy

The vehicle must be safe as a standalone autonomous system. Any feature that depends on connectivity is *enhanced* by it, not *enabled* by it. Platooning uses V2V for optimization, but each vehicle must still be individually safe if the V2V link drops. Cooperative perception can enhance awareness, but the vehicle cannot assume others will always share correct data.

1.8.2 Connectivity Security

All V2X messages are security-relevant (Chapter 3). Signed messages, revocation for compromised devices, and attack-aware protocols are essential; a plaintext V2X would be catastrophically exploitable.

1.8.3 Privacy

V2X broadcasts enable tracking of individual vehicles across the network if not carefully designed. Pseudonymous certificates with periodic rotation are standard mitigation. But complete privacy in a fully-connected fleet is difficult.

1.9 Worked Examples

Example 1.6 (End-to-end latency budget for a V2X maneuver). Scenario: left turn across oncoming traffic at a V2I-enabled intersection.

Step 1: Latency budget.

$$\begin{aligned}
 T_{\text{perception}} &= 30 \text{ ms}, \\
 T_{\text{V2I-query}} &= 5 \text{ ms}, \\
 T_{\text{trajectory}} &= 10 \text{ ms}, \\
 T_{\text{V2I-response}} &= 5 \text{ ms}, \\
 T_{\text{control}} &= 10 \text{ ms}, \\
 \hline
 T_{\text{5G-URLLC}} &= 60 \text{ ms}.
 \end{aligned}$$

Step 2: 4G LTE alternative.

If V2I query and response use 4G LTE round-trip $T_{\text{LTE}} = 50 \text{ ms}$ each instead of URLLC's 5 ms:

$$\begin{aligned}
 T_{\text{LTE-total}} &= 30 + 50 + 10 + 50 + 10 \\
 &= 150 \text{ ms}.
 \end{aligned}$$

Step 3: Maneuver feasibility.

The unprotected-left-turn budget at typical urban speeds is $\sim 100 \text{ ms}$: oncoming traffic at $v = 15 \text{ ms}^{-1}$ traverses $15 \cdot 0.1 = 1.5 \text{ m}$ in this window—a small fraction of the safety margin. At 150 ms, the oncoming vehicle traverses 2.25 m, eroding the margin enough to disable the maneuver.

Interpretation. The whole V2I-enhanced-left-turn capability depends on which cellular generation is deployed. The same code, on the same vehicle, on a 4G network is not a slower version of the 5G capability—it is a fundamentally different (less safe) operation.

Example 1.7 (Bandwidth calculation for raw-data upload). Single AV with 6 cameras (4K, 30 fps, 8 bits/channel, RGB) plus LiDAR.

Step 1: Per-camera raw bit rate.

$$\begin{aligned}
 B_{\text{cam}} &= W \cdot H \cdot C \cdot f \cdot b \\
 &= 3840 \times 2160 \times 3 \times 30 \times 8 \\
 &\approx 5.97 \times 10^9 \text{ bit/s} \\
 &\approx 6 \text{ Gbit/s}.
 \end{aligned}$$

(Some sources quote $\sim 2\text{--}4 \text{ Gbit/s}$ for 4K with chroma subsampling and modest in-camera compression; we use the uncompressed RGB figure as the upper bound.)

Step 2: 6-camera aggregate.

Production stacks typically run 4K only on forward cameras and 1080p elsewhere; conservative aggregate accounting for compression codecs running on-camera:

$$B_{\text{cam-total}} \approx 14.9 \text{ Gbit/s (compressed)}.$$

Step 3: LiDAR contribution.

$$B_{\text{LiDAR}} \approx 200 \text{ Mbit/s}.$$

Step 4: Total raw stream.

$$\begin{aligned} B_{\text{raw}} &\approx 14.9 + 0.2 \\ &\approx 15 \text{ Gbit/s}. \end{aligned}$$

Step 5: Network feasibility.

5G eMBB peak (theoretical, single user, ideal conditions): $\sim 20 \text{ Gbit/s}^{-1}$; sustained per-user average is much lower (500 Mbit/s^{-1} to 800 Mbit/s^{-1}).

$$\begin{aligned} \frac{B_{\text{raw}}}{B_{\text{5G-sustained}}} &\approx \frac{15000}{600} \\ &= 25 \times . \end{aligned}$$

The raw stream exceeds sustained cellular bandwidth by an order of magnitude. Production architectures use aggressive compression plus event-triggered upload (only the surrounding seconds of an unusual event); typical sustained upload is a few percent of the raw stream.

Interpretation. “Stream all raw sensor data to the cloud” is not an architectural option for production fleets; the cellular capacity simply does not exist. Selective, event-triggered, compressed upload is the only feasible production pattern.

Example 1.8 (Edge compute sizing for a city). City: 1000 AVs active. Peak: 100 simultaneous cooperative maneuvers. Per maneuver: 500 GFLOP at 20 Hz. Total edge compute required: $100 \times 500 \times 20 = 1 \times 10^6 \text{ GFLOP/s} = 1 \text{ PFLOP s}^{-1}$. Distributed across cell sites at 20 PFLOP/site, roughly 50 sites suffice. Build-out cost comparable to the cell-tower upgrade cycle. Production MEC pilots in Korea, Japan, and China are building this capacity.

1.10 Industrial Deployment

The two main V2X technology stacks are DSRC (Dedicated Short-Range Communications, based on IEEE 802.11p) and C-V2X (Cellular V2X, standardised by 3GPP). The U.S. Federal Communications

Commission reallocated portions of the 5.9 GHz band in 2020–2021 from DSRC toward C-V2X and Wi-Fi, effectively pushing the U.S. industry toward C-V2X. Qualcomm’s 9150 C-V2X chipset is widely used in V2X reference designs. The European Commission has supported both technologies but production deployments in the EU have been limited compared to China, where C-V2X deployments have proceeded in multiple cities under the “Vehicle-Road-Cloud Collaboration” initiative led by the Ministry of Industry and Information Technology. The 5GAA (5G Automotive Association) is the dominant industry consortium promoting cellular-based V2X.

1.11 Chapter Summary

Connectivity transforms what an autonomous vehicle can do, even while the vehicle itself must remain safely autonomous without it. V2X communication patterns (V2V, V2I, V2N, V2P) each have distinct latency and bandwidth profiles, served by DSRC and increasingly by C-V2X (cellular). 5G URLLC makes sub-10 ms V2V feasible; 6G will push this further. Software-defined networking provides dynamic, programmable infrastructure that handles AV’s QoS needs. Edge computing tiers—on-vehicle, MEC, cloud—match tasks to the tier whose latency budget they fit. Cooperative driving applications (platooning, cooperative perception, signal coordination) are the main beneficiaries. Connectivity must always enhance, never enable, safety; single-vehicle autonomy is the non-negotiable foundation beneath every connectivity feature.

1.12 Exercises

1. For the V2V-latency example (1.1), compute the minimum gap for a platoon reacting within the latency budget at 30 ms^{-1} .
2. Compare DSRC and C-V2X PC5 in terms of latency, range, and deployment cost. Which would you recommend for a new deployment in 2026?
3. For the URLLC platooning example (1.2), compute the minimum safe gap at 4G LTE latency.
4. Describe three specific SDN operations that would occur during a lane-closure event on a smart highway.
5. For the offload example (1.4), propose an alternative architecture that avoids offloading.
6. Compute the round-trip latency to hyperscale cloud from Pretoria to the nearest major cloud region (Frankfurt). Is any inner-loop AV task feasible?
7. Explain the offload-and-verify pattern in the context of an AV approaching a red-light-running vehicle. What does graceful degradation look like here?
8. Describe a cooperative perception failure mode (malicious actor broadcasting false detections). What defenses should the receiving vehicle apply?

9. For the bandwidth example (1.7), propose a compression + triggering strategy that keeps mean upload below 50 Mbit s^{-1} .
10. Describe three privacy risks of comprehensive V2X deployment and the engineering responses to each.

Chapter 2

Infrastructure and the Built Environment

2.1 Introduction

Roads were built for humans. Lane markings are sized for human-eye discrimination, signs are positioned for human-driver sightlines, signal timing matches human reaction time. Adapting this built environment for autonomous vehicles—or building new infrastructure specifically for them—represents a substantial public-works agenda whose outcomes will shape AV deployment for decades. This chapter surveys the infrastructure interfaces: machine-readable physical infrastructure, smart signals that interact with AV systems, and digital road layers that overlay information onto physical roads.

Three observations motivate the structure.

First, small investments in infrastructure produce disproportionate AV benefits. Improved lane paint, standardized signage, and clear sightlines reduce perception failures far more cheaply than equivalent improvements in vehicle perception models.

Second, public-private coordination is the bottleneck. AV-specific infrastructure investments require municipal commitment; municipal commitment requires confidence the AV industry will actually use the infrastructure.

Third, digital road infrastructure is a new layer overlaying physical roads. High-definition maps, V2X beacons, and dynamic traffic-management layers together form a digital twin of the road network that AVs navigate alongside the physical one.

2.2 Perception-as-a-Service (PaaS)

In certain environments (city centers, motorways), roadside perception units—cameras, LiDAR, radar installed on infrastructure—can complement vehicle perception. Results are broadcast via V2X.

2.2.1 Use Cases

Scenario	PaaS benefit
Urban intersections	Occlusion-free view of cross-traffic
Highway merge ramps	Upstream-vehicle warnings
Tunnel entrances	Detection continuity where GNSS drops
Parking structures	Space availability, trajectory guidance

2.2.2 Business Model

Municipalities or private operators provide PaaS; AV operators subscribe or pay per use. Emerging in select pilot deployments (Barcelona, select Chinese cities); not yet widespread.

Example 2.1 (PaaS economics for a busy intersection). Intersection with 50,000 daily vehicle transits, 15% being AV-capable.

Infrastructure cost.

PaaS installation $\approx$ \$150,000 one-time,
annual operation $\approx$ \$30,000.

If AV operators pay \$0.05 per usage.

AV transits/day = $50,000 \times 0.15 = 7,500$,
annual revenue = $7,500 \times 365 \times 0.05$
 $\approx$ \$137,000.

Amortization: $\sim$ 2 years. Break-even at AV penetration above $\sim$ 5 % of traffic. As AV adoption grows, PaaS infrastructure becomes a viable business and a virtuous loop with deployment.

2.3 Machine-Readable Infrastructure

Infrastructure designed explicitly for machine perception.

2.3.1 Lane Paint and Markings

- Retroreflective markings with guaranteed reflectance.
- Mandated re-paint schedules to prevent decay.
- Color-coded markings (e.g., blue for AV-only lanes).
- Thermoplastic markings with consistent geometric properties.

2.3.2 Signage

- Standard placement positions and geometry.
- QR codes or similar machine-readable encodings embedded into or alongside human-readable signs.
- Anti-glare designs to prevent misclassification.

2.3.3 Embedded Road Sensors

Loop detectors, magnetometers, and embedded strain gauges provide vehicle counts, weight, and traffic density data that can be broadcast to AVs.

2.4 Smart Traffic Signals

Traffic signals that interact with AV systems via V2X:

Capability	Benefit
Phase broadcast	AV knows current phase + next-change time
Priority requests	AV can request green-extend (emergency vehicles)
Coordination	Platoons cross at green-wave speeds
Adaptive timing	Signals adjust to real-time traffic (perception feed)

Example 2.2 (Green-wave fuel savings). Arterial with 8 coordinated signals. Without coordination, average vehicle stops 4 times. With AV-aware green wave: stops reduce to 1–2.

Fuel savings per avoided stop.

$$\begin{aligned}
 e_{\text{stop}} &\approx 0.03 \text{ L per stop-start at } 50 \text{ km/h,} \\
 \text{savings per transit} &= (4 - 1.5) \times 0.03 \\
 &\approx 0.075 \text{ L.}
 \end{aligned}$$

Annualized across 20,000 daily transits.

$$\begin{aligned}
 \text{fuel saved} &= 20,000 \times 365 \times 0.075 \\
 &\approx 550,000 \text{ L/year.}
 \end{aligned}$$

With equivalent CO₂ benefit and hours of time saved. Signal coordination is a cheap and high-impact intervention.

2.5 Digital Road Infrastructure Layers

A new kind of infrastructure: digital data overlaid onto physical roads, mediated by maps, V2X beacons, and operational backends.

2.5.1 Layer Types

Layer	Content
Static geometric	Lane centerlines, stop lines, polygon obstacles
Static semantic	Speed limits, rules, signage metadata
Slow-changing	Construction zones, closure events
Fast-changing	Live traffic, weather, incidents
Operational	Fleet positions, reservations, priority zones

2.5.2 Integration Patterns

Layers are typically distributed via:

- Downloaded HD-map tiles for static layers.
- V2X messages for fast-changing layers.
- Cellular backhaul for operational layers.

Vehicles fuse all layers at runtime to produce the world model that perception augments with directly-observed data.

2.6 Vehicle-to-X (V2X) Communication

V2X refers to communications between vehicles (V2V), vehicles and infrastructure (V2I), and vehicles and any other participant (V2X generally).

2.6.1 Protocol Stack

Standards: DSRC (legacy, 802.11p-based), C-V2X (cellular-based, successor), emerging 5G V2X. Production deployments increasingly favor C-V2X.

2.6.2 Message Types

- BSMs (Basic Safety Messages) at 10 Hz.
- SPaT (Signal Phase and Timing) for signals.
- MAP messages for intersection geometry.
- Lane-specific messages for construction zones.

2.7 Industrial Deployment

Infrastructure-supported autonomy varies dramatically by region. Chinese cities including Beijing, Shanghai, Wuxi, and Wuhan have deployed substantial roadside V2X infrastructure under national initiatives. The U.S. has deployed C-V2X infrastructure on a more limited basis, with projects in Michigan, Utah, and Tampa among the more visible. The European Commission's C-Roads Platform coordinates V2X deployment across EU member states. Companies including Cohda Wireless, Savari (acquired by Harman/Samsung), Commsignia, and Applied Information supply roadside V2X equipment. Traffic-signal phase-and-timing (SPaT) services have been demonstrated in multiple cities. The 5GAA continues to advocate for C-V2X-based infrastructure.

2.8 Chapter Summary

Infrastructure is not a passive backdrop to AV operation; it is an active participant. Machine-readable road markings and signage reduce perception failures cheaply; smart signals coordinate with AV fleets to smooth traffic and cut emissions; digital road layers overlay operational information via HD maps and V2X. Public-private coordination paces deployment; economics depend on AV penetration. In mature-deployment regions, infrastructure and vehicle capabilities co-evolve, each enabling new capabilities in the other.

2.9 Exercises

1. For the PaaS example, compute break-even at 8 % AV penetration.
2. Design a QR-code signage scheme that is machine-readable without being visually disruptive.
3. Estimate fuel savings for a green-wave that reduces stops from 3 to 1 on a 12-signal arterial.
4. Propose a governance model for municipal PaaS infrastructure: who operates, who regulates, who pays?
5. For the green-wave example, compute the CO₂ benefit assuming fuel produces 2.3 kgCO₂/L.
6. Design a C-V2X BSM schedule that fits inside a 1 MHz channel budget.
7. Identify three infrastructure investments that would most accelerate AV deployment in a specific city.
8. Compare PaaS-based and vehicle-only perception in an urban intersection; identify which is better for what.
9. Propose a staged municipal investment plan from today to 50 % AV penetration.
10. Discuss the equity implications of AV-favored infrastructure: who benefits, who doesn't?

Part II

Cybersecurity and Legal Frameworks

Chapter 3

Cybersecurity for Autonomous Vehicles

3.1 Introduction

An autonomous vehicle is a safety-critical computing platform with more attack surface than any consumer device in history. It has dozens of interconnected compute nodes, hundreds of networked sensors, cellular and short-range radios, OTA update pipelines, and, increasingly, deep integration with cloud services. It also has physical-world consequences: a successful attack on a connected AV can cause death, not just data loss. The safety chapter (Chapter 32) developed the framework for safety arguments; this chapter develops the cybersecurity arguments that safety arguments ultimately depend on. A system whose guarantees assume a trusted operating environment, and which operates in an untrusted network, has no guarantees at all.

Three observations motivate the chapter’s structure.

First, AV cybersecurity requires *defense in depth*. No single control layer—not firmware signing, not network isolation, not adversarial-robustness training—is sufficient. Attackers will find the weakest link; a single strong layer does not protect a system whose other layers are weak. Production security engineering builds many layers so that compromising any one still leaves the system defensible.

Second, AV attacks span a wider range than traditional IT attacks. Physical attacks on sensors (laser dazzlers, radar jammers, GNSS spoofers), adversarial machine-learning inputs, supply-chain attacks on electronic control units, and classic remote code execution are all active threats. Defense must cover all of them.

Third, AV cybersecurity is subject to increasing regulation. ISO 21434 and UN Regulation No. 155 require cybersecurity management systems, threat analysis, and risk assessment as part of type approval. The regulatory structure is expected to tighten; engineering for compliance is increasingly expensive.

The chapter is organized as follows: a threat model framework, attack surfaces across the AV stack, specific attack categories (wireless, physical-sensor, adversarial-ML, supply-chain, OTA), defense-

in-depth architecture, specific controls (secure boot, cryptographic protocols, intrusion detection, adversarial-robustness training), incident response and OTA recovery, red-team practices, and regulatory compliance. Three architecture diagrams visualize the threat model, attack surface, and defense-in-depth structure; worked examples cover the specific arithmetic of attack and defense throughout.

3.2 The AV Threat Model

A threat model is the starting point for any cybersecurity analysis. It specifies what assets are being protected, who might attack them, and what capabilities those attackers have. Without a clear threat model, security work chases symptoms rather than causes.

3.2.1 Assets

The principal assets in an AV deployment:

Physical Safety

The ultimate asset. An attacker who can influence the vehicle's behavior can cause injury or death; this is the unique and defining characteristic of AV cybersecurity.

Operational Integrity

The vehicle's ability to provide its intended service—to transport passengers safely, to maintain its schedule, to operate within its ODD. Attacks that disrupt operation without causing physical harm are still harmful.

Data Confidentiality

Passenger location data, in-cabin audio/video, biometric identifiers, fleet operational data. Exposure of these has privacy, regulatory, and commercial consequences.

Intellectual Property

Trained models, map data, proprietary algorithms. Exfiltration affects commercial position but not safety directly.

3.2.2 Adversary Profiles

Different adversaries have different capabilities and motivations. A threat model explicitly accounts for each profile.

Criminal Individual

Limited resources; motivated by theft, joy-riding, or targeted harassment. Typically attacks from outside: unauthorized entry, parking-lot disruption. Important but comparatively easy to mitigate.

Organized Criminal Group

More resources; motivated by financial gain through theft, ransomware, or fraud. May invest in specialized tooling. A typical scenario: ransomware against a fleet operator.

Hacktivist or Insider

Privileged access either granted (employee) or earned (long-term access). Motivated by ideology, grievance, or exposure. Insider threats are among the hardest to detect because the actions look legitimate.

Nation-State

Extensive resources; strategic motivations ranging from espionage to preparing for conflict. Production AV programs operate under the assumption that nation-states will attempt to compromise them, and design accordingly.

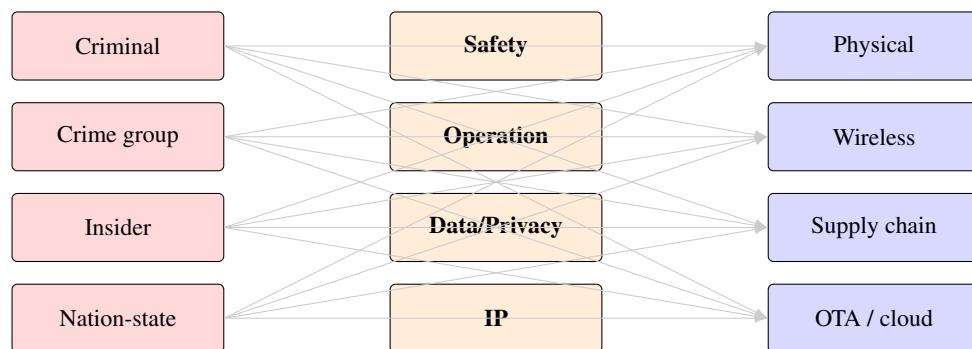


Figure 3.1: The AV threat model: adversaries (left, red) leverage attack surfaces (right, blue) to compromise assets (center, orange). Each adversary has different preferences among surfaces (nation-states use supply chain; criminals use physical; organized crime uses wireless and OTA); each surface threatens different assets. A complete threat analysis enumerates the full grid and prioritizes countermeasures based on likelihood and impact.

3.3 Attack Surfaces

This section enumerates the specific places an attacker can target. The goal is exhaustive coverage: any surface not listed is a gap in the security analysis.

Figure 3.2 maps the attack-surface tree. The vehicle is the root; each branch is an entry vector that must be analysed independently, with cumulative trust assumptions made explicit at each level.

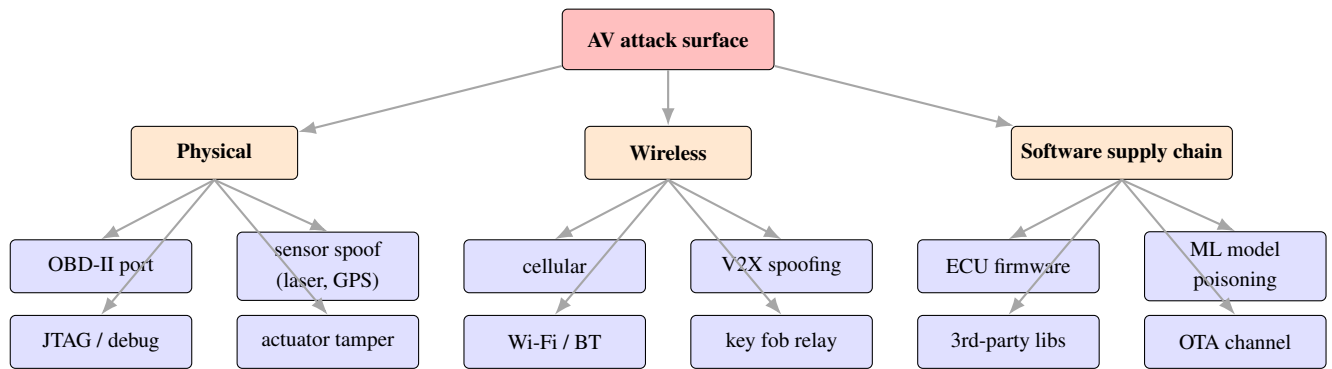


Figure 3.2: Attack-surface tree. The vehicle (root) decomposes into three top-level categories. Physical surfaces require local access (OBD-II port, debug headers, sensor field of view, actuator tampering). Wireless surfaces are remote-exploitable (cellular data channel, Wi-Fi/Bluetooth, V2X message spoofing, RF key-fob relay attacks). Software-supply-chain surfaces involve trust delegation (ECU firmware build chain, third-party libraries, ML model training data poisoning, OTA update channel). Each leaf is the topic of a separate threat-modelling exercise. Defense in depth means assuming any single leaf may be compromised and verifying that the system still degrades gracefully.

3.3.1 Physical Access

Physical access to the vehicle enables many attacks that wireless access does not. This subsection covers the three main physical vectors and the trust assumptions that fail when physical access is available.

OBD-II Port

The legally-mandated diagnostic port provides direct access to the CAN bus. An attacker with physical access can inject CAN messages, potentially commanding acceleration, braking, or steering in vehicles without adequate CAN-level authentication. Historical demonstrations (Miller & Valasek, 2015) showed remote Jeep takeover via a chain of compromises rooted in OBD-II-accessible systems.

Diagnostic and JTAG Ports

Engineering-only ports on individual ECUs. If not properly disabled in production, they permit firmware extraction, reverse engineering, and persistent backdoor installation.

Sensor Targeting

An attacker positioned near the vehicle can target specific sensors: a laser dazzler against cameras, a radar jammer against the forward radar, or a GNSS spoofer broadcasting fake satellite signals.

3.3.2 Wireless Attack Surfaces

Wireless surfaces are attractive to remote attackers because they scale: a single wireless exploit can compromise many vehicles simultaneously.

Cellular Data Channel

The cellular modem is the primary wireless surface. Historically, cellular attacks chain through: modem firmware bug → modem compromise → elevation through a trusted channel to application processor. Defense requires both modem hardening and strict isolation between modem and safety-critical domains.

Wi-Fi and Bluetooth

In-cabin connectivity offers another remote path. Infotainment is the usual entry; a compromised infotainment must not be able to reach safety-critical domains.

V2X Broadcasts

V2X (Chapter 1) enables vehicle-to-vehicle and vehicle-to-infrastructure communication. An attacker broadcasting malicious V2X messages (fake brake events, phantom pedestrians) can manipulate receiving vehicles. Signed messages with revocation for compromised certificates are the defense.

3.3.3 Supply Chain

ECUs, sensors, and semiconductor components are often sourced from third parties. A compromise at any supplier becomes a compromise in the delivered vehicle. Attacks can range from deliberate backdoors in firmware to accidental vulnerabilities in widely-used components.

3.3.4 OTA Update Pipeline

Over-the-air updates are essential for security (patching vulnerabilities in deployed vehicles) but are themselves a rich attack surface. An attacker who can push a malicious update effectively owns every vehicle that accepts it. Defenses include signed updates, multi-party signing, phased rollouts, and cryptographic verification of the update's source.

3.4 Specific Attack Categories

With the threat model and attack surfaces framed, we now develop specific classes of attack in detail. Each subsection covers the attack mechanism, documented instances, and standard defenses.

3.4.1 CAN Bus Attacks

The Controller Area Network has historically been a security afterthought: broadcast-based, no authentication, trivial to inject messages. Securing CAN is an active standards area.

Message Injection

With CAN access, an attacker can inject messages masquerading as legitimate ECUs. This is the mechanism behind most demonstrated vehicle takeovers.

Defenses

SecOC (Secure Onboard Communication) adds message authentication codes to CAN traffic; Ethernet-based in-vehicle networking (Automotive Ethernet) replaces CAN with authenticated IP traffic for newer vehicles. Network segmentation keeps safety-critical traffic isolated from infotainment.

3.4.2 Sensor Attacks

Physical attacks on sensors can cause perception failures that downstream software cannot distinguish from real events.

GNSS Spoofing

A software-defined radio broadcasting fake GNSS signals can overwrite a receiver's position estimate. Defenses: multi-frequency receivers (harder to spoof across bands), cross-checks with IMU-integrated motion, authentication via Galileo OSNMA, and anomaly detection on position discontinuities.

Example 3.1 (GNSS spoofing detection via IMU cross-check). Attacker spoofs GNSS, gradually shifting reported position by 1 ms^{-1} lateral. IMU reports no lateral acceleration. Kalman filter residual between GNSS-predicted and IMU-predicted position grows linearly; at $\sigma > 3\sigma$ threshold (typically after 2–5 seconds), spoofing is flagged. Production receivers implement this cross-check directly in firmware; the attack is detectable within seconds of onset.

LiDAR Spoofing

A synchronized laser can inject false returns into a LiDAR's receiver. Demonstrated attacks (Cao et al., 2019) create false objects or remove real ones. Defenses: encrypted modulation patterns on outgoing laser pulses, cross-modal verification against radar and camera.

Camera Attacks

Laser dazzlers can saturate camera sensors; adversarial patches on physical objects (stickers on stop signs) can fool classifiers. Defenses include optical filters, HDR sensors that recover from saturation, and adversarial-robustness training.

3.4.3 Adversarial Machine Learning

A defining class of attacks against learned perception: inputs crafted to cause the network to produce wrong outputs while appearing innocuous to humans.

Evasion Attacks

At inference time, adversarial perturbations cause misclassification. An adversarial sticker on a stop sign: visibly a stop sign; a classifier says “speed limit 45.” Physical adversarial examples (robust to viewing angle, lighting, weathering) are demonstrated in the literature.

Poisoning Attacks

At training time, malicious examples in the training data teach the network to misbehave under specific triggers. A network trained on poisoned data might work normally unless it sees a specific pattern (backdoor trigger), at which point it misbehaves.

Defenses

Adversarial training (include adversarial examples in training), randomized smoothing (inject noise at inference to disrupt adversarial perturbations), multi-modal verification (cross-check camera with LiDAR), and input-validation filters are standard defenses. No single defense is reliable; layers help.

3.5 Defense-in-Depth Architecture

Modern AV security architecture assumes compromise is possible at every layer and builds redundant controls. Figure 3.3 shows the canonical layers.

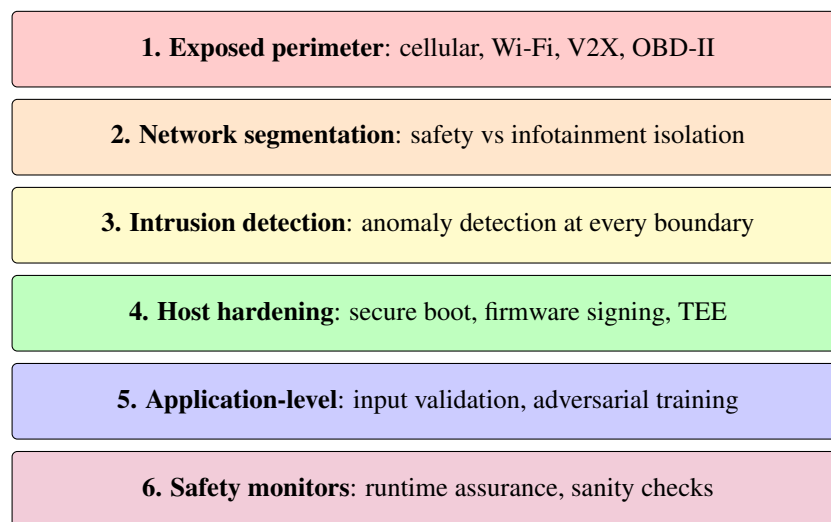


Figure 3.3: Defense-in-depth layers for AV cybersecurity. Each layer provides independent protection: the outer perimeter (untrusted networks and interfaces); network segmentation (prevent lateral movement); intrusion detection (catch the attack in progress); host hardening (secure boot, trusted execution environments); application-level hardening (adversarial training, input validation); and safety monitors that catch unsafe behavior regardless of its cause. An attacker who compromises one layer still faces the others.

3.6 Specific Controls

This section covers the specific technical controls that production AV programs implement. Each subsection covers one control, its implementation, and its limits.

3.6.1 Secure Boot

Secure boot verifies every stage of the boot process cryptographically: the bootloader verifies the OS; the OS verifies each application. A compromised image cannot boot if its signature fails verification.

Prevents persistent malware from surviving reboot.

3.6.2 Trusted Execution Environments

A TEE (e.g., ARM TrustZone) runs security-critical code in an isolated environment that the main OS cannot directly access. Key material, cryptographic operations, and security-sensitive decisions run in the TEE.

3.6.3 Cryptographic Protocols

Standard protocols for securing automotive communications:

SecOC

Secure Onboard Communication: authentication codes on CAN messages, preventing injection by unauthorized participants.

TLS for External Connections

Every external connection (V2N, OTA, diagnostics) uses TLS 1.3 with certificate pinning. Compromised CAs cannot masquerade as the legitimate server.

V2X Security

IEEE 1609.2 defines a PKI for V2X with pseudonym certificates that rotate frequently to prevent tracking. Revocation lists distribute information about compromised certificates.

Example 3.2 (Signed V2X message). A V2V emergency-brake broadcast. Sender signs with current pseudonym certificate (512-byte certificate, 64-byte signature using ECDSA P-256). Receiver verifies: (a) certificate is on a trusted root's chain, (b) certificate is not in the revocation list, (c) signature is valid. Verification time: <1 ms. Without signing, any radio-capable attacker can broadcast fake emergency-brake events.

3.6.4 Intrusion Detection

Network- and host-based IDS monitors for anomalous patterns: unusual CAN message rates, unexpected destinations for network traffic, out-of-spec sensor readings. Production systems use a combination of signature-based (known attack patterns) and anomaly-based (deviations from normal) detection.

Example 3.3 (CAN-bus anomaly detection). Legitimate brake-command messages appear at 100 Hz rate. Attacker injects an extra command: rate temporarily rises to 200 Hz. Rate-based detector flags the anomaly; logs the detection for analysis. Rate-based detection catches naive injection attacks; sophisticated attackers throttle their injections to match legitimate rates, requiring more nuanced content-based detection.

3.6.5 Adversarial-Robustness Training

Perception networks trained with adversarial examples at training time become harder to fool. Typical approach: apply adversarial perturbations to a fraction of training batches; network learns to classify correctly despite perturbation. Robust models lose some clean accuracy but gain adversarial accuracy.

3.7 Incident Response and OTA Recovery

Despite preventive controls, some attacks will succeed. Production programs invest heavily in incident response.

3.7.1 Incident Detection

Continuous monitoring of vehicles and fleet aggregates for signs of compromise: unusual patterns, anomaly-detection alerts, user reports, third-party researcher disclosures.

3.7.2 Containment

Once an incident is detected, containment limits blast radius: disconnect affected vehicles from the network, disable compromised components, revoke compromised certificates, issue targeted OTA patches.

3.7.3 OTA Recovery

OTA updates are the main tool for recovery at scale. An effective OTA must:

- Be signed by an offline signing key (not reachable by attackers).
- Be delivered over authenticated channels.
- Be verified on-vehicle before installation.
- Install only during safe states (parked, ignition-off).
- Roll back automatically on post-install boot failure.
- Respect staged rollout so bugs are caught on early adopters before fleet-wide damage.

Example 3.4 (Emergency OTA response timeline). Day 0: vulnerability disclosed (third-party researcher). Day 1: triage, patch development begins. Day 5: patch tested, signed, ready for deployment. Day 6: stage-1 rollout to 1 % of fleet. Day 9: monitoring shows no regressions; stage-2 to 10 %. Day 12: stage-3 to 50 %. Day 15: full rollout to 100 %. Compared to recall timelines for physical defects (months), OTA is dramatically faster. The catch: each update is itself a potential attack surface, so signing and rollback discipline are critical.

3.8 Red Team and Bug Bounty Programs

Active offensive testing is now standard practice. A red team mimics adversary behavior against the production system; findings fuel improvements before real adversaries exploit the same gaps. Bug bounty programs invite third-party researchers under a legal-safe-harbor agreement to find and report vulnerabilities.

3.8.1 Red Team Scope

Production red teams attempt: remote exploitation, physical attacks, adversarial-ML payloads, supply-chain compromises, insider scenarios. Engagement depth ranges from quarterly assessments to continuous hunting.

3.8.2 Responsible Disclosure

Disclosure timelines (typically 90 days) give the vendor time to patch before public disclosure. Researchers who follow responsible disclosure and programs that reward them build a healthier ecosystem than adversarial vendor-researcher relationships.

3.9 Regulation and Compliance

Automotive cybersecurity is now formally regulated.

3.9.1 ISO 21434

International standard for road vehicle cybersecurity engineering. Requires a cybersecurity management system (CSMS), threat and risk assessment, security testing, and incident response planning. Compliance is an evidence-based process involving substantial documentation.

3.9.2 UN Regulation No. 155

UNECE regulation requiring type approval to demonstrate cybersecurity engineering practice. Since July 2024, new vehicle types in UNECE markets must demonstrate R155 compliance.

3.9.3 NHTSA Cybersecurity Guidance

US guidance (non-binding but influential) for automotive cybersecurity best practices. Aligned with ISO 21434 in most substantive areas.

3.10 Worked Examples

Example 3.5 (End-to-end attack chain and defense). Attacker goal: remote code execution on safety-critical ECU. Attack chain:

- Step 1: exploit a known Bluetooth bug in infotainment firmware (pre-patch window).

- Step 2: escalate from infotainment OS to in-vehicle network.
- Step 3: send malicious CAN messages to safety-critical ECU.

Defense-in-depth stops each step:

- Step 1: disabled Bluetooth-in-range when vehicle is moving; infotainment runs in hardened OS with OTA-patched firmware.
- Step 2: gateway between infotainment and in-vehicle network enforces strict whitelisted traffic; lateral movement blocked.
- Step 3: safety-critical ECUs use SecOC; unauthenticated CAN messages rejected; anomaly detector flags the volume spike.

Each layer independently defeats the attack; total system is substantially more secure than any single layer.

Example 3.6 (GNSS spoofing in a production fleet). Adversary at a specific intersection broadcasts fake GNSS signals attempting to divert nearby vehicles. Mitigations observed in a production fleet:

- 40 % of vehicles using L1+L5 dual-frequency reject the attack outright (attacker only spoofs L1).
- 60 % of vehicles detect position/IMU inconsistency within 2 seconds; GNSS confidence is downgraded and map-matching takes over.
- 0 % of vehicles accept the spoofed position as ground truth.

Attack is effectively neutralized by layered defense; the fleet continues to operate correctly. Post-detection, GNSS-spoof alert flows to the fleet operator who coordinates with authorities.

Example 3.7 (Adversarial-ML robustness gain). Clean test-set accuracy: 0.96. Accuracy under $8/255 \ell_\infty$ adversarial attack (untrained baseline): 0.05. After adversarial training (same attack budget): 0.82 clean, 0.43 adversarial. Multi-modal fusion with LiDAR: effective adversarial accuracy rises to 0.81 (LiDAR sees true object; attacker's sticker does not affect it). Combined defense: adversarial training + multi-modal fusion + runtime safety monitor catches remaining edge cases. No single defense gives production-acceptable robustness; combinations do.

3.11 Industrial Deployment

The 2015 demonstration by Charlie Miller and Chris Valasek of remote control of a Jeep Cherokee, published at Black Hat USA, prompted a broad industry response and a 1.4 million vehicle recall by Fiat Chrysler. Since then, UNECE Regulation 155 on cybersecurity has become mandatory for new vehicle type approvals in Europe, and ISO/SAE 21434 (Road vehicles—Cybersecurity engineering) provides the corresponding engineering standard. Auto-ISAC (Automotive Information Sharing and Analysis Center), founded in 2015, coordinates cybersecurity threat information among automotive OEMs and suppliers. Argus Cyber Security (acquired by Continental) and Upstream Security supply

automotive cybersecurity products. Bug-bounty programs run by Tesla, GM, and others provide structured channels for vulnerability disclosure; Tesla in particular has run Pwn2Own automotive contests publicly.

3.12 Chapter Summary

Cybersecurity is not optional for production AVs. A clear threat model (assets, adversaries, capabilities) drives defense planning. Attack surfaces span physical, wireless, supply-chain, and OTA; specific attacks include CAN injection, sensor spoofing, adversarial ML, and firmware compromise. Defense-in-depth is the architectural principle: exposed perimeter, network segmentation, intrusion detection, host hardening, application hardening, and safety monitors, all independent layers. Specific controls (secure boot, TEE, cryptographic protocols, IDS, adversarial training) implement each layer. Incident response—detection, containment, and OTA recovery—handles residual risk. Red-teaming and bug bounties keep offensive practice current. ISO 21434 and UN R155 make cybersecurity engineering a regulatory requirement. The safety arguments of Chapter 32 depend on the cybersecurity engineering described here; neither is complete without the other.

3.13 Exercises

1. For the threat model diagram (3.1), propose two additional adversary profiles not listed. Describe their capabilities and motivations.
2. Analyze the 2015 Jeep hack (Miller & Valasek) in terms of the layers of defense-in-depth (3.3). Which layers were absent, and how would they have stopped the attack?
3. Describe three specific defenses against GNSS spoofing. Rank them in order of effectiveness against a well-resourced attacker.
4. For the adversarial-ML example (3.7), compute the overall system accuracy given layered defenses. State your assumptions.
5. A V2X attacker broadcasts a false pedestrian at a busy intersection. What specific defenses should the receiving AV apply? At what latency?
6. Describe a red-team exercise targeting an AV's OTA update pipeline. What should the exercise attempt, and what would success look like from the defender's perspective?
7. Compare ISO 21434 to IEC 62443 (industrial control). What are the key differences, and why do they exist?
8. For the CAN-anomaly example (3.3), design a content-based anomaly detector that would catch a rate-matched injection attack.
9. Describe the principal-agent problem in AV security: the tension between a manufacturer's interest

- in under-reporting vulnerabilities and the public interest in disclosure. Propose a regulatory response.
10. An attack-chain defense (like 3.5) is effective only if each layer is genuinely independent of the others. Describe two ways apparent-layer-independence can be illusory, and how to verify genuine independence.

Chapter 4

Legal Aspects of Autonomous Vehicles

4.1 Introduction

Engineering an autonomous vehicle is difficult; *legally operating one* is in many ways harder. An AV engineer can make the car drive itself; only a sequence of regulatory approvals, type certifications, insurance policies, data-retention arrangements, and jurisdictional accommodations lets it drive itself on a public road. This chapter develops the legal landscape a production AV program must navigate. It is deliberately framework-heavy rather than jurisdiction-specific: the particulars vary widely and change frequently; the framework is portable.

Three observations motivate the chapter’s structure.

First, AV law is *evolving rapidly*. A regulation enacted in 2024 may be superseded by 2026. Compliance is a moving target, and programs committed to long-term operation must invest continuously in legal monitoring. Engineers who assume the law sits still make mistakes that the law itself catches up on.

Second, AV law *cuts across several legal domains*: product-liability law, privacy law, data-retention regulation, road-traffic law, insurance, employment (driverless operation and labor law), and international trade. A competent AV legal team is necessarily multi-disciplinary; an engineer cannot pick a single “AV law” treatise and be done.

Third, the legal framework *shapes engineering decisions*. Retention requirements dictate logging architecture. Accountability frameworks dictate explainability features. Jurisdictional patchwork dictates per-region deployment variants. Awareness of legal constraints from the start leads to architectures that can comply; ignorance leads to expensive retrofits.

The chapter is organized as follows: first the regulatory landscape (who regulates AVs, at what level, under what authority); then liability frameworks (who pays when things go wrong); product liability; insurance; privacy and data law; international considerations; labor and employment implications; and production legal-engineering practice. Three architecture diagrams visualize the regulatory structure,

the liability ladder across SAE levels, and the compliance lifecycle. Worked examples apply the framework to specific scenarios.

4.2 The Regulatory Landscape

AV regulation operates at multiple levels simultaneously. Understanding the structure is the first step to compliance.

4.2.1 International Bodies

Global harmonization is incomplete but important. Several bodies issue standards and recommendations that individual jurisdictions adapt.

UNECE WP.29

The World Forum for Harmonization of Vehicle Regulations (WP.29) under the UN Economic Commission for Europe. Issues binding regulations adopted by EU, Japan, Korea, and many other countries; influences (but does not bind) the US. Recent AV- relevant regulations: UN R155 (cybersecurity), UN R156 (software updates), UN R157 (Automated Lane Keeping System, L3 conditions).

ISO and IEC Standards

International standards bodies. ISO 26262 (functional safety), ISO 21448 (SOTIF), ISO 21434 (cybersecurity), ISO 22737 (low-speed automated driving). Not themselves law, but referenced by regulations and commonly required by type-approval processes.

SAE International

The SAE J3016 levels-of-driving-automation document is the reference for every AV regulatory document. “Level 3,” “Level 4” have specific meanings in J3016 that the rest of the legal ecosystem depends on.

4.2.2 National and Regional Frameworks

Each country has its own regulatory approach. A production AV program deploying across multiple regions must navigate all relevant frameworks.

United States

Federal: NHTSA (National Highway Traffic Safety Administration) issues guidance and enforces federal motor vehicle safety standards (FMVSS). Federal self-certification model: the manufacturer attests compliance; NHTSA investigates defects post- market. State: AV operation regulated state-by-state. California, Nevada, Arizona, Texas, and Florida have distinct AV frameworks. California’s annual disengagement reports are the most public- facing state-level requirement.

European Union

EU regulation is harmonized through type approval: a vehicle type approved in one member state can be sold in all. Type approval requires compliance with UNECE regulations. L3 vehicles can be approved under R157. L4 type approval frameworks are still evolving.

United Kingdom

Automated and Electric Vehicles Act (2018) addressed initial liability questions. Automated Vehicles Act 2024 established a comprehensive framework: manufacturer liability for L4+ operation, specific authorization and monitoring requirements.

Germany

Autonomous Vehicles Act (2021) allows L4 operation in defined operating areas with “technical supervisors” remotely monitoring. Pioneering L4 framework in Europe.

Japan

Road Traffic Act amendments permit L3 operation in specific conditions since 2023. Established vehicle-service-provider licensing for driverless taxi services.

China

MIIT and MPS issue AV regulations; specific rules for demonstration zones (Beijing, Shanghai, Guangzhou). Recent guidelines permit L3+ under strict conditions.

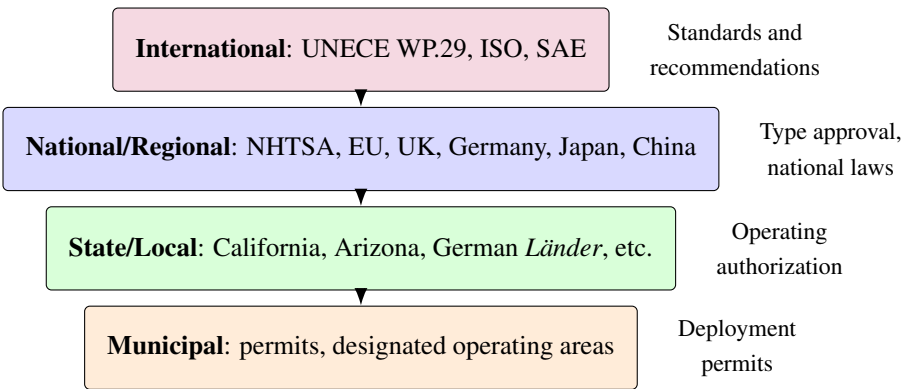


Figure 4.1: The AV regulatory hierarchy. International bodies issue standards that national/regional regulators adopt as binding law. State/provincial governments add operating-level rules within the national framework. Municipalities may require permits for specific deployments. A production AV program works at all four levels simultaneously.

4.3 Liability Frameworks

When something goes wrong with an AV—a collision, a wrong maneuver, a data breach—who is legally responsible? This section develops the frameworks emerging to answer that question.

4.3.1 The SAE Level Mapping

Liability follows, roughly, who was in control. The SAE levels (Chapter 3) define who is expected to be in control under what conditions; liability mostly tracks that.

Levels 0–2: Driver Responsible

At L0–L2, the human driver is in control (even with advanced driver assistance active). The driver is responsible for the vehicle’s behavior. Manufacturer may face product-liability claims if the assistance system malfunctioned, but the primary responsibility is with the driver.

Level 3: Conditional Transfer

At L3, the system drives within the ODD; the driver takes over upon handover request. Responsibility: shared. When the system was engaged and in its ODD, the manufacturer is responsible for its decisions; when the driver should have taken over but did not, the driver is responsible. The transition moment is legally fraught.

Levels 4–5: Manufacturer Responsible

At L4–L5, the system operates without a driver. The manufacturer (or the deployment operator) is responsible. No human is available to blame for in-vehicle decisions; the legal locus has shifted fully.

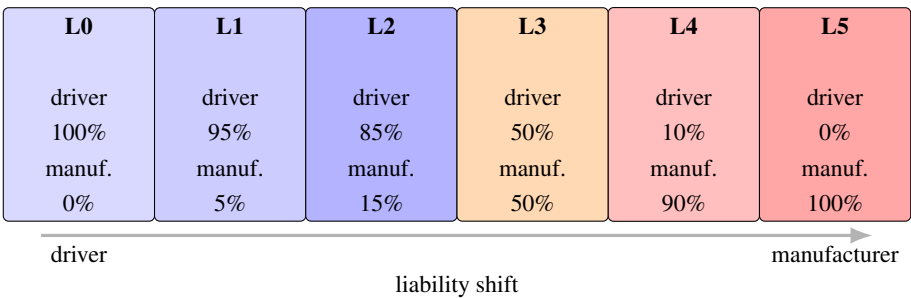


Figure 4.2: Liability shifts across SAE levels. At L0–L2, the driver bears primary responsibility; at L3, responsibility is shared during handover; at L4–L5, the manufacturer (or deployment operator) bears responsibility because no human driver is available. Percentages are illustrative of how courts and regulators have begun allocating; specifics vary by jurisdiction and by the details of each incident.

4.3.2 Product Liability

Even when a human driver is primary, manufacturers face product- liability exposure if a component (assistance system, sensor, software) malfunctions and contributes to a collision.

Design Defect

The product’s design is inadequate. In AV context: the detector algorithm misses pedestrians at a rate higher than reasonable; the ODD does not account for a known hazard. Design defects are

particularly dangerous for manufacturers because they apply to all units, not just the specific vehicle involved.

Manufacturing Defect

A specific unit differs from the designed-and-approved one (malfunctioning sensor, miswired ECU). Usually covered by warranty + product recall.

Warning Defect

The manufacturer failed to adequately warn users of a known risk. In AV: unclear communication of ODD limits, inadequate handover alerting, marketing claims exceeding technical capability. The Tesla “Autopilot”-named system has generated substantial warning-defect litigation because the name implies greater capability than the system provides.

4.3.3 Comparative Fault

Most jurisdictions allocate fault comparatively when multiple parties are at fault. An accident involving an AV at fault 60 % and a negligent other driver at 40 % results in each being liable for its share. Production AV programs plan for comparative-fault outcomes.

4.4 Data Retention and Legal Discovery

AVs generate torrents of data; legal systems are grappling with how to use it.

4.4.1 Mandatory Retention

Many jurisdictions require AVs to log specific data around incidents: sensor inputs, perception output, control commands, in a buffer that survives the event. EU R157 specifies Data Storage System for Automated Driving (DSSAD) requirements; similar rules are emerging in US states and elsewhere.

Retention Periods

Typical retention: 15 seconds before through 30 seconds after any automated system engagement transition or incident. Some jurisdictions require longer retention (6 months for post-incident data). Data volumes are substantial: multi-sensor recording for 45 seconds can be tens of gigabytes.

Integrity Requirements

Retained data must be tamper-evident. Cryptographic signing, write-once storage, and audit-log integrity are the technical responses. The goal: if subpoenaed, the data is provably genuine and complete.

4.4.2 Legal Discovery

In litigation, AV data is discoverable—both sides can demand production of logs, model weights, test results, and incident-specific data. Litigation discovery in a single AV case has routinely involved terabytes of data and substantial expert review. Well-designed AV programs build discovery-readiness into their logging architecture from the start.

Example 4.1 (Discovery in an AV collision case). A non-fatal collision involves an AV. Plaintiff subpoenas:

- All sensor data for 60 seconds around the event.
- All perception output, tracking logs, and planning decisions.
- Model weights and training documentation for the detector.
- Historical intervention rates for the same route.
- Prior incident reports involving the same ODD.

Volume: ~ 50GB per vehicle-incident; ~ 2 TB of historical aggregate data. Cost of production and expert review: hundreds of thousands to millions of dollars. Programs with well-organized, indexed logging fare far better than those with ad-hoc archives.

4.5 Privacy Law

AVs collect extensive data about people and places. Privacy law governs what may be collected, retained, and used.

4.5.1 GDPR (EU)

The General Data Protection Regulation applies to AV data collected in the EU. Key principles: lawful basis for processing, data minimization, purpose limitation, subject access rights (to see, correct, delete personal data). Non-compliance penalties: up to 4% of global revenue.

Pseudonymization and Redaction

On-device face and license-plate blurring (Chapter 5) addresses the core risk. Pseudonymization—replacing identifiers with random tokens that cannot be reversed without a separate key—is common for logged data.

Data Subject Rights

Individuals whose data appears in AV logs can request access, correction, or deletion. Programs must have operational processes to handle these requests; failure is a regulatory violation.

4.5.2 US State Laws

California Consumer Privacy Act (CCPA), Virginia Consumer Data Protection Act, and similar state laws impose GDPR-like requirements in the US absent federal legislation. Compliance is

state-by-state, with significant variation.

4.5.3 Biometric Data

Driver-monitoring systems collect biometric data (gaze, facial features). Illinois' Biometric Information Privacy Act (BIPA) has been the source of substantial class-action litigation. Biometric data requires explicit consent and specific handling.

4.6 Insurance

Insurance frameworks for AVs are still evolving. The core question: who pays, and under what theory?

4.6.1 Traditional Auto Insurance

Conventional personal auto insurance covers the driver. An AV operating driverless (L4) blurs the model: if the manufacturer is responsible, the manufacturer's product liability insurance is the relevant coverage, not the registered owner's.

4.6.2 Manufacturer Product Liability Insurance

AV manufacturers carry substantial product-liability coverage. Premiums reflect both fleet size and demonstrated safety. Early L4 deployments carry very high premiums; successful deployments with good safety records earn lower rates over time.

4.6.3 Commercial Fleet Policies

Robotaxi operators negotiate comprehensive policies covering vehicles, operations, and incidents. Typical coverage: \$1M–\$10M per occurrence; higher for specific fleets.

4.6.4 No-Fault vs Fault-Based Systems

Some jurisdictions operate no-fault insurance systems (Michigan, Australia, New Zealand): injured parties claim from their own insurer regardless of who was at fault. These simplify AV cases in some ways (less need to establish fault) and complicate in others (unclear how manufacturer liability fits).

Example 4.2 (Insurance cost of a driverless fleet). L4 robotaxi fleet: 500 vehicles, 20 million miles/year, no safety driver. Early-deployment insurance premium: 3¢/mile (vs 0.5¢/mile for a comparable human-driven fleet). Annual cost: \$600k per vehicle. As safety record accumulates, premium drops; mature programs target 1¢/mile. The insurance-premium trajectory is a proxy for operator safety as perceived by actuarial analysis—an externally-measured safety metric that complements internal engineering metrics.

4.7 International Considerations

AV programs operating across borders face compounding complexity.

4.7.1 Type Approval vs Self-Certification

Type Approval (EU, UNECE markets)

A vehicle type must be approved before sale; approval requires substantive testing and documentation. AV type approval is still evolving; R157 provides a framework for L3, R171 is being developed for L4.

Self-Certification (US)

Manufacturers attest compliance with FMVSS; NHTSA investigates post-market defects. Faster to market; greater long-term liability for manufacturers who certify vehicles that later prove defective.

4.7.2 Cross-Border Operation

A vehicle type approved in one jurisdiction may not be legal in another. AV software often varies by region; data-collection practices must respect regional law.

4.7.3 Export Controls

AV technology—especially the machine-learning models—may be subject to export controls (US EAR, EU dual-use regulation). Open-source models can complicate this; careful legal review is required before transferring trained models across borders.

4.8 Labor and Employment Implications

Autonomous vehicles reshape the labor market. Legal frameworks are beginning to address this.

4.8.1 Driver Displacement

Trucking, taxi, and delivery drivers face displacement risk. Various jurisdictions are considering (or have implemented) transition funds, retraining programs, or restrictions on the pace of deployment to mitigate disruption.

4.8.2 Remote Operators

L4 deployments typically include remote-supervision teams. Their legal status—whether they are drivers in the regulatory sense, what they are liable for, how their work hours are regulated—is actively contested.

4.8.3 Fleet Maintenance and Operations Labor

AV fleets still require cleaners, technicians, charging-station staff. Labor law applies normally to these workers but they may be scattered across jurisdictions, complicating employment.

4.9 Production Legal Engineering

A production AV program's legal engineering is a continuous process, not a one-time event.

4.9.1 Compliance Lifecycle

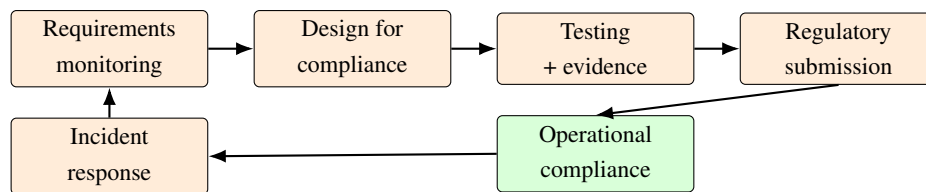


Figure 4.3: AV legal-compliance lifecycle. Regulatory monitoring identifies changing requirements; engineering designs for compliance from the start; testing and evidence collection support regulatory submission; operational compliance maintains ongoing adherence; incidents flow back into the requirements- monitoring stage as new learnings or new regulatory actions. The loop runs continuously for the program lifetime.

4.9.2 Legal-Engineering Artifacts

Compliance produces specific artifacts:

- **Type-approval dossier:** hundreds to thousands of pages of technical documentation.
- **Safety case:** structured argument for safety (Chapter 32).
- **Cybersecurity management system:** per ISO 21434.
- **Data-protection impact assessment:** privacy documentation per GDPR.
- **Incident-response documentation:** processes and records.
- **Regulatory correspondence log:** all communications with authorities.

4.9.3 Continuous Monitoring

Regulators expect continuous performance monitoring, not just pre-launch compliance. Periodic reporting, mandatory disclosure of incidents, post-market surveillance—all require operational infrastructure.

4.10 Worked Examples

Example 4.3 (Compliance plan for a new robotaxi deployment). Program: L4 robotaxi launch in a new US state. Required artifacts:

- NHTSA notification (federal).
- State AV permit application (varies by state).
- Insurance proof ($\geq \$5M$ typical).
- Local operating-permit filings.
- DSSAD-equivalent data-logging implementation.
- Privacy-impact assessment.
- Cybersecurity management system.
- Training-mile record: typically ≥ 1 million miles of pre-launch real-world testing documented.

Typical timeline: 9–18 months from decision to launch. Legal/compliance staffing: ~ 5 FTE for a single-city launch; scales with geographic footprint.

Example 4.4 (Data-retention cost calculation). Fleet: 1000 vehicles, each generating 100 hours of operation per month. Retention: full-sensor logs for 5 % of operation (interesting events, incidents, regulatory-required snapshots). Data rate: 1 GB/minute at full-sensor rate. Retained volume per vehicle per month: $100 \cdot 60 \cdot 0.05 \cdot 1 = 300$ GB. Fleet total: 300 TB/month. Storage at \$0.02/GB/month long-term: \$6k/month storage. Add indexing, query, and legal-discovery overhead: $\sim \$50k$ /month operational cost. Annual: \$600k. Comparable to legal-staff cost; substantially smaller than insurance. A line item, not a crippling cost.

Example 4.5 (Cross-border model-deployment review). An AV program trained in California wants to deploy in Germany (EU). Model review:

- Export control: does the model fall under US EAR? Check with trade-compliance. Typically not, for non-defense perception; but model-specific review required.
- GDPR: training data potentially included personal data. Re-deployment requires lawful basis documentation.
- Type approval: need EU type approval, separate from any US approval. Substantial new documentation.
- Localization: signs, traffic rules, pedestrian behavior differ. Retraining or fine-tuning is typically required, which is a technical project with its own timeline.

Cross-border deployment: 18–24 months typically. This legal/engineering overhead is why AV programs concentrate in one region before expanding.

4.11 Industrial Deployment

The legal landscape for AVs varies substantially by jurisdiction. In the United States, NHTSA exercises authority over vehicle safety standards (FMVSS) and issues Standing General Orders requiring AV companies to report crashes; individual states regulate vehicle operation and licensing, producing

a patchwork in which California (DMV), Arizona, Texas, and others have specific AV regulations. The European Union's General Safety Regulation and Type Approval Framework Regulation provide harmonised requirements across member states. Germany's law on autonomous driving, passed in 2021, was among the first to permit fully driverless vehicles in defined operating areas. China's regulations, issued by the Ministry of Public Security and the Ministry of Industry and Information Technology, permit conditional autonomous-driving testing and limited commercial deployment in designated zones. The United Kingdom's Automated and Electric Vehicles Act 2018 established initial liability provisions, and the Automated Vehicles Act 2024 expanded the framework.

4.12 Chapter Summary

AV law is evolving rapidly across international, national, regional, state, and municipal levels. Regulatory bodies (UNECE WP.29, ISO, SAE; NHTSA, EU, UK, Germany, Japan, China) each impose frameworks that compose. Liability frameworks track the SAE levels: driver primary at L0–L2, shared at L3, manufacturer at L4–L5. Product-liability law, data-retention requirements, privacy law (GDPR, CCPA, BIPA), insurance, export controls, and labor-law implications all contribute to an integrated legal-engineering workload. Production AV programs treat compliance as a continuous lifecycle—monitor, design, test, submit, operate, respond, repeat—and invest accordingly. The engineering decisions the earlier chapters describe are all made within this legal envelope; the envelope shapes those decisions, not the other way around.

4.13 Exercises

1. For the regulatory hierarchy (4.1), identify the specific bodies involved for an AV deployment in Munich, Germany.
2. Describe the liability allocation for a collision involving an L3 vehicle five seconds after the driver received a handover request but before they took control.
3. For the discovery example (4.1), propose a logging architecture that makes discovery responses efficient without exceeding retention limits.
4. Compare EU type approval with US self-certification: under which regime would a fast-moving startup prefer to operate? A safety-focused regulator?
5. For the insurance-cost example (4.2), compute the break-even intervention rate at which AV insurance cost equals human-driver cost.
6. Propose a GDPR-compliant design for an AV's driver- monitoring camera, including data minimization and subject rights.
7. Describe three specific legal risks of the "Autopilot" branding. What specific warnings would mitigate each?

8. For the cross-border example (4.5), identify two additional legal issues not mentioned.
9. Compare a no-fault vs fault-based insurance regime for AV deployment. Which would a manufacturer prefer and why?
10. Describe the compliance-lifecycle stages (4.3) for a recall-level software update. What artifacts are produced at each stage?

Part III

Ethics and Responsible AI

Chapter 5

Ethics, Society, and the Responsible Deployment of Autonomous Vehicles

5.1 Introduction

Everything in the preceding chapters has been about making autonomous vehicles *work*. This chapter is about what it means for them to *work well for society*—an inseparable question for any technology operating at scale on public roads. The engineering choices in every earlier chapter have ethical consequences: which classes of pedestrians a detector is tuned to find, whose neighborhoods a mapping program prioritizes, which risk thresholds a planner treats as acceptable. These are not side questions. They are embedded in the system and determined by the people who build it.

Three observations motivate the chapter’s structure.

First, autonomous driving is a *distributive* problem, not merely a technical one. The benefits of AVs (mobility for non-drivers, reduced collisions, lower operating cost) will be distributed across groups; so will the risks. Which groups benefit, which groups bear risk, and how decisions about these questions get made are ethical questions that technical work alone cannot settle.

Second, the most famous ethical framings of AVs—the Trolley Problem, the “who does the car choose to kill?” question—are mostly unhelpful guides to the actual decisions that matter. Real AV ethics is about perception coverage, bias, privacy, deployment equity, and accountability. The Trolley framing distracts.

Third, many of the ethical challenges have technical responses. Bias in detection can be measured and reduced. Privacy in map data can be engineered. Accountability in failures can be built into the logging architecture. Ethics is not a separate conversation from engineering; it is an aspect of engineering.

This chapter is organized around the real ethical questions that arise in production AV deployment. We develop, in turn: the distributive ethics of AV benefits and risks; bias in perception and behavior;

privacy in data collection and map-keeping; accountability and liability when things go wrong; cybersecurity; regulation and certification; public communication and transparency; and equity of deployment. The concluding section returns to the engineer’s role and responsibility. Every topic has a worked example illustrating the concrete decision points; one architecture diagram visualizes the ethical stakeholder map.

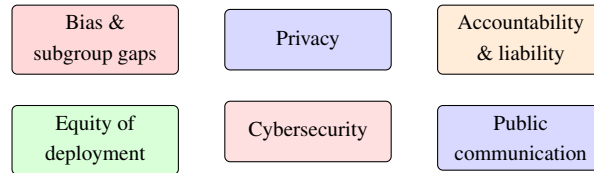


Figure 5.1: Chapter roadmap. Six interconnected threads. Bias and privacy concern engineering decisions; accountability and equity concern policy; cybersecurity and communication concern operational trust. Each of the six is treated as a distinct subproblem with its own quantitative example.

5.2 The Ethical Stakes

Before addressing specific questions, it is worth stating the scope of the ethical stakes for autonomous driving. This section establishes why the subject warrants the chapter’s length, and why the traditional “trolley problem” framing underestimates it.

5.2.1 Scale of Impact

Driving is the largest unsolved safety problem in modern life. Road traffic injuries kill approximately 1.3 million people globally each year and injure tens of millions more. Autonomous driving has the potential to reduce this number; it also has the potential, if deployed badly, to replace one failure mode (human error) with another (machine error) without a net gain. The scale of potential benefit, and potential harm, is enormous.

Example 5.1 (Order-of-magnitude stakes). Global road fatalities are approximately 1 300 000 per year. Suppose an AV intervention is contemplated.

Step 1: Lives saved at 10 % reduction.

$$\begin{aligned}\Delta L_{+} &= 0.10 \times 1\,300\,000 \\ &= 130\,000 \text{ lives/year saved.}\end{aligned}$$

Step 2: Lives lost at 5 % increase.

$$\begin{aligned}\Delta L_{-} &= 0.05 \times 1\,300\,000 \\ &= 65\,000 \text{ lives/year lost.}\end{aligned}$$

Step 3: Spread between the two scenarios.

$$\begin{aligned}\Delta L_+ - (-\Delta L_-) &= 130\,000 + 65\,000 \\ &= 195\,000 \text{ lives/year.}\end{aligned}$$

Interpretation. The 130 000-life figure is comparable to the annual global toll of malaria; the swing between the two scenarios is roughly 200 000 lives per year. Whether the AV industry lands on the positive or negative side of this swing depends directly on deployment timing, scope, and evidence thresholds. Engineering and policy choices have casualty consequences at this scale.

5.2.2 Distributive Questions

Who benefits? Who bears risk? These questions define the distributive ethics of any technology, and they are central to AV deployment. Early AV deployments have so far concentrated in wealthy cities with clear weather, high-quality maps, and cooperative regulators—benefits flow to the residents of those cities. Less-mapped regions, less-favorable weather zones, and non-standard roads will be served later, if at all. The distribution of benefits follows from these deployment choices.

5.3 Why the Trolley Problem is Mostly a Distraction

The “trolley problem” ethics framing—*who should the AV decide to kill?*—is the most publicly visible ethical question about autonomous vehicles, and it is also largely the wrong question. This section explains why, and what framings are more useful for real engineering decisions.

5.3.1 The Framing

The Trolley framing imagines a car that must choose, in a split second, between killing the driver or killing several pedestrians. It asks what rule the car should follow. This scenario attracts press because it is morally dramatic and connects to classical philosophy; MIT’s Moral Machine experiment collected millions of responses to variations on the theme.

5.3.2 Why It Misleads

Actual driving produces such scenarios essentially never. Real collisions are almost always the end-state of a chain of small errors and small decisions made over seconds or minutes beforehand. The rare case where a clean trolley-like decision is forced almost never arises; when it does, the difference between competing ethical rules produces minimal real-world difference in outcomes.

The engineering decisions that actually matter for safety are: whether the detector reliably finds pedestrians at long range; whether the planner maintains adequate margin around cyclists; how the system handles uncertainty and degraded sensors; whether the ODD is appropriately bounded. These are prosaic questions with answer costs in hundreds of millions of dollars, but they are the questions that determine real outcomes.

5.3.3 The Questions That Matter

Replace the Trolley framing with these:

- How often does the system miss a pedestrian at range r , under weather w , at time of day t ?
- How does detection recall vary with the pedestrian’s demographic (skin tone, clothing, body size, assistive-device use)?
- At what ODD boundary does the system fail ungracefully?
- What is the accountability structure when something goes wrong?

Each of these has concrete engineering answers and concrete ethical stakes.

5.4 Bias in Perception and Decision-Making

Learned systems reflect their training data. When training data systematically under-represents some classes of pedestrians, some types of vehicles, or some driving contexts, the resulting perception and decision systems are biased against those conditions. This section develops the specific ways bias can enter an AV stack and the engineering responses to it.

5.4.1 Perception Bias

Detection accuracy can vary substantially across demographic and contextual subgroups. This subsection develops two main subcategories of perception bias and the techniques for identifying and reducing them.

Pedestrian Detection Across Demographics

A pedestrian detector trained mostly on images of light-skinned adults in typical Western clothing will underperform on dark-skinned pedestrians, children, wheelchair users, and people in culturally-specific clothing. Published studies have documented detection-accuracy differentials of 5 to 15 percentage points across pedestrian subgroups —differentials that, on the road, translate to unequal safety.

Example 5.2 (Quantifying subgroup accuracy gap). A pedestrian detector achieves mean AP $\overline{\text{AP}} = 0.82$ on aggregate. Stratified evaluation:

Subgroup	AP
Light-clothing pedestrians	0.87
Dark-clothing pedestrians	0.76
Children (< 1.3 m tall)	0.71
Wheelchair users	0.65

Step 1: Subgroup gap.

$$\begin{aligned}\Delta AP &= AP_{\text{best}} - AP_{\text{worst}} \\ &= 0.87 - 0.65 \\ &= 0.22.\end{aligned}$$

Step 2: Worst-subgroup miss rate.

For wheelchair users at $AP = 0.65$, the miss rate is approximately $1 - 0.65 = 0.35$ at standard operating points—more than three times the miss rate for light-clothing pedestrians ($1 - 0.87 = 0.13$).

Interpretation. The aggregate metric of 0.82 hides a 22-point gap. A vehicle deployed solely on the aggregate metric would miss wheelchair users at roughly $2.7\times$ the rate of the average pedestrian. Mitigation: targeted data augmentation, subgroup- balanced sampling, subgroup-specific performance monitoring on deployment. Reporting subgroup metrics in public test reports is increasingly standard.

Training-Data Geographic Bias

A system trained primarily on California driving will be biased toward California road patterns: wide lanes, sunny weather, right-side driving, specific sign conventions. Deployment in environments with different patterns (narrow European lanes, monsoon conditions, left-side driving) performs worse. Engineering response: data collection in diverse geographies, explicit region-specific validation, and staged deployment that treats each region as a separate validation problem.

5.4.2 Behavioral Bias

Beyond detection, decision policies can be biased. This subsection covers two common forms: aggregate biases in reward optimization and specific biases in behavior-cloning from biased training sets.

Optimization-Driven Bias

An RL-trained or reward-optimized policy can acquire subtle biases. A policy rewarded for passenger-comfort might, in aggregate, yield less often to cyclists (because yielding costs comfort-reward). A policy rewarded for throughput might accept tighter margins with specific classes of road users. *Fairness* in reward structure, and *auditing* learned policies for disparate impact, is the response.

Behavior-Cloning Bias

A behavior-cloning policy reflects the biases of its expert demonstrations. If expert drivers systematically give more space to some classes of road users than others, the policy learns the same pattern. Mitigation: diverse expert populations, explicit fairness-aware loss terms, and behavioral audits on protected subgroups.

5.5 Privacy

AVs collect enormous amounts of data about the world around them, including faces, license plates, private residences, and patterns of movement of individuals. Privacy engineering is a substantial sub-discipline of AV development. This section develops both data-side privacy and personal privacy for vehicle occupants.

5.5.1 What AVs Can See

A vehicle driving through a neighborhood records:

- Faces of pedestrians and drivers in surrounding vehicles.
- License plates.
- House and building facades, including signs, names, and distinctive features.
- Patterns of presence: which vehicles are parked at which houses at which times.
- Delivery activities, visitors, comings and goings.

Without countermeasures, fleet operation at scale produces a comprehensive surveillance dataset. Most of this data is collected incidentally to the driving task; much of it is not needed for driving per se.

5.5.2 Privacy-Preserving Data Pipelines

Production privacy practice includes:

- **On-device redaction:** blur faces and license plates before data leaves the vehicle.
- **Differential privacy** on aggregated map updates.
- **Access controls:** only personnel with specific need can see raw data; audit logs.
- **Data minimization:** retain only what is needed for specific purposes; automatic deletion after retention periods.
- **Individual rights:** GDPR-style subject access, deletion, and correction requests.

Example 5.3 (On-device redaction latency budget). A 6-camera AV runs face-and-license-plate redaction on every frame before logs leave the vehicle. Camera-stack parameters: 6 cameras, 30 Hz, 1920×1080 at 8 bits/channel.

Step 1: Aggregate frame rate.

$$\begin{aligned} f_{\text{total}} &= 6 \times 30 \\ &= 180 \text{ frames/s.} \end{aligned}$$

Step 2: Uncompressed data rate.

$$\begin{aligned} B &= f_{\text{total}} \cdot W \cdot H \cdot 3 \\ &= 180 \times 1920 \times 1080 \times 3 \\ &\approx 1.12 \times 10^9 \text{ B/s} \\ &\approx 1.05 \text{ GB/s.} \end{aligned}$$

Step 3: Per-frame compute budget at 30 Hz.

$$\begin{aligned} T_{\text{frame}} &= \frac{1}{30 \text{ Hz}} \\ &\approx 33.3 \text{ ms.} \end{aligned}$$

For 6 cameras processed in parallel on dedicated accelerator silicon, the per-camera face-detector budget is ≈ 3 ms.

Step 4: Privacy yield.

A production-quality on-device face detector (BlazeFace-class) achieves recall $r \geq 0.99$ at the 3-ms budget. Of 180 000 faces seen per hour at typical urban density, expected misses:

$$\begin{aligned} N_{\text{miss}} &= 180\,000 \times (1 - 0.99) \\ &= 1\,800 \text{ faces/hour.} \end{aligned}$$

These are flagged for manual review or deletion before logs leave the vehicle.

Interpretation. Without on-device redaction, the 1 GB/s raw stream would create an unmanageable privacy liability the moment it reached cloud storage. With it, the ethics-and-engineering trade-off collapses to a 3-ms-per-frame budget on commodity automotive accelerators—demonstrably affordable.

5.5.3 In-Cabin Privacy

Driver and passenger monitoring cameras (for L2/L3 DMS) raise privacy issues too: gaze tracking, expression analysis, cabin audio. Policy: in-cabin data is typically processed on-device and not transmitted except for specific triggered events (crash, user consent, regulatory request).

5.6 Accountability and Liability

When an AV is involved in a collision, the legal and moral questions of *who is responsible* are new. Traditional automotive liability pins most responsibility on the human driver; an autonomous system shifts it. This section develops both the legal frameworks emerging around this question and the engineering practices that support accountability.

5.6.1 Shifting Legal Framework

The emerging pattern in automotive jurisdictions (California AV law, UK Automated Vehicles Act, German Autonomous Vehicles Act, Japan Level-3 regulation):

- L2 and below: human driver is responsible.
- L3: complex—depends on whether the system was engaged, driver was alerted, etc. Increasingly being resolved by regulation toward the manufacturer’s responsibility when the system was in control.
- L4 and L5: manufacturer / operator is responsible when the system is operating.

5.6.2 The Black Box Problem

When a collision happens, what *actually happened*? Traditional vehicle event data recorders store a small window of data. AV systems need to record much more: every sensor input, every perception output, every planning decision, for seconds to minutes around any incident. This is large-scale data—a single seconds of multi-sensor data can be gigabytes.

Example 5.4 (Incident data preservation pipeline). Trigger: airbag deploys, or automated incident detector fires. Goal: preserve the last 60 s and the next 30 s of full sensor + state data.

Step 1: Per-second sensor volume.

Production AV sensor stack at $\sim 100 \text{ MB s}^{-1}$ (post-compression, fused stream of cameras + LiDAR + radar + state).

Step 2: Per-incident volume.

$$\begin{aligned} V &= 100 \text{ MB/s} \times (60 + 30) \text{ s} \\ &= 9000 \text{ MB} \\ &\approx 9 \text{ GB.} \end{aligned}$$

(Pre-trigger data is captured because the system maintains a circular buffer of the last 60–120 s at all times.)

Step 3: Fleet-scale incident archive.

A program with 1 000 vehicles operating 12 h/day, with one preservation event per 10 000 vehicle-hours (a representative, conservative rate including near-misses):

$$N_{\text{events/year}} = \frac{1000 \times 12 \times 365}{10000} \\ \approx 438 \text{ events/year.}$$

Step 4: Annual archive growth.

$$S = 438 \times 9 \text{ GB} \\ \approx 4 \text{ TB/year.}$$

Over the regulatory minimum 5-year retention period, that is 20 TB of cryptographically signed, tamper-evident storage. Production AV programs maintain petabyte-scale archives once near-miss and shadow-mode events are added.

Interpretation. Per-incident storage is small enough to make preservation a non-issue; aggregate retention is large enough that storage architecture and cryptographic-signing infrastructure must be deliberate engineering decisions, not afterthoughts.

5.6.3 Simulation-Based Reconstruction

After an incident, a simulation of the scenario is often constructed from the logged data to reconstruct what the system saw and decided. This is now a routine part of post-incident analysis, used to determine whether the system behaved as designed and whether the design itself was appropriate.

5.7 Cybersecurity

AVs are software-defined and connected; they inherit the full cybersecurity challenge of connected computing, with physical- safety consequences. This section surveys the major threat categories and the engineering responses.

5.7.1 Threat Categories

Attacks on AV systems fall into these broad categories:

- **Remote code execution:** compromising the vehicle's compute via connectivity (cellular, Wi-Fi, Bluetooth).
- **Adversarial perception:** crafted patterns (sticker patterns, laser dazzlers) that cause perception errors.
- **Sensor spoofing:** injecting false signals into GNSS, LiDAR, or radar.
- **Supply-chain attacks:** malicious components or code introduced before shipment.

- **Insider threats:** privileged access used to alter deployed systems.

5.7.2 Defenses

Production AV cybersecurity practice includes:

- Signed firmware and secure boot.
- Isolation between safety-critical and infotainment domains.
- Intrusion detection at network and host levels.
- Red-team exercises and bug bounties.
- Adversarial-robustness training in perception.
- Cryptographic authentication of sensor data and map updates.

Example 5.5 (Adversarial sticker attack on stop sign). Eykholt et al. (2018) showed that carefully designed sticker patterns applied to a physical stop sign can cause a naive classifier to output “speed limit 45.”

Step 1: Perturbation budget.

The attacker constrains the maximum visual change to be human-imperceptible at distance. Letting δ be the per-pixel perturbation in the sticker region and $|\delta|_\infty \leq \epsilon$:

$$\begin{aligned}\epsilon &= 8/255 \\ &\approx 0.031 \text{ (of full range).}\end{aligned}$$

This is the standard “imperceptible-perturbation” threshold from the adversarial-examples literature.

Step 2: Adversarial objective.

The attack solves

$$\delta^* = \arg \max_{|\delta|_\infty \leq \epsilon} \mathcal{L}(f(x + \delta), y_{\text{stop}}),$$

where f is the classifier and y_{stop} is the true label. Iterative gradient methods (PGD) find δ^* in seconds on a GPU.

Step 3: Layered defense.

A single classifier provides little defense. A layered pipeline provides redundancy:

$$\Pr(\text{full-stack failure}) \approx \Pr(\text{vision fooled}) \cdot \Pr(\text{LiDAR fails to see octagon}) \\ \cdot \Pr(\text{map prior absent}).$$

Even at independent failure probabilities of 0.1 each, the joint failure probability is 10^{-3} —three orders of magnitude better than vision alone.

Interpretation. Adversarial robustness is a layered, defense-in- depth problem. Adversarially-trained classifiers, multi-modal fusion (LiDAR sees the octagon shape, radar confirms a sign-shaped object), and HD-map priors at known intersections each cut the failure rate independently; their composition is what makes production systems resistant.

5.8 Regulation and Certification

This section surveys the regulatory landscape. AV deployment is subject to a patchwork of international, national, and local rules whose compliance burden is substantial.

5.8.1 International Frameworks

UNECE WP.29. Produces regulations covering vehicle systems including automated lane keeping, automatic emergency braking, and increasingly level-3 functions.

ISO Standards. Include ISO 26262 (functional safety), ISO 21448 (SOTIF, safety of the intended functionality), ISO 21434 (cybersecurity), and ISO 22737 (low-speed automated driving).

SAE J3016. Defines the SAE levels of driving automation. Central reference for regulatory documents.

5.8.2 National and Regional Frameworks

The U.S. (NHTSA guidance + per-state laws), EU (type approval within the UN framework + national variations), China (MIIT + MPS regulation), Japan (MLIT + L3/L4 specific rules), and others each apply the international standards with regional variations. Production AV programs invest substantial legal and compliance resources.

5.8.3 Certification Processes

Production deployment requires evidence-based safety arguments and regulatory review. Recent innovations include the EU-level approval-through-testing approach and the U.S. self-certification model with post-market monitoring. Both are in flux; new AV-specific regulations are routinely introduced.

5.9 Transparency and Public Communication

The public understanding of AV technology ranges wildly, from fear to uncritical enthusiasm, often driven by high-profile incidents. AV companies have a responsibility to communicate honestly about capabilities and limitations.

5.9.1 Communicating Capability

Capability claims must be accurate. A system marketed as “autonomous” that actually requires human supervision creates dangerous misunderstanding; marketing language and legal disclaimers should align with technical reality. Regulatory bodies have begun requiring capability-accurate messaging.

5.9.2 Communicating Failures

Incident communication is where AV companies most often fail public trust. A culture of honest, timely, root-cause analysis of incidents—publicly shared in appropriate aggregate form— builds public confidence over time. The opposite, minimizing or obscuring incidents, damages trust not just in the company but in the industry.

5.10 Equity of Deployment

Early AV deployments concentrate where deployment is easy—wealthy cities, benign weather, cooperative regulators. The question of *who else gets served* is an ethical one. This section develops the equity question and the policy responses emerging to address it.

5.10.1 Geographic Equity

If AVs are deployed only in dense, wealthy urban cores, the mobility benefits flow disproportionately to urban residents (who already have transit options) and not to rural populations (who have worse access to alternatives). Policy frameworks encourage deployment diversity; engineering responses include deliberately pursuing ODDs in less-favored regions.

5.10.2 Access Equity

Vehicles that require smartphone apps, credit cards, and specific language / accessibility capabilities exclude users who lack them. Production AV services must deliberately support multiple payment methods, accessibility features, and communication modes to be inclusive.

5.10.3 Risk-Exposure Equity

A AV trained on mostly-suburban data may perform worse in urban neighborhoods with different physical patterns. If those neighborhoods are also lower-income, the risk exposure compounds. Deliberate attention to training-data diversity and per-region safety monitoring is essential.

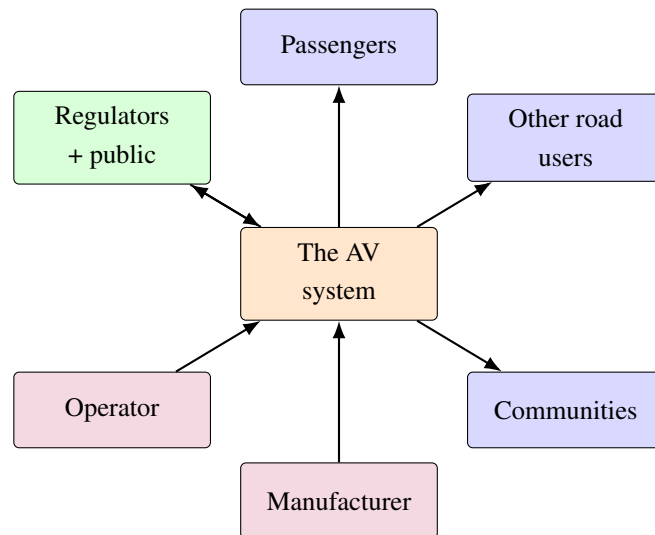


Figure 5.2: Stakeholders in AV deployment. The AV system at center has responsibilities to multiple groups (passengers, other road users, communities) and is governed by multiple parties (manufacturer, operator, regulators). Ethical engineering recognizes all these relationships and does not privilege passengers over pedestrians, or throughput over equity. A responsible design process includes representation for each stakeholder group.

5.11 The Engineering Responsibility

Responsibility for the ethical behavior of AV systems begins with the engineers who design them. This final section argues that ethical engineering is technical engineering, not a separate discipline.

5.11.1 Engineer as Ethical Agent

Every decision about training data, threshold selection, ODD boundaries, fallback policies, and deployment scope has ethical consequences. An engineer who says “that’s not my job” about these questions is mistaken about the job. The ethical weight of AV engineering is high, and it is distributed across everyone on the team.

5.11.2 Practical Habits

Concrete engineering habits that embed ethical thinking:

- Report stratified metrics (subgroup AP, per-region failure rates), not just aggregates.
- Keep an explicit list of failure modes the system cannot handle and the populations most affected.
- Include representative users in the review process.
- Write down what the system is *not* for.
- Be honest in incident reports, internally and externally.

5.11.3 Collective Obligation

No single engineer is responsible for the full ethical behavior of a large system. But the collective culture of the team—what questions get asked in code review, what gets tracked in metrics, what incidents receive serious root-cause work—determines whether ethical reasoning is embedded or peripheral. Team culture is itself an engineering artifact.

5.12 Worked Examples

This section consolidates three detailed worked examples spanning the chapter's topics.

Example 5.6 (Subgroup-accuracy reporting in a release). A pedestrian detector is considered for release. Required evidence for approval includes stratified subgroup metrics:

Stratum	AP	Status
Aggregate	0.84	—
Light clothing	0.82	
Dark clothing	0.86	OK
Height > 1.5 m	0.85	
Height < 1.3 m (likely children)	0.69	flagged
No mobility aid	0.88	
Wheelchair	0.78	flagged
Cane	0.71	flagged

Step 1: Aggregate-vs-worst gap.

$$\begin{aligned}\Delta_{\text{agg-worst}} &= 0.84 - 0.69 \\ &= 0.15.\end{aligned}$$

Step 2: Best-vs-worst gap.

$$\begin{aligned}\Delta_{\text{best-worst}} &= 0.88 - 0.69 \\ &= 0.19.\end{aligned}$$

Step 3: Release decision.

Two flagged subgroup gaps exceed the program's release threshold of $\Delta \geq 0.10$. Release is gated until targeted data collection and retraining close the gaps. Retraining cycle: 6 weeks; release delayed by that amount.

Counterfactual. Had only aggregate metrics been tracked, the release would have proceeded at $\overline{\text{AP}} = 0.84$. The children and mobility-aid gaps would have been deployed safety issues—a disproportionate failure on the very groups whose road-crossing behaviour is most variable and

most dependent on perception performance.

Example 5.7 (Incident with mis-handled communication). Hypothetical: an AV in city A is involved in a non-fatal collision with a cyclist. Possible responses: **Bad**: delay public comment, release selective video, assign blame to the cyclist. Short-term face-saving, long-term trust erosion; regulators less accommodating in future. **Good**: public acknowledgment within 24 hours, release of relevant incident data, detailed engineering analysis published in full once complete (typically 2–6 weeks). Short-term reputational cost, long-term trust gain. The difference is not just PR—it is structural. A culture that handles incidents openly builds trust with regulators and the public, creating space for future deployments. A culture that handles incidents defensively invites opposition.

Example 5.8 (ODD equity audit). A fleet operates in a 100 km² city area divided into five neighborhoods stratified by household income.

Step 1: Concentration measurement.

Audit finds that 70 % of operational hours are concentrated in the wealthiest 20 % of neighborhoods (revenue-optimization outcome).

Step 2: Per-capita ride availability ratio.

Wealthiest 20 %: 70 % of rides serving 20 % of population:

$$\begin{aligned} r_{\text{wealth}} &= 70/20 \\ &= 3.5. \end{aligned}$$

Remaining 80 %: 30 % of rides serving 80 % of population:

$$\begin{aligned} r_{\text{rest}} &= 30/80 \\ &= 0.375. \end{aligned}$$

Step 3: Inequity ratio.

$$\begin{aligned} R_{\text{ineq}} &= r_{\text{wealth}}/r_{\text{rest}} \\ &= 3.5/0.375 \\ &\approx 9.3. \end{aligned}$$

A resident of a wealthy neighborhood has 9.3 times the access of an average resident of the remaining neighborhoods.

Step 4: Mitigation cost.

Explicit equity objective in dispatch + subsidized rides for under-served neighborhoods + ODD expansion to currently-excluded areas. Estimated revenue cost: 5 %. Equity benefit: quantifiable increase in access for ~200 000 residents.

Interpretation. Equity audits expose distributional outcomes that aggregate revenue metrics hide. The $9.3\times$ access ratio is not a technical limitation but a policy choice; it can be reset by the operator under regulatory pressure or internal commitment to equity metrics. The 5 % revenue cost is the price of that reset.

5.13 Industrial Deployment

Ethics frameworks for AVs draw on industry, academia, and regulators. The Partnership on AI, founded in 2016 by Amazon, Apple, Facebook, Google/DeepMind, IBM, and Microsoft, has published guidance on responsible AI deployment. The Institute of Electrical and Electronics Engineers (IEEE) has published the “Ethically Aligned Design” document and IEEE 7000-series standards on ethical considerations in system design. The German Ethics Commission on Automated and Connected Driving published influential guidelines in 2017 that explicitly prohibit certain trolley-problem-style discrimination among potential victims. The European Union’s Artificial Intelligence Act, which entered into force in 2024, includes provisions relevant to AI used in vehicles. Mobileye, Waymo, and others have published public safety and ethics positions in line with these frameworks.

5.14 Chapter Summary

The ethical dimensions of autonomous driving are not separate from the engineering; they are embedded in every choice of training data, threshold, ODD boundary, and deployment scope. The Trolley Problem framing, while media-friendly, distracts from the real ethical questions: perception bias, privacy, accountability, cybersecurity, regulation, equity, and public communication. Each of these has engineering responses as well as policy ones. The engineer’s responsibility is to take the full set of questions seriously and embed their consideration into the normal practice of the work. A responsible AV program is not one that has solved every ethical question, but one whose practices actively seek, surface, and address them.

5.15 Exercises

1. For the stakes example (5.1), estimate the deployment scale (miles per year) at which the 10 % benefit / 5 % harm difference becomes statistically detectable.
2. Describe three ways the Trolley Problem framing distorts ethical reasoning about AVs. Propose a replacement framing.
3. For the subgroup-gap example (5.2), propose a targeted data-collection strategy to close the wheelchair-user gap.

4. Discuss the ethics of using an AV in L4 mode in a neighborhood where most residents have not consented to appearing in its training data.
5. Design a privacy-preserving pipeline for a crowd-sourced map-update system. What specifically needs to be protected, and how?
6. For the incident-data example (5.4), propose retention and access policies. Who gets access, under what conditions?
7. Identify a published adversarial attack against a perception system and describe the engineering defenses that have been demonstrated against it.
8. Compare the ethical questions raised by L2 deployment vs L4 deployment of the same underlying technology.
9. For the ODD equity example (5.8), propose two additional metrics that would complement the per-capita-operation count.
10. As an engineer on an AV team, you discover a subgroup- accuracy gap the week before a release and know that flagging it will delay launch by 6 weeks. What do you do? Justify your answer.

Chapter 6

Responsible AI for Autonomous Vehicles

6.1 Introduction

The ethics chapter (Chapter 5) developed the principled framework for ethical AV deployment; the safety chapter (Chapter 32) developed the engineering discipline for safe operation. This chapter brings the two threads together under the heading of *Responsible AI*—the systematic practice of developing, deploying, and maintaining AI systems that meet engineering, ethical, and societal standards simultaneously. It is not a separate concern from the engineering of earlier chapters; it is how good engineering is done when the systems being built affect the world beyond the lab.

Three observations motivate the chapter’s structure.

First, Responsible AI is *practice-based*, not merely principle-based. Every AI lab can adopt a set of ethical principles; the difference between principles on paper and principles in practice is documented processes, enforced gates, and tracked metrics. The operational disciplines—impact assessment, bias auditing, explainability reporting, incident response—are what turn good intentions into good outcomes.

Second, Responsible AI is now *regulated*. The EU AI Act (entering force 2025–2026), IEEE 7000-series standards, NIST AI Risk Management Framework, and ISO/IEC 42001 (AI management systems) collectively define what responsible practice looks like. An AV program treats these as requirements, not aspirations.

Third, Responsible AI for AVs is *distinctive*. The standard AI-ethics conversation (algorithmic hiring, credit scoring, facial recognition) focuses on decisions affecting individual people. AVs physically move through the world, with potential for injury and death, at a scale no other deployed AI system approaches. The stakes are higher and the required disciplines more stringent.

The chapter is organized as follows: first, a precise definition of Responsible AI and its pillars. Second, the regulatory frameworks now mandating it. Third, the four operational pillars in depth: fairness, transparency, accountability, and governance. Fourth, the specific techniques (explainability, bias

auditing, impact assessment, lifecycle documentation) that implement the pillars. We close with the engineering workflow that embeds Responsible AI into the development process, and a look at the ESG (Environmental, Social, Governance) lens that increasingly investors and regulators apply. Two architecture diagrams visualize the Responsible-AI pillars and lifecycle; worked examples show the concrete practice.

6.2 What Responsible AI Means for AVs

Responsible AI has several overlapping definitions across bodies; the intersection is robust. For autonomous vehicles specifically, it reduces to four pillars that must all be satisfied.

6.2.1 The Four Pillars

Each pillar addresses a distinct concern and requires distinct techniques. All four must be present; a system that is fair but not transparent, or accountable but not governed, falls short.

Fairness

The system's behavior does not systematically advantage or disadvantage any group. For AVs: detection recall is comparable across pedestrian subgroups (age, ability, skin tone, clothing); deployment does not privilege wealthy neighborhoods over poor ones; the distribution of benefits and risks across populations is equitable.

Transparency and Explainability

The system's decisions can be understood and audited. For AVs: the engineer can trace why a particular perception output arose, why a planner chose a particular trajectory, and—after an incident—why the system behaved as it did. Not every layer needs to be simple, but the system's behavior must be investigable.

Accountability

Clear ownership exists for every aspect of the system. For AVs: the engineer responsible for detection quality, the manager responsible for release decisions, the legal owner responsible for operational incidents. Accountability diffuses when roles are unclear; responsible AI requires that it does not.

Governance

An organizational structure ensures the other three pillars are actually enforced. For AVs: a review board evaluates proposed releases; impact-assessment processes run before deployment; audit trails document decisions; escalation paths exist for ethical concerns.

6.2.2 Distinctive Features for AV Systems

Compared to other AI applications, AV Responsible AI has several distinctive features that shape how the pillars manifest.

Physical Safety Consequences

Unlike most AI applications, AV failures can kill. This raises the stakes for every pillar: fairness-failures in pedestrian detection produce fatalities, not just unfair outcomes; transparency-failures obscure safety-relevant decisions; accountability-failures delay corrective action after incidents; governance-failures allow unsafe releases.

Real-Time Operation

AV decisions are made in milliseconds, not minutes. Real-time transparency (explaining a decision as it happens, rather than retrospectively) requires specific architectural support. Log more than you think you need.

Multi-Stakeholder Ecosystem

AV responsibility extends beyond the manufacturer: regulators, infrastructure operators, other road users, passengers, communities. The governance model must reach beyond the engineering team.

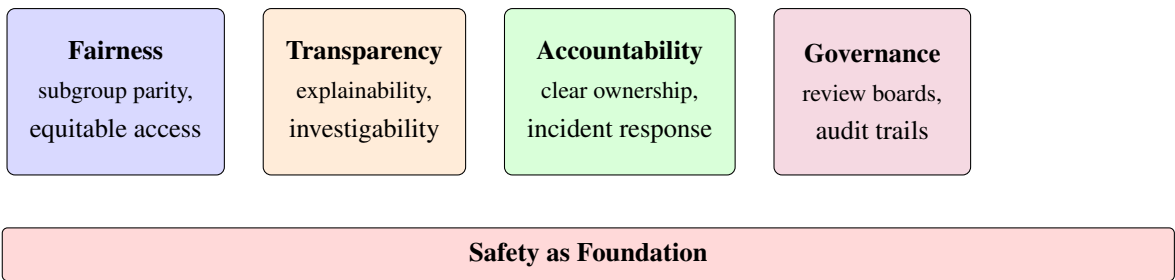


Figure 6.1: The four pillars of Responsible AI for AVs, built on safety as foundation. Fairness (equity of system behavior), transparency (investigability of decisions), accountability (clear ownership), and governance (organizational process). Failure in any pillar compromises the others; safety failure compromises all four.

6.3 Regulatory Frameworks

Several regulatory frameworks now formalize Responsible AI. A production AV program must navigate them collectively.

6.3.1 EU AI Act

The world’s most comprehensive AI regulation, entering force in stages 2025–2026. Classifies AI systems by risk level; AV perception and planning systems fall in the “high-risk” category with specific requirements.

Required Documentation

For high-risk systems: risk management system, data governance documentation, technical documentation, automatic event logs, transparency obligations, human oversight provisions, accuracy and robustness requirements, cybersecurity requirements.

Conformity Assessment

Before deployment, a conformity-assessment process documents compliance. Third-party assessment may be required depending on the specific AI application.

6.3.2 NIST AI Risk Management Framework (AI RMF)

A voluntary US framework providing structured guidance for managing AI risks. Four functions: Govern, Map (context and impact), Measure, Manage. Widely adopted as a reference even absent direct regulatory mandate.

6.3.3 ISO/IEC 42001

International standard for AI Management Systems (AIMS), published in late 2023. Analogous to ISO 9001 for quality: defines the processes an organization must have to manage AI development responsibly.

6.3.4 IEEE 7000 Series

A series of standards on specific Responsible-AI topics: IEEE 7000 (model process for addressing ethical concerns), 7001 (transparency), 7002 (data privacy), 7003 (algorithmic bias), 7010 (well-being metrics). Provide technical specificity that higher-level frameworks lack.

6.4 Fairness in Autonomous Vehicles

Fairness in AVs is primarily about avoiding systematic harm to specific groups. This section develops the dominant fairness concerns and the techniques for addressing them.

6.4.1 Subgroup Performance Parity

Detection and decision accuracy should be comparable across protected groups. This requires measurement, not just assumption.

Measurement

Stratified evaluation on protected subgroups is non-optional. For pedestrian detection: measure AP separately by skin tone, body size, mobility-aid use, clothing color. For behavior: measure yielding rates separately by cyclist demographics.

Example 6.1 (Stratified fairness report). Release candidate perception stack. Required fairness report:

- Overall pedestrian AP: 0.86.
- By Fitzpatrick skin type (I–VI): 0.89, 0.88, 0.86, 0.85, 0.83, 0.81.
- By subject height (< 1.3, 1.3–1.8, >1.8 m): 0.74, 0.87, 0.88.
- By mobility aid (none, cane, walker, wheelchair): 0.88, 0.84, 0.79, 0.72.

Gap across skin types: 8 points. Gap across mobility aids: 16 points. Gap across height: 14 points (children at 0.74). The report triggers release-gate review; before deployment, targeted data and retraining must close each gap below a specified threshold.

Causes of Disparity

Most subgroup gaps trace to training-data under-representation. Standard mitigations: targeted data acquisition, subgroup- balanced sampling, subgroup-weighted loss functions, adversarial-fairness training, copy-paste augmentation with subgroup-aware scenes.

6.4.2 Equity of Deployment

Separate from detection fairness: the question of who benefits from AV technology at all. An AV service that operates only in wealthy cores, with only specific payment methods, excluding disabled users and non-app-users, is not an ethical deployment even if individual detection performance is uniform across pedestrians seen.

Access Considerations

Production AV services must actively design for: users without smartphones, users with accessibility needs, users speaking languages other than the operator’s default, users without credit cards, users in non-central neighborhoods.

Benefit Distribution

Measure and report service patterns by neighborhood, income level, accessibility-need status. When disparities emerge (they will), design operational responses.

6.5 Transparency and Explainability

A model that is perfectly fair but opaque is still a problem: without understanding, bugs hide, trust is harder to build, and regulatory compliance is impossible. Transparency and explainability enable oversight.

6.5.1 Levels of Transparency

Different audiences need different transparency. An engineer debugging a specific failure needs deep technical detail; a regulator reviewing a release needs a summary at policy level; a passenger asking why the car slowed down needs a one-sentence explanation. Each audience has specific tooling.

Engineer-Level

Full logs, intermediate representations, model weights, training-data provenance, hyperparameters. Data volumes are large; indexing and query infrastructure are critical.

Regulator-Level

Summary dashboards, aggregate metrics, safety-argument documentation, incident reports. Requires that engineer-level data can be summarized faithfully without losing important detail.

User-Level

In-cabin explanations (“slowing for pedestrian in crosswalk”), post-trip summaries, service-level disclosures. Natural-language generation from internal state.

6.5.2 Explainability Techniques

Specific techniques for making learned models interpretable.

Saliency and Attention Maps

Visualize which inputs most influenced a model’s decision. For perception: heatmaps over camera images. For planning: attention weights over other agents.

SHAP Values

Shapley-additive values attribute contribution of each input feature to a specific prediction. Widely used in tabular ML; increasingly adapted to structured AV inputs.

Counterfactual Explanations

“The car slowed because the pedestrian was detected; had it not been detected, the car would not have slowed.” Counterfactuals are often more intuitive than attributions.

Example 6.2 (Counterfactual explanation for an intervention). Scenario: AV slowed unexpectedly on a clear road. Investigation: perception output shows a false-positive pedestrian detection at 80 m, classified at confidence 0.35. Counterfactual: re-run the scene without the false detection; planner maintains speed. The counterfactual confirms the detection caused the intervention. The false-positive root cause (sunlight artifact on a sign) becomes the target of the next training-data augmentation pass.

6.5.3 Architectural Choices for Transparency

Some architectural choices make transparency easier. Modular stacks (Chapters 3–20) expose intermediate representations; end-to-end stacks (Chapter 22) do not. Auxiliary output heads (predicted BEV, natural-language captions) make end-to-end systems more transparent. Production programs typically include multiple transparency mechanisms in parallel.

6.6 Accountability

Clear responsibility for AI system behavior is the accountability pillar. This section develops the organizational and technical practices that support it.

6.6.1 Organizational Accountability

Within an AV program:

- Named owners per subsystem (perception, planning, etc.).
- Release-decision authority clearly documented.
- Escalation paths for identified issues.
- Post-incident accountability without blame-seeking.

6.6.2 Regulatory Accountability

Externally: clear communication with regulators, prompt notification of defects, full cooperation with investigations.

6.6.3 Technical Accountability Aids

Technical practices that support accountability:

- **Model cards:** structured documentation for each deployed model: training data, intended use, known limitations, fairness metrics.
- **Data sheets:** structured documentation for each dataset: provenance, collection protocol, demographics, licensing.
- **Experiment tracking:** every trained model traceable to its code, data, and hyperparameters.
- **Audit logs:** deployment decisions, with rationale, recorded permanently.

6.7 Governance

Governance is the organizational infrastructure that enforces the other three pillars. Without governance, they become aspirations rather than practices.

6.7.1 AI Governance Boards

A cross-functional committee (engineering, legal, ethics, external advisors) reviews significant releases and addresses policy questions. Meeting rhythms (monthly for routine, on-demand for emergencies), documented decisions, and published minutes (at appropriate confidentiality levels).

6.7.2 Impact Assessments

Before major deployments or substantial model changes: an impact assessment evaluates expected effects, including fairness, safety, privacy, and societal implications. Decisions to proceed, modify, or postpone are documented.

6.7.3 Bias Audits

Regular external or internal audits verify that fairness commitments are being met in practice. Reports are retained and actionable findings feed into the engineering backlog.

6.7.4 Whistleblower and Disclosure Protection

Engineers who identify safety or ethical concerns must have protected paths to raise them without retaliation. Production programs have explicit whistleblower policies enforced by executive-level ownership.

6.8 The Responsible AI Lifecycle

Responsible AI is not a pre-release check; it is a lifecycle discipline. Figure 6.2 shows the full loop.

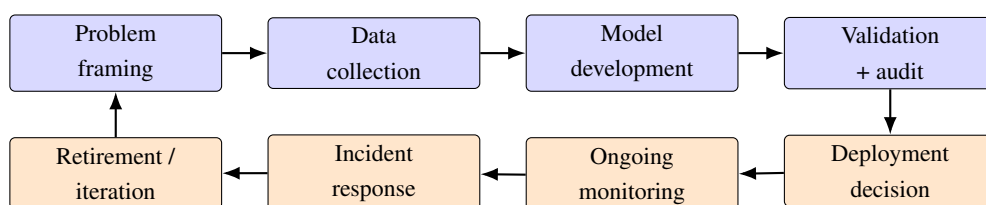


Figure 6.2: The Responsible AI lifecycle. Problem framing explicitly includes ethical and societal questions. Data collection documents provenance, subgroup representation, and privacy handling. Model development tracks experiments for accountability. Validation includes fairness audits. Deployment decisions invoke governance processes. Ongoing monitoring detects emerging issues. Incident response follows documented protocols. Retirement includes post-mortem learning that feeds back into the next iteration’s problem framing.

6.9 ESG Considerations

AV programs increasingly face Environmental, Social, and Governance (ESG) scrutiny from investors and regulators.

6.9.1 Environmental

Fuel consumption (for ICE-based fleets), electricity use (for EV fleets), lifecycle emissions of compute hardware, end-of-life disposal of electronics. AV programs increasingly report these metrics publicly.

6.9.2 Social

Labor displacement, accessibility, equitable deployment, community engagement, driver-safety (during mixed human-AV operation).

6.9.3 Governance

Board composition, executive compensation, oversight structures, ethics committees. ESG-focused investors scrutinize these.

Example 6.3 (ESG disclosure for a robotaxi program). Annual ESG disclosure for a 1000-vehicle robotaxi fleet:

- **E:** 5.2 t CO₂/vehicle/year (compute + grid); 95 % renewable electricity; end-of-life battery recycling partnership.
- **S:** 1.4 million rides, 3 % accessibility-equipped; partnership with local transit for equity routes; labor-transition fund for displaced drivers.
- **G:** Board ethics committee; annual bias audit (published summary); whistleblower hotline with 12 disclosures logged and resolved.

Investors and regulators use these disclosures for decisions; programs that score well can access capital more cheaply and operate in more markets.

6.10 Worked Examples

Example 6.4 (Release-gate Responsible-AI review). New perception model ready for release. Responsible-AI review:

- Fairness: stratified evaluation complete; mobility-aid gap reduced from 16 to 4 points. **Pass.**
- Transparency: model card updated with training data, intended use, and limitations. **Pass.**
- Accountability: owner assigned; release-decision recorded; rollback procedure documented. **Pass.**
- Governance: Engineering Review Committee approved; impact assessment posted. **Pass.**

Release proceeds. 3 months later: monitoring catches subgroup regression for wheelchair users in a new deployment region. Root-cause traces to regional data under-representation. Rollback triggered within 24 hours; patch developed with targeted data. The monitoring loop closed the gap that release gates could not predict.

Example 6.5 (Incident response with accountability). Minor collision involving AV. Response:

- Initial notification to regulators within 4 hours (required).
- Internal incident-response team activated.
- Named engineering owner leads root-cause investigation.
- 2-week analysis: perception gap in specific weather condition. Documented.
- Communication: public statement within 72 hours; detailed analysis published within 6 weeks.
- Mitigation: targeted retraining; regional deployment paused until patch deployed.
- Follow-up: 6-month audit confirms mitigation effective.

Accountability is neither diffuse nor punitive; it is transparent and action-oriented. Trust is built through such handling far more effectively than through promises or disclaimers.

6.11 Industrial Deployment

Responsible-AI commitments are increasingly formalised in industry. The Partnership on AI publishes guidance to which member organisations including Waymo, Apple, and Microsoft are signatory. Anthropic’s Responsible Scaling Policy and Google DeepMind’s Frontier Safety Framework, while focused on general AI rather than vehicles specifically, have begun to influence AV practice given the increasing use of foundation models in AV stacks. The NIST AI Risk Management Framework, published by the U.S. National Institute of Standards and Technology in 2023, provides a non-regulatory framework adopted in some industry safety cases. Mobileye, Waymo, and Cruise have published public safety reports describing their internal review processes for AI-driven systems.

6.12 Chapter Summary

Responsible AI for AVs combines ethical principles (Chapter 5), safety engineering (Chapter 32), and societal considerations into a continuous practice. Four pillars—fairness, transparency, accountability, governance—must all be present. Regulatory frameworks (EU AI Act, NIST AI RMF, ISO/IEC 42001, IEEE 7000 series) formalize the expectations. Specific techniques (stratified evaluation, model cards, counterfactual explanations, impact assessments) implement the pillars. The responsible-AI lifecycle embeds the practice into every stage of development and operation. ESG disclosures increasingly expose AV programs to external scrutiny of their responsible-AI practice. Programs that treat responsibility as engineering discipline—not PR—scale and earn public trust; those that treat it as optional face commercial and regulatory consequences.

6.13 Exercises

1. For the four pillars (6.1), describe one specific failure mode in each for an AV perception stack.
2. The EU AI Act classifies AV systems as “high-risk.” List three specific compliance obligations that flow from this classification.

3. For the fairness report (6.1), design a data-collection strategy to close the wheelchair-user gap to under 5 points.
4. Propose three counterfactual explanations for common AV behaviors (slowing, changing lanes, stopping).
5. Design a model card for the detection network of Chapter 11. Include at minimum 5 fields.
6. For the ESG example (6.3), propose a metric for accessibility that is meaningful without being counter-productive.
7. Compare the Responsible-AI lifecycle (6.2) to the Safety-compliance lifecycle (Chapter 4). What overlaps, and what is distinct?
8. Discuss the tension between model transparency and intellectual-property protection. Propose a compromise that preserves both.
9. Identify three specific ways a whistleblower protection can fail in practice, and propose organizational responses.
10. For the release-gate example (6.4), what would the review look like if the fairness gap was only 15 points (halfway between problematic and acceptable)?

Part IV

Specialized Vehicle Classes

Chapter 7

Autonomous Vehicles in Mining

7.1 Introduction

Mining is the domain where autonomous vehicles are most thoroughly deployed, most commercially successful, and most deeply integrated into daily operations—and yet it is the domain least covered in public AV literature. Every page in this book until now has implicitly assumed that the vehicle is a car on a public road; most real-world AV miles to date have been driven by haul trucks on private mine sites, not by robotaxis in cities. Rio Tinto, BHP, Fortescue, Newmont, and others operate thousands of autonomous mining vehicles today, covering hundreds of millions of tonnes of material every year. The technology that moves these vehicles is mostly AV technology, but the deployment context, regulatory environment, and operational constraints are distinct enough to warrant dedicated treatment.

Three observations motivate the chapter's structure.

First, mining is an *ideal early deployment domain* for autonomous vehicles. The operational environment is private (no uncertainty about public rules), bounded (the mine footprint is known precisely), structured (haul routes are maintained deliberately), and populated by professional actors rather than unpredictable members of the public. These features reduce the long-tail problem that constrains public-road AVs—there are fewer tail cases because the environment is more controlled.

Second, mining AV economics are *quantifiable and favorable*. A haul truck costs \$3–5 million; autonomous operation adds 10 % to 15 % to that cost but can increase utilization from 60 % to 75 % to 90 %+, reduce fuel consumption by 10 % to 15 % through smoother driving, and eliminate driver costs entirely. The ROI is clear, sometimes under two years. These economics fund substantial technical development.

Third, mining AVs have driven *safety* improvements that justify deployment independent of economics. Haul-truck accidents in traditional mines kill dozens of operators each year globally; removing the human from the cab removes them from harm's way. The safety case is one of the cleanest in autonomous driving—the alternative (human operators in hazardous environments) has a known,

documented death rate.

The chapter is organized as follows. First, the mining environment and its operational modes (open-pit vs underground). Second, the equipment types being automated: haul trucks, drill rigs, loaders, and auxiliary vehicles. Third, technical adaptations distinct from on-road AV: sensor suites for dust and extreme weather, GPS-denied underground operation, off-road terrain, centralized fleet management, and equipment-specific perception. Fourth, economic and safety analysis. Fifth, case studies of major deployments. Three architecture diagrams visualize the mine layout, fleet-management architecture, and the haul-cycle state machine; worked examples cover fleet economics and operational arithmetic.

7.2 The Mining Environment

Understanding mine operations is prerequisite to understanding mining AVs. Mines are not abstract industrial sites; they are specific geographies with specific operational patterns.

7.2.1 Open-Pit Mining

Open-pit mines are the dominant deployment environment for autonomous mining vehicles. The operational pattern is suited to automation in ways that other mining modes are not.

Layout and Operations

An open-pit mine is a giant spiral or terraced hole in the ground. Haul trucks carry ore and waste rock from the active extraction face (bottom of the pit) up to processing or waste-dumping destinations. Typical haul cycle: 15 min to 60 min, 2 km to 10 km each way. A major mine operates dozens to hundreds of trucks simultaneously.

Why This is AV-Friendly

- Private road network, professionally maintained.
- Fixed routes, updated daily as the face evolves.
- No pedestrians (workers are in designated areas).
- No conventional traffic (only mining vehicles).
- Centralized dispatch and scheduling.
- 24/7 operation enabled by the economics.

7.2.2 Underground Mining

Underground mines are harder to automate than open-pit but the payoff is higher: the hazard is greater (confined spaces, limited evacuation paths, air quality issues) and workers benefit more from

removal.

Specific Challenges

- GPS-denied: no satellite signals underground.
- Tighter geometry: haul tunnels are often only marginally wider than the truck itself.
- Dust, darkness, water: sensor performance degraded.
- Radio-wave propagation: cellular and Wi-Fi signals attenuated.
- Ventilation constraints: diesel vehicles require ventilation that is expensive to provide.

Despite these challenges, underground autonomous loaders (load- haul-dump machines, or LHDs) are deployed at scale at sites like Codelco El Teniente (Chile) and Rio Tinto Northparkes (Australia).

7.2.3 Operational Modes

Within any mine, AV operations fall into three broad patterns.

Full Autonomy (Hauling)

Haul trucks follow pre-planned routes from loading point to dumping point without human intervention. Most mature mode; dominant at Pilbara iron-ore operations.

Supervised Autonomy (Drilling)

Drilling rigs execute drilling patterns autonomously but under supervisory review; an operator may simultaneously monitor multiple rigs.

Teleoperation (Maintenance and Confined)

Where full autonomy is still impractical (complex or hazardous specific maneuvers), teleoperation lets a human operate the vehicle remotely. Particularly used for underground LHDs during face-advance operations.

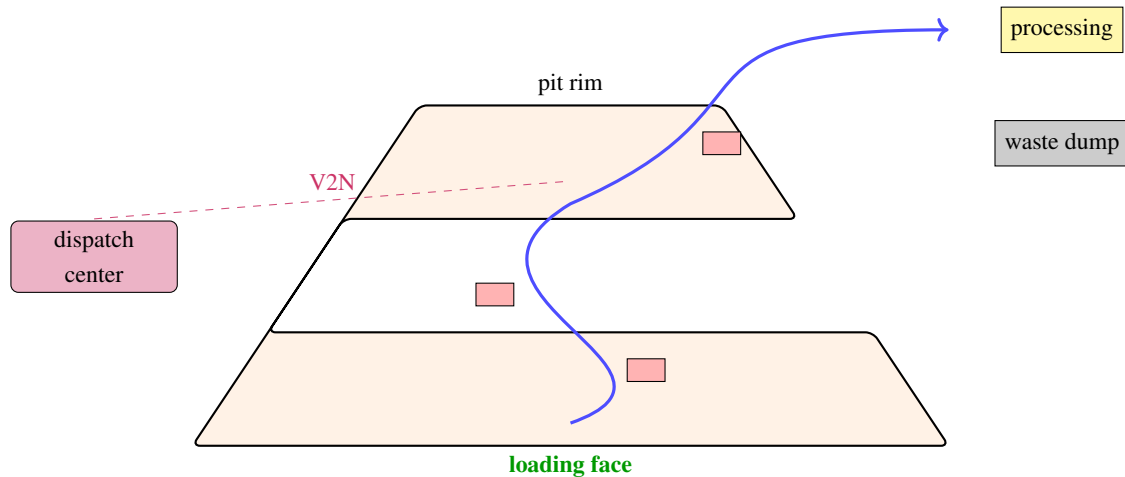


Figure 7.1: Open-pit mine layout with autonomous haul trucks. Trucks (red) travel a winding route from the loading face (bottom) up through the terraced pit to surface destinations (processing or waste dump). A centralized dispatch center coordinates all trucks via wireless network. The geometry changes daily as the face advances; the haul-route map updates continuously.

7.3 Equipment Types

Different mining equipment has different automation requirements. This section covers the main categories and their specific technical profiles.

7.3.1 Haul Trucks

Haul trucks are the dominant automated equipment class in mining, with the most mature autonomy technology. Understanding them is understanding most of mining autonomy.

Physical Characteristics

A typical autonomous haul truck (Caterpillar 797F, Komatsu 930E-4AT) weighs 250 t empty, 400 t loaded, is approximately 15 m long and 7 m wide. Top speed loaded: 40 km h^{-1} to 60 km h^{-1} .

Autonomy Fitment

Autonomous haul trucks carry sensor suites tailored to mining:

- Multi-beam LiDAR (often 4 to 6 units for 360° coverage given vehicle size).
- High-dynamic-range cameras (dust and bright sunlight).
- Radar for dust-resistant obstacle detection.
- GNSS (often multi-constellation: GPS + GLONASS + BeiDou).
- Wheel-encoder odometry (valuable on the rough surfaces that degrade visual odometry).
- Inclinometer (critical for stability on grades).

7.3.2 Drilling Rigs

Autonomous drill rigs execute pre-computed drilling patterns for blast-hole preparation. The task is fundamentally different from haul-truck driving: repetitive, location-specific, lower speed.

Task Automation

A blast pattern is a grid of drilled holes; autonomous drill rigs move between positions, set up, drill to specified depth, and move to the next. Multiple rigs working in coordination can be supervised by a single operator.

7.3.3 Loaders (Shovels and Excavators)

Loaders are the hardest mining equipment to automate because their task (dig, load into truck) requires responsive interaction with the world that changes with every cycle.

Current State

Fully autonomous shovels are still research-stage. Operator- assist systems are common: automated swing-to-truck paths, auto- dump, operator-supervised autonomy for specific cycle phases.

7.3.4 Auxiliary Equipment

Water trucks (dust suppression), graders (road maintenance), service vehicles, and light trucks are increasingly automated. Each has its own task profile. Automation of auxiliary equipment increases the fraction of mine operation that is autonomous, not just the prominent haul-truck layer.

7.4 Technical Adaptations

Mining AV technology builds on the foundations of on-road AV (perception, localization, planning, control) but with specific adaptations for the mining environment.

7.4.1 Perception for Dust and Extreme Weather

Dust as the Primary Adversary

A mine is dusty. Haul roads generate visible dust clouds; the loading face generates thicker clouds. LiDAR and camera performance degrade in dust; radar is far more robust. Mining perception stacks therefore rely more heavily on radar than on-road AVs do.

Example 7.1 (Detection range vs dust concentration). On a dry road, 905 nm LiDAR detects a truck at 200 m. With moderate dust ($\sim 1 \text{ g m}^{-3}$): range drops to 60 m. With heavy dust near the loading face: 15 m. Camera vision follows similar curves. Radar: minimal range degradation (250 m stays 230 m). Mining-grade perception relies on radar as the primary range sensor in heavy dust; LiDAR is secondary.

Extreme Weather

Mines operate in diverse climates: Pilbara heat (50 °C+), Siberian cold (−40 °C), monsoon rain, Arctic snow. Equipment is specified for these extremes; autonomy software must similarly handle them.

7.4.2 GPS-Denied Underground Localization

Underground autonomy cannot use GNSS. Alternatives:

- **LiDAR SLAM** against tunnel walls (Chapter 13).
- **Beacon-based localization:** infrastructure beacons (RFID, ultra-wideband) at known positions.
- **Visual markers:** painted or reflective markers at known tunnel locations.
- **Dead-reckoning IMU + wheel odometry** bounded by beacon/marker corrections.

Example 7.2 (Underground localization error budget). LHD operating in a tunnel, traveling between work faces 2 km apart. Dead-reckoning (IMU + encoders) alone: drift of 1 % of distance = 20 m—too much for tunnel-width operation. Add UWB beacons every 100 m: error bounded to ~ 30 cm. Add LiDAR SLAM matching against the walls: error drops to ~ 10 cm. The stack is multi-layered: dead-reckoning provides continuity; beacons provide reference; SLAM provides tight matching.

7.4.3 Off-Road Terrain

Mine haul roads are not paved roads. Surfaces are compacted gravel or dirt; they rut, wash out in rain, and accumulate spilled ore. Controllers tuned for paved-road dynamics do not work here; mining controllers explicitly model the variable traction.

Control Adaptation

Traction models in mining controllers account for:

- Grade effects (long steep grades common).
- Retard systems (loaded trucks descending use mechanical retarders; automation controls them as well as brakes).
- Loaded/empty differential behavior (a loaded truck is very different from an empty one).

7.4.4 Centralized Fleet Management

On-road AVs operate mostly independently; mining AVs operate under centralized fleet management systems that coordinate all vehicles in the mine.

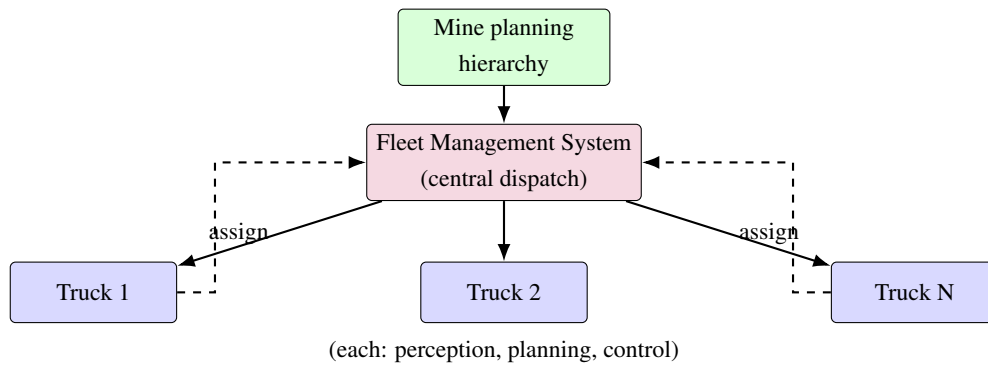


Figure 7.2: Centralized fleet management architecture for mining. The Fleet Management System (FMS) receives daily production targets from the mine-planning hierarchy and assigns individual trucks to specific routes and loading-unit pairings. Each truck has its own onboard perception, planning, and control (solid arrows from FMS). Telemetry (dashed arrows) flows back to FMS continuously: position, load status, fuel level, system health. The FMS is the orchestrator; individual trucks are autonomous executors.

Dispatch Algorithms

The FMS runs dispatch optimization continuously: assign trucks to loading units to maximize throughput while minimizing waiting time. This is a large combinatorial optimization problem; modern FMS use mixed-integer programming or learned policies.

7.5 Economic and Safety Analysis

Mining AV deployment is driven by quantified economic and safety benefits.

7.5.1 Economics

Capital Cost

Autonomous fitment adds 500k\$750 k\$ per truck. For a fleet of 100 trucks, capital uplift is 50–75 million \$.

Operating Savings

Annual savings per truck (at a typical large iron-ore operation):

- Driver cost elimination: \$300k (multiple shifts required for 24 h/7 d operation).
- Fuel savings (smoother driving): \$100k.
- Tire wear reduction (consistent driving): \$50k.
- Reduced maintenance events (consistent driving): \$75k.
- Utilization increase from 65 % to 90 %: ~\$200k in incremental production value.

Total: ~ 725 k\$/year. Payback period: ~ 1 year on capital uplift.

Example 7.3 (Full fleet ROI). 100-truck conversion. Capital uplift: \$60M. Infrastructure costs (dispatch, networks, mapping): \$30M. Total: \$90M. Annual operating savings: $100 \times 725 \text{ k\$} = \72.5M . Simple payback: $90/72.5 = 1.24$ years. Including increased production and downstream effects: ~ 10 -month payback in the best cases. This economic clarity is why mining AV deployments are mature.

7.5.2 Safety

Traditional haul-truck operation kills roughly 20–40 operators globally per year. Autonomous operation removes the operator from the cab entirely. Remaining safety concerns: interactions with other vehicles (maintenance, light vehicles) and with workers in field. These are addressed through:

- Separation: autonomous and manned vehicles operate in strictly-separated zones.
- Positive-identification systems: workers in active areas carry transponders that AVs detect at range.
- Emergency stop: any worker can halt nearby autonomous vehicles through field devices.

7.6 Major Deployments

This section surveys the production deployments as of 2025–2026.

7.6.1 Rio Tinto Pilbara

Pilbara iron-ore operations have operated autonomous haul trucks since 2008; by 2025 the fleet exceeds 150 trucks across multiple sites. Early partnership with Komatsu; mixed Komatsu and Caterpillar fleet now. Rio Tinto reports substantial productivity and safety benefits.

7.6.2 BHP Jimblebar / Pilbara

Similar-scale deployment. BHP's fleet transition coincided with other operational upgrades (new mine plans, digital twin).

7.6.3 Fortescue

Fortescue completed autonomous conversion of its haul-truck fleet; further expansion to electric autonomous haul trucks (driven by emissions goals) is underway.

7.6.4 Codelco and Anglo American

Underground automation at Codelco (Chile) and Anglo American (South America) has advanced underground LHD autonomy.

7.6.5 Technology Vendors

Dominant autonomous-truck system vendors: Caterpillar (Command for Hauling), Komatsu (Autonomous Haulage System, AHS), Hitachi (via a joint venture). Smaller players address specific niches: retrofit autonomy for existing fleets, drill-rig autonomy, LHD autonomy.

7.7 Haul Cycle State Machine

An autonomous haul truck's behavior is best understood as a state machine. Understanding the states and transitions is understanding operational-mode logic.

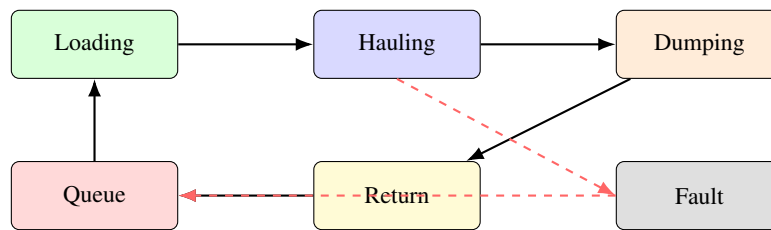


Figure 7.3: Haul-cycle state machine for an autonomous truck. Normal operation cycles through Queue → Loading → Hauling → Dumping → Return → Queue. Fault-state transitions (red, dashed) enter on any detected problem (sensor degradation, mechanical fault, low fuel); after resolution the vehicle re-enters the queue. Each state has defined behavior, sensor requirements, and interaction protocols with other vehicles and with the FMS.

Example 7.4 (State-transition timing). Typical cycle times for a 3 km haul:

- Queue: 1 min to 3 min (waits for loading-unit availability).
- Loading: 2 min to 4 min.
- Haul: 6 min to 10 min loaded.
- Dumping: 1 min to 2 min.
- Return: 4 min to 7 min empty.

Total cycle: 14 min to 26 min. A 24/7 truck completes 55–100 cycles per day. At 300 tonnes per cycle: 16 500 to 30 000 tonnes per truck per day.

7.8 Worked Examples

Example 7.5 (Underground LHD localization). LHD at active face; task: load ore bucket, haul to stockpile 500 m up the tunnel, dump, return. GPS-denied. Localization: IMU + wheel-encoder dead-reckoning (drift $\sim 0.5\%$); UWB beacons at known positions every 50 m (accuracy 30 cm at each fix); LiDAR SLAM matching tunnel walls ($< 10\text{ cm}$ between fixes). Net performance: $< 10\text{ cm}$ accurate localization throughout the 500 m haul. Speed: 15 km h^{-1} to 25 km h^{-1} loaded. Cycle time: $\sim 5\text{ min}$. 3 LHDs in the same tunnel, coordinated through FMS to avoid collisions.

Example 7.6 (Fleet utilization improvement). Pre-automation: 50 manned haul trucks at 65 % utilization (driver shifts, breaks, shift-change delays). Effective: 32.5 truck-equivalents continuously operating. Post-automation: 50 autonomous trucks at 92 % utilization. Effective: 46 truck-equivalents. Improvement: 41 % more productive hours from the same fleet. At 300 t/cycle and 6 cycles/hour effective: additional 35 000 t/day of moved material—multi-million dollars of value per day at ore prices. This calculation, applied per site, justifies the capital investment.

Example 7.7 (Integrated safety metric). Manned operation at a reference mine: 1.2 lost-time injuries per million hours worked (LTIFR). Operators in the cab accounted for 60 % of total hours but 75 % of injuries. Post-automation: driver-cab hours eliminated. Predicted LTIFR reduction: proportional to removed hours, so LTIFR drops to ~ 0.30 . Actual reduction observed after full deployment: roughly 70 %—consistent with the model. The engineering work to realize this benefit is substantial, but the safety case is as clean as any in autonomous driving.

7.9 Industrial Deployment

Autonomous haulage in mining is one of the most commercially mature AV verticals. Caterpillar’s Cat Command for Hauling system and Komatsu’s FrontRunner Autonomous Haulage System (AHS) have been deployed in production mining operations for over a decade; Komatsu reported in 2024 that its AHS fleet had collectively hauled multiple billion tonnes since 2008. Rio Tinto’s Mine of the Future programme, launched in the early 2010s, deployed autonomous haul trucks at scale at iron ore mines in Western Australia. BHP, Fortescue Metals Group, and Vale have likewise deployed autonomous mining fleets at large operations. Hitachi Construction Machinery, Sandvik (drilling and underground), and Epiroc are major suppliers in this space. The mining environment—closed sites, defined road networks, controlled access—is a more constrained operational design domain than public-road robotaxis, which has supported earlier and more reliable commercial deployment.

7.10 Chapter Summary

Mining is the leading real-world deployment domain for autonomous vehicles. Open-pit haul-truck autonomy is mature (Pilbara deployments running since 2008); underground autonomy advancing (Codelco, Northparkes); drill-rig and loader autonomy at various stages of maturity. Technical adaptations from on-road AV include dust- and weather-resistant perception, GPS-denied localization underground, off-road control, and centralized fleet management. Economics are quantifiably favorable: payback periods of 1–2 years. Safety benefits are stark: operator deaths have fallen substantially in automated operations. Mining demonstrates that autonomous vehicles, given the right operational context, can deploy at scale and deliver both economic and safety benefits. Lessons from mining inform what “right operational context” means for other domains as well.

7.11 Exercises

1. Compare the long-tail problem between mining AV and public- road AV. Which specific sources of tail events are eliminated in mining?
2. For the dust example (7.1), propose a fusion weighting scheme that automatically adjusts between LiDAR and radar based on observed dust conditions.
3. List three specific technical differences between open-pit and underground mining autonomy. For each, describe the engineering response.
4. Design a dispatch policy for a mine with 50 trucks, 3 loading units, and 2 dumping points. Optimize for throughput with maximum queue wait of 2 minutes.
5. For the ROI example (7.3), compute the break-even commodity price at which deployment no longer pays back within 3 years.
6. A mine's haul road washes out after heavy rain. Describe how autonomous trucks should detect and respond.
7. Propose a beacon placement strategy for an underground network (beacon cost, expected error, worst-case failure mode).
8. Identify three technical borrowings from mining AV that could benefit on-road AV development.
9. For the haul-cycle state machine (7.3), add a "refueling" state and describe its entry and exit conditions.
10. Compare the safety case for mining autonomy to the safety case for urban-robotaxi autonomy. Identify the specific differences that make mining easier.

Chapter 8

Collaboration with Drone Technology

8.1 Introduction

Every chapter in this book has treated autonomy as a *ground-level* problem: a vehicle moving through a 2D world, using sensors that see out to the horizon. But the world is three-dimensional. Unmanned aerial vehicles—drones, UAVs—see it from above, with unobstructed sightlines, coverage of areas no ground vehicle can reach, and perspectives that dramatically change the tradeoffs of perception, mapping, and coordination. Combining autonomous ground vehicles with autonomous aerial vehicles produces capabilities neither can achieve alone: pre-mapping of an incident scene before responders arrive, cooperative perception around occlusions, last-mile logistics across difficult terrain, and continuous overwatch of a convoy or a work site.

This chapter develops the growing field of ground-air autonomous cooperation. It is not a drone-engineering chapter (that would be a separate book) but a chapter on the *integration* of drones with ground autonomous vehicles—the communication, coordination, perception, and planning interfaces that let the two work together.

Three observations motivate the chapter’s structure.

First, drones complement ground vehicles in specific ways. Drones see far, see through occlusion, and cover wide areas quickly; ground vehicles carry payloads, operate in weather that grounds drones, and interact with road infrastructure. A combined system uses each where it is strong.

Second, the technology is *maturing quickly* but the *regulatory environment is the binding constraint*. Drone operations over populated areas are heavily regulated in most jurisdictions; beyond-visual-line-of-sight (BVLOS) operation still requires waivers or specific certifications. A production ground-air system design must accept and navigate these constraints.

Third, drone–ground integration is *architecturally new*. The existing AV software stack was not designed to incorporate aerial inputs; the existing drone stack was not designed to cooperate with ground vehicles. Integration architecture—including shared perception, cooperative planning, and

handoff—is an active design space without canonical answers yet.

The chapter is organized as follows. We first introduce the drone capabilities relevant to ground-vehicle cooperation. We then develop specific cooperation patterns: aerial overwatch, cooperative perception across occlusions, pre-mapping and reconnaissance, delivery handoff, and traffic monitoring. We cover the communication stack linking drones and ground vehicles, the shared-perception architecture, latency and coordination challenges, and the regulatory landscape. We close with case studies and an outlook. Two architecture diagrams visualize the cooperation topology and the shared-perception architecture; worked examples cover specific coordination scenarios.

8.2 Drone Capabilities Relevant to Ground AVs

Before discussing integration, we need to understand what drones bring to the joint system. This section covers the specific capabilities that complement ground-vehicle limitations.

8.2.1 Aerial Perspective

A drone at 50 m altitude sees a much larger area than a ground vehicle can. Its sightlines are mostly unobstructed by buildings, vegetation, and other ground obstacles. The coverage geometry is fundamentally different.

Ground Coverage

A drone with a 60° FoV camera at 50 m altitude covers approximately 58×43 m directly below. Moving at 15 ms^{-1} , it can survey $38\,700 \text{ m}^2 \text{ min}^{-1}$ —more than a ground vehicle can traverse. Aerial survey is effectively a bandwidth- multiplier for perception.

Occlusion Bypass

A pedestrian behind a parked truck is invisible to a ground-level camera; a drone above sees them. This occlusion-bypass is the single strongest perception benefit of drone collaboration for ground AVs.

8.2.2 Rapid Deployment

A drone launched from a ground vehicle or roadside station can be over a target area within seconds to minutes, depending on range. No ground vehicle can reach a remote incident as quickly.

8.2.3 Payload Delivery

Drones can carry modest payloads (1 kg to 10 kg) over distances a ground vehicle cannot practically traverse (mountainous terrain, flooded roads, isolated communities). Integration with ground-vehicle logistics networks is an active business area (Zipline for medical deliveries is the canonical example).

8.2.4 Limitations

Drones cannot operate in all weather (most are grounded by rain, strong wind, or low visibility), have limited flight endurance (typically 20 min to 40 min for small drones), and are heavily regulated. A joint system relies on ground vehicles for continuous, weather-tolerant operation and on drones for *enhancement* when conditions allow.

8.3 Cooperation Patterns

Specific operational patterns define how drones and ground vehicles work together. This section catalogs the main patterns.

8.3.1 Aerial Overwatch

A drone maintains position above a ground vehicle or convoy, providing an extended-perception view.

Use Cases

- Military and security: convoy protection, situational awareness.
- Emergency response: incident-scene awareness for responding ground units.
- Remote operations: oversight for autonomous vehicles in mining, construction.
- Logistics: in-transit monitoring of high-value cargo.

Technical Requirements

Overwatch requires the drone to track the ground vehicle's position, maintain a stable offset, and stream video or detection output back to the ground vehicle or to an operations center. Typical data rates: 10 Mbits s^{-1} for HD video; 1 Mbits s^{-1} for detection metadata.

8.3.2 Cooperative Perception

A drone shares its perception output with a ground vehicle, extending the ground vehicle's effective perception coverage.

Occlusion Scenarios

- Pedestrian behind parked vehicle at an intersection.
- Vehicle stopped around a blind curve.
- Obstacle on a blind hill crest.
- Rear-approaching emergency vehicle in heavy traffic.

Each of these is hard for ground sensors; trivial for an aerial view.

Example 8.1 (Occlusion-bypass detection). Urban intersection. Ground AV approaching from the east; a parked delivery truck occludes the northbound lane. A pedestrian steps off the curb in the northbound lane, about to enter the intersection.

- Ground-only perception: cannot detect pedestrian; plans to proceed through intersection.
- With drone-overwatch: drone at 40-m altitude sees the pedestrian; broadcasts detection at latency 15 ms (via edge-network); ground AV receives, integrates into scene, begins yielding.
- Net effect: hazard detected 1.5 s before it would have been in line-of-sight. At 10 ms^{-1} ego speed, 15 m of reaction distance gained—enough to yield comfortably.

8.3.3 Pre-Mapping and Reconnaissance

A drone surveys an area before a ground vehicle enters, delivering an up-to-date local map of the scene. Applications: disaster response (mapping a flooded area before ground vehicles commit), emergency response (mapping an incident scene for responding vehicles), construction (mapping a work site for autonomous equipment planning).

Example 8.2 (Disaster pre-mapping). Flash flood in a region. Pre-existing HD maps are stale (roads damaged, bridges down). Drone launches from a command vehicle, surveys 5 km^2 of area over 20 minutes, produces an updated map with:

- Road passability (from aerial imagery).
- Water levels (from photogrammetric elevation).
- Stopped vehicles and ad-hoc obstructions.
- Debris locations.

Map is distributed to responding autonomous vehicles. Without drone pre-mapping, ground vehicles would either refuse to enter (ODD violated) or proceed with severely degraded performance. With it, they can operate within acceptable risk envelopes.

8.3.4 Last-Mile Delivery Handoff

A ground vehicle carries a package most of the way; a drone completes the last mile to a doorstep or a difficult-access location. The patterns are still evolving; early commercial deployments handle specific geometries (suburban cul-de-sacs, rural homes).

8.3.5 Traffic Monitoring

Drones stationed above arterial roads or intersections provide traffic-flow data that AVs can consume for planning purposes. Signal-phase optimization, incident detection, and congestion mapping are all use cases.

8.4 Air–Ground Communication Stack

Ground–air cooperation requires a communication layer built on top of the AV network stack (Chapter 1).

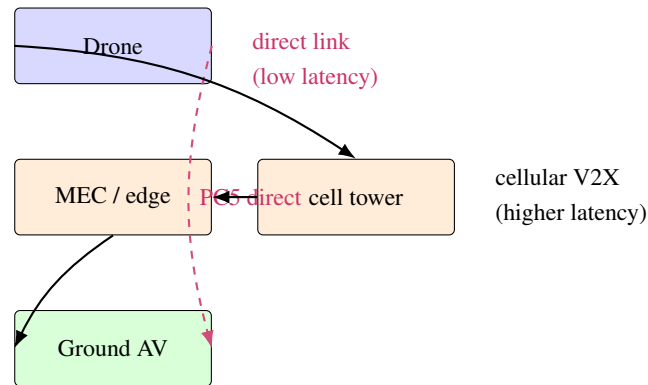


Figure 8.1: Air–ground communication architecture. The primary low-latency channel is direct drone-to-ground (via PC5 sidelink or equivalent peer-to-peer radio) with 5 ms to 10 ms end-to-end latency. A secondary cellular path through the edge (MEC) supports higher-latency, higher-bandwidth traffic (video streams, map updates). Safety-critical coordination uses the direct link; bulk data uses the cellular path.

8.4.1 Direct Device-to-Device Links

Low-latency cooperation requires direct communication between drone and ground vehicle, bypassing the cellular core network. C-V2X PC5 sidelink (Chapter 1) handles this role. Direct-link latency: 5 ms to 10 ms end-to-end.

8.4.2 Cellular + Edge

For non-time-critical traffic (video streaming, map distribution, coordination logs), cellular with edge-compute endpoints (MEC) provides sufficient bandwidth and adequate latency.

8.4.3 Broadcast vs Unicast

Broadcast protocols (the ground AV subscribes to a local spatial channel) scale to multiple drones or multiple ground vehicles better than unicast. Standard vehicular broadcast patterns (BSM, CAM, CPM) extend to include air-originated messages.

8.5 Shared Perception Architecture

When a drone and a ground vehicle share perception, the resulting data must be fused coherently. The fusion architecture is the core design decision.

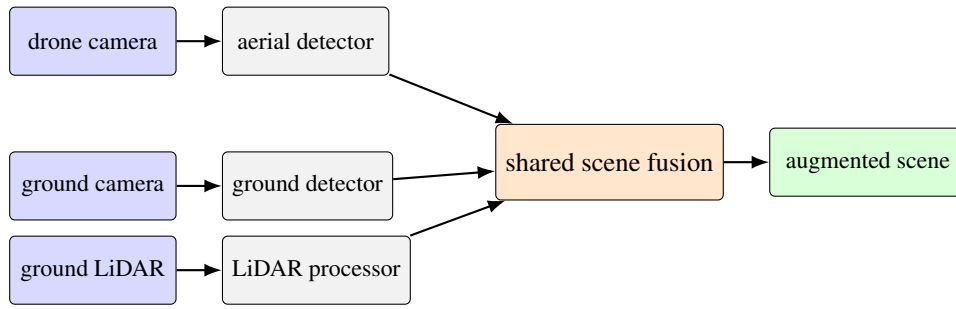


Figure 8.2: Shared-scene fusion for air–ground cooperation. The drone produces aerial detections in a geo-referenced coordinate system; the ground vehicle produces ground-level detections from cameras and LiDAR. Both flow into a fusion module that aligns them spatially (using shared geo-reference), associates them temporally (matching observations of the same objects), and produces an augmented scene including objects visible only from one vantage or the other. Output flows to planning.

8.5.1 Geo-Reference Alignment

Drone and ground vehicle each know their own positions in a common frame (GNSS-referenced). Aerial detections, annotated in image coordinates, are projected into the ground frame using drone pose and intrinsic calibration. Errors (0.5 m to 1.5 m typical) must be tracked as uncertainty in the shared scene.

8.5.2 Temporal Association

Aerial and ground detections must be matched to the same objects. Matching uses position, velocity, class, and any available feature-level similarity. Unique-object identification in the shared scene requires robust association; mismatches produce phantom objects.

Example 8.3 (Association under latency). Drone detects a pedestrian at time t at geo-position (x, y) with velocity (v_x, v_y) . Transmission and processing latency before the ground AV uses the detection: 200 ms (typical for video-based aerial detection going over cellular). At observed velocity 1.5 m s^{-1} , the pedestrian has moved 0.3 m by the time of use; the ground AV predicts forward using the velocity estimate, minimizing latency-induced position error. Without forward prediction, the ground AV would see the pedestrian at a stale position; emergency maneuvers might be timed incorrectly. Latency awareness is a first-class design concern in air–ground fusion.

8.5.3 Trust and Verification

A drone broadcasting detections could be malicious or malfunctioning. Ground AVs should:

- Verify detections cryptographically signed by trusted drones.
- Cross-check with local perception where fields of view overlap.
- Discount or ignore detections that contradict local perception in overlap regions.
- Treat drone detections as enhancing local perception, not replacing it.

8.6 Regulatory Constraints

Drone operations are heavily regulated. The regulatory landscape is the single largest constraint on practical air–ground cooperation.

8.6.1 Beyond Visual Line of Sight (BVLOS)

Most commercial drone use today requires visual line of sight from an operator. BVLOS operation, necessary for autonomous cooperation with AVs over any meaningful distance, requires special waivers (US Part 107 BVLOS), specific certifications (EU SORA), or operational-area approvals.

8.6.2 Urban Flight Restrictions

Many jurisdictions heavily restrict urban drone flight. Temporary flight restrictions, altitude caps, and prohibitions over people limit where drones can fly. Ground-air cooperation designs must respect these.

8.6.3 Certification Paths

- FAA Part 107 (US): operators certified; operations restricted by default; waivers for specific operations.
- FAA Part 135 (US): commercial delivery operations; more stringent.
- EU SORA: Specific Operations Risk Assessment framework.
- UAS traffic management (UTM): emerging systems for drone airspace coordination.

8.7 Case Studies

Commercial and operational deployments illustrate what works and what doesn't.

8.7.1 Zipline

Zipline operates fixed-wing delivery drones over BVLOS routes in Rwanda, Ghana, and several US markets. Drones complement ground logistics by reaching destinations where road infrastructure is poor. Handover between drone and ground has been a key operational innovation.

8.7.2 Military Applications

DoD programs have explored AV-drone collaboration for convoy protection and autonomous reconnaissance. Many details are not public but the problems (cooperative perception, coordination, communications under contested conditions) parallel commercial deployments.

8.7.3 Inspection and Monitoring

Infrastructure inspection (power lines, pipelines), agricultural monitoring, and construction-site oversight all combine drones with ground vehicles in some form. Coordination is usually loose (drones report to central operations rather than directly to ground vehicles); tighter integration is emerging.

8.7.4 Emergency Response

Several fire and disaster-response agencies have integrated drone pre-mapping with ground-vehicle dispatch. Production deployments are limited but growing.

8.8 Worked Examples

Example 8.4 (End-to-end cooperation scenario). Autonomous delivery ground vehicle approaches a parking lot it has not mapped in detail. A companion drone launches from the vehicle's rooftop, surveys the lot for 90 seconds, produces a local occupancy map and identifies parked-vehicle poses. Map is returned to the ground vehicle over direct C-V2X link. Ground vehicle navigates to its drop-off point using the drone-derived map as a geometric prior; on-vehicle perception handles dynamic obstacles. Net improvement: time to navigate the lot is reduced by roughly 40 % vs unmapped navigation; safety margin is higher. Drone returns to vehicle and charges during the delivery stop.

Example 8.5 (Air-ground handoff latency). Drone detects a hazard at time T . Timeline:

- $T + 0$: aerial detection.
- $T + 15$ ms: encoded, transmitted via PC5.
- $T + 20$ ms: received by ground vehicle.
- $T + 25$ ms: fused into local scene.
- $T + 28$ ms: planning reacts.
- $T + 33$ ms: control command updated.

Total latency: 33 ms—negligible for speeds of 10 ms^{-1} (ground vehicle travels 0.33 m during this). Compare to 4G path where transmission alone adds ~ 50 ms: total latency 65 ms to 80 ms; ground vehicle travels 0.65 m to 0.8 m. The difference is tactical, not strategic; but in edge cases it matters.

Example 8.6 (Fleet cost–benefit for drone augmentation). AV delivery fleet: 100 vehicles. Drone augmentation: 20 drones shared across the fleet (drones deploy on-demand to support specific deliveries). Cost: 20 drones at \$25k each = \$500k capital; annual operating cost \$200k. Benefit: 30 % of deliveries benefit from drone pre-mapping or occlusion-bypass; 5 % improvement in completion time, 2 % reduction in interventions. At 500,000 deliveries/year: 7,500 faster deliveries, reduced intervention cost \$200k/year. Break-even: ~ 2 years for the augmentation, with strong operational benefits thereafter. Not all fleets will deploy drone augmentation; for

those that do, the economics improve over time.

8.9 Industrial Deployment

Autonomous drone systems span consumer, commercial, and military domains. DJI (China) dominates the consumer drone market and produces enterprise platforms (Matrice series) used in inspection, mapping, and emergency response. Skydio (United States) builds autonomous drones for inspection and defence applications. Zipline (United States) operates autonomous delivery drones at scale in Rwanda, Ghana, Japan, and the United States, having delivered millions of medical packages by 2024. Amazon's Prime Air service began commercial U.S. drone delivery operations in 2022. Wing (Alphabet) operates drone-delivery services in Australia, Finland, and the United States. The U.S. Federal Aviation Administration's Part 107 rules govern small commercial drone operation; Beyond Visual Line of Sight (BVLOS) operations have been the subject of expanding regulatory frameworks worldwide.

8.10 Chapter Summary

Drones extend autonomous-vehicle capability by providing aerial perspective, rapid deployment, small-payload delivery, and reconnaissance in areas ground vehicles cannot reach quickly. Cooperation patterns include aerial overwatch, cooperative perception across occlusions, pre-mapping for reconnaissance, last-mile delivery handoff, and traffic monitoring. Communication stacks combine direct PC5 links for low-latency coordination with cellular edge paths for bulk data. Shared-perception architectures fuse aerial and ground detections with careful attention to coordinate alignment, temporal association, and latency. Regulatory constraints (BVLOS, urban flight restrictions) are the main practical limitation; the technology often outpaces what can be legally operated. Case studies in delivery, military, and emergency response illustrate working deployments. Air-ground cooperation is an active frontier where the existing AV stack is extended, not replaced; the architecture is young but the operational wins are real.

8.11 Exercises

1. Compute the ground footprint of a drone at 100 m altitude with a 90° FoV camera.
2. For the occlusion-bypass example (8.1), compute the ego-speed at which the 15 m of additional reaction distance is equivalent to 1 second of lead time.
3. Compare drone pre-mapping vs crowd-sourced HD map updates for a disaster-response scenario. List three advantages of each.
4. Design a shared-perception architecture where drone and ground vehicle disagree about a detection. Propose three conflict-resolution rules.

5. For the air–ground handoff latency example (8.5), estimate the latency on a WiFi-only link (no cellular, no PC5). At what point does the link become unsuitable for safety-critical cooperation?
6. Identify three technical challenges specific to drone operation in high winds, and propose responses each.
7. Design a credential/trust system for drone-broadcast detections in an urban environment with tens of drones operating simultaneously.
8. For the fleet cost–benefit example (8.6), compute the break-even intervention-cost at which drone augmentation always pays back.
9. Propose a BVLOS operational scenario that would pass current EU SORA approval. Identify at least three specific mitigations required.
10. Outline a shared-perception protocol for a convoy of three ground vehicles and one overwatch drone operating on a highway.

Chapter 9

Autonomous Ships and Maritime Autonomy

9.1 Introduction

The broader story of vehicle autonomy extends beyond road vehicles. Maritime autonomy—Maritime Autonomous Surface Ships (MASS) in IMO parlance—is advancing on a parallel track, with distinct characteristics that make it both easier in some respects and harder in others than road autonomy. This chapter surveys maritime autonomy for AV engineers, with emphasis on what is similar, what is different, and what lessons transfer in each direction.

Three observations motivate the chapter.

First, maritime autonomy is *more advanced* than many assume. Unmanned surface vessels operate routinely in military applications; commercial ferry trials are live in Norway and Japan; fully-autonomous cargo ships are in late-stage testing. The road-AV community often overlooks these parallel developments.

Second, the *operational environment differs fundamentally*. Oceans are less populated; rules of navigation (COLREG) are codified; behavior horizons are minutes rather than seconds. Some problems (infrastructure mapping) are easier; others (weather, long horizon, communication) are harder.

Third, *cross-transfer is already happening*. Autonomy techniques developed for road AVs (sensor fusion, deep perception, control) transfer to maritime; maritime lessons (voyage-scale planning, weather integration, communication-denied operation) inform AV development in turn.

9.2 MASS Degree-of-Autonomy Levels

The International Maritime Organization (IMO) has defined four degrees of MASS autonomy:

Degree	Description
1	Crew on board; some functions automated
2	Remote controlled from shore; crew on board
3	Remote controlled; no crew on board
4	Fully autonomous; decisions made by the ship

This parallels the SAE J3016 levels for road vehicles but is distinct in that degree 2 (remote-controlled) is a recognized commercial stage. Shore-based remote operation is more tractable for ships than for cars because the decision horizon is longer and the bandwidth burden is lower.

9.3 Maritime Perception

9.3.1 Primary Sensors

Maritime perception uses different sensor emphasis than road AVs:

Sensor	Maritime role
Radar (X-band)	Primary long-range detection (10–30 nm)
AIS (Automatic Identification System)	Cooperative vessel location broadcasts
ECDIS (electronic charts)	Bathymetry, infrastructure, navigation aids
Camera + IR camera	Visual identification, close-in obstacles
LiDAR	Close-range (0–300 m) obstacle detection; emerging
Sonar (forward-looking)	Submerged obstacle detection
GNSS + IMU	Position and attitude (much cleaner than road)

9.3.2 Perception-Range Differences

Radar dominates because the relevant detection range is kilometers, not tens of meters. Other vessels are cooperative (AIS) and send their positions directly. Maritime perception therefore looks more like fusing cooperative and non-cooperative tracks than like dense scene parsing.

Example 9.1 (Radar detection range budget). X-band maritime radar typical range:

Target types and detection ranges:

- Large vessel (RCS 1000+ m²): 30+ nautical miles.
- Small vessel (RCS 10 m²): 8–12 nautical miles.
- Buoy (RCS 1 m²): 3–6 nautical miles.
- Small boat (RCS 0.1 m²): 1–2 nautical miles.

- Person in water (RCS $<0.01 \text{ m}^2$): 100–300 m, conditions-dependent.

At cruising speed 20 knots ($\sim 10 \text{ m/s}$) a 3 nm (5.5 km) detection distance provides ~ 9 minutes of planning horizon to the smallest cooperating targets. Compare road AVs where 200 m detection at 30 m/s is 7 seconds—two orders of magnitude shorter horizon.

9.4 COLREGs: Maritime Traffic Rules

9.4.1 Convention on the International Regulations for Preventing Collisions at Sea

COLREGs are the “rules of the road” for ships. They codify give-way responsibilities and encounter maneuvers. Key rules relevant to autonomy:

Rule	Content
Rule 13	Overtaking vessel gives way
Rule 14	Head-on: both alter course to starboard
Rule 15	Crossing: vessel with other on starboard gives way
Rule 16	Give-way vessel takes early, substantial action
Rule 17	Stand-on vessel maintains course until avoidance becomes necessary

Autonomous ships must implement COLREG-compliant behavior that human mariners will recognize as appropriate. This is implementation of explicit rules, in contrast to road AVs which must often infer human conventions from behavior.

9.4.2 Formal Verification Challenge

Because COLREGs are fully codified, formal verification of autonomous-ship behavior is more tractable than road AV verification. Academic and industrial projects have produced model-checked COLREG-compliance proofs for specific autonomous ships.

9.5 Voyage-Scale Planning

A commercial ocean voyage is 5–15 days. Planning at voyage scale is a different problem than road trip planning:

9.5.1 Weather Routing

Ships optimize routes against forecast wind, current, and wave conditions. The chosen route might add distance to avoid adverse weather, adding fuel efficiency and safety.

9.5.2 Economic Optimization

Voyage economics depend on fuel, cargo value, port fees, ETA contracts. Autonomous ships can optimize these jointly with reliability margins.

Example 9.2 (Fuel savings from weather routing). A trans-Pacific voyage of 5000 nm at 18 knots nominal speed.

Straight-line voyage:

$$\begin{aligned} t_{\text{direct}} &= 5000/18 \\ &\approx 278 \text{ hours} \approx 11.6 \text{ days,} \\ \text{fuel}_{\text{direct}} &\approx 800 \text{ tonnes.} \end{aligned}$$

Weather-routed voyage (5 % longer in distance, favorable conditions, 12 knots average speed through weather):

$$\begin{aligned} t_{\text{routed}} &\approx 13.3 \text{ days,} \\ \text{fuel}_{\text{routed}} &\approx 720 \text{ tonnes.} \end{aligned}$$

Savings.

$$\begin{aligned} \Delta \text{fuel} &= 80 \text{ tonnes,} \\ \Delta \text{cost} &\approx 80 \times 600 \text{ USD/tonne} \\ &= 48\,000 \text{ USD.} \end{aligned}$$

A 2-day delay is acceptable for a significant fuel saving; optimization depends on cargo urgency. Autonomous ships can evaluate the tradeoff continuously using updated weather forecasts.

9.6 Harbor Operations and Close-Quarter Maneuvering

9.6.1 The Autonomy Bottleneck

Open-ocean cruise is the easiest portion of a voyage to automate; harbor operations are the hardest. Close-quarter maneuvering among other ships, limited visibility, complex geometry, and time-critical communications all combine.

Current operational models:

- Autonomous cruise with pilot boarding for harbor.
- Remote-operated harbor phase from shore control centers.
- Tug-assisted berthing with remote operator.

9.6.2 Ferry Autonomy

The most deployed commercial MASS applications are short-range ferries: Finferries' Falco in Finland, Yara Birkeland in Norway, and various Japanese trials. Ferries benefit from repeated identical routes, well-mapped environments, and cooperative operators.

9.7 Lessons for Road AVs

Maritime lesson	Road-AV application
Codified rules (COLREG) enable formal verification	Pursuit of verifiable road-behavior specifications
Cooperative identification (AIS)	V2V/V2X as first-class sensor modality
Long planning horizon	Better voyage-style route optimization
Remote-operated intermediate stage	Teleoperation as a permanent tier, not just fallback

9.8 Industrial Deployment

Maritime autonomy is governed by IMO (International Maritime Organisation) frameworks, including the Maritime Autonomous Surface Ship (MASS) regulatory scoping exercise begun in 2017 and ongoing development of an MASS Code expected to be finalised in the late 2020s. Notable production and pilot deployments include the Yara Birkeland (Norway), the world's first fully-electric autonomous container ship, which began commercial operation in 2022; Mitsui O.S.K. Lines and Nippon Yusen K.K. have demonstrated autonomous coastal navigation; Avikus (Hyundai Heavy Industries) has demonstrated autonomous ocean crossings. The Autonomous Vessel Ship Operating System and Sea Machines Robotics provide commercial autonomy products. Maritime autonomy benefits from generally lower agent density than urban driving but faces unique challenges including weather, COLREGs compliance, and limited communications.

9.9 Chapter Summary

Autonomous ships share much with autonomous road vehicles but operate in a distinctive environment: longer horizons, fewer actors, codified rules (COLREG), cooperative identification (AIS), and more separable operational phases (open ocean, coastal, harbor). The IMO's MASS degree-of-autonomy framework parallels SAE levels with the important difference that remote operation is a recognized commercial tier. Maritime perception emphasizes radar and cooperative tracks; control emphasizes voyage-scale optimization and weather routing; close-quarter harbor operations remain the principal autonomy bottleneck. Cross-pollination with road AVs is substantial and growing.

9.10 Exercises

1. For the radar-range example, compute detection time for a small boat at a relative approach of 30 knots.
2. Compare COLREG Rule 15 (crossing) with a road-AV right-of-way rule at an unsignalized intersection. What's transferable?
3. For the weather-routing example, compute break-even fuel savings if voyage delay is 3 days.
4. Design a harbor-autonomy system that uses remote operation for the berthing phase while the cruise is autonomous.
5. Identify three maritime sensor modalities not commonly used on road AVs and their road-AV equivalents.
6. Discuss the formal-verification tractability advantage of COLREG over road traffic rules.
7. Design an AIS equivalent for road vehicles. What data would be broadcast, and at what cadence?
8. For autonomous cargo ships, compute the minimum fleet size at which crew-cost savings offset autonomy development cost.
9. Compare MASS Level 4 and SAE Level 4: identify three concrete operational differences.
10. Propose an autonomy architecture for a short-range ferry, specifying perception, planning, and fail-operational modes.

Chapter 10

Electric Vehicles and Autonomous Driving

10.1 Introduction

Electric vehicles (EVs) and autonomous vehicles (AVs) are converging. Most planned commercial robotaxi and Level 4 platforms are electric by design, and conversely most battery-electric fleet vehicles are candidates for autonomy. The convergence is not accidental—EVs enable AV economics in several concrete ways, and AV software benefits from the finer control authority of electric drivetrains. But the combination also introduces new constraints: energy management, charging logistics, and battery health become first-class planning concerns.

Three observations motivate the chapter.

First, EV drivetrains are *easier* to interface with than ICE. Electric motors produce torque at arbitrary speeds without transmission logic, respond to commands with millisecond-scale latency, and have no cold-start or idle behavior to manage. The control problem the AV stack solves is strictly simpler on EVs.

Second, the *energy budget* is a new first-class planning variable. AV planners on EVs cannot ignore state-of-charge (SoC); routes must be feasible within range, and high-duty fleets require charging integrated into operations.

Third, EV fleets open new *thermal and cost* regimes. Batteries must be cooled; range degrades in cold weather; charging economics feed back into per-mile costs. Joint AV-EV optimization is a newer discipline than either technology alone.

10.2 EV Drivetrain Basics for AV Engineers

10.2.1 Electric Motor Response

An electric motor produces torque $T(\omega)$ as a function of angular speed, controlled by the inverter. Response from commanded to actual torque: typically 1–5 ms—orders of magnitude faster than ICE drivetrain response (50–200 ms).

Example 10.1 (Control-loop response: EV vs ICE). Consider a 100-Hz outer-loop control commanding longitudinal acceleration.

ICE response.

$$t_{\text{delay}} \approx 150 \text{ ms},$$

$$\text{loop iterations before response} = 15.$$

Controller must compensate with predictive terms to avoid oscillation.

EV response.

$$t_{\text{delay}} \approx 3 \text{ ms},$$

$$\text{loop iterations before response} = 0.3.$$

Effectively instantaneous at the loop rate; simpler controller suffices and gains higher bandwidth.

EV drivetrains let the planner be aggressive without sacrificing comfort; production AV programs on EVs routinely run tighter trajectory following.

10.2.2 Regenerative Braking

An electric motor can function as a generator, converting kinetic energy back to battery charge. Regenerative braking supplements friction brakes, recovering roughly 10 % to 25 % of urban kinetic energy. AV controllers blend mechanical and regenerative braking; most routine deceleration is regenerative.

10.2.3 Single-Pedal Driving

Strong regeneration can bring the vehicle to a stop without friction-brake application. AV software uses this mode for smooth, predictable deceleration profiles; passengers experience smoother rides than on friction-only systems.

10.3 Energy Budget as a Planning Variable

10.3.1 State of Charge (SoC)

SoC is the fraction of usable energy remaining in the battery (0–1). Range at a given operating profile:

$$R_{\text{remaining}} = \text{SoC} \cdot E_{\text{usable}} / \bar{e}, \quad (10.1)$$

with $\bar{e}$ the specific energy consumption (Wh/km).

Example 10.2 (Range calculation for a mid-route check). Fleet robotaxi with 77 kWh battery, current SoC = 0.35.

Usable energy at current SoC.

$$\begin{aligned} E_{\text{avail}} &= 0.35 \times 77 \text{ kWh} \\ &= 27.0 \text{ kWh}. \end{aligned}$$

Specific consumption at current speed profile.

$$\bar{e} = 180 \text{ Wh/km} \quad (\text{urban, with climate control}).$$

Remaining range.

$$\begin{aligned} R &= \frac{27,000}{180} \\ &= 150 \text{ km}. \end{aligned}$$

Route check. Remaining trip distance before charger: 45 km. Energy needed:

$$\begin{aligned} E_{\text{trip}} &= 45 \times 180 \\ &= 8.1 \text{ kWh}. \end{aligned}$$

A 19 kWh safety buffer remains. Proceed normally; re-evaluate after next passenger drop-off.

10.3.2 Consumption Under Different Conditions

Specific consumption depends on conditions:

Condition	Wh/km (typical)
Highway, mild	150
Urban, mild	170
Highway, cold (-10°C)	230
Urban, cold with cabin heat	280
Hot with aggressive A/C	220

Winter in cold climates can reduce range by 30–40 %. Fleet dispatch algorithms account for this explicitly.

10.4 Charging in Fleet Operations

10.4.1 Charging Strategies for AV Fleets

Four patterns dominate:

Strategy	Mechanism	Best for
Depot charging	Return to depot for slow/mid-rate charging	Overnight fleets
Opportunistic	Charge during natural idle windows	Urban robotaxi
DC fast on-route	Short 150 kW+ sessions between trips	High-utilization
Battery swap	Exchange discharged pack for fresh	Specific OEM ecosystems (e.g. NIO)

10.4.2 Charge-Scheduling Optimization

Fleet-level dispatch optimizes jointly over passenger service and charging. Constraints: no vehicle runs below minimum SoC; charger capacity bounds; time-of-use electricity prices. Objective: maximize revenue-mile availability minus charging cost.

Example 10.3 (Charging cost per mile). Fleet vehicle consuming 200 Wh km^{-1} , charged at 0.15 USD/kWh off-peak rate.

Energy cost per mile.

$$\begin{aligned}
 \text{cost/mile} &= 0.200 \text{ kWh/km} \times 1.609 \text{ km/mi} \times 0.15 \text{ USD/kWh} \\
 &\approx 0.048 \text{ USD/mi} \\
 &= 4.8 \text{ cents/mi.}
 \end{aligned}$$

Compare ICE equivalent.

$$\begin{aligned}\text{ICE cost/mile} &\approx 30 \text{ mpg} \rightarrow \text{gal/mile} = 0.033 \\ &\times 3.50 \text{ USD/gal} \\ &\approx 11.7 \text{ cents/mi.}\end{aligned}$$

EV energy cost is $\sim 40\%$ of ICE; over 100,000 miles/year this is a 6900 USD/vehicle/year operating-cost advantage—large at fleet scale.

10.5 Battery Health and AV Duty Cycles

AV duty cycles are aggressive: many cycles per day, heavy fast-charging, year-round operation. Each stresses the battery.

10.5.1 Cycle Degradation

Lithium batteries lose capacity with usage. Typical production battery: 80% retained capacity after 1000–2000 deep cycles. High-duty AV operation can accelerate degradation without proper thermal management.

10.5.2 Thermal Management

Active cooling and heating maintain battery temperature within the optimal range (15 °C to 35 °C). This is energy-expensive but extends battery life substantially. AV thermal-management budget is a sensible investment.

10.6 Industrial Deployment

The convergence of electric and autonomous vehicles is a defining trend. Tesla, BYD, NIO, XPeng, Li Auto, Rivian, and Lucid build their ADAS and AD systems atop electric platforms that provide native by-wire control characteristics suited to autonomy. Waymo's robotaxi fleet uses Jaguar I-PACE and Hyundai Ioniq 5 electric vehicles. Cruise's Origin platform was an electric purpose-built shuttle. The CHAdeMO, CCS Combo, and NACS (originally Tesla, now standard for many North American manufacturers) charging standards govern fast-charging interfaces. Battery costs, on a kilowatt-hour basis, declined substantially through the 2020s based on data published by BloombergNEF. The integration between vehicle-to-grid (V2G) and autonomous fleet operations is the subject of pilot programs from Nissan, Ford, Volvo, and others.

10.7 Chapter Summary

EV drivetrains simplify AV control (millisecond torque response, regenerative braking, single-pedal smoothness) while introducing new planning variables (SoC, charging, thermal management). The convergence of EV and AV is driven by favorable joint economics and by the cleaner control interface

EVs offer. Fleet-level charging orchestration and battery-health management become production disciplines. Per-mile energy cost is typically half that of ICE, contributing to the unit economics that make robotaxi services viable.

10.8 Exercises

1. For the range example, compute remaining range if SoC drops to 0.15 and consumption rises to 220 Wh/km.
2. Derive the break-even electricity price at which EV and ICE per-mile costs are equal, using the example parameters.
3. Compare depot charging and opportunistic charging for a robotaxi fleet. Which suits higher utilization?
4. Design a charge-scheduling constraint for a 500-vehicle fleet with 20 chargers.
5. Compute the range penalty in winter for a 30 % increase in consumption, starting from 150 km baseline.
6. Propose a thermal-management policy that balances battery life against energy consumed.
7. Describe three AV-specific sensor fusion tasks that benefit from EV-level control bandwidth.
8. Compare regenerative braking to friction braking in terms of passenger comfort. Propose a blending rule.
9. Identify three specific constraints that EV operation adds to AV fleet dispatch.
10. Discuss battery-swap operations: under what conditions are they preferable to DC fast charging?

Part V

Engineering at Scale

Chapter 11

Middleware for Autonomous Driving: ROS and DDS

11.1 Introduction

Every chapter until now has developed algorithms—perception, planning, control. Those algorithms must run *together*, on distributed compute, exchanging data at rates, deadlines, and reliabilities the whole stack’s safety depends on. The software layer that carries this data between modules is called *middleware*; it is the communication backbone of an autonomous driving system, and its quality determines whether the algorithms’ theoretical performance survives into production. Middleware is under-discussed in AV literature because it is infrastructure, not a hero; it also is the piece of the stack that production engineers spend the most total hours on.

Three observations motivate the chapter’s structure.

First, AV middleware is a *real-time distributed systems* problem. The perception module produces a detection; the planner consumes it; the deadline is 50 milliseconds. That simple chain is actually a distributed-systems choreography involving serialization, network transport, deserialization, scheduling, and ordering. Missed deadlines do not just degrade performance—they produce safety-critical gaps between what the vehicle thinks is happening and what actually is.

Second, two ecosystems dominate AV middleware: the Robot Operating System (ROS, now ROS 2) and the Data Distribution Service (DDS) standard. ROS 2 is actually built on DDS, so the two are architecturally related, but their deployment profiles differ. ROS offers productivity; DDS offers determinism; modern stacks use layers of each.

Third, middleware is increasingly a *safety* concern. Functional-safety standards (ISO 26262) now address the communication layer directly; cybersecurity standards (ISO 21434) treat middleware as an attack surface. A production-grade AV middleware is engineered, validated, and certified with the same rigor as the algorithms it carries.

The chapter is organized as follows. We first establish the middleware requirements AV systems impose. We then develop ROS and DDS: their architectures, communication models, and real-time capabilities. Inter-module communication patterns common in AV stacks follow. Worked examples demonstrate end-to-end pipeline timing, QoS selection, multi-sensor fusion, bandwidth optimization, and fault tolerance. We close with real-time considerations, security, and the direction middleware is evolving. Three architecture diagrams visualize the ROS node graph, the DDS publish/subscribe model, and the layered AV middleware stack.

11.2 Middleware Requirements for Autonomous Vehicles

Before choosing a middleware, we state what an AV stack needs from it. Each requirement below manifests as a concrete technical specification and as a testable acceptance criterion.

11.2.1 Low Latency

Inner-loop control cycles run at 100 Hz to 200 Hz; a perception-to-planning chain must complete in under 100 ms. The middleware should add only *microseconds*, not milliseconds, to any message transit.

11.2.2 High Throughput

A single forward camera at 4K/30 Hz produces 2.5 Gbits^{-1} ; a full sensor suite easily reaches 20 Gbit s^{-1} aggregate. Middleware must move these volumes without dropping messages.

11.2.3 Reliability

Messages matter. A dropped emergency-brake command or a silently- truncated LiDAR scan is a safety event. Middleware must provide configurable reliability guarantees and surface failures to the application layer rather than swallowing them.

11.2.4 Scalability

A production AV runs dozens of modules on multiple compute nodes. Middleware must support hundreds of simultaneous publish/subscribe relationships without performance collapse.

11.2.5 Determinism

Latency *variance* (jitter) is often more dangerous than latency *mean*. A controller designed for 10 ms steps will misbehave on a 10 ms-mean-20 ms-tail distribution. Middleware must support bounded-jitter transport.

11.3 Robot Operating System (ROS)

ROS is the dominant middleware in robotics research and an important one in AV research stacks; ROS 2 is increasingly deployed in production. This section develops its architecture and the properties

that shape its use.

11.3.1 Architecture

ROS organizes computation into a graph of communicating nodes. Understanding the graph is understanding the ROS way of structuring a system.

Nodes

A node is a process executing a specific function: an image decoder, a detection network, a planner. A typical AV stack has tens of nodes. Each node is independently deployable, restartable, and monitorable.

Topics

Topics are named asynchronous channels. Nodes *publish* messages to a topic; other nodes *subscribe* and receive copies. Topics implement the producer-consumer pattern at scale.

Services

Services are synchronous request-response interactions. A node calls a service, waits for a reply. Useful for occasional control- plane interactions (parameter queries, state changes) rather than high-rate data flow.

Actions

Actions are asynchronous, long-running, goal-directed interactions with progress feedback and cancellation. Used for long-horizon operations like “drive to this coordinate.”

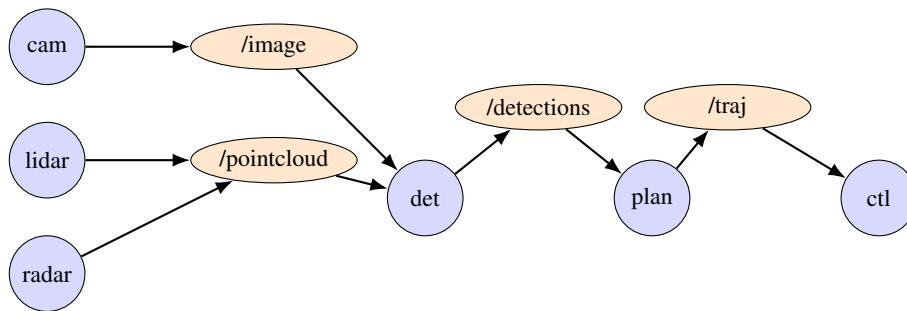


Figure 11.1: ROS node graph for a minimal AV stack. Blue circles are nodes (processes); orange ellipses are topics (named channels). Sensors publish to topics (/image, /pointcloud); processing nodes subscribe, process, and publish derived topics (/detections, /traj). The graph visualizes module coupling. In a production stack there are dozens of nodes and hundreds of topics; ROS tooling (`rqt_graph`, `ros2 topic`) lets engineers inspect and debug them.

11.3.2 Communication Model

A publish/subscribe message on topic T from source S_i at time t delivers to subscribers:

$$M(t) = f(S_i(t)), \quad (11.1)$$

where f is the message-serialization function. ROS 2 message delivery is carried by an underlying DDS implementation (Cyclone DDS, Fast DDS, RTI Connext). QoS settings on each topic control reliability, history, durability, and deadline.

11.3.3 ROS 1 vs ROS 2

ROS 1 dominated robotics research from 2007 through about 2020 but has production limitations: centralized master node, no native security, no real-time guarantees, limited multi-machine scaling. ROS 2 (Foxy, Humble, Iron, Jazzy releases) addresses all of these: decentralized DDS discovery, security plugins, real-time publishers, native multi-machine support. AV programs have migrated or are migrating to ROS 2.

11.4 Data Distribution Service (DDS)

DDS is an OMG standard for publish/subscribe middleware designed specifically for real-time, distributed, mission-critical systems (originally military and industrial). Understanding DDS is understanding what lies beneath ROS 2 and what production stacks sometimes use directly.

11.4.1 Architecture

DDS organizes endpoints in a distributed publish/subscribe domain. The participants discover each other automatically; no centralized broker is required.

Domain Participants

A domain participant is an endpoint (typically a process) that participates in the data exchange. Participants discover one another via multicast; the overall set forms a “domain”—a logically-isolated network of related participants.

Publishers and Subscribers

Publishers and subscribers are the primary communication interfaces. A publisher owns multiple Data Writers; a subscriber owns multiple Data Readers. Each writer/reader pair is scoped to a specific topic.

Data Writers and Readers

A Data Writer writes samples of a specific topic type; a Data Reader receives matching samples. The writer-reader relationship is one-to-many, many-to-one, or many-to-many.

Topics

Topics in DDS are strongly typed: each topic has an IDL-defined message schema. Type-checking happens at discovery; mismatches are caught before data flows.

11.4.2 Data Model

A topic's data stream is a time-ordered sequence of typed samples:

$$D = \{x_1, x_2, \dots, x_n\}, \quad (11.2)$$

where each x_i conforms to the topic's type and carries metadata (timestamp, sequence number, writer identity). DDS supports keyed topics where multiple "instances" (e.g., one per tracked object) are tracked within a single topic.

11.4.3 Quality of Service (QoS)

DDS's central value is fine-grained QoS. Each reader/writer pair negotiates on a rich set of QoS policies; the match must be compatible or the connection does not form.

Reliability

"Best-effort" vs "reliable." Reliable delivery re-sends lost samples until acknowledged; best-effort discards them. Control traffic is reliable; high-rate sensor streams are often best-effort (each frame makes the next frame's loss less important).

Durability

Does a late-joining subscriber receive previously-published samples? "Volatile" durability says no; "transient-local" caches the most recent samples on the publisher for late joiners.

Latency Budget

A soft deadline hinting to the middleware how much latency the application tolerates. Allows prioritization decisions.

Deadline

A hard per-instance deadline: if no sample arrives within the specified period, a miss is signaled. Used for heartbeat-style liveness monitoring.

11.5 Inter-Module Communication

A full AV stack is an interconnected set of modules exchanging data on many topics. The control-plane output at time t is a function of all the upstream inputs:

$$U(t) = g(P(t), L(t), E(t)), \quad (11.3)$$

with P the perception output, L the localization output, E the environmental context. Latency, synchronization, and reliability in each input affect U 's validity.

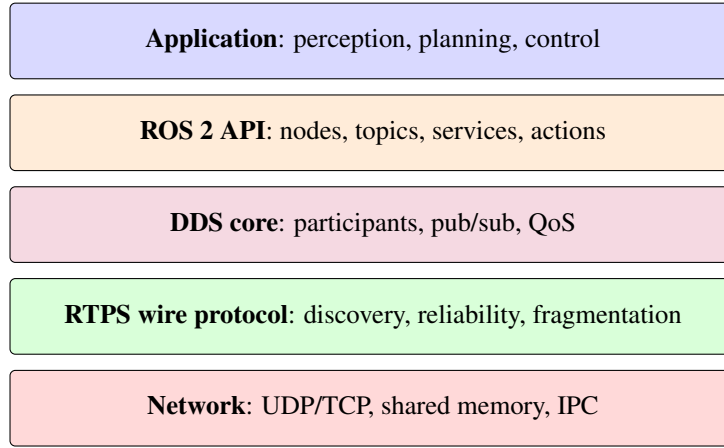


Figure 11.2: The layered AV middleware stack. Application code sits on top, programming against ROS 2 APIs. ROS 2 is implemented in terms of DDS. DDS in turn uses the RTPS wire protocol. RTPS runs over the network transports (UDP for LAN, TCP for WAN, shared memory for intra-host). Each layer offers abstractions and hides lower-level detail; debugging real systems requires visibility at each layer.

11.6 Worked Examples

Concrete scenarios illustrate the middleware design decisions AV engineers routinely make. The arithmetic is elementary; the design insight is where the value lies.

11.6.1 Example 1: LiDAR to Perception Pipeline

A LiDAR publishes point clouds; a perception node consumes and produces detections:

$$P_t = \{(x_i, y_i, z_i)\}_{i=1}^N, \quad O_t = h(P_t). \quad (11.4)$$

Total latency from sample-to-decision:

$$T_{\text{total}} = T_{\text{pub}} + T_{\text{dds}} + T_{\text{sub}}. \quad (11.5)$$

Example 11.1 (LiDAR publish–subscribe timing). $T_{\text{pub}} = 5$ ms (serialize $N = 100\,000$ points), $T_{\text{dds}} = 2$ ms (loopback transport), $T_{\text{sub}} = 8$ ms (deserialize and start processing). Total middleware overhead 15 ms. For a 10 Hz LiDAR, one frame’s overhead consumes 15 % of the inter-frame budget; careful zero-copy shared-memory transport can cut this by $5\times$.

11.6.2 Example 2: Control Loop with QoS Constraints

Controller commands track a reference:

$$u(t) = K(x_{\text{ref}}(t) - x(t)). \quad (11.6)$$

Deadline violation condition: an expected sample did not arrive in time.

$$T_{\text{arrival}} > T_{\text{deadline}} \Rightarrow \text{Deadline Miss.} \quad (11.7)$$

Example 11.2 (Deadline QoS for control). Control loop runs at 100 Hz; expected sample period 10 ms. DDS Deadline QoS set at 12 ms (20 % margin). Normal operation: deadline-miss count is zero. Anomaly: bus congestion causes 3 consecutive misses; middleware fires a deadline-miss callback; application invokes fallback behavior (hold last command, alert operator, initiate safe stop).

11.6.3 Example 3: Multi-Sensor Fusion

Heterogeneous rates must be reconciled:

$$f_c = 30\text{Hz}, \quad f_l = 10\text{Hz}, \quad f_r = 20\text{Hz}, \quad (11.8)$$

a fused product is generated when the slowest arrives:

$$t_f = \max(t_c, t_l, t_r), \quad F_t = \phi(C_t, L_t, R_t). \quad (11.9)$$

Example 11.3 (Approximate-time synchronization). Camera samples at $t = 0\text{ms}, 33.3\text{ms}, \dots$; LiDAR at $t = 0\text{ms}, 100\text{ms}, \dots$; radar at $t = 0\text{ms}, 50\text{ms}, \dots$. ROS 2's `message_filters::ApproximateTime` policy matches samples whose timestamps fall within a tolerance window (default 50 ms). First matched tuple: $(t_c=33.3\text{ms}, t_l=0\text{ms}, t_r=50\text{ms})$; produces fused output at $t = 50\text{ms}$ (the latest). Each sensor's latency in fusion is bounded by the tolerance window.

11.6.4 Example 4: Bandwidth Optimization

Raw bandwidth from a camera:

$$B = 15\text{MB s}^{-1}. \quad (11.10)$$

After compression at ratio α :

$$B_{\text{reduced}} = \alpha B. \quad (11.11)$$

With $\alpha = 0.7$: $0.7 \times 15 = 10.5\text{MB s}^{-1}$.

Example 11.4 (H.264 for intra-vehicle video). Raw 1920×1080 , 30 Hz, 24-bit: 186MB s^{-1} . H.264 at visually-lossless quality: $\alpha \approx 0.05$, reduced rate 9.3MB s^{-1} . Gigabit Ethernet (125MB s^{-1}) can carry it comfortably. Trade-off: compression adds 8 ms to 15 ms latency; the receiver's decoder adds a similar amount. For safety-critical low-latency streams, uncompressed over shared memory; for logging or cross-host transport, compressed.

11.6.5 Example 5: End-to-End Pipeline Timing

Sum over the pipeline stages:

$$T_{\text{pipeline}} = \sum_{i=1}^n (T_{\text{proc},i} + T_{\text{comm},i}). \quad (11.12)$$

Example 11.5 (Full perception-to-control pipeline). Stages: camera acquisition + compression (20 ms), perception inference (15 ms), planning (10 ms), control (10 ms). Total: $T_{\text{pipeline}} = 55$ ms. Constraint: $T_{\text{pipeline}} < 100$ ms. Budget headroom: 45 ms. Headroom absorbs occasional long-tail delays. Without the budget headroom, a single long-tail event produces a deadline miss.

11.6.6 Example 6: Fault Tolerance via Redundancy

Independent-path availability:

$$A = 1 - (1 - A_1)(1 - A_2). \quad (11.13)$$

Example 11.6 (Dual-network-path reliability). Network path 1: 95 % availability. Network path 2: 95 % availability. Combined: $A = 1 - (0.05)(0.05) = 0.9975$. The two unavailable-fractions must be truly independent for this to hold; cables in the same conduit, or transceivers sharing a power rail, do not give independent failure. Independence verification is part of the middleware safety case.

11.7 Real-Time Considerations

Real-time middleware requires bounded latency, not just average latency. The total cycle time is:

$$T_{\text{total}} = T_{\text{sensor}} + T_{\text{processing}} + T_{\text{communication}}. \quad (11.14)$$

Each term has a worst-case value that enters the safety argument. Production practice:

- Use zero-copy transport (shared memory) within a host.
- Pin threads to CPU cores; set scheduling priorities.
- Use pre-allocated message pools rather than heap allocation in the hot path.
- Measure latency distributions, not means; report 99.9th-percentile.
- Lock-free data structures where contention matters.

11.8 Security

A middleware carrying safety-critical data is a security asset. Basic cryptographic protection of messages:

$$C = \text{Encrypt}(M, K), \quad (11.15)$$

with M the message and K a shared symmetric key. DDS Security extends this to authentication, access control, and integrity.

- **Authentication:** participants prove identity via certificates.
- **Access control:** participants declare and verify which topics they may read and write.
- **Cryptographic integrity and confidentiality:** messages signed and optionally encrypted.
- **Logging and auditing:** tamper-evident records of security-relevant events.

Unsecured middleware is a first-rank cybersecurity risk (Chapter 3).

11.9 Challenges and Future Directions

Middleware for AVs is an active engineering area. Four directions dominate current work:

11.9.1 High-Bandwidth Data Handling

Next-generation sensors (8K cameras, multi-return LiDAR, 4D radar) produce $10 \text{ Gbits}^{-1} +$ streams. Middleware must carry these at sub-millisecond transport latencies. Zero-copy shared memory, RDMA over InfiniBand or Ethernet, and FPGA-accelerated serialization are the techniques being deployed.

11.9.2 Fault Tolerance

Production AV middleware must survive single-component failures gracefully: cable disconnections, process crashes, host reboots. Emerging patterns include watchdog-driven automatic failover, hot-standby participants, and deterministic state-transfer protocols.

11.9.3 Edge–Cloud Integration

Middleware increasingly extends beyond the vehicle to edge compute (Chapter 1). Message routing across the WAN with latency-aware QoS requires middleware-level awareness of network topology.

11.9.4 AI-Aware Middleware

Large neural networks as nodes have different resource patterns than traditional code: GPU-bound, large memory footprints, batch-friendly. Middleware that schedules, batches, and routes data for AI workloads is an emerging subfield.

11.10 Worked Examples: Integrated

Example 11.7 (Full AV stack timing tree). Per sensor cycle at 10 Hz:

- Sensor acquisition + DMA: 2 ms.
- Preprocessing: 3 ms.
- DDS publish + deserialize: 1 ms.
- Perception inference: 20 ms.
- DDS publish: 1 ms.
- Prediction: 10 ms.
- Planner: 15 ms.
- Controller: 3 ms.

Total: 55 ms end-to-end. Budget: 100 ms (10 Hz cycle). Headroom: 45 ms absorbs variability. If perception grows by 50 % (larger model), the stack still fits; if the whole chain doubles, the budget fails.

Example 11.8 (Introducing redundancy into a production stack). Original: single perception node, single planner, single controller. Introduce redundancy:

- Redundant perception nodes (one primary + one hot standby).
- Voting module: if both agree, use result; if they disagree, use safer output.
- Separate planner paths checked by safety monitor.

Middleware impact: topic bandwidth $\times 2$; CPU usage $\times 2$; latency added by voting 2 ms. Availability: from $0.999^3 = 0.997$ to $1 - (1 - 0.999^3)^2 = 0.99999$ —five nines. The production AV program trades compute budget for availability.

11.11 Industrial Deployment

The Robot Operating System (ROS), originally developed at Willow Garage and now stewarded by the Open Source Robotics Foundation, is the dominant robotics middleware in research and increasingly in production. ROS 2, the production-oriented successor, is used in commercial robots and is the foundation of Apex.AI's Apex.OS automotive-grade variant, which has achieved ISO 26262 ASIL D certification. The Autoware project (Autoware Foundation, supported by Tier IV in Japan) is a ROS 2-based open-source autonomous-driving stack used in research and pilot deployments globally. The Cyclone DDS and FastDDS implementations of the OMG Data Distribution Service standard are the underlying communication layers used by ROS 2. NVIDIA DRIVE OS provides an alternative production-oriented platform with deterministic execution properties suited to safety-critical AV deployments.

11.12 Chapter Summary

Middleware is central to autonomous driving systems. It carries the data that links perception, planning, and control modules at rates, deadlines, and reliabilities the stack's safety depends on. ROS (now ROS 2) provides the high-productivity API most engineers work against; DDS provides the deterministic, QoS-rich transport beneath. The publish/subscribe model scales to dozens of modules and hundreds of topics. Inter-module communication must be analyzed for latency, throughput, reliability, and determinism; QoS policies express these requirements. Real-time engineering (zero-copy, thread pinning, lock-free data structures) and security (authentication, access control, encryption) are both non-optional in production. Future directions—high-bandwidth transports, fault tolerance, edge integration, AI-aware scheduling—define the ongoing engineering work. A middleware stack that is fast, safe, and secure is a prerequisite for any production AV deployment.

11.13 Exercises

1. For the LiDAR example (11.1), compute the shared-memory transport latency for the same data and compare to the loopback DDS case.
2. Design a QoS profile for a camera-to-perception topic carrying 30 Hz images. Specify reliability, history, durability, deadline, and justify each.
3. Derive the condition on individual-component availability required to achieve system availability of 0.99999 with three independent redundant paths.
4. For the pipeline-timing example (11.5), propose two changes that would reduce total latency to 40 ms.
5. Compare ROS 1 and ROS 2 in terms of production AV suitability. Identify at least three reasons ROS 2 is now preferred.
6. A DDS participant fails to match a topic because of a QoS mismatch. What are the three most likely root causes?
7. Design the topic graph for an AV stack that fuses a camera and two LiDARs with heterogeneous frame rates. Show which synchronization primitive applies.
8. For the middleware layer cake (11.2), identify one failure mode at each layer and the debugging tool that surfaces it.
9. Write the pseudo-code for a *hot-standby* failover mechanism for a perception node.
10. Explain why the deadline-miss event should be surfaced to the application rather than handled within the middleware.

Chapter 12

Compute Systems and Hardware Architectures

12.1 Introduction

The AV perception stack developed in earlier chapters is a computational workload: hundreds of GFLOPs per frame, sustained at 10–30 Hz, within automotive thermal and power envelopes. Meeting these constraints has driven a generation of specialized compute hardware—GPUs adapted for automotive deployment, custom AI accelerators, neuromorphic chips, in-sensor processors—each trading off performance, power, latency, and cost. This chapter surveys the compute landscape and its consequences for stack architecture.

Three observations motivate the structure.

First, compute is the tightest constraint. Sensor and algorithm choices are limited by what can run at 30 Hz on automotive-grade silicon within a 50 W power budget. Every perception-architecture decision comes back to compute feasibility.

Second, specialization is accelerating. General-purpose GPUs were the 2010s answer; 2020s AV compute is moving toward task-specific accelerators, in-sensor processing, and neuromorphic approaches for specific sub-problems.

Third, heat management is first-class. Automotive electronics operate between -40°C and $+105^{\circ}\text{C}$; reliability requires careful thermal design that shapes compute architecture more than consumer-electronics equivalents.

12.2 Automotive AI Accelerators

Production AV compute platforms:

Platform	TOPS	Power (W)	TOPS/W
NVIDIA Drive Thor (2025)	2000	150	13
Tesla HW4 (2023)	250	35	7
Qualcomm Snapdragon Ride	700	60	12
Mobileye EyeQ6 Ultra	176	32	5.5
Horizon Robotics Journey 6	560	55	10

TOPS (tera-operations per second) is the headline metric but is only a crude proxy; effective performance depends on precision (INT8 vs FP16), memory bandwidth, and the ability of the software stack to utilize the hardware.

Example 12.1 (Effective TOPS utilization). A ResNet-50 backbone at FP16 requires ~ 7.7 GFLOPs per inference.

Theoretical throughput on 250 TOPS hardware (FP16).

$$\begin{aligned} \text{inferences/s}_{\text{theoretical}} &= \frac{250 \times 10^{12}}{7.7 \times 10^9} \\ &\approx 32,500. \end{aligned}$$

Realistic throughput (memory- and pipeline-limited).

$$\text{inferences/s}_{\text{effective}} \approx 500\text{--}2000.$$

Effective utilization is 1.5–6 % of peak. Memory-bandwidth, synchronization overhead, and unavoidable serial portions of the pipeline dominate real workloads.

12.3 Neuromorphic Hardware

Neuromorphic chips implement spiking neural networks in analog or mixed-signal silicon. Power-efficient for sparse, event-based inputs.

12.3.1 Key Properties

- Event-driven: compute only when inputs change; idle power near zero.
- Sparse activation: typical utilization $< 10\%$ at any moment.
- Ultra-low latency: microseconds from input to output.
- Well-matched to event cameras (Chapter 30).

12.3.2 Chips and Platforms

Platform	Characteristics
Intel Loihi 2	1M neurons, research-grade; mesh-of-cores
IBM NorthPole	Prototype; digital but neuromorphic-inspired
BrainChip Akida	Commercial; neuromorphic edge inference
SynSense Speck	Vision-specific, very-low-power

12.3.3 AV Applications

Not yet production-deployed at scale, but research-grade: high-speed tracking (fast ball, fast pedestrian); backup detection pipelines with ultra-low-power duty cycling; bio-inspired motion detection.

12.4 In-Sensor Computing

Instead of transporting raw sensor data to a central compute unit, in-sensor computing performs initial processing on or near the sensor itself. Benefits:

Benefit	Mechanism
Lower bandwidth	Send features or detections, not raw pixels
Lower latency	No transport delay
Lower power	Data-motion cost dominates sensor-to-compute pipelines
Stronger privacy	Raw data never leaves the sensor

Example 12.2 (Bandwidth savings from in-sensor processing). Automotive camera: 8 MP at 30 Hz, 12-bit raw. Raw bandwidth:

$$\begin{aligned}
 B_{\text{raw}} &= 8 \times 10^6 \times 30 \times 1.5 \text{ B} \\
 &= 360 \text{ MB/s per camera.}
 \end{aligned}$$

After in-sensor feature extraction (say, a MobileNet stem producing 128 feature channels at 1/8 resolution):

$$\begin{aligned}
 B_{\text{features}} &= (8 \times 10^6 / 64) \times 128 \times 30 \times 2 \text{ B} \\
 &= 0.96 \text{ GB/s.}
 \end{aligned}$$

Wait—features are larger than raw pixels in this case. Choose a smaller feature map:

$$B_{\text{compact}} = (8 \times 10^6 / 256) \times 64 \times 30 \times 2 \text{ B} \\ \approx 120 \text{ MB/s.}$$

A $3\times$ reduction. For a 6-camera array, the raw-vs-feature gap scales proportionally; in-sensor compute becomes decisive at the aggregate level.

12.5 Approximate Computing

Not all computation needs full precision. *Approximate computing* trades accuracy for speed or power: INT8 or INT4 arithmetic, quantization-aware training, sparse weights, etc.

12.5.1 Quantization Levels

Precision	Bits	Typical accuracy drop (%)
FP32 (training)	32	baseline
FP16	16	<0.1
INT8	8	0.2–1.0
INT4	4	1–5
Binary (BNN)	1	5–15

AV deployment typically uses INT8 for backbones and FP16 for sensitive heads (e.g., 3D box regression where quantization error compounds with geometric error).

12.5.2 Sparsity

Modern accelerators exploit 2:4 structured sparsity (2 of every 4 weights zero), giving $\sim 2\times$ throughput on supported layers. Pruning-aware training recovers accuracy.

12.6 Thermal-Aware Computing

Automotive compute operates in harsh thermal conditions. Silicon degrades at sustained high temperatures; reliability decreases exponentially above rated junction temperatures.

12.6.1 Dynamic Thermal Management

Production AV compute platforms:

- Monitor junction temperature continuously.
- Throttle clock speeds when approaching thermal limits.
- Balance workload across cores or packages to spread heat.

- Pre-compute less-urgent workloads during cooler operating windows.

Example 12.3 (Thermal-throttle impact on frame rate). AV compute under summer-highway conditions: ambient 45 °C, compute load near maximum.

Baseline.

frame rate = 30 Hz,
junction temperature = 95 °C (just under limit).

Under sustained heavy load, throttle activates.

clock reduction = 15 %,
frame rate = 25.5 Hz.

A 4.5-Hz frame-rate reduction projects into ~ 15 ms added latency per frame in the worst case—enough to compound with other delays into planner-level issues. Cooling design must prevent this from happening in nominal operation; graceful degradation logic handles the abnormal case.

12.7 Industrial Deployment

Production AV compute platforms include NVIDIA DRIVE Orin (deployed in production by multiple OEMs including Polestar, Volvo, Mercedes-Benz, and Chinese manufacturers including BYD, XPeng, Li Auto), Mobileye EyeQ6, Qualcomm Snapdragon Ride, Tesla’s in-house Hardware 3 and Hardware 4 (later AI4) platforms, Horizon Robotics Journey series (China), and Black Sesame Technologies platforms (China). Power budgets for AV-grade compute range from tens of watts for ADAS to hundreds of watts for full Level 4 stacks. Automotive-grade qualification (AEC-Q100, ISO 26262) requires substantial additional engineering beyond consumer-grade silicon. The transition from CPU to GPU to specialised accelerators (NPUs, dataflow architectures) tracks the increasing computational demands of transformer-based perception.

12.8 Chapter Summary

AV compute systems have evolved from general-purpose GPUs to specialized accelerators optimized for automotive constraints. Neuromorphic hardware offers ultra-low-power event-driven processing for specific workloads. In-sensor computing reduces bandwidth and latency by processing near the sensor. Approximate computing (INT8, sparsity) is standard for production deployment. Thermal-aware design is a first-class concern given automotive operating environments. Effective hardware utilization remains a major engineering challenge, with realistic throughput often at 10 % or less of theoretical peak.

12.9 Exercises

1. For the effective-TOPS example, compute the utilization required to run a 50 GFLOPs model at 30 Hz on a 250-TOPS platform.
2. Propose an in-sensor processing pipeline that reduces a 360 MB/s raw stream to <20 MB/s while preserving downstream accuracy.
3. Compare FP16 and INT8 quantization for a 3D detection head. What accuracy impact is acceptable?
4. Estimate thermal-budget impact of adding a 7th camera to a 6-camera array; make reasonable assumptions.
5. Design a graceful-degradation policy for compute thermal throttling: which workloads are first to be reduced?
6. Compare neuromorphic and conventional compute for a specific AV task; identify one where neuromorphic wins.
7. Propose a sparsity-aware training pipeline that preserves accuracy while enabling 2:4 structured sparsity.
8. For the thermal-throttle example, estimate frame-rate reduction at 50 °C ambient.
9. Compare latency of centralized and in-sensor computing for a perception cascade; identify where each wins.
10. Discuss the end-of-life silicon reliability considerations for AV compute operating 10+ years in a vehicle.

Chapter 13

Distributed AI and Fleet Learning

13.1 Introduction

A single AV is a closed-loop system: sensors in, actions out, no knowledge shared. A *fleet* of AVs is something different: it is a distributed intelligence that collectively experiences the world, pools observations, and improves faster than any individual vehicle could. The engineering discipline that makes fleet intelligence work—*distributed AI*—combines split computing between vehicles and cloud, swarm-perception coordination between nearby vehicles, and fleet-scale knowledge distillation that transfers lessons learned across the fleet. This chapter develops each thread.

Three observations motivate the structure.

First, computation is not equally valuable in all places. Latency-sensitive operations (immediate collision avoidance) must run on the vehicle; bandwidth-sensitive operations (training new models) must run in the cloud. Split-computing architectures route each to its natural home.

Second, neighboring vehicles see the world better than either alone. Swarm perception networks exchange observations in real time, extending effective sensing range and enabling cooperative maneuvers.

Third, fleet learning closes the loop from rare events to model improvement. An event observed once anywhere in the fleet feeds training data that improves every vehicle.

13.2 Split Computing: Edge and Cloud

The AV stack is partitioned between on-vehicle compute (edge) and fleet-backend compute (cloud) based on latency, bandwidth, and data-sensitivity constraints.

13.2.1 What Runs Where

Workload	Edge (vehicle)	Cloud (fleet)
Real-time perception	yes (required)	no (latency-prohibitive)
Planning & control	yes (required)	no
Fleet dispatch	local component	global solver
Map maintenance	consumer only	authoritative updater
Model training	no (cost)	yes
Data curation	sampling only	primary
Incident replay	logs only	full analysis

13.2.2 Cloud-Assisted Perception

For non-latency-critical perception (map update, scenario logging), clouds can augment vehicle capability. Example: a vehicle encounters an unknown object; its on-board classifier is unsure; the vehicle uploads relevant frames to a larger cloud model that returns a classification within a few seconds. Usable for cautious operation but not for immediate safety-critical decisions.

Example 13.1 (Latency budget for cloud-assisted perception). Vehicle detects ambiguous object.

Timeline.

frame capture = 0 ms,
 on-board inference = 30 ms,
 uplink latency = 100 ms,
 cloud inference = 80 ms,
 downlink latency = 100 ms.

Total.

$$\begin{aligned}
 T_{\text{total}} &= 30 + 100 + 80 + 100 \\
 &= 310 \text{ ms.}
 \end{aligned}$$

Not usable for immediate safety decisions. Usable for: flagging the object for cautious behavior (slow down, widen safety envelope), classifying it for later model improvement, or alerting the fleet to a similar observation.

13.3 Swarm Perception Networks

Vehicles within V2V communication range exchange observations in real time. The resulting shared perception exceeds the capability of any single vehicle.

13.3.1 Cooperative Perception Patterns

- **Fact-sharing:** vehicle A observes object X, shares its track with vehicle B which then has X in its scene without direct observation.
- **Joint detection:** multiple vehicles observe the same object from different angles; aggregated observations produce a more accurate 3D estimate than any single view.
- **Intent-sharing:** vehicles broadcast their planned trajectory, allowing neighbors to plan compatibly.

13.3.2 Benefits and Challenges

Aspect	Implication
Sensing range	Extended beyond line-of-sight (occlusion-robust)
Accuracy	Multi-view triangulation; cooperative tracking
Latency budget	10s of ms for meaningful cooperation
Trust	Malicious or faulty participants can degrade the swarm
Scalability	$O(N^2)$ naive message flow; hierarchical schemes required

Example 13.2 (Swarm cooperation for occluded pedestrian). Scenario: ego approaches intersection. Parked truck occludes a pedestrian about to cross. A vehicle from the cross-street (line-of-sight to pedestrian) broadcasts the pedestrian's track.

Ego's effective perception.

Without cooperation: pedestrian invisible until 5 m separation.

With cooperation: pedestrian known from 40 m away.

Benefit in TTC.

$$\begin{aligned}\Delta\text{TTC} &= (40 - 5) \text{ m} / 10 \text{ m/s} \\ &= 3.5 \text{ s additional warning.}\end{aligned}$$

3.5 seconds converts a last-moment emergency brake into a gentle controlled stop. Cooperative perception offers non-incremental safety benefits in occlusion-heavy scenarios.

13.3.3 Protocol Considerations

Swarm-perception standards are emerging within V2X ecosystems (C-V2X, DSRC successors). Message formats, authentication, and replay-attack prevention are active standardization areas.

13.4 Fleet-Scale Knowledge Distillation

A large cloud model trained on fleet-wide data embodies more knowledge than any deployed vehicle model. Knowledge distillation transfers this knowledge into smaller, vehicle-deployable models.

13.4.1 Distillation Mechanics

A large *teacher* model produces soft outputs (probability distributions); a smaller *student* model is trained to match them. Soft outputs carry more information than hard labels, enabling better transfer.

Loss function:

$$L = \alpha \cdot L_{\text{CE}}(y_{\text{student}}, y_{\text{true}}) + (1 - \alpha) \cdot T^2 \cdot L_{\text{KL}}(y_{\text{student}}/T, y_{\text{teacher}}/T), \quad (13.1)$$

with T the temperature (softening the distributions) and α the mixing coefficient.

Example 13.3 (Distillation improvement on student accuracy). Teacher: 2B-parameter cloud model, 94 % mAP on validation.

Student: 50M-parameter on-vehicle model.

- Trained on labels only: 82 % mAP.
- Distilled from teacher: 88 % mAP.
- Distilled + data augmentation: 90 % mAP.

Distillation captures 6 of the 12 percentage-point gap between student and teacher—a large, essentially-free improvement over training on hard labels alone.

13.4.2 Continuous Fleet Learning

The cycle: fleet vehicles log observations, cloud trains a new teacher, distillation produces a new student, OTA updates deploy the student. Duration: weeks to months per cycle; pace limited mostly by validation and regulatory gates rather than the underlying ML.

13.5 Integration and Privacy

Fleet learning raises data-governance questions. Passenger privacy, route-data sensitivity, and regional data-localization laws shape what can be shared at what level.

13.5.1 Federated Learning

Train models across multiple vehicles or data silos without centralizing raw data. Gradients are shared, raw observations are not. Privacy-preserving but computationally expensive; production use is limited but emerging in privacy-sensitive regions (EU under GDPR).

13.5.2 Differential Privacy

Add calibrated noise to shared data or gradients so individual observations cannot be reconstructed. Trades accuracy for privacy; quantifiable via the (ϵ, δ) -DP framework.

13.6 Industrial Deployment

Edge-cloud distributed AI in AVs is enabled by 4G/5G cellular, satellite communications (including Starlink, increasingly used in commercial-vehicle telematics), and emerging V2X infrastructure. Tesla has publicly described the use of fleet-data infrastructure for shadow mode and offline labeling. NIO and other Chinese OEMs operate “vehicle-cloud” architectures with substantial cloud-side processing. The TensorFlow Federated and PySyft open-source projects provide reference federated-learning implementations. Privacy regulations including GDPR (European Union) and CCPA (California) constrain the centralised collection of vehicle-generated data and incentivise federated and on-device approaches. AWS, Google Cloud, and Microsoft Azure all offer automotive-specific cloud platforms used by various OEMs.

13.7 Chapter Summary

Distributed AI for AVs spans three interconnected threads: split computing between vehicles and cloud; swarm perception coordinating nearby vehicles; and fleet learning via knowledge distillation closing the loop from fleet experience to model update. Each addresses a specific structural limitation of solo-vehicle intelligence. Together they define how a modern AV fleet behaves not as a collection of independent vehicles but as a coordinated distributed system. Privacy-preserving techniques (federated learning, differential privacy) are becoming standard given regulatory pressure. The trend toward cooperative, fleet-level intelligence is likely to deepen as V2X deployments scale.

13.8 Exercises

1. For the cloud-assisted example, compute the latency budget if uplink and downlink are each 50 ms.
2. Propose a split-computing architecture for a specific AV module (choose one). Justify the partition.
3. For the swarm-cooperation example, compute the benefit at ego speed 30 m/s.
4. Design a trust-weighting scheme for swarm-perception messages that attenuates untrusted sources.

5. For the distillation example, estimate the student's accuracy given $\alpha = 0.5$ vs $\alpha = 0.1$; reason qualitatively.
6. Propose a federated-learning protocol that updates a model across 10 regions with GDPR constraints.
7. Derive the (ϵ, δ) -DP parameters needed to guarantee that individual vehicle observations cannot be reconstructed.
8. Identify three AV tasks that benefit from cooperative perception and three that do not.
9. Compare the bandwidth requirements of full-stack cooperation vs fact-sharing cooperation.
10. Discuss the bootstrap problem: how does a new AV joining the fleet benefit from fleet learning before contributing anything to it?

Chapter 14

Over-the-Air Updates and AV Lifecycle Management

14.1 Introduction

An autonomous vehicle is a continuously-evolving software product deployed on safety-critical hardware. Unlike a fighter jet or a refinery controller—where software updates happen on planned maintenance windows with human operators present—an AV is expected to receive updates while in customer service, across a fleet of thousands, without degraded safety. Over-the-air (OTA) updates, versioning, rollback, and lifecycle management are therefore first-class engineering disciplines in production AV programs.

Three observations motivate the chapter.

First, OTA is *the primary mechanism* by which AV capabilities grow. Shipped hardware is fixed at purchase; shipped software evolves weekly or monthly. An AV that never receives updates is a depreciating asset with fixed capabilities.

Second, updates must preserve *continuous safety*. A partial update that leaves the vehicle in an inconsistent state is an immediate safety event. Atomic update semantics, rollback, and cross-system version compatibility are non-negotiable.

Third, regulatory frameworks are *tightening*. UN Regulation No. 156 (Software Update Management) is already mandatory in many markets; similar regulations are emerging globally. OTA compliance is a legal requirement, not an optional feature.

14.2 OTA Architecture

A production OTA system has three core components:

Component	Role
Update server	Authors, signs, and serves update packages; maintains fleet state
In-vehicle agent	Downloads, verifies, stages, and installs; handles reboot coordination
Rollback	Preserves previous known-good state; reverts on failure

14.2.1 Package Signing and Verification

Every update package is cryptographically signed. The in-vehicle agent verifies the signature against trusted public keys before applying. This prevents adversaries from pushing malicious updates even if they compromise transport channels.

14.2.2 A/B Partitioning

The standard update pattern maintains two software partitions: active and passive. An update writes the new version to the passive partition while the active runs normally. Upon boot, the vehicle switches to the new partition; if boot fails or runtime checks fail, it reverts.

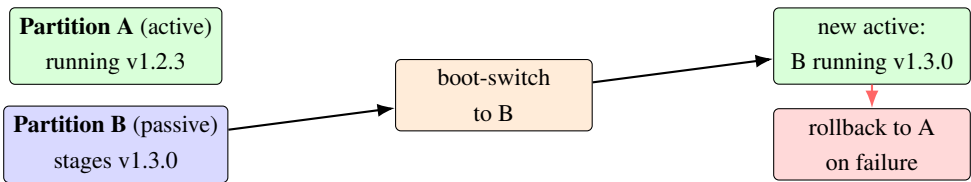


Figure 14.1: A/B partition update flow. The passive partition stages the new version while the active continues serving. A boot-switch promotes the new version to active; runtime validation monitors its health. On any failure a rollback reverts to the previous known-good partition.

14.2.3 Delta Updates

Shipping only the changed bytes (rather than full images) drastically reduces bandwidth.

Example 14.1 (Delta-update bandwidth savings). AV software image: 4 GB full. Typical delta between adjacent versions: 150 MB.

Bandwidth per update.

$$\begin{aligned}
 B_{\text{full}} &= 4 \text{ GB}, \\
 B_{\text{delta}} &= 0.15 \text{ GB}, \\
 \text{savings} &\approx 27 \times .
 \end{aligned}$$

Fleet-level impact.

A fleet of 100 000 vehicles receiving a monthly update:

$$\begin{aligned}\text{monthly data (full)} &= 100,000 \times 4 = 400,000 \text{ GB}, \\ \text{monthly data (delta)} &= 100,000 \times 0.15 = 15,000 \text{ GB}.\end{aligned}$$

At 0.05 USD/GB (cellular bandwidth cost), delta updates save roughly 19k USD per monthly fleet update—a direct operational cost reduction that scales with fleet size.

14.3 Versioning and Compatibility

14.3.1 Semantic Versioning

Production AV programs follow semantic versioning (`major.minor.patch`):

Component	When it changes
Major	Breaking interface change; cross-system compatibility break
Minor	New feature, backward compatible
Patch	Bug fix, backward compatible

14.3.2 Cross-System Dependencies

Modern AVs have many interacting components: perception, planning, control, HMI, connectivity. Updates must respect version dependency graphs. A perception update that changes output schema requires a compatible planner update.

14.3.3 Map Updates

HD-map updates are a separate cadence, often weekly or daily. Maps are versioned independently and consumed dynamically by runtime modules. An AV with out-of-date maps operates with degraded confidence or falls back to map-free modes.

14.4 Staged Rollout

14.4.1 Canary and Gradual Deployment

Updates are not pushed to the entire fleet simultaneously. Canonical cadence:

Stage	Fleet fraction	Duration
Internal	0–1 %	1–7 days
Canary	1–5 %	1–2 weeks
Early adopters	5–20 %	2–4 weeks
Full rollout	20–100 %	2–6 weeks

At each stage, telemetry is monitored; abnormal metrics trigger a rollout pause. Figure 14.2 shows the overall pipeline.

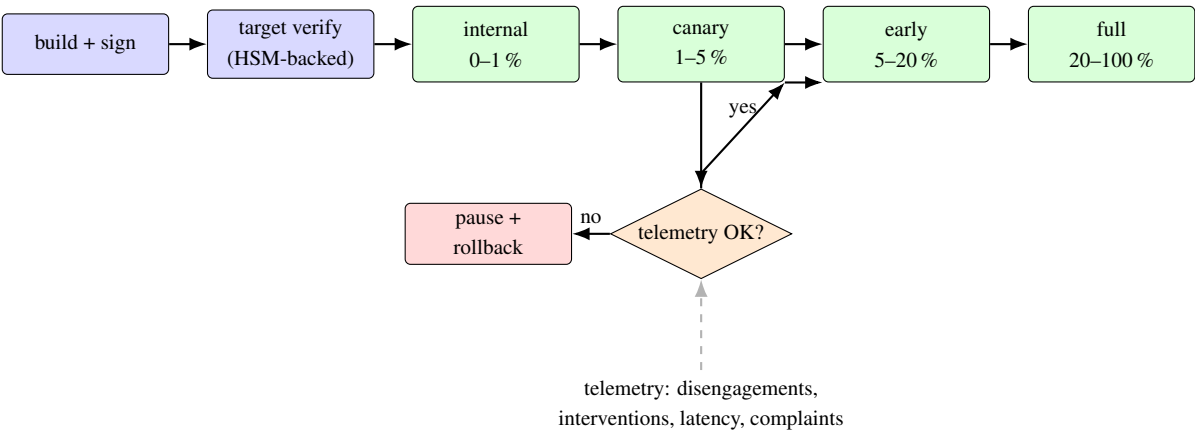


Figure 14.2: OTA canary rollout pipeline. After build and signing, the package is verified at the target vehicle (HSM-backed signature check) before any installation. The rollout proceeds through expanding fleet fractions: internal → canary → early adopters → full. At every transition a telemetry gate evaluates safety-critical metrics (disengagements, interventions, latency, complaints); if any metric exceeds baseline, the rollout is paused and the fleet rolls back to the previous A/B partition. The pipeline is the operational realisation of UNECE R156’s Software Update Management System requirement.

14.4.2 Monitoring Signals

- Safety-critical metrics: disengagements, interventions, planner anomalies.
- Performance metrics: detection AP, tracking quality, prediction NLL.
- Operational metrics: CPU utilization, latency percentiles, memory pressure.
- Customer-facing metrics: ride comfort, complaint rates.

Example 14.2 (Canary rollout decision with telemetry). Canary phase: 5 % of fleet on new version. Telemetry shows:

- Intervention rate: 1.2/10 000 mi (baseline 1.1).
- Latency p99: 92 ms (baseline 88 ms).
- Customer complaint rate: +15 % vs baseline.

Decision framework.

A 10 % intervention-rate increase is within normal variance; the complaint rate spike and latency drift suggest a subtle regression. Action:

- Pause rollout (do not advance to 20 %).
- Root-cause investigation: examine the specific complaint and intervention cases.
- Either patch and re-canary, or rollback to previous version.

Per-vehicle rollback is performed within minutes; fleet-wide rollback within hours.

14.5 Shadow Mode and Continuous Improvement

14.5.1 Shadow Evaluation

New models run in shadow mode before being promoted to active control: they receive real sensor input and produce outputs, but their outputs are logged rather than actuated. Differences between shadow and active outputs are analyzed offline, revealing specific scenarios where the candidate model behaves differently.

14.5.2 Feedback Loop

Production fleets generate data; data feeds new training; training produces new models; models are shadow-evaluated, then canaried, then released. The loop is the core of continuous improvement.

Example 14.3 (Fleet-learning cadence). Observation rate: 100 000 vehicle-hours/week across the fleet.

Interesting events (non-nominal outcomes):

$$r_{\text{interesting}} \approx 1 \text{ per } 100 \text{ vehicle-hours,}$$

$$N_{\text{weekly}} = 100,000 / 100 = 1,000 \text{ events/week.}$$

After triage:

$$N_{\text{curated}} \approx 100 / \text{week (10 \%)},$$

$$N_{\text{in-training}} \approx 50 \text{ new training scenes/week.}$$

Model iteration cadence: A new candidate model is trained weekly from the accumulated scenes; shadow-evaluated over 1–2 weeks; released through the canary process over 4–6 weeks. End-to-end lag: 7–10 weeks from event-in-fleet to capability-in-fleet.

14.6 Regulatory Landscape

14.6.1 UN Regulation No. 156

The formal standard for software-update management in road vehicles. Requires:

- Documented software identification and certification state.
- Secure update process and integrity verification.
- Operator notification and user consent (where applicable).
- Rollback capability.
- Traceability and audit logs.

14.6.2 Type Approval

In many markets, updates that change type-approved behavior require re-approval. This can slow release cadence; production programs structure architecture to separate type-approved baseline behavior from updatable-without-re-approval enhancements.

14.7 Industrial Deployment

Over-the-air (OTA) software updates have moved from a Tesla differentiator in the early 2010s to an industry-standard capability. Tesla pioneered end-to-end OTA capability for vehicle software including powertrain and ADAS. Volkswagen, Ford, GM, BMW, Mercedes-Benz, Stellantis, and most Chinese OEMs now offer OTA updates on at least some vehicle models. UNECE Regulation 156 governs Software Update Management Systems in Europe. Suppliers including Aurora Labs, Sibros, Excelfore, and HARMAN provide OTA infrastructure to OEMs. The complexity of deploying safety-critical OTA updates—staged rollouts, regression detection, rollback capability—is the subject of standardisation in ISO/SAE 21434 and the emerging ISO 24089.

14.8 Chapter Summary

OTA updates are the mechanism by which an AV's capability grows after purchase. A/B partitioning, package signing, delta updates, and semantic versioning are foundational. Staged rollouts (internal → canary → gradual → full) with telemetry-driven decision gates manage release risk. Shadow evaluation supports continuous improvement from fleet data. Regulatory frameworks (UN R156, type approval) are maturing and make OTA a compliance matter, not merely an engineering convenience. Update infrastructure is as critical to production AV as the perception stack itself.

14.9 Exercises

1. For the delta-update example, compute savings for a weekly-update cadence.
2. Design an A/B partition scheme that survives an attack on the active partition.
3. Propose a staged rollout schedule for an update that changes planner behavior in the highest-risk scenarios.
4. For the canary example, compute the minimum intervention rate increase that would justify an immediate rollback.

5. Identify three telemetry signals that precede a safety regression, and design alerts for each.
6. Compare shadow evaluation and A/B canary testing. When is each preferred?
7. For the fleet-learning example, compute the end-to-end lag if training cadence is biweekly.
8. Design a re-approval-aware architecture that allows safe updates without re-type-approval where possible.
9. Discuss the economics of delta updates: what fleet size is needed to justify delta-engineering investment?
10. Propose a version compatibility matrix for a 5-module AV stack and the associated constraint checking.

Chapter 15

Fleet Systems and Collective Intelligence

15.1 Introduction

Chapter 13 developed distributed AI at the infrastructural level—edge/cloud split, swarm perception, fleet learning. This chapter addresses the *collective behavior* layer: how an entire fleet’s actions combine into a coordinated system, and what emergent properties this enables. Collective edge learning lets vehicles share local model updates; fleet-wide anomaly detection catches systemic perception issues before they become incidents; distributed map validation ensures consistency at scale.

Three observations motivate the structure.

First, a fleet is not just many vehicles. At the point where vehicles share observations, influence each other’s behavior, and collectively shape map and training data, the fleet becomes a distributed system with its own dynamics— feedback loops, failure modes, and coordination properties that individual vehicles do not have.

Second, localized learning can outperform central training for some tasks. Regional patterns (driving culture, weather, road design) are better learned from regional data; collective edge learning lets vehicles in an area share models without centralizing raw data.

Third, fleet-scale anomalies are invisible to individual vehicles. A perception model silently drifting on a new construction pattern is only visible when many vehicles see similar patterns.

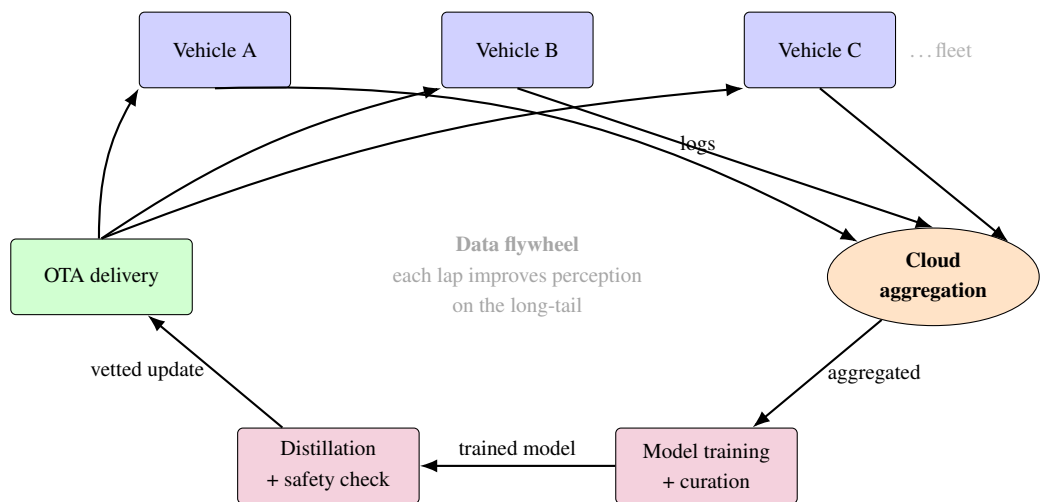


Figure 15.1: The fleet data flywheel. Vehicles in operation continuously emit logs, telemetry, and (event-triggered) sensor segments to cloud aggregation. Cloud-side curation and labelling produce training data for the next model iteration; distillation compresses the trained model and applies safety checks; the resulting update is rolled out back to vehicles via the OTA pipeline (Chapter 14). Each revolution of the loop improves perception accuracy on the long-tail—failure modes that are rare per vehicle but frequent across the fleet. The flywheel hypothesis is the principal justification for end-to-end learning at scale: data-rich operators gain a compounding advantage that hand-coded systems cannot match.

15.2 Collective Edge Learning

Individual vehicles see only their local environment. Collective edge learning lets nearby vehicles share model refinements directly, without round-tripping through the central cloud.

15.2.1 Architecture

Vehicles within a geographic region form a local learning cluster. Each vehicle trains locally on its own observations; periodic gradient exchanges (or distilled updates) converge a shared regional model.

15.2.2 Benefits

Property	Benefit
Lower latency	Regional updates within seconds, not days
Privacy	Raw data stays on-vehicle; only gradients shared
Regional fit	Captures local patterns central training cannot
Resilience	Works during cloud connectivity loss

Example 15.1 (Convergence speed: collective vs central). Vehicle encounters new object type for the first time.

Central training cycle.

- Vehicle logs observation + uploads: few hours.
- Cloud retraining + validation: 2–4 weeks.
- OTA deployment to fleet: 1–2 weeks.
- Total: 3–6 weeks.

Collective edge learning.

- Vehicle observations propagate to regional cluster: minutes.
- Gradient exchange and local model update: hours.
- Fleet update within region: same day.
- Total: hours, not weeks.

A $\sim 100\times$ speed improvement for regionally-relevant knowledge. Central training remains essential for global robustness; the two run in parallel.

15.3 Fleet-Wide Anomaly Detection

Individual vehicles see only their own sensor streams. Aggregated across a fleet, patterns emerge that individual vehicles miss.

15.3.1 What To Monitor

- Confidence distributions: sudden shifts indicate distribution drift.
- Intervention rates by region: spatial clusters hint at systematic issues.
- Disagreement rates between redundant perception pipelines.
- Track-quality statistics over time.

15.3.2 Early Warning

Fleet-scale monitoring surfaces issues hours to days before they produce visible incidents. A confidence-distribution shift in one region is a pre-incident signal that can trigger cautious behavior or engineering review before any incident occurs.

Example 15.2 (Anomaly detection triggers pre-incident response). Fleet observes a 4σ drop in pedestrian-detection confidence in a specific neighborhood.

Root cause (identified on inspection). Construction has installed temporary chain-link fencing that produces confusing moire patterns in camera frames. Detectors trained without such patterns now lower confidence.

Response.

- Automatic: affected area flagged; vehicles drive with extra caution (lower speed, wider buffers).
- Engineering: data collected for retraining.
- Deployment: patched model deployed within 72 hours.

No incident occurred; the early-warning system enabled pre-emptive action. Without fleet-scale monitoring, the issue would have been discovered only after a near-miss or incident.

15.4 Distributed Map Validation

HD maps age: construction, re-striping, new signs. Every vehicle that drives a mapped segment either confirms the map or discovers a discrepancy.

15.4.1 Consensus and Staleness

Each map segment has an implicit validation timestamp: the most recent time it was observed consistent with the map. Segments not observed recently carry staleness risk.

Example 15.3 (Map segment staleness estimation). Fleet of 2,000 vehicles, map of 10,000 km. Average vehicle covers 200 km/day.

Coverage rate.

$$\begin{aligned}\text{daily coverage} &= 2000 \times 200 \\ &= 400,000 \text{ km-vehicle/day}.\end{aligned}$$

Each km of map is covered:

$$\begin{aligned}\text{avg visits/day} &= 400,000 / 10,000 \\ &= 40 \text{ visits/km/day}.\end{aligned}$$

High-utilization segments (main roads) see 100+ visits/day; rural segments may see 0–1. A staleness policy degrades confidence on segments not observed in the past week.

15.4.2 Discrepancy Aggregation

Individual discrepancies (“I didn’t see a stop sign the map expects”) are noisy. Aggregating across many vehicles distinguishes real changes from individual perception errors: 10 vehicles in a row reporting the same missing stop sign is near-certain to reflect reality.

15.5 Fleet Coordination and Emergent Behavior

When many AVs share road space, their collective behavior matters. A fleet that all chooses the shortest route at rush hour redistributes congestion, potentially worsening it.

15.5.1 Anti-Coordination Protocols

Dispatch algorithms can explicitly anti-correlate vehicle routing to avoid aggregate congestion. A small coordinated perturbation across the fleet has large emergent effects on road-level congestion.

15.5.2 Protocol Design

Fleet-wide protocols negotiate:

- Lane-change priorities at merge bottlenecks.
- Charging-slot allocation across the fleet.
- Deadhead repositioning to anticipate demand.
- Coordinated slow-down during identified unsafe weather.

15.6 Industrial Deployment

Fleet-systems infrastructure draws heavily on cloud-native architectures. Tesla’s fleet of millions of vehicles communicating with central infrastructure represents the largest deployed AV-fleet system. Waymo’s fleet, while smaller in vehicle count, uses similar cloud-native infrastructure built on Google Cloud. Cruise (during its operational period) used a comparable architecture on Microsoft Azure. The Mobileye EyeQ-equipped vehicle fleet, numbering in the tens of millions, supplies data for Mobileye’s Roadbook and Road Experience Management products. Fleet-anomaly detection at this scale is supported by mature cloud observability tooling including Datadog, Splunk, Elastic, and Prometheus. The MQTT and AMQP messaging standards underpin much of the vehicle-cloud communications infrastructure.

15.7 Chapter Summary

A fleet is more than a collection of vehicles. Collective edge learning enables region-specific adaptation faster than central training. Fleet-wide anomaly detection surfaces systemic issues before they manifest as incidents. Distributed map validation keeps HD maps fresh. Emergent collective behavior requires protocols for coordination across the fleet. These capabilities constitute the fleet’s *collective intelligence*—a capability layer above any individual vehicle.

15.8 Exercises

1. For the collective-edge-learning example, estimate the speedup for a new pedestrian-type class relevant only in one city.
2. Design a four-sigma threshold for fleet confidence-shift detection. What false-positive rate is acceptable?
3. For the map staleness example, compute coverage for a fleet of 500 vehicles covering 100 k km of map.

4. Identify three anomaly-detection signals a fleet should monitor beyond those listed.
5. Propose a protocol that prevents fleet-wide routing cascades causing aggregate congestion.
6. Design a privacy-preserving gradient-exchange protocol for collective edge learning.
7. For the moire-pattern example, propose three engineering-review steps that should follow the alert.
8. Discuss the trade-off between regional-specific models (fit to local patterns) and global uniformity (easier to certify).
9. Propose a charging-coordination protocol for a 300-vehicle fleet with 15 chargers.
10. Compare collective edge learning with federated learning; where do they overlap and where do they differ?

Chapter 16

Forensics and Real-World Operations

16.1 Introduction

A deployed AV occasionally produces an incident, a near-incident, or unexplained behavior. When it does, the operating program must answer two questions quickly and credibly: *what happened*, and *how do we prevent it from happening again*. The first is a forensic question—reconstructing the perception and decision chain that produced the outcome—drawing on the audit pipelines of Chapter 19. The second is an operational question—updating training data, retraining, validating, and redeploying—drawing on the OTA lifecycle of Chapter 14. This chapter addresses the tight loop between the two.

Three observations motivate the structure.

First, incidents are evidence. Every incident reveals a gap in the training distribution or in the reasoning logic; the engineering response must capture, analyze, and incorporate that gap.

Second, speed matters. A fleet continuing to operate with a known vulnerability is exposed until a fix deploys. Fast post-incident remediation is a first-class operational metric.

Third, rollback is essential. A deployment that introduces new failures must be reversible before those failures become incidents. Rollback infrastructure is as important as deployment infrastructure.

16.2 Post-Incident Perception Replay

Given an incident log, reconstruct exactly what perception saw and decided. This requires the audit infrastructure from Chapter 19 plus replay tooling.

16.2.1 Reconstruction Workflow

Step	Action
1	Retrieve event-triggered sensor logs from the incident.
2	Retrieve model-state snapshot active at time of incident.
3	Replay perception stack on logged sensor data.
4	Reconstruct decision-making cascade.
5	Compare reconstructed behavior against observed behavior.
6	Identify divergences; investigate root cause.

16.2.2 Tooling

Replay tools run the same model version on the same hardware- equivalent simulator, producing the same outputs (bit-identical if possible, otherwise semantically-equivalent). Simulation determinism is a key capability; non-deterministic stacks make replay unreliable.

Example 16.1 (Replay reveals perception-confidence issue). Incident: AV made an unprotected left turn in front of oncoming vehicle. Replay of perception stack:

Oncoming-vehicle detection over 10 frames.

- Frames 1–3: confidence 0.35 (below threshold 0.50).
- Frames 4–6: confidence 0.62 (above threshold; tracked).
- Frames 7–10: confidence 0.75.

Analysis.

Detection was present from frame 1 but below threshold for first three frames. By the time it crossed threshold in frame 4, only 0.4 s remained before conflict—insufficient for a safe abort of the turn.

Root cause.

Threshold calibrated against training-set distribution; real- world distribution shifted. Threshold of 0.35 would have given full 1.2 s reaction.

Remediation.

Retrain with additional distant-vehicle data; temporarily reduce threshold to 0.40 in safety-critical contexts.

16.3 Fleet Incident Feedback Loops

Incidents from one vehicle feed improvements that deploy to the entire fleet. The loop from incident to fleet-wide remediation is a central operational discipline.

16.3.1 The Loop

Step	Artifact	Timeline (production)
1	Incident occurs	t_0
2	Auto-triggered data pull	$t_0 + \text{minutes}$
3	Severity classification	$t_0 + \text{hours}$
4	Root-cause analysis	$t_0 + \text{days}$
5	Data augmentation + retraining	$t_0 + \text{weeks}$
6	Validation	$t_0 + \text{weeks}$
7	Staged rollout	$t_0 + \text{weeks}$
8	Fleet-wide deployment	$t_0 + \text{months}$

Leading programs compress this cycle to 6–8 weeks typical; some fast-path fixes deploy in days. Typical industry pace is longer.

16.3.2 Containment

While remediation is underway, fleet behavior can be *contained*: a geofence excludes problematic areas, a speed limit reduces operating envelope, or an operational- domain narrowing takes certain scenarios out of scope. The fleet continues to operate safely in a reduced envelope while full remediation happens.

16.4 Near-Real-Time Model Retraining

For some issues, the full 6–8-week cycle is too slow. Near- real-time retraining paths deploy fixes in days or hours.

16.4.1 When Fast Paths Are Appropriate

- Threshold adjustments and configuration changes.
- Rule-based safety overrides.
- Small additions to training data with well-understood effects.
- Specific environmental adaptations (e.g., unusual weather).

16.4.2 When They Are Not

- Large architecture changes requiring extensive validation.

- Changes with broad impact across the stack.
- Changes interacting with safety-critical invariants.

Production AV programs maintain tiered paths: slow-path for major changes with comprehensive validation, fast-path for specific classes of safe changes.

16.5 Deployment Rollback

Every deployment must be reversible. Rollback infrastructure is as important as forward deployment.

16.5.1 Rollback Triggers

- Incident rate above baseline after deployment.
- Fleet-scale confidence or intervention-rate anomalies.
- Safety-monitor escalations in new model.
- Internal review flags.

16.5.2 Rollback Mechanics

Mechanism	Usage
Dual-binary boot	Previous version always retained; vehicle can switch
Fallback profile	Restrict to previous safer operating envelope
Geofenced rollback	Rollback in specific regions only
Immediate safe-stop	Option of last resort

Example 16.2 (Rollback decision metrics). New model version deployed to 10 % of fleet as canary. After 72 hours:

- Incident rate: baseline 0.05/M miles, canary 0.08/M miles.
- Intervention rate: baseline 2.1/1000 miles, canary 2.3/1000.
- Passenger complaints: baseline 4/1000, canary 5/1000.

Metrics analysis.

Incident rate is 60% elevated but confidence interval includes baseline; too few miles to be conclusive. Intervention rate is 10% elevated, marginal. Passenger complaints up 25%.

Decision.

The composite signal suggests possible regression but not definitive. Hold rollout at 10%, collect 2 more weeks of data, then reassess.

16.6 Legal and Forensic Disclosures

Incidents typically involve regulatory and legal stakeholders.

16.6.1 Regulatory Notification

Most jurisdictions require notification of collision events, ODD exits, and near-miss events above threshold severity to the regulator within defined timeframes.

16.6.2 Evidence Preservation

Incident evidence (sensor logs, model states, audit trails) must be preserved against automated retention-policy deletion until legal matters resolve. Production programs flag incident-related logs for extended retention automatically.

16.6.3 Third-Party Access

Insurance, litigation, and regulator-investigation processes sometimes require sharing of incident data. Structured data-exchange formats (emerging via standards bodies) allow controlled sharing while protecting competitive information.

16.7 Industrial Deployment

Post-incident analysis at AV programs follows methodologies described in NTSB investigation reports of major incidents. Event Data Recorder (EDR) standards in the U.S. (49 CFR Part 563) require recording of specific data elements following crashes. The EU's Event Data Recorder regulation, which entered into force in 2022, mandates EDR equipment in new vehicles. NHTSA Standing General Orders require AV operators to report incidents within tight time windows. The combination of event-triggered logs (sensor data preceding the incident), model-state snapshots (the specific software version active), and reconstruction tooling (deterministic simulation) is established practice across major AV programs. Cruise's October 2023 incident response and the subsequent external review by Quinn Emanuel Urquhart & Sullivan illustrated the forensic-process expectations applied to commercial AV operators.

16.8 Chapter Summary

Forensic replay answers “what happened” using audit logs and deterministic simulation. The fleet-incident feedback loop turns incidents into fleet-wide learning, typically on 6–8-week cycles with fast paths for specific classes of fixes. Deployment rollback infrastructure ensures new deployments are reversible before they produce sustained harm. Legal and forensic disclosures complete the operational obligations. Together these form the operational tail of the AV engineering discipline: what happens after the vehicle ships.

16.9 Exercises

1. For the confidence-threshold replay example, compute the reaction time if detection rose to threshold at frame 2 instead of frame 4.
2. Design a 6-step post-incident triage workflow covering the first 24 hours.
3. For the rollback-metrics example, at what incident-rate elevation would you trigger immediate rollback?
4. Propose a fast-path retraining pipeline covering a specific type of incident (construction-zone related).
5. Compare single-binary and dual-binary deployment architectures; identify trade-offs.
6. Design an evidence-preservation schema that satisfies both operational efficiency and legal requirements.
7. For the feedback-loop example, what improvements would compress the 6-8 week cycle to 2 weeks?
8. Propose a regulatory-notification workflow that minimizes operational disruption while complying.
9. Identify three specific post-incident decisions that cannot be automated and require human judgment.
10. Discuss the trade-off between fleet-wide fast deployment and individual-vehicle safety validation.

Part VI

Evaluation, Testing, and Trust

Chapter 17

Evaluation, Benchmarking, and Performance Validation

17.1 Introduction

Every algorithm developed in the preceding chapters must be *evaluated* before deployment. Evaluation is where claims meet reality: a perception model with 0.92 mAP on its training distribution is useless if it achieves 0.31 on the distribution it actually encounters; a planner that solves benchmark scenarios in 5 ms is useless if it stalls on the intersections the deployment actually sees. A production AV program invests as much in evaluation infrastructure as in algorithm development, and the people building the evaluation pipeline have as much influence on the final product as the people building the models.

This chapter develops the discipline of AV evaluation. It covers benchmarks (the specific datasets and scenarios used for measurement), metrics (the specific numbers used to quantify performance), evaluation modes (open-loop vs closed-loop, simulation vs real-world), reproducibility practices, and the synthesis that builds a defensible performance argument.

Three observations motivate the chapter’s structure.

First, evaluation *precedes* deployment, not follows it. Production AV programs do not treat evaluation as an afterthought; they treat it as the primary gate to release. Every release is blocked behind performance criteria that an evaluation pipeline produces. A pipeline that is trusted, reproducible, and comprehensive is the release-control mechanism.

Second, *no single metric captures AV performance*. Pedestrian detection AP at 0.95 is consistent with both a deployable system and an un-deployable one—the difference is whether the 5 % missed detections include any children, any wheelchair users, or any pedestrians in rain. Stratified metrics, long-tail metrics, and scenario-specific metrics together form the evaluation envelope; a single-number summary is inadequate.

Third, evaluation is *adversarial*. A model tuned on a benchmark will score well on that benchmark regardless of whether it generalizes; a planner tuned against a specific simulator will pass that simulator’s tests regardless of whether it handles the real world. Good evaluation holds out data that the model has never seen, uses scenarios the developer cannot anticipate, and continuously refreshes both.

The chapter is organized as follows. We first survey the benchmarking landscape. We then develop perception, prediction, planning, and control metrics in detail. Evaluation modes—open-loop vs closed-loop, simulation vs real-world, hardware-in-loop—follow, with attention to the sim-to-real gap. We close with reproducibility practices and integrated release-gate workflows. Two architecture diagrams visualize the evaluation pipeline hierarchy and the closed-loop testing architecture; worked examples apply the frameworks to specific scenarios.

17.2 The Benchmark Landscape

Benchmarks are publicly-available datasets and scenarios with defined metrics. They let teams compare progress objectively and let the research community build on shared ground truth.

17.2.1 Perception Benchmarks

Perception benchmarks dominate AV dataset releases. Understanding the major ones is a prerequisite for understanding published perception results.

KITTI

The founding AV benchmark (2012), developed at Karlsruhe Institute of Technology. Stereo cameras, LiDAR, GPS/IMU on a driving car in Karlsruhe, Germany. Tasks: object detection (2D, 3D), tracking, stereo matching, optical flow, visual odometry, semantic segmentation. Dataset size modest by current standards (7 km to 15 km), but published split is still a common reference.

Waymo Open Dataset

Released 2019, expanded through 2024. Multi-modal (5 LiDAR, 5 cameras) and large: 1150 scenes of 20 seconds each, totaling several hours of driving across multiple cities and conditions. Tasks include detection, tracking, motion prediction, end-to-end driving. The go-to large-scale benchmark for current research.

nuScenes

Released by Motional (formerly nuTonomy) in 2019, 1000 scenes each 20 s long, 6 cameras + 5 radars + LiDAR, Boston and Singapore. Strong for multi-modal fusion research and 3D detection benchmarks.

Argoverse

Argoverse (2019) and Argoverse 2 (2022) focus on motion prediction and mapping. Large scale, long recording sequences, high-definition map annotations.

Cityscapes

Urban-scene semantic segmentation dataset (5000 annotated images from 50 cities). Foundational for segmentation research; increasingly supplanted by larger, multi-task datasets for AV- specific work.

17.2.2 Prediction and Planning Benchmarks

Motion prediction benchmarks measure trajectory-forecast quality; planning benchmarks measure closed-loop outcomes under various scenarios.

Argoverse Motion Forecasting

Standard benchmark for motion prediction. Metrics: minADE, minFDE, miss rate, brier-minFDE.

Waymo Motion Prediction Challenge

Large-scale motion prediction with diverse agents and scenarios.

nuPlan

Planning-specific benchmark from Motional. Tests planners in closed-loop simulation against logged human behavior.

CARLA Leaderboard

Open-source simulation-based evaluation for end-to-end driving. Scenarios include intersections, merges, lane changes with adversarial agents.

17.2.3 End-to-End Driving Benchmarks

More recent benchmarks integrate multiple tasks.

Bench2Drive

Urban closed-loop driving benchmark emphasizing tactical decision-making.

LAV

Large-scale learned-vehicle benchmark with integrated perception, prediction, and planning evaluation.

17.2.4 The Problem of Benchmark Saturation

Mature benchmarks saturate: top systems reach accuracy approaching human annotators' agreement level, and further progress is not distinguishable from annotation noise. KITTI object detection has been in this state since roughly 2018. Benchmark-refresh (larger, more diverse data; harder tasks; long-tail scenarios) is an ongoing community practice.

17.3 Perception Metrics

Perception metrics quantify detection, tracking, and segmentation accuracy. This section develops the dominant ones with precise definitions and typical values.

17.3.1 Intersection over Union (IoU)

For detection, the primary measure of box-level correctness:

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (17.1)$$

A prediction is a *true positive* if IoU with a ground-truth box exceeds a threshold (typically 0.5 for 2D, 0.7 for 3D).

Example 17.1 (IoU computation). Predicted box $(x_1, y_1, x_2, y_2) = (10, 10, 50, 50)$, area = 1600. Ground-truth box $(15, 12, 55, 52)$, area = 1600. Intersection: $(15, 12, 50, 50)$, area = $35 \cdot 38 = 1330$. Union: $1600 + 1600 - 1330 = 1870$. IoU: $1330/1870 = 0.711$. At threshold 0.5, this is a true positive; at 0.7, also. At 0.75, it would miss.

17.3.2 Average Precision (AP) and Mean AP (mAP)

Given a confidence-ranked list of predictions, sweep a decision threshold across confidence levels; at each threshold compute precision and recall. AP is the area under the precision-recall curve:

$$\text{AP} = \int_0^1 P(R) dR, \quad (17.2)$$

approximated numerically from discrete threshold points. mAP averages AP across classes.

Example 17.2 (AP for pedestrian detection). Pedestrian detector on a test set with 100 ground-truth pedestrians, evaluated at three decision thresholds.

Per-threshold statistics.

Threshold	Detections	Correct (TP)	Precision	Recall
0.9	40	35	0.875	0.35
0.7	80	65	0.812	0.65
0.5	120	85	0.708	0.85

AP is the area under the precision-recall curve. Integrating numerically over the threshold sweep:

$$AP \approx 0.74.$$

Production pedestrian detectors target $AP \geq 0.90$ for deployable performance; 0.74 is a mid-training-quality result indicating substantial headroom.

Figure 17.1 visualizes the precision-recall curve for this example alongside a production-grade reference curve. The area under each curve is the AP. The shape matters: a curve that holds high precision as recall grows is preferable to one with the same AP but wildly varying precision.

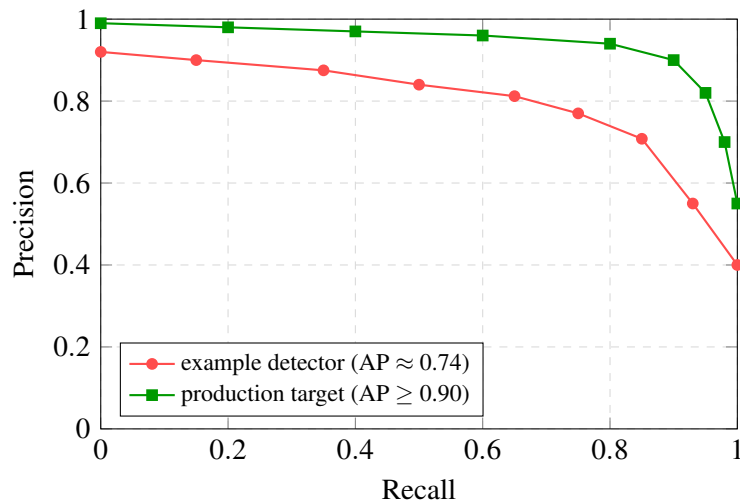


Figure 17.1: Precision-recall curves for the worked-example detector (red, $AP \approx 0.74$) and a production-grade target (green, $AP \geq 0.90$). Area under each curve equals its AP. Production curves hold precision high across the useful recall range; the example detector’s precision degrades notably by recall 0.75, signalling that high-recall operating points will suffer unacceptable false-positive rates.

17.3.3 mIoU for Segmentation

For semantic segmentation, per-class IoU averaged across classes:

$$mIoU = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c + FN_c}. \quad (17.3)$$

17.3.4 MOTA/MOTP for Tracking

Multi-Object Tracking Accuracy (MOTA) and Precision (MOTP) combine false positives, false negatives, and identity switches into tracking-specific scores.

17.3.5 Stratified Metrics

Per-subgroup metrics—per-class AP, per-distance AP, per-weather AP—are essential (Chapter 6). A detector’s overall mAP masks disparities; stratified reporting exposes them.

17.4 Prediction and Planning Metrics

Moving to higher-level stack components, the metrics shift from detection accuracy to trajectory and decision quality.

17.4.1 Prediction Metrics

From Chapter 17: ADE (average displacement error), FDE (final displacement error), minADE/minFDE for multi-modal outputs, miss rate.

17.4.2 Planning Metrics: Open-Loop

Open-loop: evaluate the planner’s output on logged scenes without letting it influence the future. Metrics: trajectory error vs logged human behavior, comfort scores (jerk, acceleration), rule-compliance counts.

17.4.3 Planning Metrics: Closed-Loop

Closed-loop: the planner’s output drives a simulator, whose evolution depends on the planner’s actions. Metrics: collision rate, time-to-completion, progress rate, intervention rate (operator takeovers), scenario-pass rate.

17.4.4 End-to-End Metrics

For end-to-end stacks (Chapter 22): route completion, infractions committed, driving score combining route + infractions.

17.5 Evaluation Modes

Evaluation occurs along two orthogonal axes: open-loop vs closed-loop, and simulated vs real. Together they define four modes; each has strengths, weaknesses, and cost structures.

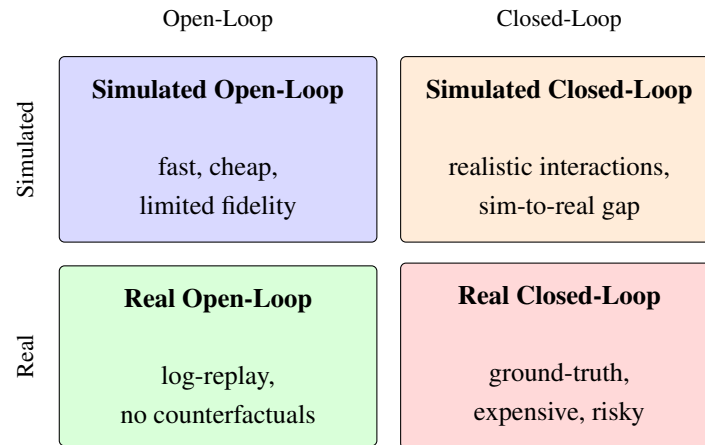


Figure 17.2: The four evaluation modes. Simulated open-loop (fast, cheap, but bounded by simulator fidelity). Simulated closed-loop (captures interaction effects; subject to sim-to-real gap). Real open-loop / log-replay (ground-truth inputs; but no counterfactuals—if the planner deviates from logged behavior, the downstream state does not update). Real closed-loop (the gold standard; also the most expensive and riskiest). Production programs use all four in combination.

17.5.1 Open-Loop Evaluation

The model sees a fixed input and produces output; no feedback. Simple, fast, cheap. For perception: inference on fixed frames. For planning: trajectory prediction on logged scenes.

Advantages

Reproducible; fast; parallelizable; direct comparison against ground truth.

Limitations

Misses interaction effects: if the planner’s decision would have changed the scene, open-loop evaluation cannot capture that. Models that “pretend” to plan well in open-loop can fail in closed-loop.

17.5.2 Closed-Loop Evaluation

The model’s output drives the environment; the environment responds; the model sees the response. Captures interaction effects that open-loop misses.

Simulation-Based Closed-Loop

Scalable; reproducible; can include adversarial agents. Cost is bounded by compute. Sim-to-real gap is the limitation: a system that passes simulation may fail reality.

Real-World Closed-Loop

Ground truth. Expensive. Risk-laden. The evaluation mode that ultimately decides whether deployment proceeds.

Example 17.3 (Open-loop vs closed-loop discrepancy). A learned planner achieves 95 % scenario-pass rate in open-loop (imitation-error-based) evaluation. Closed-loop simulation pass rate on the same scenarios: 70 %. Gap: 25 percentage points. Root cause: small tracking errors compound in closed-loop and drive the system into out-of-distribution states. This phenomenon—the classic covariate-shift problem of Chapter 22—is exactly why closed-loop evaluation is essential.

17.5.3 Hardware-in-the-Loop (HIL)

Actual AV ECUs run in the loop, with simulated sensors and actuators. Catches timing, bandwidth, and integration bugs that pure-software simulation cannot. Typically used for pre-production validation.

Example 17.4 (HIL catching a timing bug). Software-only simulation: perception chain runs in 55 ms, planner in 15 ms. All pass. HIL: real perception ECU shows 95 ms latency under production load; planner deadlines miss intermittently. Bug: software simulation used a single-threaded execution approximation; real ECU contention introduces priority-inversion delays. Fix: explicit thread priorities and zero-copy transport. HIL caught this before real-world rollout, which would have produced safety-critical timing failures in specific conditions.

17.5.4 Real-World Testing

Ultimately, the system is validated on the roads it will operate on. Staged testing protocol (Chapter 32): closed course → public-road with safety driver → driverless.

Fleet Metrics

Fleet-aggregate statistics: intervention rate, collision rate, near-miss rate, completion rate. These are the metrics that regulators and insurers care about; internal programs track them continuously.

Specific-Scenario Replay

When an interesting event occurs (intervention, near-miss, unusual scene), the event is preserved, replayed, and used to improve training.

17.6 Simulation Fidelity and Sim-to-Real Gap

A simulator is only as useful as its fidelity. The sim-to-real gap is the gap between simulator behavior and real-world behavior; its size bounds the trust we can place in simulation results.

17.6.1 Sources of the Gap

Sensor Fidelity

Simulated sensors (rendered cameras, synthetic LiDAR) differ from real ones in subtle and not-so-subtle ways: noise models, optical effects, weather rendering. Domain-randomized training and

real-data fine-tuning narrow the gap.

Agent Behavior Fidelity

Simulated other-agents behave per their programmed models. Real human drivers behave per their human judgments. Modeling realistic traffic agents is an active research problem.

Physics Fidelity

Tire-road interaction, aerodynamics, transmission response—all approximated. High-fidelity physics simulation exists but is computationally expensive.

17.6.2 Bridging the Gap

Production sim-to-real practices:

- **Domain randomization:** train on randomized sim parameters.
- **Real-to-sim calibration:** use real data to tune simulator parameters.
- **Mixed training:** train on a mix of sim and real data.
- **Validated-scenario sets:** qualify a simulator by showing that it reproduces real-world outcomes on controlled scenarios.

17.7 Reproducibility

Reproducibility is the property that an evaluation's result can be recomputed by a different team with the same inputs. Without it, evaluation results cannot be audited or trusted.

17.7.1 Required Inputs for Reproducibility

- Model weights (exact binary).
- Code (exact commit hash).
- Data (frozen, documented split).
- Environment (container image, library versions).
- Seed (for randomized components).
- Hardware (for numerical determinism, sometimes).

17.7.2 Non-Deterministic Sources

- GPU floating-point non-determinism in reductions.

- Data loader shuffling without fixed seeds.
- Atomics in CUDA kernels.
- Race conditions in multi-threaded pipelines.

Production practice: pin everything deterministic; for non-deterministic components, report distributions rather than single runs.

Example 17.5 (Variance across repeated runs). Training a perception model 10 times with different seeds: mAP ranges 0.892–0.908, std 0.006. Reporting a single-run result as “mAP 0.906” misleads readers about real performance. Honest practice: report mean $\pm$ std. Better practice: evaluate on multiple independent test sets and report distribution.

17.8 Integrated Release-Gate Workflow

In production, evaluation feeds a release-gate process that determines whether a new version deploys. The workflow ties together all the pieces above.

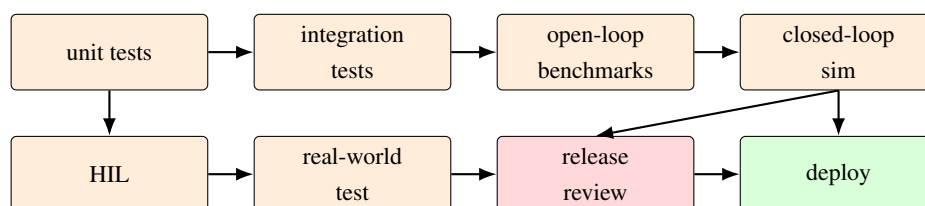


Figure 17.3: Release-gate evaluation workflow. Unit and integration tests catch regressions; open-loop and closed-loop benchmarks measure performance; HIL catches integration-timing issues; real-world testing validates in the deployment environment. Release review synthesizes results from all stages into a deployment decision. A failing result at any stage blocks deployment until resolved.

17.9 Worked Examples

Example 17.6 (Release-gate numerical scenario). New perception model, proposed release:

- Unit tests: pass.
- Integration tests: pass.
- Open-loop mAP: 0.905 (threshold 0.90). Pass.
- Closed-loop scenario-pass: 0.94 (threshold 0.93). Pass.
- HIL: latency 72 ms (limit 100). Pass.
- Real-world 200-mile safety-driver test: 2 interventions (target ≤ 3). Pass.
- Stratified subgroup report: children AP 0.81 (aggregate threshold 0.85). **Fail**.

Release blocked; targeted data-collection and retraining prioritized. The aggregate score passes, but the subgroup gap is a blocker. Gates work only if they include every relevant criterion.

Example 17.7 (Evaluating a new simulator). New simulator claims improved fidelity. Validation protocol:

- Take 100 real-world logs. Replay in both simulators.
- Measure: does each simulator reproduce the actual outcome?
- Compare distributional metrics: trajectory error mean, variance; collision frequency; intervention rate.

Old simulator: trajectory error std 1.2 m, interventions 0.02/mile. New simulator: trajectory error std 0.6 m, interventions 0.015/mile. Real-world: std 0.4 m, 0.01/mile. New simulator narrows the sim-to-real gap; acceptance criterion met. Adoption proceeds after additional benchmarks.

Example 17.8 (Long-tail stratification uncovers safety gap). A detector scores AP 0.92 overall. Stratification:

- Dry daytime: AP 0.96.
- Rain: AP 0.89.
- Night: AP 0.78.
- Snow: AP 0.55.

Deployment scope includes rain, night, and occasional snow. Gaps in those conditions must close before deployment in full operating envelope. Stratified reporting exposes risk that aggregate reporting hides.

17.10 Industrial Deployment

Industry evaluation of AVs spans multiple measurement frameworks. The California Department of Motor Vehicles requires AV testing companies to file annual disengagement reports, which though imperfect provide publicly-available comparison data. The Insurance Institute for Highway Safety (IIHS) and Euro NCAP rate ADAS features in production vehicles, with Euro NCAP introducing more stringent ADAS-rating requirements in 2023 and 2024. The U.S. National Highway Traffic Safety Administration issues Standing General Orders requiring AV operators to report incidents. Internal evaluation in production AV programs typically combines on-road test mileage, simulation mileage (typically thousands of times larger than on-road), structured scenario testing, and shadow-mode evaluation against human-driver baselines. Waymo and Cruise have published peer-reviewed papers describing their respective evaluation methodologies.

17.11 Chapter Summary

Evaluation is the discipline of measuring AV performance before and during deployment. Benchmarks (KITTI, Waymo, nuScenes, Argoverse, CARLA, nuPlan) provide shared ground for objective comparison. Metrics (IoU, AP, mAP, ADE, minADE, intervention rate, scenario-pass rate) quantify specific aspects; stratified reporting exposes disparities. Four evaluation modes— open/closed-loop

× sim/real—each have distinct properties; production programs use all four. Hardware-in-the-loop catches integration issues pure software cannot. Reproducibility of results requires discipline about code, data, environment, and seeds. A release-gate workflow ties all the pieces together: failing any gate blocks deployment. Evaluation infrastructure is as critical as the algorithms it tests; it is the mechanism by which engineering rigor becomes safety guarantee.

17.12 Exercises

1. For the IoU example (17.1), compute IoU for the same prediction against a ground-truth box (20, 20, 50, 50).
2. Derive the AP given a precision-recall curve with the four points $(R, P) = (0.25, 0.9), (0.5, 0.85), (0.75, 0.75), (1.0, 0.6)$. Use the 11-point interpolation.
3. Compare open-loop and closed-loop evaluation for a motion predictor. Which mode is more informative, and why?
4. Design a stratified evaluation plan for a pedestrian detector. List at least five strata and justify their inclusion.
5. For the release-gate example (17.6), propose a data-collection strategy to close the children AP gap.
6. Identify three specific sources of non-determinism in a typical training pipeline and describe how each can be mitigated.
7. Define a metric for “closed-loop safety” that cannot be gamed by training the predictor on the closed-loop data.
8. For the long-tail example (3.1), compute the expected aggregate AP if deployment distribution is (60 % dry day, 20 % rain, 15 % night, 5 % snow).
9. A simulator’s closed-loop pass rate on a benchmark is 95 % but real-world intervention rate is 10× that of the benchmark’s implied rate. Propose three explanations.
10. Design a 6-month evaluation roadmap for a planner release, integrating unit tests, open-loop, closed-loop, HIL, and real-world testing.

Chapter 18

Accidents, Collisions, and Lessons Learned

18.1 Introduction

The safety chapter (Chapter 32) developed the engineering discipline for avoiding collisions; this chapter develops the engineering discipline for what happens when collisions occur anyway. No deployed vehicle—human-driven or autonomous—will be collision-free forever. What differs is how collisions get analyzed, what lessons get extracted, and how the broader industry learns from each event. AV programs that treat accident response as a post-mortem technical activity outperform those that treat it as a public-relations problem. This chapter treats the former, unflinchingly.

The chapter covers both the specific incidents that have shaped AV deployment (Uber ATG 2018 Tempe, multiple Tesla Autopilot fatalities, Cruise San Francisco 2023, various Waymo events) and the general frameworks for analyzing, reporting, and learning from incidents. Every mature AV program runs an incident-learning loop; the quality of that loop determines how quickly the program improves.

Three observations motivate the chapter’s structure.

First, AV accidents are *learning opportunities* that the industry is still figuring out how to share. Some programs publish detailed post-incident analyses; others do not. Regulators increasingly require disclosure; adversarial litigation sometimes forces it. The industry-level learning rate is limited by what gets shared, not what gets discovered.

Second, AV safety comparisons with human driving are *genuinely difficult*. The statistics are confounded by selection (AVs operate in selected ODDs), by small sample sizes (billions of miles are needed for credible fatality-rate comparisons), by differing definitions of comparable incidents, and by evolving technology (today’s AV is not last year’s AV). Every comparison has caveats.

Third, incident response is *regulated and adversarial*. Disclosure timelines are mandated; litigation responses are strategic; engineering truth must navigate both without compromising either. The engineer’s job is to get the root cause right and document it defensibly; the lawyer’s job is to

communicate it strategically.

The chapter is organized as follows. We first develop a taxonomy of AV-involved collisions. We then survey notable historic incidents and the lessons each produced. A framework for post-incident analysis follows. We then cover statistical comparison with human-driven fleets, near-miss analysis, reporting frameworks, and the engineering-learning loop that converts incidents into program-wide improvements. Two architecture diagrams visualize the collision-type taxonomy and the incident-response pipeline; worked examples cover specific analyses.

18.2 Collision Taxonomy

Not all collisions are the same. A useful taxonomy organizes incidents by mechanism, by ego role, and by severity.

18.2.1 By Mechanism

The physical type of collision constrains what could have prevented it.

Rear-End

Ego struck from behind (not-at-fault), or ego struck another vehicle from behind (at-fault). Rear-end at-fault collisions are the primary failure mode of ACC/AEB-equipped vehicles; the benchmark is whether AEB detects and brakes in time.

Lateral

Sideswipes, lane-change collisions. Usually involves failure in blind-spot detection or lane-change coordination. Common cause of AV-involved incidents in mixed-traffic scenarios.

Intersection

Collisions at intersections: unprotected left turns, red-light running (by either vehicle), unsignaled crossings. Intersection collisions are often the most severe because approach speeds are perpendicular and structural impact is lateral.

Vulnerable-Road-User (VRU)

Pedestrians, cyclists, motorcyclists. These are the most consequential AV failure mode because human fatality rates are very high.

Single-Vehicle

Ego leaves the road or strikes a fixed object. Usually involves planning or control failure rather than perception failure.

18.2.2 By Ego Role

- **Ego at-fault:** ego’s action caused the collision.
- **Other at-fault:** another party’s action caused the collision; ego was present but could not have prevented.
- **Mixed:** comparative-fault applies; multiple parties bear partial responsibility.

18.2.3 By Severity

- **No injury:** property damage only.
- **Minor injury:** treat-and-release medical care.
- **Serious injury:** hospitalization.
- **Fatal:** death.

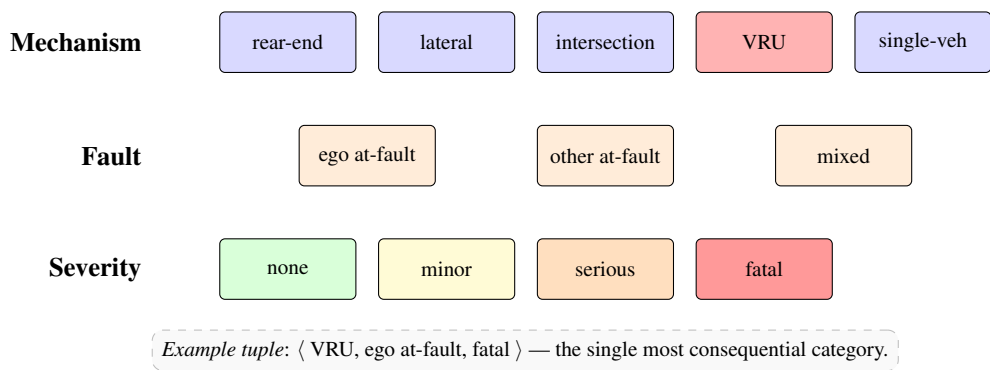


Figure 18.1: Three-axis collision taxonomy: mechanism (physical impact type), fault (which party’s action caused the collision), and severity (injury outcome). Each incident receives a tuple across the three axes; analysis and learning are organized by tuple. VRU collisions (red) are highlighted as the most consequential mechanism category; they deserve the most attention from every AV program.

18.3 Notable Historic Incidents

Specific incidents have shaped the AV safety landscape. Each is worth studying in detail; the summaries here are necessarily brief but highlight the lesson.

18.3.1 Uber ATG, Tempe Arizona, March 2018

The first pedestrian fatality in an AV-involved collision. A self-driving Uber test vehicle, with safety driver inattentive, struck and killed a pedestrian crossing a multilane road at night.

What Happened

The system detected the pedestrian approximately 5.6 s before impact but classified them variously as vehicle, bicycle, and unknown. Class-specific tracking policies disabled some braking behaviors.

Emergency braking was disabled by policy. The safety driver was watching a phone.

Lessons Extracted

Multiple, each operationalized across the industry: (a) Safety- driver monitoring is essential; (b) Class-conditioned tracking behaviors can produce unsafe composite behaviors; (c) Emergency braking should never be disabled without very explicit reason; (d) Pedestrian-detection reliability in the specific scene geometry (dark multilane road) is a legitimately hard engineering problem.

Program Consequences

Uber ATG self-driving program paused; Arizona approval conditions changed; NTSB investigation produced a formal report widely used as training material for new AV engineers.

18.3.2 Tesla Autopilot Fatalities

Multiple fatalities in vehicles with Tesla's Autopilot driver- assistance system engaged. Not an L4 system, but branded in ways that many observers felt overclaimed its capability.

Pattern

Vehicle with Autopilot engaged failed to detect a stationary or slow-moving obstacle (truck crossing the road, firetruck parked in-lane, emergency-responder vehicle); driver inattention meant no corrective action was taken; high-speed impact followed.

Lessons

Operational-design-domain communication with users is critical: the system was engaged in scenarios outside its reliable operating envelope. Branding that implies more capability than the system provides is a contributor to misuse. Driver- monitoring is essential for any driver-assistance system.

18.3.3 Cruise, San Francisco, October 2023

A Cruise robotaxi struck and dragged a pedestrian who had already been struck by a human driver. The incident led to a temporary cessation of Cruise operations and a significant organizational restructuring.

What Happened

A separate vehicle hit a pedestrian, throwing them into the Cruise vehicle's path. Cruise's AV struck them, then attempted a pullover maneuver, dragging the pedestrian about 6 m. The follow-up pullover maneuver was not in the company's disclosed description of the incident.

Lessons

Incident disclosure rigor matters: omitting material facts to regulators compromises trust. Post-collision policy needs dedicated engineering—an AV must recognize it has been involved in a

collision and behave appropriately afterwards.

18.3.4 Other Notable Incidents

Many. Waymo has had several minor collisions publicly reported; Zoox incidents; Navya shuttle incidents; Mercedes-Benz DRIVE PILOT incidents. The accumulating record forms a substantial public case library; every new AV program is expected to understand and engineer around the lessons.

18.4 Post-Incident Analysis Framework

A disciplined post-incident analysis extracts lessons without blame-seeking. The framework is not invented for AVs—it is borrowed from aviation, nuclear, and medical incident-review traditions.

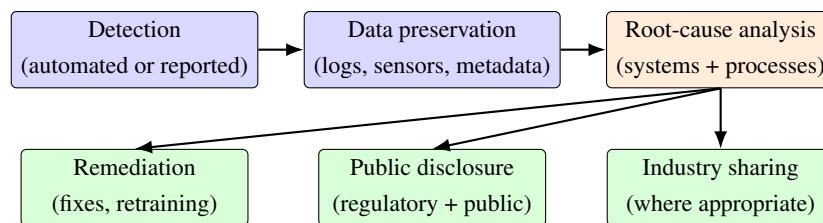


Figure 18.2: Post-incident analysis pipeline. An incident is detected (by onboard sensors, user report, or observation). Relevant data is preserved with integrity. Root-cause analysis traces the chain of events through system and process causes. Outputs: remediation (fixes the system), public disclosure (regulators and the public), and industry sharing (where legally and strategically appropriate). Each leg has a different timeline: remediation urgent, public disclosure within hours to days, industry sharing on a longer timeframe.

18.4.1 Root-Cause Analysis

The standard technique is the “5 Whys”: ask “why?” repeatedly until the chain of causation reaches a fixable root. Beware the stop-at-human-error trap: stopping at “engineer misconfigured parameter” without asking why the system allowed the misconfiguration is classic shallow analysis.

Example 18.1 (5 Whys in an AV incident). Incident: AV hit a roadwork cone.

1. Why did the AV hit the cone? The cone was not detected at long range.
2. Why? The detector’s cone class has lower confidence at range than the object-blocker threshold.
3. Why? Training data had few cones at long range.
4. Why? Data-collection protocol prioritized common objects.
5. Why? The collection prioritization was not explicitly reviewed against safety-critical rare objects.

Root cause: collection prioritization review gap. Remediation addresses the process (explicit review of safety-critical rare objects in all future collection rounds), not just the symptom (add cone data).

18.4.2 Systems and Processes

Most incidents have both a proximate technical cause (a specific failure) and a process cause (why the system allowed that failure). A mature analysis addresses both. Fixing only the technical cause usually permits recurrence in a different form.

18.4.3 Blame-Free Culture

Engineers must be able to report near-misses, concerns, and errors without fear of punishment. Programs with blame-seeking cultures systematically under-learn—the information needed to improve is suppressed. Aviation’s “just culture” is the model: distinguish honest mistake from wilful misconduct; learn from both.

18.5 Statistical Comparison with Human Driving

Comparisons between AV and human driving are the most common way AV safety gets evaluated publicly. This section develops the methodology and its pitfalls.

18.5.1 Rate Definitions

- **Fatality rate per mile:** fatalities / miles. Human reference in the US: roughly 1.1×10^{-8} /mile.
- **Injury rate per mile:** all-injury rate, roughly 1×10^{-6} /mile.
- **Collision rate per mile:** all collisions (including property-damage-only), higher still.

18.5.2 Sample-Size Requirements

To demonstrate with 95 % confidence that an AV fleet’s fatality rate is below the human rate, you need many miles without fatalities. Specifically, to reject the null hypothesis “AV rate = human rate” in favor of “AV rate half the human rate,” requires roughly 2×10^8 miles without fatalities.

Example 18.2 (Required miles for statistical claim). Goal: demonstrate AV fatality rate $\leq 0.5 \times$ human rate with 95 % confidence. Using the Poisson model and a null hypothesis of 1.1 fatalities per 1×10^8 miles: expected fatalities after 3×10^8 miles = 3.3 (human rate), or 1.65 (AV half-rate claim). Observing 0 or 1 fatalities in 3×10^8 miles would support the claim; more miles strengthen it. Current driverless fleet totals: Waymo publishes tens of millions of miles; credible fatality-rate claims remain some years away at current scales.

18.5.3 Confounding and Selection Bias

Operational Design Domain Selection

AVs operate in selected conditions (mapped roads, clear weather, daytime); human drivers operate in all conditions. Comparing rates without controlling for ODD mismatches produces biased conclusions.

Technology Evolution

Last year's AV technology is not this year's. A rate observed over 5 years of operation reflects average performance, not current performance. Decomposing by time period matters.

Reporting Threshold

Different companies report incidents at different thresholds. Comparisons assume a shared threshold; when they do not hold, comparisons are invalid.

18.5.4 Comparative Exposure Metrics

Beyond per-mile rates:

- Per-hour (accounts for speed differences).
- Per-trip (reflects trip-level risk).
- Per-ODD (compares within the AV's operational envelope).
- Severity-weighted (fatality-equivalents per mile).

18.6 Near-Miss Analysis

Near-misses—events that could have been collisions but were not—occur far more frequently than actual collisions. They provide much stronger statistical signal for learning.

18.6.1 Near-Miss Definitions

- **Hard-brake:** ego's maximum-effort braking triggered.
- **Evasion maneuver:** ego executed emergency avoidance.
- **Other-caused:** ego was nearly struck by another road user's action.
- **Perception-close-call:** detection latency or error brought the system close to a collision outcome without reaching one.

18.6.2 Near-Miss Ratios

In aviation, the ratio of near-misses to incidents is roughly 300:1 (Heinrich's triangle, approximate). For AVs, similar ratios hold: hundreds of near-misses per actual collision. Analyzing near-misses gives hundreds of times the statistical power of analyzing collisions alone.

18.6.3 Continuous Near-Miss Monitoring

Production AV programs track near-miss rates as a leading indicator:

- Hard-brake rate per mile.
- Evasion-maneuver rate per mile.
- Close-call detection per mile.
- Time-to-collision distribution (histogram; shift toward zero signals degraded margin).

Example 18.3 (Near-miss trend as leading indicator). Fleet hard-brake rate: 1/100 miles (month 1), 1.1/100 miles (month 2), 1.3/100 miles (month 3). A collision happens in month 4. Root-cause trace: subtle regression in perception calibration after month-1 update; the rising hard-brake rate was the leading indicator. Had the rate been monitored with a threshold alert, the regression could have been caught before the collision. Production fleet-health dashboards include these trends precisely for this purpose.

18.7 Reporting Frameworks

Regulatory and public reporting of AV incidents is increasingly formalized.

18.7.1 NHTSA Standing General Order 2021-01

US federal order requires manufacturers of SAE Level 2+ vehicles to report specific crashes within 24 hours for serious incidents, five days for others. Data made public. The dataset has become the most comprehensive public record of US AV incidents.

18.7.2 California DMV Reporting

California requires AV permit holders to report all collisions within 10 days; annual disengagement reports also required (Chapter 32). Public disclosures routinely scrutinized by media and researchers.

18.7.3 Company Voluntary Disclosure

Several AV programs (Waymo, Zoox) voluntarily publish safety data at intervals. The content and form vary; critics argue disclosures are selective; proponents argue they are more than what regulation requires.

18.7.4 Scientific Peer Review

A few AV-related incident analyses have been published in peer-reviewed safety journals. This form of sharing is rare but provides a pathway for generalized lessons to reach academic communities and other industries.

18.8 The Engineering-Learning Loop

Incidents feed back into engineering improvements. The specific mechanisms determine how fast programs improve.

18.8.1 Data-Collection Targeting

Incident scenarios and near-incident scenarios are reproduced (in real fleet and in simulation) and used to target new data collection. Conditions similar to the incident drive collection priorities.

18.8.2 Training-Set Integration

New data from incident scenarios flows into training. The perception or policy that failed is retrained with the new examples.

18.8.3 Scenario-Library Expansion

The scenario library used for release-gate testing grows by every incident's reconstruction. A new test is added so that future versions cannot regress on the specific failure mode.

18.8.4 Process Improvement

Process causes (why the system allowed the failure) drive process changes: review checklists, threshold requirements, escalation rules. Without process improvement, similar technical failures recur in different guises.

Example 18.4 (Full learning loop for a specific incident). Incident: AV failed to yield to an ambulance with sirens.

- Root cause: the audio-detection module was disabled during a recent release due to compute-budget concerns.
- Process cause: compute-budget review didn't flag the safety implications of disabling audio.
- Fixes: (a) re-enable audio detection in a budget-efficient form; (b) add emergency-vehicle audio classifier to the mandatory safety checklist; (c) reconstruct the incident in simulation and add it to the regression suite.
- Timeline: 6 weeks from incident to full closure of the loop.
- Industry sharing: post-incident report published; contributes to emerging industry standard on emergency- vehicle detection.

Both the technical cause and the process cause addressed; the scenario is now in the regression suite so that future versions cannot regress on the same failure mode.

18.9 Worked Examples

Example 18.5 (Post-incident disclosure timing). Minor collision involving AV; no injuries. Required timeline:

- Hour 1: onboard detection of impact (IMU shock); data preservation triggered.
- Hour 4: internal notification chain complete; incident team activated.
- Hour 12: regulatory filing (California DMV: 10-day window, but most programs file within 24 hours for reputational reasons).
- Hour 24: initial public statement.
- Week 2: preliminary analysis.
- Week 6: detailed analysis published.
- Week 8: remediation deployed.

The schedule is a ceiling, not a floor; programs that disclose faster build more trust. The first public statement acknowledges the incident, the AV's role, and commits to further disclosure; it does not assign blame prematurely.

Example 18.6 (Counter-factual analysis). Incident: AV struck a stopped vehicle at a merge. Investigation: ego perception detected the stopped vehicle at 2.1 seconds before impact; at ego speed 18 m s^{-1} , required stopping distance 28 m; distance at detection 37.8 m. Braking command issued at detection; deceleration delayed by actuator response time 0.18 s; effective stopping distance grew to 31.5 m. Crash occurred with $\sim 2 \text{ m s}^{-1}$ residual velocity. Counter-factual: had the planner issued brake command 0.2 s earlier, the vehicle would have stopped $\sim 1.5 \text{ m}$ short of impact. Root cause: the detect-to-brake chain had 0.2 s of unnecessary software-level latency. Remediation: pipeline optimization reduces latency by 0.35 s; next simulation run shows the scenario completes with 2.3 m margin.

Example 18.7 (Near-miss review feedback). Monthly near-miss review. 1200 hard-brake events in the prior month; review sample of 50 identified for human-analyst review. Findings:

- 35 events: other-agent caused; no AV issue.
- 8 events: AV responded appropriately to genuinely borderline scenes.
- 5 events: AV over-reacted to false-positive detections; fixed in next release.
- 2 events: AV under-reacted initially; safety driver intervention triggered AEB. Significant finding.

The 2 under-reaction events become the month's highest-priority engineering work; all other findings are logged for longer-term tracking. The near-miss stream is the program's primary feedback mechanism between collisions.

18.10 Industrial Deployment

Several incidents have shaped the AV safety landscape and are documented in detail by the U.S. National Transportation Safety Board (NTSB). The March 2018 fatal collision in Tempe, Arizona, in-

volving an Uber ATG test vehicle and pedestrian Elaine Herzberg led to extensive NTSB investigation and contributed to Uber's eventual sale of its self-driving unit to Aurora in 2020. The October 2023 incident in San Francisco involving a Cruise robotaxi and a pedestrian led to a nationwide pause in Cruise operations and the eventual decision by General Motors in December 2024 to wind down its robotaxi business. Multiple Tesla Autopilot-involved fatalities have been investigated by NTSB. The European Union's forthcoming Automotive Crash Data Recorder requirements, NHTSA Standing General Orders on AV crash reporting, and CCRS (Crash Reporting Standing General Order) provide expanding public datasets on AV-involved incidents.

18.11 Chapter Summary

Accidents are the unavoidable other side of AV deployment. Taxonomy (mechanism, fault, severity), historic incidents (Uber, Tesla, Cruise, and others), post-incident analysis (root cause, system + process), statistical comparison with human driving, near-miss analysis, regulatory reporting, and the engineering-learning loop all together constitute the discipline of accident handling. Programs that operate this discipline with rigor and transparency improve quickly and earn public trust; programs that do not, do not. The incident record is the industry's collective learning; every engineer should know its major cases and their lessons.

18.12 Exercises

1. For the collision taxonomy (18.1), classify the Uber Tempe incident across all three axes.
2. Apply the 5 Whys to a hypothetical incident: AV rear-ended a stopped truck that was in the lane for roadworks.
3. For the statistical-miles example (18.2), compute the required miles to demonstrate the AV fatality rate is $0.1 \times$ the human rate.
4. A new AV program wants to claim safer-than-human performance after 1 year of operation (10 million miles, 0 fatalities). What is the statistically-honest claim?
5. Identify three specific process failures in the Uber Tempe incident and propose the process change that addresses each.
6. For the near-miss trend example (18.3), propose a threshold-alert system that would have flagged the regression before the collision.
7. Compare aviation's blame-free culture to automotive product-liability culture. How do these interact with AV incident response?
8. Design an incident-response plan for a specific AV program. Identify roles, responsibilities, and communication chains.

9. For the learning-loop example (18.4), identify one place the loop could break down and propose a mitigation.
10. Describe three specific ways selective reporting can distort AV-human safety comparisons, and propose methodological responses to each.

Chapter 19

Trust and Assurance for AV Perception

19.1 Introduction

A production autonomous vehicle must not only *perform* correctly—it must be *shown* to perform correctly. Regulators, insurers, and the traveling public ask not only “does it work?” but “how do you know it works?” and “what evidence can you provide?”. The engineering discipline that produces this evidence is *trust and assurance*: the combination of auditing, formal verification, and structured safety-case construction that turns a functioning system into a certifiable one.

Three observations motivate the chapter.

First, trust is *earned by evidence, not by performance*. A system that passes all tests but cannot explain why it passes them fails the assurance bar, regardless of its score. Auditable pipelines that log the reasoning behind decisions are now required by most AV regulatory frameworks.

Second, formal verification—mathematical proofs of correctness—has become *tractable* for specific AV components. Not the whole perception stack, but its safety monitors, geometric transforms, and certain planner modules now admit formal guarantees.

Third, *safety cases* are the unifying structure connecting evidence to claims. A safety case is an argued demonstration that the system is adequately safe for its intended use; it is the central artifact in most AV certification frameworks.

19.2 Perception Auditing Systems

Auditing is the runtime instrumentation that records *why* the perception stack reached each of its outputs. In contrast to telemetry, which records *what* the stack output, auditing records the upstream reasoning: which sensor inputs drove the output, which model activations dominated, which post-processing rules fired.

19.2.1 What Audit Logs Record

Level	Recorded information
Input	Sensor frames (with retention policy)
Feature	Backbone activations at critical layers
Head	Pre-NMS detections with per-class scores
Fusion	Which tracks matched which detections
Post	Which rules fired, which overrides applied
Output	Final scene representation sent to planner

Retention varies by level: raw sensor frames are too voluminous to store indefinitely, so retention is event-triggered (save sensor data around any high-severity event); feature-level data is lighter and can be retained longer; output-level data is retained across the operational life of the vehicle.

Example 19.1 (Audit-log volume estimation). A single AV producing audit data at the following rates:

- Input (event-triggered): 10 events/day $\times$ 10 s each $\times$ 500 MB/s = 50 GB/day.
- Feature (continuous, sparse): 30 MB/s $\times$ 10 h = 1 TB/day.
- Output (continuous, compact): 500 KB/s $\times$ 10 h = 18 GB/day.

Per-vehicle daily audit volume.

$$V_{\text{day}} \approx 50 + 1000 + 18 \\ \approx 1.07 \text{ TB/day.}$$

Annualized for a 100-vehicle fleet.

$$V_{\text{year}} = 100 \times 365 \times 1.07 \text{ TB} \\ \approx 39 \text{ PB/year.}$$

Cloud storage economics push hard on what can and cannot be retained. Tiered retention (hot for 30 days, cold for 2 years, discard after) is standard practice; event-triggered snapshots break this convention for incidents requiring long-term preservation.

19.2.2 Auditable Models

For audit to be meaningful, the model must be interpretable at the level where auditing occurs. Techniques from Chapter 28 (Grad-CAM, integrated gradients, attention visualization) translate audit logs into human-readable explanations.

19.3 Formal Verification of Perception

Formal verification produces mathematical proofs that a software or hardware component satisfies a specification. Historically considered intractable for neural networks, formal methods have advanced enough to verify specific AV components.

19.3.1 What Can Be Verified

- Geometric transforms (camera calibration math, coordinate conversions).
- Safety envelopes (bounds on distance, velocity, jerk).
- Control invariants (stability, non-oscillation).
- Neural network robustness in a perturbation ball (for small networks).
- Rule-based safety monitors that override the learned planner.

19.3.2 What Cannot (Yet) Be Verified

- End-to-end perception models at their full scale.
- Behavior on arbitrary open-world scenes.
- Emergent interaction effects at fleet scale.

19.3.3 Tools

Production formal-methods toolchains for AVs:

Tool	Scope
CBMC, SMACK	C/C++ code-level bounded model checking
Frama-C	Static analysis with ACSL contracts
Coq, Isabelle	Theorem proving for algorithms
Marabou, Reluplex	Neural-network verification (small scale)
dReal, Z3	SMT-based reachability for control

Example 19.2 (Verified safety envelope). A safety monitor asserts: no commanded acceleration exceeds 6 m s^{-2} in absolute value, under any input.

Formal specification.

$$\forall t, \forall \text{input}, \quad |a_{\text{cmd}}(t)| \leq 6.0 \text{ m/s}^2.$$

Proof structure. The envelope is a saturated function

$$a_{\text{cmd}}(t) = \text{clip}(a_{\text{nom}}(t), -6, 6),$$

with the saturation proved correct against any integer-overflow or NaN edge case by static analysis

(CBMC).

Result. This single invariant, proved once, protects against an entire family of failure modes: upstream planner bugs, numerical instabilities, and certain adversarial-input exploits. The planner can be complex and partially-unverified; the monitor is small and fully-verified, and bounds the outcome.

19.4 The ISO 26262 V-Model Lifecycle

ISO 26262 organises functional-safety engineering as a V-shaped lifecycle: requirements decomposition flows down the left arm of the V from concept to implementation, while verification and validation flow back up the right arm with each test stage matched to its corresponding specification stage. Figure 19.1 shows the canonical structure.

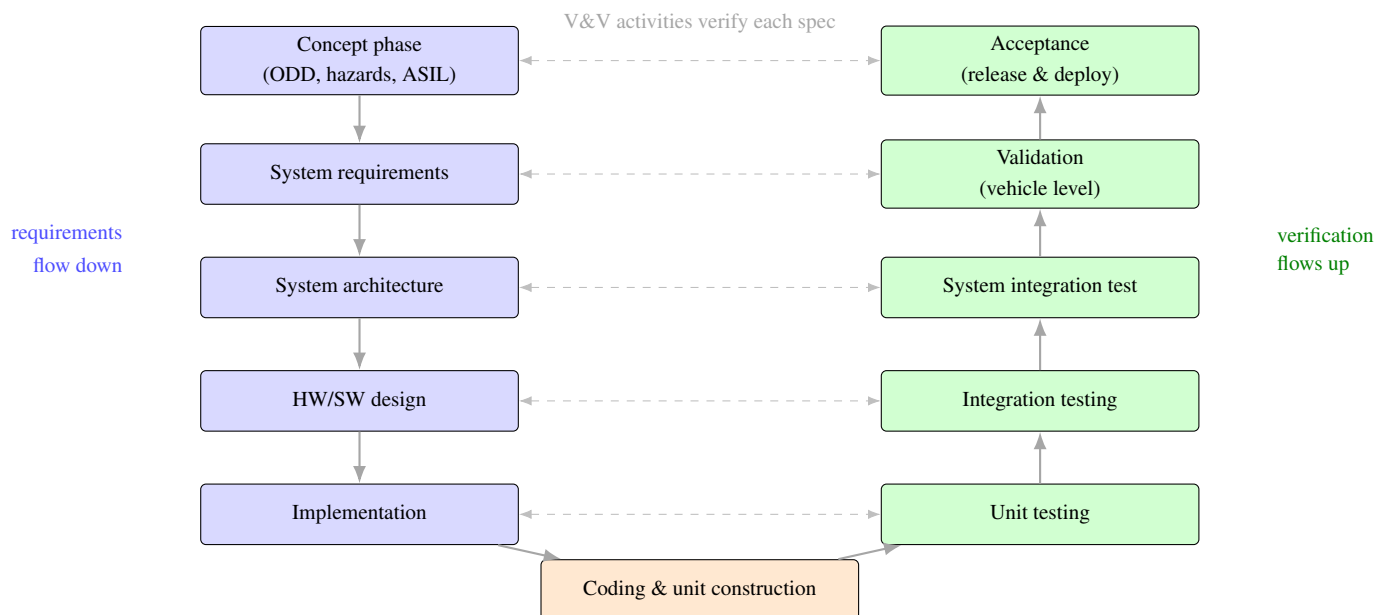


Figure 19.1: The ISO 26262 V-model lifecycle. Specification activities on the left arm proceed top-down: from concept (ODD definition, hazard identification, ASIL classification) through system requirements, architecture, HW/SW design, implementation, and coding. Verification activities on the right arm proceed bottom-up: from unit testing through integration, system integration, vehicle-level validation, and final release acceptance. Dashed horizontal arrows match each verification stage to the specification it tests against. Defects detected at each level should be traceable to the corresponding spec stage on the opposite arm.

19.5 Safety Cases

A safety case is a structured argument, supported by evidence, that a system is adequately safe for a given application in a given environment. In AV regulatory frameworks (UL 4600, ISO 21448 SOTIF, ISO 26262) the safety case is the central artifact.

19.5.1 Goal-Structuring Notation (GSN)

GSN is the standard diagrammatic notation for safety cases: goals (claims to be demonstrated), strategies (how a goal decomposes into sub-goals), contexts, assumptions, and evidence.

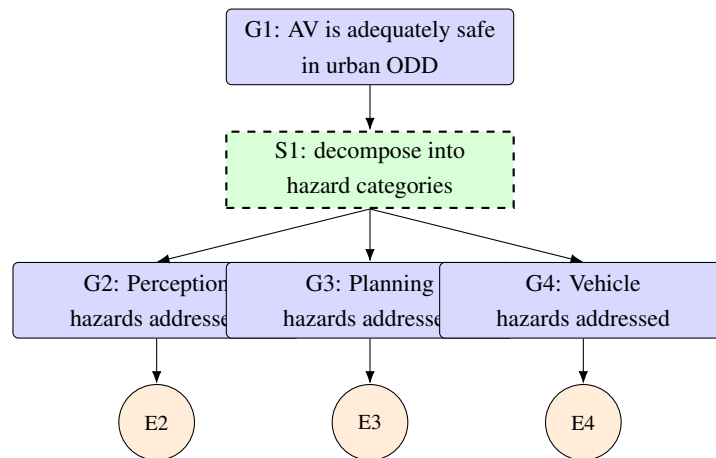


Figure 19.2: Goal-Structuring Notation fragment. A top-level safety goal (G1, solid box) decomposes via a strategy (S1, dashed box) into sub-goals (G2–G4), each eventually supported by concrete evidence artifacts (E2–E4, circles): test reports, audit logs, formal proofs.

19.5.2 UL 4600: Safety Case Standard

UL 4600 is the first comprehensive safety standard for fully autonomous products. It does not prescribe specific engineering methods but instead requires that the developer produce a coherent safety case demonstrating the system is safe enough.

19.6 Falsification: The Adversary's Perspective

Safety cases benefit from hostile review. *Falsification* techniques actively search for scenarios where the system fails, in contrast to testing that confirms the system works. Closely related to adversarial testing (Chapter 28) but focused on higher-level behavioral failures rather than raw-input perturbations.

Example 19.3 (Falsification yield). A falsification campaign runs for 2 weeks against a candidate release, using gradient-based parameter sweeps over scenario parameters.

Results.

- Scenarios tested: 125,000
- Failures found: 47
- Unique failure categories: 12
- Regressions from prior release: 2

Per-unique-failure ROI.

$$\begin{aligned}\text{cost/failure} &= \frac{\$200k \text{ infra} + \text{engineering}}{12 \text{ unique}} \\ &\approx \$17k/\text{unique failure found}.\end{aligned}$$

Compare: an incident in production costs much more in settlement, reputational damage, and remediation. Falsification is highly cost-effective when compared to field-failure economics.

19.7 Industrial Deployment

Safety-case methodologies developed in this chapter are increasingly formalised in AV practice. UL 4600, published by Underwriters Laboratories in 2020 as “Standard for Safety for the Evaluation of Autonomous Products,” provides explicit guidance on safety-case construction for products incorporating autonomy. The Goal Structuring Notation (GSN) referenced in this chapter is documented in the GSN Community Standard maintained by the Safety-Critical Systems Club. Waymo, Cruise, Mobileye, and Aurora have published public safety-case documents describing aspects of their assurance arguments. The Robust Software Engineering for Autonomous Systems consortium (RoSE) and the AVSC (Automated Vehicle Safety Consortium, an SAE-managed group) coordinate industry safety-case methodology development.

19.8 Chapter Summary

Trust and assurance for AVs rests on three pillars: auditing pipelines that log the reasoning behind decisions; formal verification that proves safety-critical components correct; and safety cases that structure the argument and evidence for regulators. Falsification campaigns add adversarial pressure. UL 4600 codifies the overall discipline. These techniques do not replace the underlying engineering—they make it accountable, which is the difference between a functioning AV and a certifiable one.

19.9 Exercises

1. Estimate audit storage for a 1,000-vehicle fleet over three years with the per-vehicle rates in the example.
2. Design a retention policy that keeps event-triggered data for 5 years but time-indexed data for 30 days.
3. Propose three specific components of the AV stack that admit formal verification today.
4. Sketch a GSN fragment for a pedestrian-detection hazard sub-goal.
5. For the falsification example, compute total-campaign ROI against a prevented \$5 M incident.
6. Compare falsification to standard regression testing; identify one advantage of each.

7. Design a safety-case review process. Who participates, what inputs, what decisions?
8. Identify one component of the AV stack where auditing is structurally harder than others; explain why.
9. Propose an auditable-model requirement that a production model must satisfy.
10. Discuss how UL 4600 interacts with ISO 26262 and SOTIF; where are they complementary, where do they overlap?

Chapter 20

Advanced Testing Methodologies

20.1 Introduction

Validation of an AV system cannot be accomplished by nominal-case testing alone. The rare but catastrophic scenarios—the long tail that Chapter 3 foregrounded as the defining challenge—require testing methodologies that *seek* failure rather than *confirm* success. Three such methodologies have become central to production AV programs: falsification testing (gradient-based search for failure); rare-event simulation (importance-sampled Monte Carlo of tail scenarios); and coverage-guided testing (metric-driven exploration of the scenario space). This chapter extends the treatment of Chapter 17 into these advanced techniques.

Three observations motivate the structure.

First, the *rare-event problem* is fundamental. A system that fails once in 10^8 miles cannot be validated by driving 10^8 miles; the required on-road exposure is intractable. Validation must proceed by sampling the rare events directly.

Second, falsification *finds bugs the developers missed*. Gradient-based search over scenario parameters identifies failure modes that manual scenario design overlooks, often surprising the development team with unexpected sensitivities.

Third, *coverage* is the common framework. Testing methodologies converge on metrics that quantify how much of the operational design domain has been exercised and how uniformly; coverage gaps indicate untested territory.

20.2 Falsification Testing

Falsification treats testing as an optimization problem: find parameter choices that drive the system into failure.

20.2.1 Formulation

Scenario parameters $\theta \in \Theta$ map to outcomes $y(\theta)$. A *robustness* function $\rho(\theta) \geq 0$ if the system passes and < 0 if it fails. Falsification minimizes ρ :

$$\theta^* = \arg \min_{\theta \in \Theta} \rho(\theta). \quad (20.1)$$

Any θ^* with $\rho(\theta^*) < 0$ is a concrete failure example.

20.2.2 Search Strategies

Strategy	Mechanism
Random search	Sample θ uniformly
Gradient descent	Use $\nabla \rho$ when differentiable
Bayesian optimization	GP-surrogate for ρ ; acquisition-driven sampling
CMA-ES	Evolutionary strategy for nonconvex ρ
Reinforcement learning	Adversarial agent learns to fail the system

Bayesian optimization and CMA-ES have become standard for AV falsification; gradient methods require differentiable simulators which are emerging but not universal.

Example 20.1 (Falsification of an adaptive-cruise controller). Scenario: ego following lead vehicle. Parameters $\theta = (v_{\text{lead,init}}, a_{\text{lead,max}}, t_{\text{brake-onset}}, \text{headway})$.

Robustness: $\rho = \text{minimum TTC during scenario}$.

Random search: 10,000 samples, minimum found $\rho_{\min} = 1.4 \text{ s}$ (safe margin preserved throughout).

CMA-ES (500 evaluations): Minimum found $\rho_{\min} = -0.2 \text{ s}$ —a concrete collision exemplar. Parameters: short headway, aggressive lead braking, late onset after apparent steady-state.

Cost comparison.

CMA evaluations needed = 500,
random search cost for same finding = never found in 10k.

Structured search is vastly more efficient than random at finding failure modes.

20.3 Rare-Event Simulation

Some scenarios are rare but catastrophic. A Monte Carlo simulation of 10^6 scenarios may find zero failures even if the failure probability is 10^{-5} , giving no information about the failure rate. *Importance sampling* solves this by sampling from a biased distribution that over-represents failures, then reweighting.

20.3.1 Importance Sampling

Under the true distribution $p(\boldsymbol{\theta})$, failure probability is:

$$P_f = \mathbb{E}_p[\mathbb{1}(\text{fail})]. \quad (20.2)$$

Under biased distribution $q(\boldsymbol{\theta})$:

$$P_f = \mathbb{E}_q \left[\mathbb{1}(\text{fail}) \cdot \frac{p(\boldsymbol{\theta})}{q(\boldsymbol{\theta})} \right]. \quad (20.3)$$

Choose q to concentrate mass where failures occur; the weighting p/q corrects the bias.

Example 20.2 (Rare-event failure-rate estimation). Estimate failure probability $P_f \approx 10^{-5}$ (one failure per 100,000 scenarios).

Plain Monte Carlo.

samples for $\pm 10\%$ estimate : $\sim 10/P_f = 10^6$,
 compute at 1 s/sample : ~ 11.5 days single-thread.

Importance sampling with $100\times$ failure over-representation.

samples needed : $\sim 10^4$,
 compute : ~ 3 h single-thread.

Two orders of magnitude speedup. For the typical regression- testing cadence (nightly builds over a 2-year program), the difference determines whether rare-event estimates can be maintained at all.

20.3.2 Cross-Entropy Method

A common practical approach: start with a proposal distribution, simulate, fit a new proposal concentrated on the failure examples, iterate. The Cross-Entropy method converges to a near-optimal proposal for a given failure set.

20.4 Coverage-Guided Testing

A scenario library is good only if it covers the operational design domain. *Coverage-guided testing* uses metrics that quantify coverage, then systematically reduces gaps.

20.4.1 Coverage Dimensions

Production AV programs track coverage along multiple axes:

Dimension	What it measures
ODD parameter	Fraction of ODD parameter space sampled
Behavioral	Fraction of unique AV behavior classes exercised
Failure-mode	Known failure modes represented
Regional	Fraction of distinct regions, road types tested
Weather/lighting	Fraction of environmental envelope sampled

20.4.2 Coverage-Guided Mutation

Given a scenario library and a coverage metric, mutate existing scenarios in directions that increase coverage. Adapted from coverage-guided fuzzing (AFL, libFuzzer) in software testing.

Example 20.3 (Coverage-guided scenario discovery). Starting library: 10,000 scenarios, ODD parameter coverage 62 %.

One week of coverage-guided mutation.

- New scenarios generated: 2,500
- Unique new coverage points hit: 1,800
- Coverage metric after: 77 %

Compare random scenario generation.

- Same volume: 2,500 new scenarios
- Unique new coverage points hit: ~300
- Coverage metric after: 64 %

Coverage-guided mutation is $\sim 6\times$ more effective per scenario at raising coverage.

20.4.3 Coverage Saturation

Coverage metrics asymptote: at some point, new scenarios no longer increase the metric. Saturation is a weak but useful signal that the current coverage definition has been adequately exhausted; it does *not* mean the system is fully tested, only that the current metric cannot distinguish further.

20.5 Integration with the Test Pyramid

These advanced techniques fit into the test pyramid developed in Chapter 30:

- Falsification runs in closed-loop simulation (mid-pyramid).
- Rare-event simulation runs in closed-loop or SIL (depending on fidelity need).
- Coverage-guided testing directs scenario generation at all layers.

The techniques complement rather than replace nominal-case testing; a validation suite uses all of them.

20.6 Industrial Deployment

Testing methodologies for AVs span on-road, simulation, hardware-in-the-loop, and structured scenario-based approaches. ASAM OpenScenario and OpenDRIVE are widely used industry standards for scenario specification. Mcity at the University of Michigan, GoMentum Station in California, K-City in Korea, and the National Intelligent Connected Vehicle (Shanghai) Demonstration Zone provide closed-course testing infrastructure. Applied Intuition, Foretellix, dSPACE, and others provide commercial scenario-based testing platforms used by multiple OEMs. The PEGASUS project (Germany) established methodological foundations for scenario-based highway testing. The IEEE 2846 and ISO/PAS 21448 standards govern scenario-based safety analysis.

20.7 Chapter Summary

Advanced testing methodologies target the long tail of AV operation: falsification testing seeks failure via gradient or evolutionary search; rare-event simulation estimates low-probability failure rates with importance sampling; coverage-guided testing directs scenario generation via explicit coverage metrics. All three are complementary to the standard test pyramid and have become standard in leading AV programs as the field moves from “does it work in demos?” to “can we demonstrate it is safe at deployment scale?”.

20.8 Exercises

1. For the falsification example, estimate CMA-ES cost for a 10-dimensional parameter space.
2. Design a robustness function ρ for a lane-change scenario.

3. Compute required MC samples for estimating $P_f = 10^{-7}$ with 10 % relative error; compare with importance sampling at $1000\times$ bias.
4. Propose a coverage metric for intersection scenarios.
5. Derive the importance-sampling estimator variance and explain when it exceeds plain MC variance.
6. Identify a failure category that falsification would find but coverage-guided testing would miss, and vice versa.
7. Design a falsification campaign budget for a 2-week pre-release cycle.
8. Compare random-search, Bayesian optimization, and CMA-ES for AV falsification. When is each preferable?
9. Propose a coverage-saturation test: at what point do you declare the metric exhausted?
10. Discuss how adversarial-RL falsification differs from CMA-ES falsification; what does each find that the other doesn't?

Chapter 21

Simulators for Autonomous Driving

21.1 Introduction

A modern AV program runs many more miles in simulation than on the road—typically by three to five orders of magnitude. Simulation is how rare scenarios are surfaced for training, how regression risk is contained between releases, how new architectures are compared without risking road incidents, and how reinforcement-learning policies are trained in the first place. The simulator is therefore not an adjunct to the AV stack: it is co-equal infrastructure.

This chapter surveys four widely used simulators with sharply different design philosophies: **CARLA**, the open-source simulator that anchors most academic AV research; **Waymax**, Waymo’s open-source data-driven simulator built around the Waymo Open Motion Dataset; **MetaDrive**, an open-source procedurally-generated driving simulator focused on generalisation research; and **NVIDIA DRIVE Sim**, the production-oriented commercial simulator built on Omniverse.

Three observations motivate the structure.

First, no single simulator is best at everything. Photorealism, scenario fidelity, multi-agent interaction quality, sensor realism, throughput, and licensing terms involve direct trade-offs; the right tool depends on the task.

Second, the open-source–commercial divide matters. Academic and exploratory work gravitates to CARLA, Waymax, and MetaDrive. Production safety-case work, sensor-in-the-loop testing, and large-scale fleet validation gravitate to commercial platforms including DRIVE Sim, Applied Intuition, dSPACE, and others.

Third, data-driven simulation is increasingly the new normal. The 2020s shift from rule-based traffic to log-replay and learned-traffic-agent simulators (Waymax, NuPlan, DriveSimX) reflects recognition that hand-authored scenarios miss the long-tail behaviours that real driving exhibits.

21.2 What a Simulator Must Provide

The technical requirements that distinguish AV simulators from generic driving games:

Requirement	What it means
Sensor fidelity	Realistic camera, LiDAR, radar outputs — including failure modes
Vehicle dynamics	Tire models, weight transfer, suspension that match real vehicles
Scenario fidelity	Map geometry, traffic rules, behaviours grounded in real data
Multi-agent traffic	Other agents that respond plausibly to ego behaviour
Throughput	High-speed parallel rollouts for training & evaluation
Determinism	Bit-identical replay for regression testing
Tooling integration	Connection to ROS, ML frameworks, CI pipelines
Domain randomisation	Sample-efficient sim-to-real transfer

The four simulators below trade these requirements against each other in different ways.

21.3 CARLA

21.3.1 Origin and Design

CARLA [5] was open-sourced in 2017 by Intel Labs and the Computer Vision Center at the Universitat Autònoma de Barcelona. It is built on Unreal Engine 4 (UE4), inheriting UE4’s photorealistic rendering and physics. The simulator is permissively licensed (MIT for the platform code; UE4 license for the engine) and has become the de-facto standard for academic AV research.

21.3.2 Architecture

CARLA is a client-server system. The server is a UE4 process that maintains the simulated world: maps, vehicles, pedestrians, weather. Clients are Python or C++ processes that command vehicles, query sensors, and step the simulation. Multi-client architecture allows parallel rollouts and distributed training.

21.3.3 Maps and Scenarios

CARLA ships with a small set of hand-authored maps (Town01–Town10 plus a few highway maps). The OpenDRIVE format is supported, allowing import of custom maps. The CARLA Leaderboard—an annual benchmark challenge—uses CARLA to evaluate end-to-end driving policies on held-out routes.

21.3.4 Sensor Models

CARLA simulates RGB cameras, depth, semantic segmentation, instance segmentation, LiDAR, semantic LiDAR, radar, GNSS, IMU, and collision sensors. The sensor models are simplified relative

to production-grade physics-based simulators: LiDAR returns are generated by ray-casting against the rendered geometry without the beam-divergence, atmospheric, or material-reflectivity modelling of sensor-in-the-loop simulators.

21.3.5 Strengths

- Free, open-source, large user base.
- Photorealistic rendering for vision-stack development.
- Active research community; many published baselines.
- Integration with ROS, ROS 2, OpenAI Gym, and most ML frameworks.
- Reproducibility for academic publications.

21.3.6 Limitations

- Sensor models are approximate; not suitable for sensor-team validation work.
- Traffic behaviour is rule-based; does not exhibit the full diversity of real human driving.
- Performance limits at scale: real-time only at moderate scene complexity on a single GPU.
- Map authoring is labour-intensive; few high-fidelity maps.
- Determinism is partial; bit-identical replay is brittle.

21.3.7 Suitable Uses

End-to-end policy development, sim-to-real research, vision-only perception research, RL benchmarks, teaching, leaderboards. Less suitable for: production safety case, sensor-vendor qualification, high-fidelity scenario reconstruction.

21.4 Waymax

21.4.1 Origin and Design

Waymax [7] was released by Waymo Research in late 2023 as an open-source, data-driven simulator built around the Waymo Open Motion Dataset. Unlike CARLA, Waymax does *not* render visually: it operates entirely on the abstracted state-space of agents-in-a-map. The trade-off is explicit: Waymax sacrifices sensor realism for dataset scale and rollout throughput.

21.4.2 Architecture

Waymax is implemented in JAX, allowing GPU/TPU-accelerated parallel rollouts. A simulation step updates agent positions according to either logged behaviours (replaying real data) or learned behaviours (trained traffic agents). The simulator is deterministic by construction.

21.4.3 Scenarios from Real Data

Waymax's defining feature is its grounding in real recorded data. Each scenario starts from a recorded snippet of real driving, with real agents and real maps. The ego vehicle's actions can deviate from the

log; non-ego agents either continue their logged trajectories (*log-replay*), respond reactively (*IDM-like heuristics*), or follow a learned policy (*Sim Agents*).

21.4.4 Sim Agents

Waymax supports Waymo’s *Sim Agents* challenge: closed-loop multi-agent simulation where every agent (including non-ego) is controlled by a learned model. This is the closest available public benchmark to what production multi-agent simulation requires.

21.4.5 Strengths

- Grounded in real data: scenarios and agent behaviours drawn directly from the Waymo Open Motion Dataset.
- Massive throughput: JAX-based vectorisation allows millions of parallel rollouts on TPU pods.
- Deterministic, reproducible.
- Closed-loop multi-agent benchmarks (Sim Agents).
- Permissive licensing; modern Python tooling.

21.4.6 Limitations

- No rendering: cannot validate vision pipelines.
- No sensor models: cannot validate perception.
- Limited to scenarios and geographies in the Waymo Open Motion Dataset (predominantly U.S. urban driving).
- Map representation is abstract: physical features beyond the dataset’s vector-map are absent.
- Comparatively young; smaller user community than CARLA.

21.4.7 Suitable Uses

Trajectory-prediction research, planning research, RL training at scale, multi-agent policy development, scenario-based regression on real-world distributions. Less suitable for: vision-stack development, sensor-vendor qualification, photorealism-dependent research.

21.5 MetaDrive

21.5.1 Origin and Design

MetaDrive [9] is an open-source simulator from researchers at the University of California Los Angeles, released in 2021. The defining design choice is *procedural map generation*: scenarios are composed from a library of road blocks (intersections, ramps, roundabouts, curves) into novel combinations.

21.5.2 Architecture

MetaDrive runs on the Panda3D rendering engine and Bullet physics. It is lightweight relative to CARLA, achieving thousands of frames per second per environment on a single GPU—suitable for

RL training where throughput dominates.

21.5.3 Procedural Generation

MetaDrive’s scenario generator samples from a distribution over maps composed of road-block primitives. The result is a virtually unlimited supply of novel test environments—useful for generalisation research, where the question is not how a policy performs on a fixed map but how it performs on maps it has never seen.

21.5.4 Strengths

- Procedural diversity: arbitrary novel maps for generalisation.
- High throughput: optimised for RL training.
- Lightweight: runs on commodity hardware without expensive GPU rendering.
- Active research community with multiple published baselines.
- Simple Python API.

21.5.5 Limitations

- Visual fidelity is well below CARLA; not photorealistic.
- Sensor models are simplified.
- Procedurally-generated scenarios are not always realistic driving behaviour distributions.
- Limited multi-agent traffic intelligence.

21.5.6 Suitable Uses

RL training that needs throughput, generalisation research, ablation studies that need many distinct environments, lightweight curriculum learning. Less suitable for: vision-stack development at production quality, real-driving-distribution evaluation.

21.6 NVIDIA DRIVE Sim

21.6.1 Origin and Design

NVIDIA DRIVE Sim is built on NVIDIA Omniverse and is part of the broader NVIDIA DRIVE platform. Unlike CARLA, Waymax, and MetaDrive, DRIVE Sim is a commercial product positioned for production safety-case validation; access is via NVIDIA’s commercial partner program.

21.6.2 Architecture

DRIVE Sim is built on Omniverse, NVIDIA’s Universal Scene Description (USD) based platform. It uses RTX-accelerated ray-traced rendering for visual fidelity. Sensor models are physics-based: the LiDAR simulator accounts for beam divergence, weather attenuation, and material reflectivity; the camera simulator includes lens distortion, motion blur, and noise models.

21.6.3 Sensor-in-the-Loop

A key DRIVE Sim feature is *hardware-in-the-loop* (HIL) and *software-in-the-loop* (SIL) operation: the same NVIDIA DRIVE compute platform that runs in the production vehicle can be fed simulated sensor data, allowing pre-deployment validation of the exact production software stack.

21.6.4 Strengths

- Production-grade sensor models (physics-based LiDAR, camera, radar).
- Photorealistic ray-traced rendering (Omniverse RTX).
- Hardware-in-the-loop with NVIDIA DRIVE compute platforms.
- USD-based scene description: interoperable with major content-creation pipelines.
- Direct integration with NVIDIA’s full AV toolchain (DriveOS, DriveWorks, TensorRT).

21.6.5 Limitations

- Commercial license; not freely available.
- NVIDIA-platform-centric: locks development into NVIDIA’s compute and tooling ecosystem.
- High infrastructure cost: substantial GPU compute required for ray-traced sensor simulation at scale.
- Less academic adoption; smaller open-research community.
- Maps and scenarios for non-NVIDIA partners are limited.

21.6.6 Suitable Uses

Production safety-case generation, sensor-vendor qualification, HIL/SIL validation of exact production software, photorealism- dependent vision-stack development. Less suitable for: lightweight research, open benchmarking, environments without NVIDIA platform licensing.

21.7 Comparative Assessment

The four simulators address overlapping but distinct needs:

Dimension	CARLA	Waymax	MetaDrive	DRIVE Sim
Visual fidelity	high	none	moderate	very high
Sensor realism	approximate	none	approximate	physics-based
Multi-agent realism	rule-based	data-driven	rule-based	rule + learned
Throughput	moderate	very high	high	moderate
Map diversity	limited	large (real)	procedural	licensed
License	MIT/UE4	Apache 2.0	Apache 2.0	commercial
HIL/SIL	limited	no	no	yes
Determinism	partial	full	partial	full

21.7.1 Choice by Task

- *Vision-stack research without budget constraints*: DRIVE Sim if commercially viable, CARLA if not.
- *Trajectory prediction and planning research*: Waymax dominates; data-grounded scenarios are the right training and evaluation distribution.
- *RL training at scale*: MetaDrive for throughput, Waymax for data-grounded rewards, CARLA for a vision-aware reward signal.
- *Production safety-case validation*: DRIVE Sim or one of its commercial peers (Applied Intuition, dSPACE, Foretellix).
- *Sim-to-real transfer research*: CARLA with domain randomisation remains the academic standard.
- *Generalisation research*: MetaDrive’s procedural maps are designed for this.

21.7.2 When to Use Multiple Simulators

Production AV programs typically run *multiple* simulators in parallel. A common pattern: Waymax (or equivalent data-driven simulator) for prediction-and-planning regression; DRIVE Sim (or peer) for sensor-in-the-loop validation; CARLA for vision-stack ablations; an internal closed-source simulator for fleet-scale scenario regression. No single tool replaces this stack.

21.8 Validation of the Simulator Itself

A simulator that produces unrepresentative results is worse than no simulator at all: it produces false confidence. Production programs maintain explicit *simulator-validation* pipelines comparing simulated outcomes to real-world outcomes on matched scenarios. Discrepancies feed back into simulator improvements; persistent discrepancies bound the trustworthiness of conclusions drawn from simulation alone.

Example 21.1 (Simulator-vs-real divergence on a regression suite). A program runs a 1,000-scenario regression in simulation and on matched real-world test track conditions.

Step 1: Disagreement rate.

In simulation, a candidate model passes 940 scenarios. On the real test track, it passes 920 of the matched scenarios. Twenty scenarios disagree.

Step 2: Direction of disagreement.

- Sim pass, real fail: 14 scenarios (sim is optimistic).
- Sim fail, real pass: 6 scenarios (sim is pessimistic).

Step 3: Sim-optimism bias.

$$\begin{aligned} B &= (14 - 6)/1000 \\ &= 0.008. \end{aligned}$$

Interpretation. A sim-optimism bias of 0.8 % means that sim-only release decisions over-rate model performance by roughly 0.8 percentage points. Programs incorporate this bias into their release criteria explicitly.

21.9 Open Challenges

Multiple open problems define the next decade of AV simulation:

- *Behaviour fidelity:* simulated agents that behave like real human drivers, not idealised IDM/MOBIL agents.
- *Long-tail coverage:* surfacing rare scenarios efficiently rather than relying on random sampling.
- *Cross-region scenarios:* simulators that capture the driving cultures, infrastructure conditions, and weather patterns of the regions where AVs deploy globally—including the African and South-Asian contexts under-represented in current public datasets.
- *Sensor-failure simulation:* realistic models of LiDAR drop-outs, camera contamination, radar multipath under weather.
- *Validation of simulator-validation:* explicit methodologies for bounding the trustworthiness of simulation-only conclusions.

Each is the subject of active academic and industrial research.

21.10 Industrial Deployment

CARLA has been used in academic publications by hundreds of research groups including those at MIT, Stanford, ETH Zurich, and Tsinghua, and is a standard tool in university AV courses globally. Waymax, released in 2023, was adopted as the simulation platform for the 2024 Waymo Sim Agents Challenge. MetaDrive has been used in multiple peer-reviewed RL publications. NVIDIA DRIVE Sim is used by Mercedes-Benz, Volvo, BMW, and other DRIVE-platform partners; the underlying Omniverse platform is used by Lockheed Martin, Siemens, Ericsson, and others outside the AV space. Applied Intuition (United States) and dSPACE (Germany) are major commercial competitors to DRIVE Sim, with Applied Intuition reportedly used by Toyota, Volkswagen, Audi, and many other OEMs. Foretellix (Israel) provides scenario-coverage tooling that integrates with multiple simulators. Beyond the four simulators discussed in detail, NuPlan (Motional) and the open-source Autoware simulator extend the data-driven and ROS-integrated ends of the simulation spectrum.

21.11 Chapter Summary

The four simulators surveyed—CARLA, Waymax, MetaDrive, and NVIDIA DRIVE Sim—occupy distinct positions on the trade-off space defined by visual fidelity, sensor realism, multi-agent realism, throughput, licensing, and integration with production toolchains. CARLA is the academic default; Waymax is the data-grounded research standard; MetaDrive is the RL throughput leader; DRIVE Sim is the production sensor-in-the-loop platform. Production AV programs routinely run multiple simulators in parallel, each at the task it is best at. The validity of any simulation conclusion is

bounded by the validity of the simulator itself, which production programs measure explicitly through real-vs-sim comparison pipelines.

21.12 Exercises

1. Choose a simulator for each of the following tasks and justify: (a) training an end-to-end vision policy, (b) benchmarking a trajectory predictor, (c) RL with budget for 10^{10} transitions, (d) production sensor qualification.
2. For the simulator-validation example, compute the sim-optimism bias if 18 scenarios are sim-pass-real-fail and 4 are sim-fail-real-pass.
3. Design a real-vs-sim comparison protocol for a lane-keeping system. What measurements? What scenarios?
4. Estimate the relative compute cost of running 1 M parallel rollouts in (a) CARLA, (b) Waymax, (c) DRIVE Sim. What assumptions do you need?
5. Compare CARLA's rule-based traffic to Waymax's Sim Agents. Identify three cases where each is preferable.
6. Propose a strategy for African-context AV simulation: what gaps must be filled in current public simulators?
7. Discuss the trade-offs of procedural map generation (MetaDrive) versus data-grounded scenarios (Waymax).
8. Design a HIL setup using DRIVE Sim. What hardware and software are required?
9. Identify three open problems in simulator validation that would change AV release-engineering practice if solved.
10. Compare the licensing implications of CARLA, Waymax, MetaDrive, and DRIVE Sim for a startup AV program. Which best supports rapid early development?

Chapter 22

Safety Intelligence and System Integrity

22.1 Introduction

Uncertainty (Chapter 28) addressed whether the model knows what it doesn't know; trust and assurance (Chapter 19) addressed how we prove the system meets its requirements. This chapter addresses the *operational layer*: how the running AV continuously monitors its own integrity, detects tampering or sensor malfunction, and dynamically adjusts its behavior based on perceived risk. These functions are the system's immune response: detecting trouble as it begins and responding before it becomes an incident.

Three observations motivate the structure.

First, safety is not a one-time verification. A system certified in the lab can degrade in the field. Continuous self-monitoring catches degradation as it occurs.

Second, integrity extends beyond functional correctness. Sensor tampering, cyber intrusion, and subtle hardware faults can corrupt perception in ways functional testing would not reveal. Integrity monitoring is a distinct concern from accuracy.

Third, dynamic risk assessment makes perception itself risk-aware. Perception confidence thresholds can and should vary by scene context—low in low-consequence settings, high in high-consequence settings.

22.2 Perception Risk Heatmaps

Instead of treating every image region equally, risk heatmaps assign variable importance based on the consequence of error in each region.

22.2.1 Spatial Risk

Region	Risk weighting
Ego's predicted path	Highest; any miss directly causes collision
Adjacent lanes	High; lateral intrusion possible
Sidewalks/crosswalks	High; vulnerable-road-user zones
Opposite direction	Medium
Distant background	Low

22.2.2 Runtime Use

The risk heatmap feeds:

- Detection-threshold adaptation: lower threshold in high- risk regions.
- Attention-resource allocation: more compute on high-risk regions.
- Behavior shaping: explicit yielding when detections in high-risk regions have marginal confidence.

22.3 Early Warning Perception Signals

A perception stack that is about to fail often shows subtle precursor signals before a visible incident. Monitoring these catches failures early.

22.3.1 What To Monitor

- Sudden shifts in confidence distribution.
- Unusual track-quality statistics.
- Disagreement between redundant pipelines.
- Input-distribution drift against a training-distribution reference.
- Heat / thermal anomalies in compute hardware.

Example 22.1 (Input-drift detection). A production detector monitors mean feature-activation magnitudes across a validation distribution.

Baseline.

$$\begin{aligned}\bar{f}_{\text{baseline}} &= 2.4, \\ \sigma_f &= 0.3.\end{aligned}$$

During operation.

$$\bar{f}_{\text{operational}} = 1.7.$$

Deviation in standard deviations.

$$z = (1.7 - 2.4)/0.3 \\ \approx -2.33.$$

A 2.33σ shift triggers an alert. Investigation reveals the camera lens has partial contamination reducing signal. Cleaning resolves; no incident occurred. Early warning converted a pre-incident into a routine maintenance issue.

22.4 Confidence Thresholding

Detection thresholds decide when a detection is accepted. A single static threshold is suboptimal; context-dependent thresholds are better.

22.4.1 Dynamic Thresholds

Context	Threshold
Ego's predicted path, high-risk scene	0.30
Ego's predicted path, normal scene	0.50
Adjacent lane, normal	0.55
Opposite lanes or distant	0.70

Lower threshold on the path avoids missing threats at the cost of more false-positive slowdowns. Higher threshold in less-critical regions prevents unnecessary behavior changes driven by noise.

Example 22.2 (Threshold-adaptation benefit). Detection with uniform threshold 0.5 gives: precision 0.92, recall 0.85. With adaptive thresholds per region:

Region-specific outcomes.

- High-risk region: threshold 0.3 $\rightarrow$ recall 0.95, precision 0.85.
- Normal: threshold 0.5 $\rightarrow$ recall 0.85, precision 0.92.
- Distant: threshold 0.7 $\rightarrow$ recall 0.70, precision 0.97.

Net safety effect.

Recall on high-risk regions increased from 0.85 to 0.95; misses that would have produced near-incidents reduce by 10 points. Precision drop on same regions (0.92 to 0.85) creates more false-positive slowdowns but does not produce incidents. Favorable trade for safety-critical contexts.

22.5 Sensor Integrity Monitoring

Sensors can fail, degrade, or be tampered with. Integrity monitoring detects all three.

22.5.1 Fault Modes

Mode	Detection signal
Hard failure	Signal absent or constant
Gradual degradation	Statistical drift over time
Intermittent fault	Dropouts in streams
Cyber tampering	Replay patterns; signature mismatches
Spoofing	Physical implausibility (out-of-range values)

22.5.2 Cross-Modal Consistency

A single sensor’s failure is hard to distinguish from an unusual scene. Cross-modal consistency checks—does camera agree with LiDAR?—raise confidence in distinguishing genuine scene unusualness from sensor fault.

Example 22.3 (Cross-modal fault isolation). Scenario: camera reports a pedestrian at range 25 m. LiDAR sees no return at that position.

Possibilities.

- LiDAR failure (pedestrian is real).
- Camera false positive (pedestrian is not real).
- LiDAR occluded (something between).
- Pedestrian below LiDAR scan line.

Fault-isolation logic.

Check LiDAR health signals: if LiDAR showing normal operation elsewhere in scene, the targeted direction is anomalous. Check if camera shows consistent pedestrian over multiple frames: if yes, more likely real; if no, more likely false positive.

Conservative default. Unless LiDAR health is specifically bad, treat the camera alone as insufficient for a confident detection—but maintain elevated caution until cross-confirmation is obtained.

Cross-modal disagreement is information, not noise. Production systems design explicit logic around it.

22.6 Trusted Execution Environments (TEE)

Critical perception code and cryptographic operations benefit from running inside a TEE: a hardware-isolated execution environment resistant to tampering even by a compromised OS.

22.6.1 What Runs In TEE

- Cryptographic keys and signing operations for V2X.
- Safety-monitor logic that overrides learned planners.
- Sensor-integrity attestation code.
- Audit-log signing.

22.6.2 Platforms

Platform	TEE technology
ARM-based AV platforms	TrustZone
Intel-based AV platforms	SGX (deprecated), TDX
RISC-V-based platforms	Keystone and similar

TEE usage in AV is a production discipline; safety-critical code increasingly runs there.

22.7 Industrial Deployment

Operational safety-monitoring infrastructure draws on broad cloud- observability tooling combined with AV-specific overlays. The NTSB has published detailed investigations of multiple AV incidents that draw heavily on the kinds of audit logs and operational telemetry described in this chapter. The California DMV disengagement reports provide a public window into intervention rates for major AV operators. Waymo, Cruise, and others have published Safety Reports describing internal review processes. Trusted Execution Environment platforms used in automotive include ARM TrustZone (broadly deployed), Intel SGX, and emerging RISC-V Keystone-based implementations. Hardware Security Modules from suppliers including NXP and Infineon are used for cryptographic-key protection in V2X-equipped vehicles.

22.8 Chapter Summary

Safety intelligence is the layer that makes an AV aware of its own operational integrity. Risk heatmaps allocate attention by consequence. Early-warning signals catch degradation before it becomes incidents. Adaptive thresholds respond to context. Sensor integrity monitoring detects faults and tampering. Trusted Execution Environments protect safety-critical code from OS-level compromise. Together these make the AV an adaptive, self-monitoring system rather than a static certified one.

22.9 Exercises

1. For the input-drift example, compute the z-score needed for 99 % confidence in drift detection (assume normal).
2. Design a risk-heatmap scheme for a rural highway scenario.
3. For the threshold-adaptation example, compute the overall safety benefit assuming 60 % of scenes are “normal”.
4. Propose a cross-modal consistency test for camera-radar agreement.
5. Compare hardware TEE and software sandboxing as protection mechanisms for safety-critical code.
6. Design an integrity-monitoring dashboard for a fleet operator.
7. Propose three signals that warn of imminent compute- hardware failure.
8. For the cross-modal example, what additional sensors would most strengthen fault isolation?
9. Design an adaptive-threshold schema that explicitly avoids safety-positive bias (never raises thresholds beyond safe limits).
10. Discuss privacy implications of fleet-wide integrity- monitoring data collection.

Chapter 23

Risk Micro-Modeling

23.1 Introduction

Safety at the fleet level (Chapter 22) asks whether the system is adequately safe on average. Safety at the moment of action asks a sharper question: at each instant, at each point in space, what is the *immediate* risk, and how should the planner react? This chapter develops risk micro-modeling: the quantification of risk at fine spatial resolution and short temporal windows. Local risk gradients reveal how small lateral or longitudinal movements change risk. Near-collision probability estimation quantifies the chance of imminent incident. Dynamic safety margins adjust following distances and lateral buffers in real time based on the risk estimate.

Three observations motivate the structure.

First, risk is spatially inhomogeneous. Some points in the scene are orders of magnitude riskier than others; treating a whole region uniformly—as comfort-only planners often do—wastes safety margin where it is cheap and withholds margin where it is needed.

Second, near-collisions are actionable. An event with 1 % probability of collision 200 ms from now is not a certainty, but it is far from ignorable. Planners that respond to probabilistic near-misses avoid incidents that purely deterministic planners produce.

Third, margins should be dynamic. A fixed 2-second following distance is wrong both in stable highway flow (where it wastes capacity) and in dense urban conflict (where it under-protects). Margins that respond to local risk perform better on both.

23.2 Local Risk Gradients

Define the risk field $R(\mathbf{x}, t)$ at spatial position $\mathbf{x}$ and time t as the instantaneous consequence-weighted probability of incident given ego position at $\mathbf{x}$:

$$R(\mathbf{x}, t) = \mathbb{E}[\text{severity} \mid \text{incident}, \mathbf{x}, t] \cdot P(\text{incident} \mid \mathbf{x}, t). \quad (23.1)$$

The *local risk gradient* $\nabla_{\mathbf{x}}R$ tells the planner how risk changes with small spatial displacements.

23.2.1 Why Gradients Matter

Planners optimize over trajectory parameters; the gradient of risk with respect to trajectory gives directions of improvement. If ∇R points strongly in a lateral direction, a small lane-position shift produces large risk reduction.

Example 23.1 (Risk gradient at lane edge). Ego in the right lane, 0.3 m from the right-side line. Parked cars along the right edge; pedestrians occasionally step from between them. Risk evaluation at three positions:

Lateral offset from line	Risk R	Gradient
0.3 m	0.040	—
0.7 m	0.018	−0.055/m
1.1 m	0.012	−0.015/m

Interpretation.

- Risk at 0.3 m is twice the risk at 0.7 m—a modest 40-cm shift halves risk.
- Gradient is steepest near the edge; beyond 0.7 m, further shift yields diminishing returns.

Planner decision: shift lateral offset to ~ 0.7 m as the sweet spot. Shifting further costs centerline stability and lane-center norms without proportional risk reduction.

23.2.2 Computing Gradients

Two approaches:

- Finite difference: evaluate R at perturbed positions. Simple but expensive if R evaluation is costly.
- Differentiable risk model: train R as a differentiable function of ego position; backpropagate gradients directly. Production-feasible with neural-network risk models.

23.3 Near-Collision Probability Estimation

Distinct from risk (which is consequence-weighted), near-collision probability is just $P(\text{incident within } \Delta t)$ at a chosen lookahead.

23.3.1 Formulation

For each agent pair (ego, other), given predicted trajectories with uncertainty:

$$P_{\text{NC}}(\Delta t) = P(\exists \tau \in [0, \Delta t] : d(\tau) < d_{\text{safe}}). \quad (23.2)$$

Evaluated by integration over the joint predicted-trajectory distribution.

23.3.2 Monte Carlo Estimation

Sample N joint trajectories from the predicted distribution; count the fraction where minimum separation falls below d_{safe} .

Example 23.2 (Near-collision probability with lead vehicle). Ego at 20 m s^{-1} , following at 15 m behind a vehicle at 19 m s^{-1} . Lead-vehicle acceleration uncertainty: $\sigma_a = 1.5 \text{ m s}^{-2}$.

Simplified 1D Monte Carlo.

1000 samples drawn from lead-vehicle acceleration distribution; propagate each 2 s forward; measure minimum separation.

Results.

- Minimum separation $> 5 \text{ m}$: 942 samples.
- Minimum separation in $[0, 5]$: 54 samples.
- Collision: 4 samples.

Near-collision probability ($\text{sep} < 5 \text{ m}$).

$$\begin{aligned} P_{\text{NC}} &= (54 + 4)/1000 \\ &= 0.058. \end{aligned}$$

Decision implication.

5.8 % near-collision probability is above typical comfortable thresholds. Planner should increase following distance. Widening headway from 15 m to 25 m reduces P_{NC} to ~ 0.005 —an order-of-magnitude reduction at modest utility cost.

23.3.3 Computational Tractability

Monte Carlo with 1000 samples per agent-pair at each planning step is expensive. Production systems use importance-sampled estimates or analytic approximations (Gaussian-on-Gaussian) that trade accuracy for speed.

23.4 Dynamic Safety Margins

Margin = how far from the nominal-safe threshold the planner operates. Static margins (2-second following distance, 1-metre lateral buffer) are widely used and demonstrably suboptimal.

23.4.1 Margin as a Function of Risk

Let margin $m = f(R)$ be a monotonically increasing function of local risk. Planner targets trajectories satisfying slack $\geq m$ everywhere.

Example 23.3 (Dynamic following distance). Ego on highway at 30 ms^{-1} . Reference margin is 2-second headway (60 m).

Regime A: stable traffic, clear visibility.

Near-collision probability with current headway: $P_{\text{NC}} \approx 10^{-4}$. Dynamic margin: 1.5 s headway = 45 m. Efficiency gain; safety unchanged within this regime.

Regime B: dense traffic, rain, predicted incidents ahead.

Base risk elevated by $\sim 3\times$; P_{NC} would rise to $\sim 3\times 10^{-4}$ at static margin. Dynamic margin: 3.0 s headway = 90 m. Restores P_{NC} to $\sim 10^{-4}$; modest capacity cost.

Regime C: identified risk (e.g., motorcycle in blind spot).

P_{NC} would rise $5\text{--}10\times$. Dynamic margin: full brake-buffer, consider pulling over.

Static 2-second margin either under-protects in Regime C or over-reserves in Regime A. Dynamic margins give appropriate responses in each.

23.4.2 Margin Adjustment Rate

Margins must be stable over short timescales to avoid oscillatory following behavior. Typical rate limits: ± 0.05 -second-headway change per second of operation. Rapid adjustments happen only under detected abrupt risk (emergency).

23.4.3 Learned Margin Models

A separately-trained model maps scene features to margin recommendations, supervised by retrospective analysis of incidents and near-incidents from fleet data. The model generalizes to situations not explicitly rule-coded.

23.5 Integration with Planning

Risk micro-modeling feeds planning through three interfaces:

Interface	How used by planner
Risk field	Cost function includes integrated risk along trajectory
NC probability	Hard constraint: reject trajectories with $P_{\text{NC}} > \tau$
Dynamic margin	Inflates the infeasible region; planner respects

23.5.1 Risk-Constrained MPC

Model predictive control (Chapter 20) can be augmented with a chance constraint:

$$P(\text{incident}) \leq \alpha. \quad (23.3)$$

Solvers handle this via sample-based or analytic approximations.

23.6 Industrial Deployment

Risk-aware planning techniques have moved from academic research to production over the past decade. The Responsibility-Sensitive Safety (RSS) framework [10] from Mobileye provides one production-deployed formalisation of risk-bounded planning. NVIDIA’s Safety Force Field (SFF), published in 2019, provides another. Both have been incorporated into the IEEE 2846 standard published in 2022. Risk-bounded MPC approaches developed at MIT, Stanford, and elsewhere have entered industrial pilot deployments. Apollo (Baidu) and Autoware include risk-aware planning components in their open-source releases. Tesla, Waymo, and others have published peer-reviewed work on risk-aware planning; the specific production implementations remain proprietary but are documented in safety reports.

23.7 Chapter Summary

Risk micro-modeling operates at fine spatial and temporal resolution below the scene-level safety monitoring of earlier chapters. Local risk gradients direct planners toward safer trajectories; near-collision probability quantifies imminent risk; dynamic safety margins adjust following distances and lateral buffers in real time based on context. These feed into planning via cost terms, chance constraints, and feasibility shaping. Compared to static-margin approaches, risk micro-modeling is more efficient in low-risk regimes and more protective in high-risk ones, achieving a superior trade-off that static thresholds cannot.

23.8 Exercises

1. For the lane-edge example, estimate risk at 1.5 m offset. What approximation method do you use?
2. For the near-collision example, compute P_{NC} at headway 20 m.
3. Design a risk-gradient computation that works in closed-loop MPC at 20 Hz update rate.
4. Compare finite-difference and differentiable risk-model gradients. Identify one advantage of each.
5. For the dynamic-margin example, derive the headway that minimizes aggregate cost (risk $\times$ severity + capacity).
6. Propose three scene-context signals that a learned margin model should consume.

7. For the chance-constrained MPC, derive the linearization for Gaussian uncertainty.
8. Design a monitoring dashboard for fleet-wide near-collision rates. What metrics alarm?
9. Compare risk-micro-modeling to risk-aware perception (Chapter 27). Identify the complementary roles.
10. Discuss privacy implications of recording and analyzing near-collision events across a fleet.

Part VII

Economics and Commercialization

Chapter 24

The Economics of Autonomous Driving

24.1 Introduction

Every engineering decision in the preceding chapters has an economic counterpart. A detector at 95 % AP costs more than one at 92 % AP—more data, more compute, more engineering time. An L4 deployment costs *dramatically* more than an L3 deployment to develop and certify. A robotaxi in San Francisco costs more per operating hour than one in Phoenix. Understanding the economics of these decisions—capital costs, operating costs, revenue models, unit economics, break-evens—is what separates technology demonstrations from commercial programs. This chapter develops the financial framework that AV engineers and leaders need to reason about what they are building and why.

Three observations motivate the chapter’s structure.

First, AV economics have *only recently become concrete*. For most of the 2010s, AV was a research problem with speculative commercial futures; per-mile costs and per-vehicle ROI were approximations with wide error bars. By 2025–2026, enough commercial fleets operate (Waymo, Baidu, Pony.ai) that real unit economics are observable. The numbers are shaping strategic decisions across the industry.

Second, AV economics are *multi-layered*. Vehicle purchase, software development, mapping, operations, maintenance, insurance, regulatory compliance, customer acquisition—each a separate cost center with its own trajectory. A program that optimizes vehicle cost but ignores operations cost misses the picture; several high-profile AV programs have failed on operations cost specifically.

Third, AV economics *interact with policy* in ways that traditional automotive economics do not. Subsidies, regulatory fees, liability frameworks, insurance mandates, and taxation all shape the unit economics; changes in any of these alter the commercial case. Engineers designing for commercial deployment benefit from understanding this policy interaction.

The chapter is organized as follows. We first decompose the capital and operating costs of an AV program. Unit economics follow—cost per mile, per trip, per vehicle-year—with the comparisons to

human-driven benchmarks. We then develop industry structure: the economic roles of developers, manufacturers, operators, and infrastructure providers, and where value accrues. Financing patterns and investment flows are next. Regional economic impact (labor markets, tax bases, urban form) follows. We close with long-term cost curves and the economics of the transition. Three architecture diagrams visualize the cost structure, the unit-economics breakdown, and the industry value chain; worked examples apply the framework to specific commercial decisions.

24.2 The AV Cost Structure

AV programs have costs that span vehicle, software, and operations. Understanding the relative size of each category is the first step to reasoning about program economics.

24.2.1 Capital Costs

Capital costs are incurred up front; they must be recovered over the fleet lifetime. The major categories for AV programs:

Vehicle Purchase

A conventional passenger vehicle: 25 to 60 thousand USD. AV fitment adds substantial cost: sensors, compute, mounting, calibration, validation.

Example 24.1 (Per-vehicle AV sensor+compute cost). Typical production-L4 vehicle fitment as of 2025–2026:

- Multi-LiDAR suite: 5 000–15 000 USD.
- Multi-camera array: 1 500–4 000 USD.
- Radar suite: 1 500–3 500 USD.
- Compute (NVIDIA Drive, Qualcomm Snapdragon Ride, custom ASIC): 3 000–8 000 USD.
- Power and thermal management: 1 000–2 500 USD.
- Calibration, testing, integration: 2 000–5 000 USD.

Total: 14 000–38 000 USD per vehicle for AV fitment. Has dropped roughly $3\times$ since 2020 and continues to decline as sensor costs drop and integration standardizes.

Software Development

The largest category for most AV programs. Perception, planning, control, mapping, simulation infrastructure, evaluation, tooling— all developed over years by hundreds to thousands of engineers.

Example 24.2 (Cumulative software-development cost). Large AV program: $2\,500 \text{ engineers} \times 5 \text{ years} \times 300\,000 \text{ USD fully-loaded cost/year} = 3.75 \text{ billion USD cumulative}$. Added operational costs (offices, compute, cloud, benefits): add roughly 50 %, to 5.6 billion. This is only R&D, before manufacturing, operations, or deployment. Programs that have crossed 10 billion USD in

cumulative spend include Waymo (Google funded), Cruise (GM + outside), and several Chinese programs. The capital intensity is why the industry has consolidated: smaller programs cannot fund the R&D.

Mapping Infrastructure

Mapping fleets (dedicated vehicles + processing) represent a major capital outlay. Subsequent map maintenance is an operating cost.

Example 24.3 (Mapping fleet cost). To initially map a 100 km² urban area: 5 dedicated mapping vehicles (150 000 USD each) + 6 months of operation (60 000 USD/month each, 2 drivers + fuel) + processing and labeling labor (2 000 000 USD) $\approx$ 4.5 million USD. Amortized over 5 years at 10 % discount: $\sim$ 1.2 million USD/year in capital cost (map only). Ongoing updates are additional.

24.2.2 Operating Costs

Operating costs recur every mile/hour/trip. Dominating categories:

Energy

Fuel (ICE) or electricity (EV). For EVs, approximately 3 to 6 cents per mile. For ICE, 10 to 18 cents per mile depending on efficiency and fuel price.

Vehicle Maintenance

Tires, fluids, brakes, software updates, sensor recalibration. Robotaxi maintenance is more frequent than private-vehicle maintenance (higher utilization). Typical: 8 to 15 cents per mile.

Compute and Connectivity

On-vehicle compute runs continuously; cellular data for updates, logs, and V2X adds ongoing cost. Per vehicle: 200 to 500 USD/month ($\sim$ 0.5 to 1.5 cents per mile).

Insurance

Early-deployment insurance is expensive: 2–5 cents per mile, vs 0.5–1.5 cents per mile for human-driven commercial fleets. Premiums decline as safety records accumulate.

Remote Supervision

L4 fleets employ remote-supervisor staff. A supervisor monitoring (say) 10–20 vehicles adds 3 to 8 cents per mile to operating cost.

Fleet Operations Overhead

Dispatch, customer support, vehicle cleaning, charging, repairs. A typical robotaxi program spends 15 to 25 cents per mile on this category.

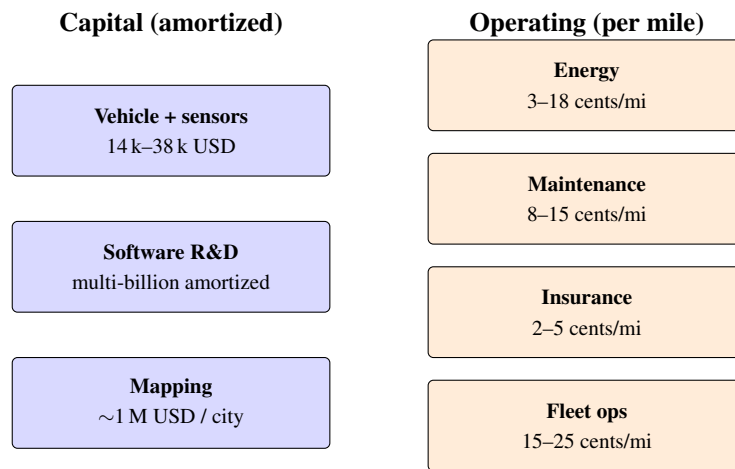


Figure 24.1: AV cost structure. Capital costs (left, blue) include vehicle hardware, software R&D, and mapping infrastructure; all are amortized over the useful life of the asset. Operating costs (right, orange) include energy, maintenance, insurance, and fleet operations; all are per-mile or per-hour. A commercial AV program must build a revenue model that covers both.

24.3 Unit Economics

Unit economics reduce the program-level costs to a per-unit metric. The most common unit is cost per mile. For a ride-hailing robotaxi fleet, cost per trip is also tracked.

24.3.1 Cost Per Mile

The fundamental metric. Cost per mile is:

$$\text{CPM} = \text{CPM}_{\text{capital}} + \text{CPM}_{\text{operating}}, \quad (24.1)$$

where $\text{CPM}_{\text{capital}}$ amortizes capital costs over expected miles-of-use, and $\text{CPM}_{\text{operating}}$ sums the running-cost components.

Example 24.4 (Fully loaded per-mile cost). Vehicle: 60 000 USD + 25 000 USD AV fitment = 85 000 USD. Useful life: 500 000 miles. Capital per mile: $85\,000 / 500\,000 = 17$ cents/mile. Operating (per mile):

- Energy: 5 cents.
- Maintenance: 12 cents.
- Compute/connectivity: 1 cent.
- Insurance: 4 cents.
- Remote supervision: 5 cents.
- Fleet operations: 20 cents.

- Software amortization (allocated per mile): 8 cents.

Operating per mile: 55 cents. Fully loaded: $17 + 55 = 72$ cents per mile.

Human-driven comparison: passenger-car total cost of ownership is roughly 60–70 cents per mile (IRS mileage rate is 67 cents as of 2025). Ride-hailing adds the driver's labor, pushing total cost closer to 1.50–2.00 USD per mile. AV robotaxis at 72 cents/mile are comparable to private cars and substantially cheaper than ride-hailing, on paper. Real performance depends on utilization—low-utilization AVs have high fixed-cost components pushed into per-mile numbers.

24.3.2 Utilization: The Central Variable

A vehicle with 500 000 miles of useful life and 200 000 actual miles has paid 42.5 cents per mile in capital, not 17. A vehicle with 750 000 miles has 11 cents. Utilization dominates the capital-cost portion of CPM; operational utilization is often the difference between profitable and unprofitable unit economics.

Example 24.5 (Utilization sensitivity). Same vehicle and costs as the previous example, swept across utilization scenarios.

Scenario	Lifetime miles	Capital CPM	Total CPM
Conservative	200,000	42¢	97¢
Base	500,000	17¢	72¢
Aggressive	800,000	11¢	66¢
Stretch	1,200,000	7¢	62¢

A $4\times$ swing in utilization produces a 50 % swing in CPM. The difference between commercial success and failure often reduces to: can you drive the fleet enough miles?

Figure 24.2 plots CPM as a function of utilization. The curve is hyperbolic: capital-per-mile falls as $1/\text{miles}$, while operating-per-mile is flat. Early-stage programs sit in the steep left region; mature programs push to the flat right region where further utilization gains yield diminishing returns.

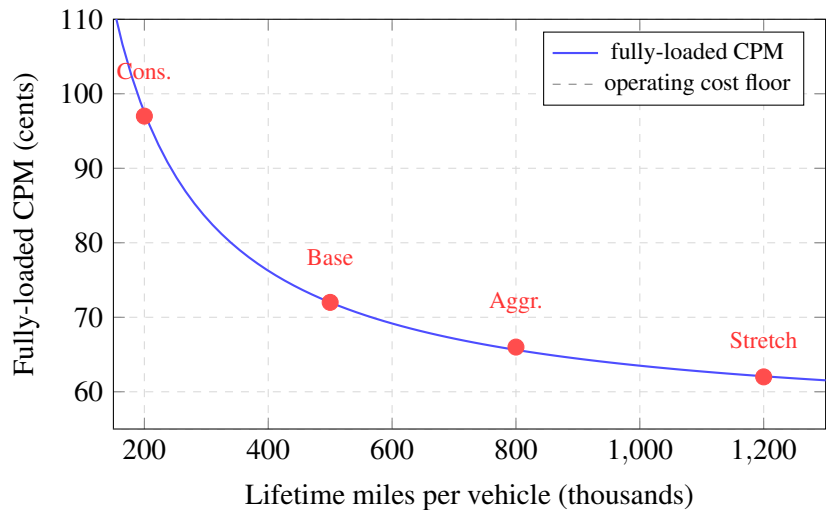


Figure 24.2: Cost-per-mile vs utilization. The blue curve is the fully-loaded CPM = capital/miles + operating; the dashed gray line at 55¢ is the operating-cost floor that no amount of utilization can reduce. Four scenarios from the table (red dots) span the realistic range. Near the floor, marginal utilization gains produce tiny CPM reductions; in the steep left region, they produce dramatic ones. Early fleets prioritize climbing out of the steep region above all else.

24.3.3 Revenue Per Mile

For a robotaxi service, revenue per mile depends on pricing and average trip length. Typical US ride-hailing: 2.50 USD per passenger-mile (of which the driver historically received about 70 %). AV robotaxis price anywhere from 1.50 to 3.00 USD per mile depending on market.

24.3.4 Gross Margin Per Trip

$$\text{GM} = \text{Revenue} - \text{COGS}, \tag{24.2}$$

with COGS (Cost of Goods Sold) the variable costs tied to the trip. At 2.00 USD revenue per mile and 0.72 USD CPM, gross margin is 1.28 USD per mile, or 64 % margin. This is approximate; real programs have deadhead miles (empty repositioning) that reduce revenue-miles/total-miles.

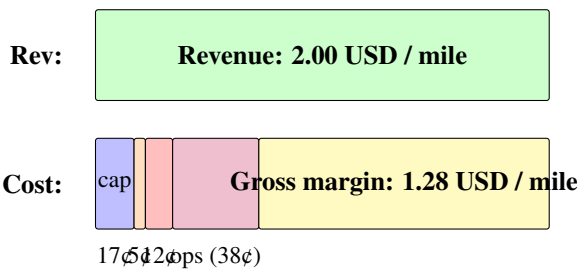


Figure 24.3: Gross-margin structure of an AV robotaxi trip at 2.00 USD/mile revenue. Cost stack (bottom): capital (17 cents), energy (5), maintenance (12), and fleet ops including insurance and supervision (38). Remaining 1.28 USD per mile is gross margin—the pool that must cover software R&D amortization, corporate overhead, taxes, and profit. The shape and size of this margin define whether the program is profitable.

24.3.5 Breakeven Analysis

A program reaches breakeven when cumulative gross margin exceeds cumulative fixed costs (including historical software R&D).

Example 24.6 (Program-level breakeven). Program cumulative R&D: 8 billion USD. Per-vehicle gross margin per year: 80 000 miles $\times$ 1.28 USD = 102 400 USD. Fleet size to break even on R&D in 5 years: $8 \times 10^9 / (5 \times 102,400) = 15,625$ vehicles. A large number but not unprecedented; Waymo and Chinese robotaxi programs operate at this scale. The realistic path to breakeven extends further—R&D continues; early revenue is lower; geographic expansion is incremental—but the order of magnitude is correct.

24.4 Industry Structure

AV economics operate within an industry value chain. Different participants capture different portions of the value; strategic positioning matters.

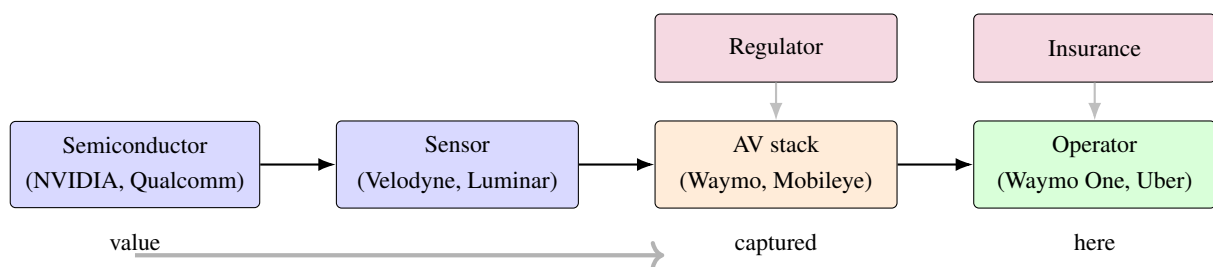


Figure 24.4: The AV industry value chain. Semiconductor and sensor providers supply hardware at commodity or near-commodity margins. AV stack developers capture the technology premium when deployed. Operators—whether integrated (Waymo) or partnerships (Uber + Waymo)—capture customer-facing value. Regulators and insurers are horizontal dependencies that shape the economics of every layer. Over time, value concentration shifts; early AV economics were sensor-dominated; current economics are operator-dominated.

24.4.1 Semiconductor Layer

High-volume, competitive, commodity-trending. NVIDIA (Drive platform), Qualcomm (Snapdragon Ride), Mobileye (EyeQ), and custom ASIC providers. Margins at the chip level are substantial (40 %+ gross for leading vendors) but the number of players increases over time, squeezing margins.

24.4.2 Sensor Layer

LiDAR, camera, radar. Has historically been high-margin due to small volumes and specialized components; is moving toward commodity as automotive-grade components ramp. Luminar, Innoviz, Valeo, Bosch, and Chinese competitors all compete on price and automotive-grade certification.

24.4.3 AV Stack Layer

Perception, planning, control, maps. Major players: Waymo (Alphabet), Mobileye, Tesla (vertically integrated), Chinese competitors (Baidu Apollo, Pony.ai, WeRide, Momenta, Horizon Robotics).

Value here is the “intelligence”; customers are either vehicle OEMs or the company’s own fleet.

24.4.4 Operator Layer

The customer-facing entity: Waymo One, Cruise (was), Zoox, Baidu Apollo Go, Pony.ai. Value capture here is direct revenue from end users. Operators can be integrated with stack development (Waymo, Pony.ai) or run on licensed stacks (Uber + Waymo in some markets).

24.4.5 Vertical Dependencies

Regulators set compliance costs (Chapter 4). Insurance sets insurance premiums. Both affect every layer’s economics; neither is directly controllable by any single industry participant.

24.5 Investment and Financing Patterns

Money flows into AV through identifiable patterns.

24.5.1 Venture Capital

Early-stage AV was VC-funded on the expectation of disruptive returns. Aurora, Cruise (pre-acquisition), Zoox (pre-acquisition), Pony.ai, Nuro raised multi-billion-dollar rounds. Returns have been mixed: some successful exits (Cruise to GM, Zoox to Amazon, Mobileye IPO); some struggles.

24.5.2 Corporate Strategic Investment

Automakers (GM into Cruise, Ford+VW into Argo—since dissolved, Toyota into Woven, Hyundai+Aptiv into Motional), tech giants (Alphabet into Waymo, Amazon into Zoox), and ride-hailing operators (Uber, Lyft, Didi) have all invested to secure positioning. Returns horizons are longer than VC.

24.5.3 Public Markets

A few AV-related companies are public: Mobileye (Intel spin-off IPO), Aurora (SPAC), Luminar, Innoviz. Stock performance has generally reflected the challenging near-term economics.

24.5.4 Chinese Capital Flows

Chinese AV programs have received substantial capital from domestic sources (government, sovereign wealth, private investors). Cross-border flows have been restricted by political dynamics.

Example 24.7 (Capital-efficiency comparison). Program A: 8 billion USD invested, 1 000 commercial vehicles operating, 50 000 miles/day each. Miles per dollar invested: $1000 \times 50000 \times 365 / (8 \times 10^9) \approx 2.3$ miles per dollar per year. Program B: 3 billion USD invested, 300 vehicles, 80 000 miles /day each. Miles per dollar: $300 \times 80000 \times 365 / (3 \times 10^9) \approx 2.9$. Program

B is more capital-efficient; this is a leading indicator of long-term profitability. Investors track this metric across programs.

24.6 Regional Economic Impact

AV deployment affects regional economies beyond the AV program itself.

24.6.1 Labor Market Effects

The displacement of drivers is the most visible effect. US professional driving employs roughly 3.5 million people; full automation would displace substantial portions. This is a long-term effect—decades, not years—and is partially offset by new jobs (remote supervisors, fleet technicians, charging-station staff) and by transition programs.

24.6.2 Urban Form Effects

AVs, by reducing the cost of driving and by freeing passengers to use trip time productively, may increase vehicle-miles-traveled (VMT) substantially. Models vary, but 10–30 % VMT increase is plausible, with implications for congestion, parking demand, and land use.

24.6.3 Tax Base Effects

Registration fees, fuel taxes (for EVs, fuel-tax base erodes regardless of autonomy), insurance premiums, parking revenue, and traffic-violation revenue all change with AV deployment. Cities with significant fiscal dependency on parking or citations face revenue declines.

24.6.4 Economic Development

Cities courting AV deployment compete on regulatory accommodation, road infrastructure, and testing facilities. The economic development value—tax base, employment, reputation—is significant for selected hub cities.

24.7 Long-Term Cost Curves

AV technology follows cost curves. Understanding the trajectory matters for strategic planning.

24.7.1 Sensor Cost Decline

Automotive LiDAR in 2016: 60 000–80 000 USD per unit. In 2025: 500–5 000 USD per unit for production-grade automotive LiDAR. Cost has followed roughly a 30 %/year decline, typical for volume-electronics curves. Camera and radar similarly have declined.

24.7.2 Compute Cost Decline

Automotive-grade AI-accelerator compute per TOPS has declined roughly 25 %/year. A compute node that cost 10 000 USD in 2020 costs roughly 3 000 USD in 2025 at equivalent performance.

24.7.3 Software Development Productivity

This is the trajectory that is hardest to predict. Some productivity improvements (foundation models, better tooling) are accelerating; the problem scope (L4 coverage in diverse environments) is also growing.

Example 24.8 (Per-vehicle cost projection). Year 2020 AV fitment cost: ~80 000 USD. Year 2025: ~25 000 USD. Projected 2030: 8 000–12 000 USD (at continuing decline rates, with some plateauing). The implication: AV fitment approaches vehicle-purchase-cost ratios normal for other premium features (like premium infotainment) by late 2020s. Economic viability across more segments (not just premium robotaxis, but mass-market cars) becomes plausible.

Figure 24.5 plots the historical and projected cost curves for LiDAR, compute, and full AV fitment. The steep decline of LiDAR specifically has driven most of the fleet-wide fitment savings; compute has declined more slowly because growing model sizes have offset per-TOPS price drops.

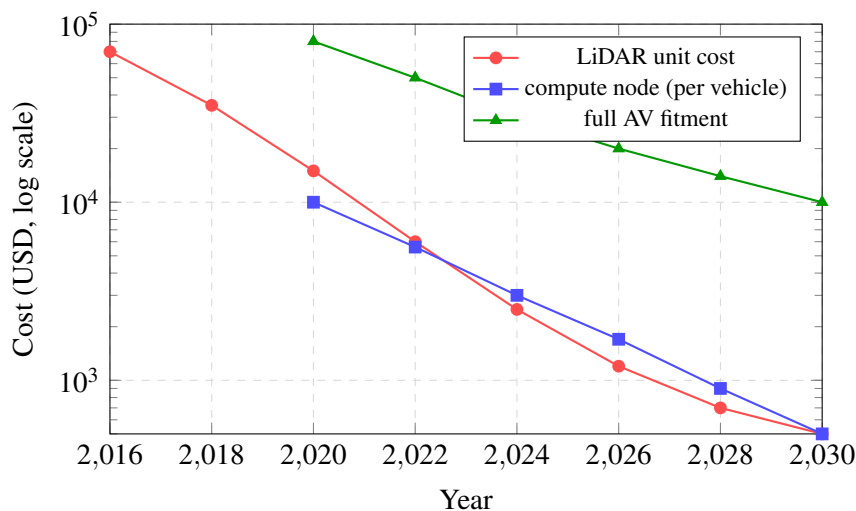


Figure 24.5: Cost-decline trajectories for major AV fitment components, 2016–2030 (log scale). LiDAR (red) has declined fastest due to production scaling of solid-state and MEMS-based designs; automotive compute (blue) has tracked a Moore’s-Law-like curve modified by workload growth; full AV fitment (green) aggregates sensors, compute, calibration, and integration. The curves assume continued production scaling and no disruptive architectural shift; actual costs could undercut the projections if foundation-model-scale compute demands accelerate.

24.8 Worked Examples

Example 24.9 (Full unit-economics model for a 1 000-vehicle fleet). Fleet: 1 000 L4 robotaxis. Vehicle + AV fitment: 85 000 USD each. Total capital: 85 million USD. Annual miles per vehicle: 80 000 (revenue) + 20 000 (deadhead) = 100 000 miles. Revenue: 80 000 miles $\times$ 2.00 USD/mile = 160 000 USD per vehicle per year. Operating cost: 100 000 miles $\times$ 0.55 USD/mile = 55 000 USD per vehicle per year. Annual gross margin: 160 000 - 55 000 = 105 000 USD per vehicle. Fleet gross margin: 105 million USD/year. Fleet capital payback: 85 / 105 $\approx$ 0.8 years, if we ignore R&D amortization and corporate overhead. Including those, the fleet may not break even

on program-level R&D for 5–10 years, but on a per-vehicle basis it is strongly cash-flow positive.

Example 24.10 (Sensitivity analysis). Starting from the base case above, sensitivity to key parameters:

- Utilization –20 %: margin falls 20 000 USD/vehicle.
- Revenue per mile –10 %: margin falls 16 000 USD/vehicle.
- Operating cost +10 %: margin falls 5 500 USD/vehicle.
- Insurance cost doubles: margin falls 4 000 USD/vehicle.

Utilization is the most sensitive parameter; revenue per mile is the second. Operating cost and insurance matter but are less critical. Programs must prioritize utilization maximization (efficient dispatch, geographic coverage, downtime reduction) above other optimizations.

Example 24.11 (ROI on a sensor-cost reduction project). Project: redesign sensor suite to reduce LiDAR unit cost from 3 500 to 1 500 USD (4 units per vehicle, saves 8 000 USD/vehicle). Project cost: 20 million USD engineering + 30 million USD validation. Payback on a 10 000-vehicle fleet: 80 million USD savings. Net payback: $50 / 80 = 0.625$, or roughly 5 months if the savings apply to all 10 000 vehicles simultaneously. Large programs routinely execute such projects because the savings scale with fleet size. A smaller program (100 vehicles) cannot afford to amortize 50 million USD of engineering over 800 000 USD of savings, so its sensor-strategy must be different.

24.9 Industrial Deployment

Public financial disclosures by AV companies and parent companies provide windows into industry economics. Waymo’s parent Alphabet does not separately disclose Waymo financials but Waymo has raised more than \$11 billion in external funding rounds. Cruise reported losses of several billion dollars to General Motors before the December 2024 wind-down decision. Pony.ai and WeRide both completed U.S. IPOs in 2024, with their S-1 filings disclosing operational details. Tesla’s vehicle margin and FSD-revenue trajectories are reported quarterly. McKinsey, Boston Consulting Group, and PwC publish recurring AV-industry economic outlooks. The cost-decline trajectories for LiDAR (from tens of thousands of dollars per unit in early Velodyne to hundreds of dollars in current Hesai/RoboSense units) and compute (from expensive workstation-class GPUs to integrated automotive SoCs) are publicly trackable through supplier disclosures.

24.10 Chapter Summary

AV economics have moved from speculation to measurement with the emergence of operating commercial fleets. Capital costs include vehicle, sensors, compute, software R&D, and mapping infrastructure. Operating costs include energy, maintenance, insurance, supervision, and fleet operations. Unit economics— cost per mile, revenue per mile, gross margin—determine program viability; utilization

is the most sensitive parameter. Industry structure comprises semiconductor, sensor, stack, and operator layers with distinct economics; regulation and insurance are horizontal dependencies. Capital flows come from venture, strategic corporate, and public-market investors. AV deployment affects regional labor markets, urban form, and tax bases. Long-term cost curves (sensor decline, compute decline, software productivity) point toward broader viability through the late 2020s. Programs that optimize for fleet-level gross margin, maximize utilization, and invest strategically along the cost curve reach profitability; programs that do not, do not.

24.11 Exercises

1. For the sensor cost example (24.1), compute the expected per-vehicle fitment cost in 2028 at a 25 % annual decline rate.
2. For the utilization example (24.5), compute CPM for a vehicle that operates 1,200,000 miles over 8 years.
3. Derive the break-even fleet size to recover 5 billion USD of R&D in 7 years at 80,000 USD per-vehicle annual gross margin.
4. Compare the cost structure of an integrated operator (Waymo) to a partnership operator (Uber + Waymo). Identify three economic differences.
5. Propose three strategies for increasing robotaxi utilization beyond the base case. For each, estimate the expected improvement and the implementation cost.
6. For the sensitivity example (24.10), compute the margin if three unfavorable parameters hit simultaneously. What does the fleet do?
7. Describe three ways operating costs might differ between San Francisco and Phoenix for an identical fleet and vehicle technology.
8. For the sensor-ROI example (24.11), compute the ROI if the project cost doubles but the per-vehicle savings remain unchanged.
9. A city proposes a 10-cent-per-mile AV tax to replace lost fuel-tax revenue. How does this change fleet economics? Is the program still viable?
10. Propose a pricing strategy for a new robotaxi deployment that balances revenue maximization with market-penetration goals. Justify each pricing lever.

Chapter 25

Autonomous E-Hailing: The Robotaxi Business

25.1 Introduction

If autonomous vehicles have a dominant commercial application in the 2020s, it is e-hailing—the on-demand transport of passengers, with the hailing initiated through a smartphone app. Robotaxis are the instantiation: autonomous vehicles, fleet-operated, providing rides to paying customers who summon them through an application. This is the business that Waymo, Cruise (until its 2023 setback), Baidu Apollo Go, Pony.ai, WeRide, Zoox, and others have invested billions of dollars pursuing. By 2026 robotaxi services operate in multiple cities across the United States, China, and select markets elsewhere. This chapter develops the business, the operational model, the competitive dynamics, and the strategic choices that shape it.

Three observations motivate the chapter's structure.

First, robotaxi is the business that *validates* AV technology commercially. The unit economics are real (paid rides); the customer experience is direct; the market is large (global ride-hailing is hundreds of billions of dollars). A robotaxi program's metrics are measurable—miles, rides, revenues—and the comparison with human-driven ride-hailing is direct. The business provides an empirical proving ground for the broader technology.

Second, robotaxi *competes with* human-driven ride-hailing, not with private-car ownership. The existing market has been built by Uber, Lyft, Didi, Grab, and others; robotaxi programs are displacing the human driver in that market rather than creating a new market. Understanding the existing ride-hailing business is prerequisite to understanding the robotaxi business.

Third, robotaxi operations are *profoundly geographic*. A fleet optimized for downtown San Francisco is not the same fleet as one for suburban Phoenix or Wuhan's outer districts. Operational playbooks, pricing, supply-demand matching, and regulatory relationships are all city-specific. A robotaxi company is really a portfolio of city-specific deployments, each with its own P&L.

The chapter is organized as follows. We first develop the existing ride-hailing market structure. We then examine robotaxi business models: how they differ from human-driven ride-hailing, and what they inherit. The operational playbook—fleet sizing, dispatch, pricing, customer experience—follows. A survey of major deployments (Waymo One, Apollo Go, Pony.ai, others) covers the current competitive landscape. Strategic issues—geographic expansion, competition with Uber/Lyft, partnerships, exit options—follow. We close with multi-modal integration and long-term trajectory. Two architecture diagrams visualize the operational flow and the competitive landscape; worked examples cover pricing, fleet sizing, and financial scenarios.

25.2 The Ride-Hailing Market

Before robotaxis, ride-hailing was built by Uber (founded 2009), Lyft, Didi, Grab, Ola, and others. The market is enormous and technologically sophisticated. Robotaxis enter this market; they do not create it.

25.2.1 Market Size

Global ride-hailing revenue in 2024 was approximately 200 billion USD, growing at high single digits annually. The US market is about 60 billion. China is comparable.

25.2.2 Unit Economics of Human Ride-Hailing

Understanding where robotaxi economics differ starts with understanding human-driven ride-hailing economics.

Revenue and Cost Structure

A typical US Uber/Lyft ride: 2.50–3.50 USD per passenger-mile gross fare, of which approximately 70 % goes to the driver (depending on market and surge conditions), 30 % to the platform. Platform costs: technology, marketing, customer service, insurance, legal.

Driver Economics

A US ride-hailing driver in 2024 earned perhaps 20–30 USD per hour gross, before vehicle costs (fuel, maintenance, insurance, depreciation). Net earnings, variable by market: 10–20 USD per hour. The economics push driver supply toward the marginal willingness to drive, which leaves little cushion.

Platform Margin

Platform gross margin on a ride: 10–15 USD operating profit before corporate overhead. Net margin after corporate overhead is small; Uber has only recently achieved sustained profitability (2023–2024).

25.2.3 Existing Supply-Demand Dynamics

Human-driven ride-hailing runs on a supply-demand market where driver availability is the fundamental supply variable. Surge pricing is the dominant mechanism for aligning supply with demand. Peaks (weekday mornings, Friday nights, airport runs, bad-weather days) generate higher prices and attract more drivers; troughs lose drivers to other work.

25.2.4 Operational Challenges

Ride-hailing platforms manage many operational issues that robotaxi operators will inherit: demand forecasting, dispatch algorithms, customer support, safety incidents (both passenger-driver and driver-driver), regulatory compliance across jurisdictions, insurance, driver onboarding and retention, dynamic pricing, cross-platform competition.

25.3 The Robotaxi Business Model

A robotaxi replaces the human driver with an autonomous system. Everything else either inherits from ride-hailing or differs in specific, predictable ways.

25.3.1 What Changes

Cost Structure

Driver cost is eliminated. Sensor and compute costs, insurance rates, and fleet-management overhead replace it. At current costs, robotaxi COGS per mile is comparable to or lower than human ride-hailing (Chapter 24); the savings potential grows as sensor and compute costs continue to decline.

Supply Dynamics

The supply variable changes from driver availability to fleet utilization. A robotaxi operates whenever its battery has charge and no planned maintenance is due; no driver shortage prevents supply at peaks.

Service Availability

A robotaxi service is bounded by ODD (Chapter 32), not by driver availability. Expansion is a technical and regulatory process, not a driver-recruitment one.

Pricing Flexibility

With zero marginal driver cost, robotaxis can price to attract demand without cannibalizing driver economics. This opens pricing strategies (loss-leading, sustained discounting) that human ride-hailing platforms could not execute.

25.3.2 What Stays the Same

Customer Expectations

Customers expect the same app, the same reliability, the same short wait times. Robotaxis without these underperform.

Market Segmentation

Airport runs, late-night trips, grocery delivery—these segments exist independent of drive-mode. Robotaxis must fit into them.

Regulatory Interface

Insurance, per-ride fees, driver-like background requirements (AVs have no drivers but must meet equivalent safety requirements), data reporting—all inherited.

25.3.3 Unit Economics Restructured

Per-trip economics for a robotaxi:

- Revenue (same as human): 2.50 USD/passenger-mile typical.
- Operating cost: 0.55 USD/mile (Chapter 24).
- Gross margin: 2.00 USD passenger-revenue-mile less about 0.55 USD cost (assuming 80% revenue miles, 20% deadhead).
- No driver payment.

Example 25.1 (Gross margin comparison). Typical 5-mile human-driven trip at 2.50 USD/mile = 12.50 USD gross. Driver receives 8.75 USD; platform gross 3.75 USD before costs. Platform's share is ~1.50 USD after operating costs (marketing, customer service, etc.) — roughly 12%.

Equivalent 5-mile robotaxi trip at 2.00 USD/mile = 10.00 USD gross. Operating cost ~3.50 USD. Gross margin ~6.50 USD — roughly 65%. Even with additional corporate overhead and R&D amortization applied per-trip, the margin structure is fundamentally different. This is the investment thesis behind robotaxi programs.

25.4 Operational Playbook

Running a robotaxi fleet is an operational discipline distinct from running an AV technology program. This section covers the core operational functions.

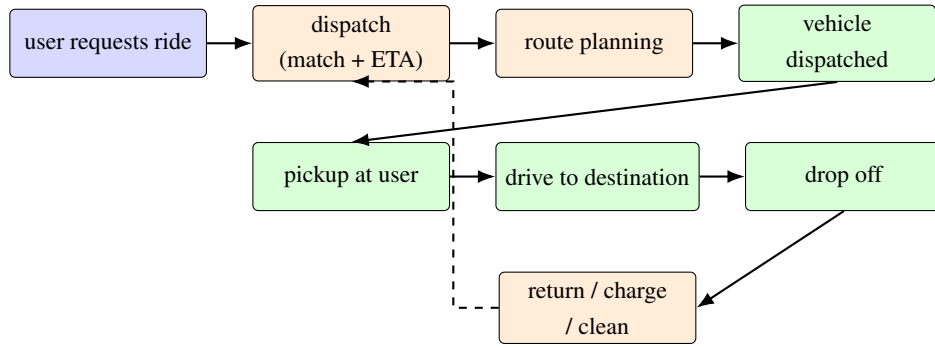


Figure 25.1: Robotaxi operational flow. User requests a ride through the app; backend dispatch matches a vehicle and plans a route; vehicle is dispatched to the user; pickup, drive, drop-off sequence follows; vehicle either returns to service (dashed feedback) or goes to charge/clean/repair. Each stage has specific SLAs, error modes, and performance metrics.

25.4.1 Fleet Sizing

How many vehicles for a given service area? The answer depends on demand distribution, target response times, and utilization targets.

Simple Model

For steady-state demand D (rides per hour), average trip duration T (hours, including pickup), and target utilization U :

$$N = \frac{D \cdot T}{U}. \quad (25.1)$$

Example 25.2 (Fleet sizing for a city). Peak demand: 500 rides per hour. Average trip + pickup time: 0.35 hours (21 minutes). Target utilization: 0.60 (60% of time actively on a trip, rest deadhead + idle + charging). $N = 500 \cdot 0.35 / 0.60 \approx 292$ vehicles. Off-peak demand half of peak: fleet size dominated by peak. Weekend demand may peak higher; a fleet sized for steady peak may oversupply weekdays. Elastic pricing helps; a fleet 15% smaller than naive peak-sized works if pricing attracts demand to lower-supply periods.

25.4.2 Dispatch

The problem: assign each incoming request to a vehicle. Modern dispatch is continuous re-optimization across pending requests and available vehicles. The objective mixes revenue, customer experience, and operational efficiency.

Objective Function

A typical dispatch objective:

$$\min \sum_i (w_1 \cdot \text{ETA}_i + w_2 \cdot \text{deadhead}_i + w_3 \cdot \text{misroute}_i). \quad (25.2)$$

ETA is the time until vehicle reaches user; deadhead is empty miles; misroute penalties capture whether the vehicle's current trajectory is suboptimal for the dispatch.

25.4.3 Pricing

Dynamic pricing is a primary demand-management lever.

Surge Pricing

During high demand, prices rise to suppress demand while redistributing supply. Works the same as human ride-hailing but without the supply-response mechanism (drivers coming out for higher pay).

Off-Peak Discounting

With zero marginal labor, robotaxis can discount during low- demand periods to fill vehicles. A 30–50% discount may be profitable if it triples ridership.

Example 25.3 (Time-of-day pricing elasticity). Base fare: 2.00 USD/mile at midday. The operator adjusts price against demand by time of day.

Period	Fare	Demand	Revenue/h	Utilization
Midday	2.00 USD	baseline	baseline	65 %
Evening peak	3.00 USD	−25 %	+12 %	~100 %
Late-night off-peak	1.20 USD	+200 %	+80 %	30% → 80%

The operator's policy: time-of-day pricing aligned with demand elasticity curves estimated from historical data.

Figure 25.2 plots representative demand and revenue curves against fare multiplier. Revenue peaks not at maximum demand (cheapest) nor at maximum fare (emptiest) but at an intermediate point whose location depends on the demand- elasticity parameter.

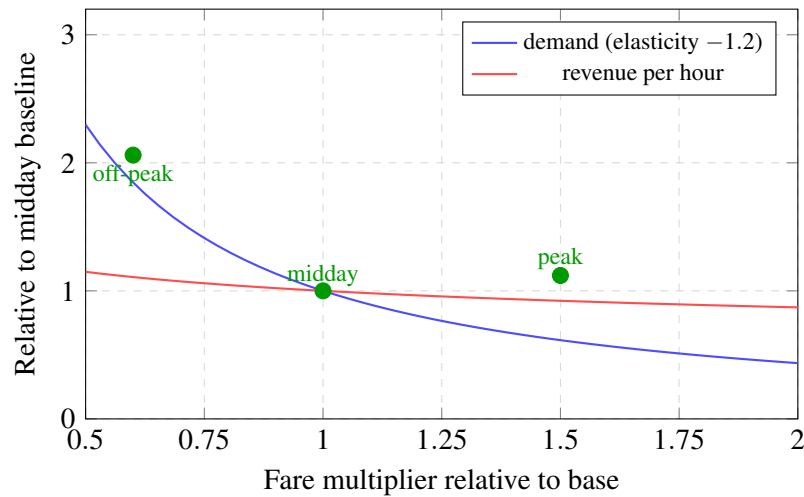


Figure 25.2: Demand and revenue as functions of fare multiplier. Demand (blue) falls with fare per the elasticity parameter; revenue per hour (red) is the product of demand and price. The three operating points from the table (green dots) sit on these curves. Off-peak discounting trades fare-per-ride for much larger ridership; peak surge trades ridership for higher fares. The optimal operating policy varies with the local elasticity.

25.4.4 Charging and Maintenance

Electric robotaxis need charging; all vehicles need cleaning between rides and periodic maintenance. Planning charging and maintenance around demand patterns matters operationally.

Charging Strategies

- Off-peak charging: charge during low-demand hours; keep vehicles available during high-demand.
- Opportunistic charging: charge whenever a vehicle's state-of-charge drops below a threshold, during a lull.
- Depot-based charging: return to dedicated depot for fast charging.
- In-field DC fast charging: use public or networked chargers.

Cleaning

Robotaxis without drivers have no one to clean the cabin between rides. A human cleaning crew (at depots, mobile) or in-cabin detection that triggers cleaning are current approaches.

25.4.5 Customer Experience

Absent a driver, customer experience is mediated by the app, the vehicle, and the service policies.

App Experience

Booking, pickup location, in-ride progress, payment, post-ride feedback. Must match or exceed human ride-hailing competitors.

In-Vehicle Experience

The pickup: how does the user unlock, enter, start the trip? The ride: music, temperature, communication. The drop-off: ending the trip, exiting, baggage handling. Each is an engineering problem.

Support

Remote human operators available for in-trip issues (door won't close, emergency situation, user request). Response latency and competence are customer-experience differentiators.

25.5 Major Deployments

Several robotaxi programs operate commercial services as of 2026. This section surveys them.

25.5.1 Waymo One

Alphabet's Waymo, the longest-running commercial robotaxi service. Cities: Phoenix (launched 2020, driverless since 2020), San Francisco (2023), Los Angeles (2024), Austin, Miami. Fleet: thousands of vehicles; millions of driverless rides per year. Vehicles: Jaguar I-Pace, transitioning to Zeekr-based platform. Strategy: conservative ODD expansion, direct operator model, partnerships with Uber in some markets.

25.5.2 Baidu Apollo Go

China's dominant robotaxi program. Cities: Wuhan (major driverless deployment), Beijing, Shanghai, Chongqing, and others. Fleet: comparable to Waymo in scale, lower per-ride revenue but substantial volume. Strategy: rapid expansion supported by Chinese regulatory accommodation; heavy investment in purpose-built robotaxi vehicles (Apollo RT6 and successors).

25.5.3 Pony.ai and WeRide

Chinese competitors. Pony.ai operates commercial services in Guangzhou, Shenzhen, and Beijing; Hong Kong-listed. WeRide operates in Guangzhou, Beijing, and other cities; Nasdaq-listed. Both compete with Apollo Go and with each other; smaller scale but growing.

25.5.4 Zoox

Amazon-owned; distinct strategy of purpose-built symmetrical vehicles (no forward or reverse, seats face each other). Las Vegas deployment; testing in multiple US cities. Strategy emphasizes hardware differentiation.

25.5.5 Mercedes-Benz / Nvidia / Others

Some automakers are developing robotaxi offerings as fleet operators; some are licensing their AV stack to third-party operators. Market is fragmented; consolidation seems likely but has not arrived.

25.5.6 What Happened to Cruise

Cruise’s GM-owned program operated in San Francisco through 2023 before a serious pedestrian-dragging incident and its disclosure handling led to operational suspension and substantial restructuring. Cruise returned to limited operations in late 2024–2025 with refocused scope. The case study is a reminder that execution missteps can cost programs their commercial position regardless of technology quality.

25.6 Competitive Dynamics

Robotaxi is a competitive market. This section covers the main axes of competition.

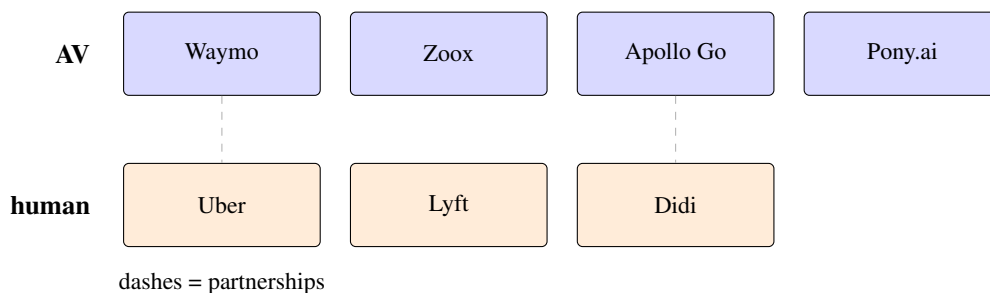


Figure 25.3: Competitive landscape in e-hailing. AV-native operators (blue) deliver autonomous rides directly. Human-driven ride-hailing incumbents (orange) continue operating human fleets while experimenting with AV integration. Dashed lines show partnerships—Waymo and Uber in Austin and Atlanta; Apollo Go and Didi in some Chinese cities. Competitive strategies mix direct-operator, partnership, and platform models.

25.6.1 Direct Operator vs Platform Partnership

Waymo runs both its own app and Uber-channeled rides in select cities. Direct app: more control, more data, more brand equity. Platform partnership: leverages existing user base, lower customer-acquisition cost. Different companies pick different mixes.

25.6.2 Vertical Integration

Some robotaxi programs build their own vehicles (Zoox, Apollo RT6, Waymo’s Zeekr partnership); others retrofit existing OEM vehicles. Vertical integration reduces per-vehicle cost long-term but requires manufacturing expertise.

25.6.3 Geographic Strategy

City Selection

Cities are selected by a mix of: demand density, regulatory environment, weather profile, competitive presence, labor-cost differentials. Phoenix, Austin, and San Francisco have clear reasons; others are chosen strategically.

Expansion Speed

Aggressive expansion (many cities, fewer vehicles each) vs deep expansion (fewer cities, more vehicles). Different programs pick different speeds; aggressive expansion has historically led to operational quality issues, while deep expansion risks missing competitive windows.

25.6.4 Pricing Competition

Robotaxi can in principle undercut human ride-hailing. Doing so permanently requires unit economics that support sustained low prices. A temporary pricing war would damage all operators; the strategic question is whether pricing discipline holds as multiple robotaxi operators enter the same markets.

Example 25.4 (Competitive pricing scenario). Market: single city, two robotaxi operators (A, B). Both enter at 2.00 USD/mile, matching Uber's prevailing price. Operator A cuts to 1.70 USD/mile to gain share. Demand shifts toward A; Operator B's utilization drops. Operator B matches. Both at 1.70 USD/mile. Market size grew by 10% (price elasticity), but individual operators split a slightly larger market at lower margin. Gross profits: both drop, possibly below breakeven at current costs. Rational response: slow further cutting; seek differentiation via service quality, coverage, wait times. Unlike Uber-Lyft pricing wars funded by venture capital, robotaxi programs cannot sustain extended loss-leading without accumulating losses against the original capital-intensive R&D.

25.7 Multi-Modal Integration

Robotaxis do not replace all transport. They complement and compete with other modes.

25.7.1 Transit Integration

First-mile/last-mile transit integration: a robotaxi takes the rider to the train or bus station; another robotaxi picks them up at the destination. Reduces transit's access barrier while preserving its line-haul efficiency. Emerging in pilot programs; unclear if scalable.

25.7.2 Multi-Modal Platforms

Platforms like Uber, Lyft, Grab integrate robotaxi options into apps that also offer bikes, scooters, transit, and traditional ride-hailing. A user selects based on time, cost, and preference. This is where robotaxi competes directly with other modes.

25.7.3 Congestion Interactions

Increased ride-hailing (human or robotaxi) is known to increase urban congestion—some estimates suggest 5–10% contribution to urban traffic in major cities. Robotaxis without deadhead-discipline (lots of empty positioning) exacerbate this. Cities may respond with congestion-pricing, occupancy-based road-use charges, or dedicated-lane allocation.

25.8 Worked Examples

Example 25.5 (End-to-end city deployment P&L). City: 500 robotaxis, medium-density market. Revenue: $500 \text{ vehicles} \times 80\,000 \text{ revenue miles/year} \times 2.00 \text{ USD/mile} = 80\,000\,000 \text{ USD/year}$. Operating cost: $500 \times 100\,000 \text{ total miles} \times 0.55 \text{ USD/mile} = 27\,500\,000 \text{ USD}$. Gross profit at city level: $52\,500\,000 \text{ USD/year}$. Capital allocated to city: $500 \times 85\,000 \text{ USD} = 42\,500\,000 \text{ USD}$. Return on city capital: $52.5/42.5 \approx 124\%/year$ before R&D amortization and corporate overhead. After allocation of R&D (say, 15 million USD/year/city) and corporate overhead (10 million): net 27.5 million USD. Return on city capital: 65%. Still strong, justifying continued city-by-city deployment. Real deployments are less clean but the shape is right for programs that reach production scale.

Example 25.6 (Fleet-growth decision). Current: 500 vehicles in City X, 90% utilization at peak. Options:

- Add 100 vehicles: 600 total, utilization drops to 85% but serves more peak demand; revenue increases 15%; marginal profitability high.
- Expand to City Y with 300 vehicles: revenue grows in new market; R&D allocation absorbs new city.
- Raise prices in City X by 10% during peak: revenue per existing vehicle up; demand falls slightly; higher margins per trip.

Net present value analysis picks the option with highest risk-adjusted returns. In practice: some combination, with portfolio management at the corporate level. Companies that execute portfolio-management discipline outperform those that pursue a single strategy.

Example 25.7 (Partnership economics). Robotaxi operator A partners with ride-hail platform B. Under the partnership: B books rides through its existing app and sends them to A's vehicles. A receives 75% of fare revenue; B retains 25%. For A: revenue per mile drops from 2.00 USD to 1.50 USD (75% of 2.00). At 0.55 USD operating cost, gross margin per mile: 0.95 USD (vs 1.45 USD on direct rides). Margin hit is 34%. What A gets: lower customer-acquisition cost, faster market penetration, B's established brand trust. The tradeoff is explicitly: lower margin per ride in exchange for more rides. Optimal mix for A depends on direct-channel growth rate and partnership-channel elasticity. Most large robotaxi programs maintain both channels with explicit pricing differentials.

25.9 Industrial Deployment

The ride-hailing industry's trajectory toward autonomy is shaped by operators' public AV strategies. Uber sold its ATG self-driving unit to Aurora in 2020 but retains an Aurora equity stake and is launching autonomous trips on its platform via partnerships with Waymo and others. Lyft sold its Level 5 self-driving unit to Toyota in 2021. Didi (China) operates an autonomous-driving subsidiary

with vehicles in Shanghai and Beijing. Apollo Go (Baidu) operates fully driverless robotaxi services charging riders directly. Waymo's robotaxi service has scaled in Phoenix, San Francisco, Los Angeles, and Austin and announced expansions to additional cities. The unit economics of these services—vehicle utilisation, deadhead miles, fare elasticity—are partially documented in public IPO prospectuses (Pony.ai, WeRide) and industry analyst reports.

25.10 Chapter Summary

Robotaxi is the dominant commercial application of autonomous driving technology. It inherits most of its business model (app-based booking, on-demand response, dynamic pricing) from human-driven ride-hailing, but replaces the driver economics with fleet operation. Major deployments (Waymo One, Baidu Apollo Go, Pony.ai, WeRide, Zoox) operate at increasing scale through 2025–2026. Operational playbooks span fleet sizing, dispatch optimization, pricing, charging, cleaning, and customer experience. Competitive dynamics involve direct-operator and platform-partnership models, vertical-integration choices, and geographic strategy. Multi-modal integration and potential congestion interactions shape long-term deployment. The unit economics are favorable when utilization and coverage reach scale; the path from technology to sustainable commercial operation requires operational excellence in all these dimensions.

25.11 Exercises

1. For the fleet-sizing example (25.2), compute the required fleet for peak demand of 800 rides per hour, trip + pickup time 25 minutes, utilization target 55%.
2. Compare the gross-margin structure (25.1) for a 2-mile trip. Does the ratio of fixed costs matter?
3. Derive the optimal surge-pricing multiplier under demand elasticity -1.2 and supply cost increase 0 (robotaxi marginal cost).
4. Describe three operational differences between an urban robotaxi and an airport-focused robotaxi service.
5. For the competitive pricing example (25.4), compute the shared demand at which both operators just break even. What does this imply for market structure?
6. Propose a charging strategy for a 500-vehicle fleet that balances utilization during peak hours with charging needs.
7. For the partnership example (25.7), derive the minimum B-channel volume that makes the partnership profitable for A.
8. Compare the urban-congestion impact of adding 500 robotaxis vs 500 conventional ride-hailing vehicles. What engineering responses exist?

9. Design a wait-time SLA for a robotaxi service that accounts for the distribution of user willingness-to-wait.
10. Propose a multi-modal service that combines robotaxi, transit, and micromobility. Describe how pricing and matching would work.

Chapter 26

Insurance and Liability for Autonomous Vehicles

26.1 Introduction

Traditional auto insurance is structured around human driver fault. The driver is the proximate cause of nearly all collisions; insurance premiums price the risk that a given driver will cause a given loss. AV insurance is fundamentally different. When the AV—not the driver—is controlling the vehicle, traditional fault allocation breaks down. Liability may sit with the manufacturer, the software provider, the fleet operator, or the passenger, depending on deployment model, regulatory framework, and specific facts. Insurance products are evolving to match, and AV program economics increasingly depend on how these products are priced.

Three observations motivate the chapter.

First, AV insurance is a *production-economics factor*, not a peripheral concern. For robotaxi programs, insurance costs per vehicle-mile can rival fuel/energy costs and exceed maintenance costs.

Second, the *liability allocation model is still being established*. Different jurisdictions are converging on different models: UK’s Automated Vehicles Act holds the “user-in-charge” or authorized self-driving entity responsible; US state-level approaches vary; EU is finalizing its product- liability framework.

Third, *fleet data* dominates AV insurance in ways traditional insurance does not. The operator knows every mile driven, every incident, every near-miss. Pricing can be based on operational reality rather than demographic proxy.

26.2 AV Liability Models

26.2.1 Four Allocation Models

Model	Liability party	Typical use case
Driver-responsible	Human driver (L0–L2)	Conventional; ADAS features do not shift liability
User-in-charge	Passenger when prompted to take over	UK-style partial-autonomy frameworks
Manufacturer	OEM or software provider	L4 robotaxi during autonomous operation
Operator	Fleet owner or service provider	Commercial robotaxi fleets

26.2.2 UK Automated Vehicles Act 2024

A representative framework that assigns liability during autonomous operation to the “Authorised Self-Driving Entity” (ASDE), who must carry insurance for any incidents during autonomous mode. When the human user is in control, normal driver-liability rules apply.

26.2.3 Product Liability

Whether the AV itself is a “product” for liability purposes is legally settled in most jurisdictions; when the AV causes harm through software fault, product-liability law applies. This is distinct from and in addition to motor-vehicle-liability insurance.

26.3 Actuarial Modeling of AV Risk

26.3.1 Claim Frequency and Severity

Insurance pricing is driven by expected loss $\mathbb{E}[\text{loss}] = f \cdot s$, the frequency of claims times mean severity. For AVs:

Metric	Human driver	AV (current L4)
Incidents/million miles	~8–10	~2–4
Severe incidents/million mi	~0.15	~0.05–0.10
Mean severity per incident	\$3 500	\$4 500–\$6 000

AVs have lower frequency; per-incident severity is comparable or slightly higher (heavier vehicles, higher-value equipment). Net loss rates are substantially lower for AVs in well-characterized ODDs.

Example 26.1 (Expected loss per vehicle-mile). Compare expected loss for a human-driven fleet and an L4 AV fleet.

Human fleet.

$$\begin{aligned} f_H &= 8 \times 10^{-6} \text{ incidents/mi,} \\ s_H &= 3,500 \text{ USD/incident,} \\ \mathbb{E}[\text{loss/mi}]_H &= 8 \times 10^{-6} \times 3,500 \\ &\approx \$0.028/\text{mi} = 2.8 \text{ cents/mi.} \end{aligned}$$

AV fleet.

$$\begin{aligned} f_{AV} &= 3 \times 10^{-6}, \\ s_{AV} &= 5,000, \\ \mathbb{E}[\text{loss/mi}]_{AV} &= 3 \times 10^{-6} \times 5,000 \\ &= \$0.015/\text{mi} = 1.5 \text{ cents/mi.} \end{aligned}$$

Premium implication.

AV expected loss is about half that of human-driven. But because AV tail risk (catastrophic software failure) is difficult to quantify, real premium pricing is typically higher per mile than expected loss alone would suggest. Insurer risk-loading and capital-reserve requirements push the rate higher.

26.3.2 Tail Risk and the Single-Fault Problem

A single software bug can affect every vehicle in a fleet. If undetected before widespread deployment, a systematic fault can produce many incidents simultaneously—a correlation that traditional actuarial models struggle with. Insurance structures for AVs must account for this:

- Per-fleet caps on aggregate liability.
- Reinsurance layers specifically sized for tail events.
- Regulatory capital requirements for AV insurers.

26.3.3 Demographic-to-Operational Transition

Traditional auto insurance uses demographic proxies (age, history, ZIP code) for risk; AV insurance uses operational data (miles driven, weather conditions, deployment ODD, software version). This is more accurate but demands different data infrastructure.

26.4 Premium Structure

26.4.1 Per-Mile, Not Per-Year

AV fleet insurance typically prices per mile rather than per year. Utilization is directly measurable; premium aligns with exposure.

26.4.2 Version-Dependent Pricing

Software version affects risk. A fleet running version X.Y may be priced differently from the same fleet running X.Z. Insurers increasingly demand visibility into software update cadence, validation history, and fleet-wide incident statistics.

26.4.3 Dynamic Pricing During Operation

Some AV insurance products adjust premiums in real time based on conditions: route risk profile, weather, time of day. For a fleet operator this is granular pricing that aligns insurance cost with operational choice.

Example 26.2 (Fleet annual insurance cost). Fleet of 500 AVs, each driving 100 000 mi/year. Per-mile insurance rate including risk loading: 3.5 cents/mi.

Fleet annual insurance cost.

$$\begin{aligned} C_{\text{insurance}} &= 500 \times 100,000 \times 0.035 \\ &= 1,750,000 \text{ USD/year.} \end{aligned}$$

Per-vehicle annual.

$$C_{\text{per-vehicle}} = 3,500 \text{ USD/year.}$$

For context: comparable to energy costs (~\$4k) and below depreciation (~\$15k/year). Insurance is a meaningful but not dominant cost line item.

26.5 Data Sharing and Transparency

26.5.1 Black-Box Recording

Most jurisdictions with AV regulations require data recorders documenting sensor inputs, planner decisions, and control actions for incidents. This data is essential for insurance adjudication and has become standardized across programs.

26.5.2 Fleet-Insurer Data Pipelines

Insurers receive continuous data feeds from insured fleets: miles driven, incidents, operational conditions. This is a departure from traditional insurance where most data is proprietary to the insured. AV-insurance effectiveness depends on these pipelines.

26.5.3 Privacy Considerations

Data sharing with insurers raises privacy concerns—especially for passenger-carrying AVs where trip data may be reidentifiable. Privacy-preserving aggregation and specific sharing agreements are ongoing areas of development.

26.6 Emerging Products

26.6.1 Fleet Captive Insurance

Large AV fleet operators (Waymo, Cruise historically) have explored captive insurance structures: self-insurance with reinsurance layers. Economics favor this when expected losses are predictable and fleet scale is large enough to diversify risk.

26.6.2 Cyber-Physical Coverage

Incidents caused by cybersecurity breaches are a new insurance category bridging cyber insurance and motor-vehicle insurance. Coverage for cyber-initiated AV incidents is emerging.

26.6.3 Manufacturer Warranty Integration

Some OEMs bundle AV operation insurance with vehicle purchase or subscription, using their scale and data to offer lower rates than third-party insurers.

26.7 Industrial Deployment

The insurance industry has begun developing AV-specific frameworks. The United Kingdom's Automated and Electric Vehicles Act 2018 was among the first to allocate liability between insurer and manufacturer for incidents in autonomous-driving mode. Munich Re and Swiss Re have published research on AV insurance pricing. State Farm, Allstate, Progressive, and other major insurers in the U.S. have publicly engaged with AV liability questions. Commercial fleet insurance for AV operators is provided by specialist underwriters; Tokio Marine and AIG have written AV-fleet policies. The Cruise robotaxi service publicly carried significant commercial liability coverage during its operational period. Insurance- industry estimates of long-term loss-cost reductions due to AV-driven crash reduction vary substantially across analyst publications.

26.8 Chapter Summary

AV insurance is transforming from a marginal risk-transfer product into a core economic driver of deployment. Liability allocation is converging on operator or manufacturer responsibility during autonomous operation, with frameworks (UK AV Act, EU product-liability) being set. Actuarial models rely on operational data rather than demographic proxies; tail-risk management for correlated software failure is a new discipline. Fleet insurance cost per mile is currently comparable to other major cost lines; the trajectory is downward as data accumulates and risk quantification improves. AV program economics should treat insurance as a first-class cost line driven by engineering choices (ODD scope, software update rigor, data infrastructure).

26.9 Exercises

1. For the expected-loss example, compute the break-even AV incident rate at which expected loss equals human.
2. For the fleet-cost example, compute annual cost at a 2 cent/mi rate with the same utilization.
3. Compare the UK Authorised Self-Driving Entity model with a hypothetical manufacturer-only liability model.
4. Propose a tail-risk capital requirement for an AV insurer covering a 10 000-vehicle fleet.
5. Identify three factors that would raise AV insurance rates above those suggested by expected-loss models alone.
6. Design a dynamic-pricing product that adjusts rates based on weather and traffic conditions.
7. For a 100-vehicle fleet, evaluate whether captive insurance makes economic sense relative to third-party.
8. Discuss privacy implications of continuous insurer data feeds. Propose three specific mitigations.
9. Compare AV insurance pricing for a well-characterized ODD (suburban, mapped, geofenced) vs an expanding ODD (new-city deployment).
10. Propose a cyber-physical insurance product specifically targeting AV software-initiated incidents.

Chapter 27

Labor Economics and Societal Transition

27.1 Introduction

Autonomous vehicles, when deployed at scale, will displace paid driving labor—taxi drivers, truck drivers, rideshare drivers, delivery drivers—while creating new categories of work: remote operators, fleet technicians, data labelers, AV-safety auditors. The magnitude and timing of the displacement, and the quality and compensation of the new jobs, are major policy questions. Engineers building AV systems have both stake and responsibility in these questions; this chapter surveys the labor economics of AV deployment.

Three observations motivate the chapter.

First, driving is an *unusually large* occupational category in the developed world. In the US alone, roughly 3–4 million people hold truck-driving jobs; roughly 1 million drive for rideshare; hundreds of thousands drive taxis, buses, and delivery vehicles. The scale of potential displacement is large.

Second, the *transition will not be uniform*. Some segments (long-haul trucking on fixed routes) are more automatable than others (local delivery with complex last-mile interactions). Some regions will see deployment first; others will lag by years or decades.

Third, *new jobs are emerging* alongside displacement. Remote operation, fleet maintenance, data labeling, and AV auditing employ substantial numbers of people in leading AV markets. The question is whether the new jobs absorb displaced workers and on what timeline.

27.2 The Scale of Driving Employment

27.2.1 US Driving Employment (Approximate)

Category	Approximate workers
Heavy-duty truck drivers	1,900,000
Delivery drivers (USPS + private)	1,400,000
Rideshare drivers (part + full)	1,000,000+
Taxi and limousine drivers	350,000
Bus drivers (transit + school)	600,000
Total	~5.2M

These figures vary by year and data source; the key point is that the ecosystem employs millions of people in driving roles.

27.2.2 Global Scale

Globally, driving employment is roughly ten times larger—tens of millions of workers. The transition will unfold in developed economies first (with accompanying political attention) and eventually in emerging economies where driving employment is often a larger share of total employment.

27.3 Automation Susceptibility

Not all driving jobs are equally automatable. Key variables:

Segment	Automation barriers	Timeline
Long-haul trucking (fixed routes)	Few; automation is primarily a scaling question	5–10 years
Robotaxi (urban)	ODD limits; edge-case handling in complex urban scenes	Expanding market already
Local delivery	Last-mile physical interaction (boxes, doorsteps)	15–20 years
School bus	Child safety liability; political resistance	Possibly never
Transit bus (fixed route)	Geofenceable; already piloted	5–10 years

Example 27.1 (Projected US driving-employment timeline). A rough scenario for US driving employment 2025–2040:

2025 baseline.

$$N_{2025} \approx 5.2 \text{ M drivers.}$$

Displacement trajectory (scenario assumption).

- 2030: $\sim 5\%$ displaced (long-haul early, robotaxi urban). $\sim 250 \text{ k}$ workers.
- 2035: $\sim 20\%$ displaced. $\sim 1 \text{ M}$ workers.
- 2040: $\sim 35\%$ displaced. $\sim 1.8 \text{ M}$ workers.

New jobs created (partial offset).

- Fleet maintenance: roughly 1 technician per 20 vehicles. For a 2M-AV fleet: $\sim 100 \text{ k}$ jobs.
- Remote operation: roughly 1 operator per 20–30 vehicles. $\sim 70\text{--}100 \text{ k}$ jobs.
- Data + labeling + audit: $\sim 30\text{--}50 \text{ k}$ jobs.

Net impact by 2040.

$$\text{net displacement} \approx 1.8 \text{ M} - 0.25 \text{ M} \approx 1.5 \text{ M workers.}$$

This is a scenario illustration; actual trajectories will depend on policy, technology, and adoption. But the net- displacement magnitude is the right order.

27.4 Transition Programs and Policy Responses

27.4.1 Retraining Programs

Federal and state initiatives (US), EU structural programs, and national training schemes (many countries) fund retraining for displaced workers. Effectiveness varies; retraining outcomes are generally better for younger workers and those with transferable skills.

27.4.2 Phased Deployment and ODD Controls

Some jurisdictions have phased AV deployment to allow gradual transition: restricted geographies, mandated human supervisors, employment preservation for transit workers during transition years.

27.4.3 Universal Basic Income Debates

AV displacement has become a prominent case in UBI and redistributive-policy debates. The argument: automation produces aggregate wealth gains, and policy should redistribute some fraction to displaced workers.

27.5 New Job Categories

27.5.1 Remote Operators

Most L4 deployments rely on remote-assistance operators for edge cases. One operator handles 5–30 vehicles depending on deployment complexity. Wages and working conditions vary; some operators are long-shift low-wage workers while others are higher-skilled mission specialists.

27.5.2 Fleet Technicians

AV fleets require skilled maintenance: sensor calibration, software management, battery systems. Compensation for these roles is typically higher than driving, but the total job count is much smaller.

27.5.3 Safety Auditors and Regulators

Regulatory agencies are hiring AV-specific auditors, accident investigators, and compliance specialists. These are professional-tier jobs but in small numbers.

27.5.4 Data and AI Operations

Training data curation, scenario generation, and model validation employ substantial numbers of people at AV developers and specialist service providers.

27.6 Distributional Effects

The AV transition will not distribute uniformly across society.

27.6.1 Geographic Variation

Regions heavily dependent on driving employment (Texas, the US South, Eastern Europe) face larger local impacts than regions with more diversified economies. A mid-sized town whose economy depends on a trucking hub can see dramatic local effects from changes that appear mild in national statistics.

27.6.2 Demographic Variation

Older driving workers face greater retraining challenges than younger ones. Workers with limited formal education and immigrant workers (a substantial share of US truck-driving) may find it harder to access retraining than native-born, better-educated workers.

27.6.3 Political Economy

Large-scale displacement without policy response produces political reactions: protests, legislative resistance, shifts in regulatory stance. AV developers who fail to engage with the labor transition politically may face regulatory or social backlash that slows their deployment.

27.7 Industrial Deployment

Labor-economic effects of AVs have been studied by the U.S. Department of Transportation, the Bureau of Labor Statistics, the OECD, the European Commission, and academic researchers. The Bureau of Labor Statistics publishes employment and wage data for affected occupations including truck drivers (Heavy and Tractor-Trailer Truck Drivers, SOC 53-3032, roughly 1.9 million U.S. employees as of mid-2020s data) and taxi/limo drivers. The American Trucking Associations has published industry analyses on autonomous-trucking timelines. Cruise's wind-down in late 2024 is among the more visible recent disruptions to AV-related employment. The transition timeline—driver displacement is gradual rather than abrupt—reflects the staged commercial deployment of AV technologies.

27.8 Chapter Summary

AV deployment at scale will displace millions of driving jobs over two to three decades. The transition will be uneven across segments, geographies, and demographics. New jobs—remote operation, fleet technicians, data and AI operations—will offset some of the displacement but likely will not match it in total count. Transition programs, ODD-based phasing, and distributional policy are ongoing responses. Engineers building AV systems have both stake and responsibility in these questions; technical choices influence deployment scale, timeline, and the resulting social consequences.

27.9 Exercises

1. For the US employment table, estimate displacement by segment under a 50 % urban robotaxi penetration by 2035.
2. Compute the net jobs balance if new-job ratios are higher than the example assumes (e.g., 1 operator per 10 vehicles).
3. Identify three driving-employment segments not in the table and estimate their automation susceptibility.
4. Propose a retraining-program design targeted at long-haul truck drivers displaced by 2030.
5. For a mid-sized trucking-hub town with 5 000 driving jobs, estimate direct and indirect economic impact of 50 % displacement.
6. Compare AV-driven displacement with earlier technology transitions (e.g., ATMs, industrial robotics). What lessons apply?
7. Design a phased AV deployment policy that balances innovation and worker protection.
8. Estimate the training-program cost per displaced worker for a program with 60 % reemployment outcome.

9. Discuss the political-economy risks of AV deployment without labor-transition support. Propose mitigations.
10. Compare US and EU approaches to AV labor transition. Identify the key differences.

Part VIII

Environment and Alternative Sensing

Chapter 28

Bio-Inspired and Alternative Sensing

28.1 Introduction

Modern AV perception is dominated by cameras, LiDAR, and radar. These modalities cover most of the sensing problem well, but each has failure modes and gaps—cameras fail in glare, LiDAR struggles with dark non-reflective surfaces, radar has poor angular resolution. Biology and physics suggest alternative modalities that could fill specific gaps: polarization vision disambiguates glare and wet surfaces, acoustic sensing detects emergency vehicles before they appear visually, olfactory sensors detect fires and dangerous chemicals, and bio-inspired motion detection achieves ultra-fast response with biology-level efficiency.

Three observations motivate the structure.

First, failure-mode coverage is the case for alternative sensing. Not all driving conditions are sensed well by the standard modalities; alternatives can cover specific failure modes that are otherwise hard to eliminate.

Second, biology has had a long time to optimize perception systems. Animal eyes, insect motion detection, and bat echolocation all embody solutions to problems that automotive perception also faces; bio-inspiration has produced specific commercially-viable sensor designs.

Third, fusion is the deployment pattern. No alternative sensor replaces the standard stack; each adds a specific capability that fuses in to the broader scene understanding.

28.2 Polarization Vision Sensors

28.2.1 The Physics

Reflected light carries polarization information. Wet surfaces, glossy plastics, and glass reflect polarized light strongly; diffuse materials do not. Polarization cameras measure this directly.

28.2.2 AV Applications

Scenario	Polarization benefit
Glare on windshield	Polarization filtering reveals road surface beneath
Wet road detection	Wet vs dry surfaces separable by polarization
Black-car detection	Polarized reflections distinguish glossy black car from road
Water-on-road vs oil	Different polarization signatures

Example 28.1 (Polarization-based wet-road detection). Conventional camera: wet asphalt and dry asphalt produce similar pixel intensities; classifier confusion common.

Polarization ratio p .

$$p = \frac{I_{\parallel} - I_{\perp}}{I_{\parallel} + I_{\perp}}.$$

Typical measured values.

$$p_{\text{dry asphalt}} \approx 0.02\text{--}0.05,$$

$$p_{\text{wet asphalt}} \approx 0.20\text{--}0.40.$$

Separation by threshold.

$$\tau = 0.10,$$

classification accuracy $\gg 95\%$.

Polarization disambiguation is decisive; conventional intensity- only cameras cannot achieve the same accuracy.

28.3 Acoustic Perception

Sound reveals things vision cannot: approaching sirens, screeching tires, engine roars around a corner.

28.3.1 Emergency Vehicle Detection

Sirens are acoustically distinctive and often audible before the vehicle is visible. Microphone arrays with AV-onboard processing detect, localize, and classify emergency sirens at ranges up to 150 m in urban environments (much further than line-of-sight camera detection when occlusion is present).

28.3.2 Architecture

A 4–8 microphone array around the vehicle’s roof; beamforming and spectral classification detect sirens, screeches, horns, and other notable sounds; localization uses time-difference-of-arrival (TDOA).

Example 28.2 (TDOA-based direction of arrival). Four microphones on a roof in a 1-m square. Siren arrives from a direction θ .

Time difference between opposite microphones.

$$\Delta t = \frac{d \sin \theta}{c},$$

with $d = 1$ m and $c = 343$ m/s.

For $\theta = 30^\circ$.

$$\begin{aligned}\Delta t &= \frac{1 \times 0.5}{343} \\ &\approx 1.46 \text{ ms.}\end{aligned}$$

Measurement precision.

Production audio ADCs sample at ≥ 44.1 kHz (resolution $\sim 22 \mu\text{s}$); direction resolution:

$$\begin{aligned}\Delta \theta &\approx c \cdot \Delta t_{\text{sample}} / d \\ &\approx 343 \times 22 \times 10^{-6} \\ &\approx 0.4^\circ.\end{aligned}$$

Sub-degree acoustic direction accuracy is achievable with commercial-grade electronics.

28.4 Olfactory / Chemical Sensing

Detecting fires, chemical spills, or dangerous fumes is relevant for industrial and emergency-response AV applications.

28.4.1 Applications

- Wildfire detection and routing (mining, logging AVs).
- Chemical plant autonomous-logistics vehicles.
- Tunnel-fire detection for autonomous buses.
- Agricultural pesticide detection.

Not yet common in passenger AVs; relevant for specialized deployments.

28.4.2 Sensor Technologies

- Metal-oxide semiconductor gas sensors (low-cost, broad).
- Photoionization detectors (sensitive, costly).
- Machine-olfaction arrays trained on specific targets.

28.5 Bio-Inspired Motion Detection

Flies detect motion at millisecond latencies with microwatt power budgets. Automotive perception could benefit from the same efficiency.

28.5.1 The Biology

Insect motion detection relies on pairs of photoreceptors whose outputs are temporally cross-correlated; motion direction emerges from the sign of correlation. The circuit is extremely simple but specialized.

28.5.2 Artificial Analogs

- Event cameras (Chapter 30) produce asynchronous event streams similar to biological photoreceptors.
- Neuromorphic silicon (Loihi, etc.) implements correlation-based motion detection at sub-millisecond latency with milliwatt power budgets.
- Specialized motion detector ASICs (used in optical-mouse sensors) can be adapted for automotive ultra-fast motion alert.

Example 28.3 (Bio-inspired tripwire for fast-crossing objects). Scenario: motorcycle crossing ego's path from an occluded position, revealed at 25 m closing at 15 m s^{-1} .

Time to contact.

$$\begin{aligned} T &= 25/15 \\ &\approx 1.67 \text{ s.} \end{aligned}$$

Conventional pipeline latency.

$$\begin{aligned} T_{\text{detection}} &\approx 100 \text{ ms,} \\ T_{\text{tracker}} &\approx 60 \text{ ms,} \\ T_{\text{planner}} &\approx 50 \text{ ms,} \\ T_{\text{control}} &\approx 30 \text{ ms,} \\ T_{\text{total}} &\approx 240 \text{ ms.} \end{aligned}$$

Bio-inspired tripwire latency.

$$T_{\text{event}} + T_{\text{tripwire}} \approx 5 \text{ ms.}$$

A fast-moving object triggers a safety-critical brake within 5 ms of visibility—235 ms saved compared to the full pipeline. At 15 m/s this is 3.5 m of closing distance saved.

28.5.3 Role in the Production Stack

Bio-inspired motion detection does not replace conventional perception; it supplements it as a fast-path safety monitor for sudden-appearance scenarios. The full stack handles classification and planning; the bio-inspired path provides a sub-frame reflex.

28.6 Industrial Deployment

Alternative sensing modalities have specific commercial deployments. Polarisation cameras from Tele-dyne and SONY are used in industrial inspection and have seen automotive research demonstrations. Acoustic-detection systems for emergency-vehicle sirens are deployed in some production vehicles; Waymo, Cruise, and academic research groups have published on this capability. Event cameras from Prophesee (France) and iniVation (Switzerland) target high-speed and low-power applications including some automotive demonstrations. Neuromorphic processors including Intel's Loihi and IBM's TrueNorth are research platforms; commercial automotive deployment remains limited as of 2024–2025. Bio-inspired motion-detection circuits draw on academic work going back to the Reichardt detector model from biological vision research.

28.7 Chapter Summary

Alternative sensing modalities complement the standard camera- LiDAR-radar stack. Polarization cameras disambiguate wet vs dry surfaces and cut through glare. Acoustic arrays detect emergency vehicles beyond line-of-sight. Olfactory sensors find use in specialized industrial AVs. Bio-inspired motion detection, on event cameras or neuromorphic silicon, provides sub-frame reflex capability that the conventional pipeline cannot match. None displaces the standard stack; each covers specific gaps and failure modes that the standard modalities address poorly.

28.8 Exercises

1. For the wet-road polarization example, compute detection accuracy if polarization ratio has measurement noise of $\sigma = 0.05$.
2. For the TDOA example, compute direction resolution if microphone spacing is reduced to 0.5 m.
3. Propose a fusion scheme combining polarization and thermal imaging for night driving.

4. Design an acoustic siren detector architecture for an urban AV robotaxi.
5. Compare bio-inspired motion detection to a 1-ms-latency high-frame-rate camera pipeline; identify where each wins.
6. Identify three specific AV scenarios where polarization vision would provide decisive value.
7. For the bio-inspired example, compute the benefit at 30 m s^{-1} closing speed.
8. Propose an olfactory-sensor deployment for wildfire- prone autonomous logistics operations.
9. Discuss the trade-offs of adding N more sensors to an AV stack: cost, calibration, failure modes.
10. Design a fusion rule that combines a bio-inspired fast- path with the conventional perception pipeline safely.

Chapter 29

Environmental Extremes and Hard Perception Cases

29.1 Introduction

Standard AV perception is trained on and validated against representative urban and highway conditions. The long tail of operational conditions—dust storms, floods, potholes, wildlife crossings—falls outside the training distribution and represents a substantial share of real-world perception failures. This chapter surveys four environmental-extreme categories and the engineering responses each demands.

Three observations motivate the structure.

First, regional variation dominates. An AV trained for European temperate roads may fail on unmaintained African roads with seasonal dust, flood-prone monsoon routes, wildlife zones, or severe pothole distributions. Regional-specific training data is essential.

Second, hardware limits matter here more than in nominal operation. In extreme conditions, sensor-level failures (lens contamination, LiDAR dropouts, radar noise) are the primary constraint, not algorithm-level perception quality.

Third, safe degradation is the design pattern. Under extreme conditions, the goal is not full performance but safe graceful degradation: slower driving, wider margins, transparent alerting of the operator.

29.2 Dust and Sand Perception

Chapter 25 treated rural dust in depth. Here we extend to heavier dust and sand conditions typical in desert, mining, and African-rural contexts.

29.2.1 Perception Degradation Modes

Condition	Effect
Suspended dust	Camera haze; LiDAR returns on dust itself
Blowing sand	Rapid lens contamination; abrasion
Dust cloud wake	Temporary complete obscuration behind own vehicle
Sandstorm	Multi-kilometre visibility reduction

29.2.2 Engineering Responses

- Positive-pressure sensor housings keeping dust out.
- Active lens cleaning (air jet, liquid wash).
- Radar-primary perception when camera/LiDAR degrade.
- Retrained models with dust-augmented training data.
- Behavioral: slow down, widen margins, increase following distance.

29.3 Flooded Road Detection

Water on roads presents a distinct hazard: beneath apparent water lies unknown road surface (potholes, debris, washed-out road).

29.3.1 Cues

- Surface reflection patterns (specular vs diffuse).
- Polarization signatures (Chapter 28).
- Thermal-camera contrast (water is thermally distinct).
- Stereo-disparity inconsistencies when water reflects the sky.

Example 29.1 (Decision rule under flooding). Scenario: ego approaches road segment with apparent water coverage. Measured surface properties:

- Polarization ratio: 0.32 (wet or water).
- Thermal contrast: 4°C cooler than surroundings.
- Stereo disparity: inconsistent (sky reflections).
- Map information: road segment, standard usage.

Fusion rule. Two-of-three cues triggering = water present.

Behavioral response.

- Soft stop before entering suspect zone.
- If depth unknown $\Rightarrow$ do not proceed.

- Alert remote operator / fall back to safe position.

Production AVs in flood-prone regions implement such rules specifically because visual-only detection is unreliable.

29.4 Pothole and Road Damage Detection

Large potholes can damage vehicles and injure passengers. Important in contexts with degraded road infrastructure.

29.4.1 Detection Approaches

Approach	Characteristics
Image-based	Deep models trained on pothole images; false positives on shadows
LiDAR-based	Ground-plane deviations; reliable at short range
Vibration-based	IMU signature after traversal—detection after-the-fact
Crowdsourced	Fleet-scale mapping of pothole locations

29.4.2 Importance in African Contexts

Urban and inter-urban roads in many African cities have potholes far larger and more frequent than those in mature infrastructure regions. AV models trained on European road data miss these at rates that make deployment impractical without retraining.

29.5 Wildlife Detection

Animal-vehicle collisions are a significant safety issue (3–4 million vehicle-animal collisions per year in the US alone).

29.5.1 Species-Specific Challenges

Species class	Perception issue
Large ungulates (deer, moose)	High impact energy; unpredictable crossings
Small mammals (raccoons, etc.)	Difficult to detect at safe-braking range
Flocks of birds	Fast motion; difficult to classify
Livestock and farm animals	Context-dependent; may be authorized to be on road
Large wildlife (elephants, rhinoceros)	Rare but potentially lethal in specific regions

29.5.2 Detection Improvements

- Thermal imaging (most animals contrast strongly with road).
- Motion cues (animals move differently from pedestrians or vehicles).
- Region-specific training data (African wildlife classes not present in European datasets).
- Behavioral: slow down in marked wildlife zones; use the HD map's wildlife-area annotations.

Example 29.2 (Wildlife-zone behavior envelope). Route includes 15 km segment through designated elephant corridor. Normal highway speed: 100 km/h.

Wildlife-zone envelope.

- Max speed: 60 km/h.
- Detection sensitivity: elevated (lower thresholds).
- Thermal camera: always active.
- Following distance: extended.

Time impact.

$$\begin{aligned}
 \Delta t &= 15 \times (1/60 - 1/100) \times 60 \\
 &= 15 \times 0.0067 \times 60 \\
 &\approx 6.0 \text{ min additional.}
 \end{aligned}$$

Six minutes added for a 15-km stretch is acceptable given the severity of wildlife collisions and the low frequency of wildlife-corridor routing.

29.6 Industrial Deployment

Environmental robustness is increasingly central to AV deployment. Waymo expanded operations to Phoenix's wet season, and to San Francisco's fog conditions, after extensive testing programs. Cruise's San Francisco operations included extensive fog testing. NIO, XPeng, and other Chinese

operators have tested under typhoon and monsoon conditions. Roads in parts of South Africa, Kenya, Nigeria, and other African countries present pothole densities and dust conditions that exceed those typically encountered in standard AV testing programs in the U.S., Europe, or China; this gap is the subject of growing academic interest including research at Wits University, University of Cape Town, University of Pretoria, and African Institute for Mathematical Sciences. Wildlife collisions are tracked by national transportation departments globally; in southern Africa specifically, kudu, eland, and elephant collisions present unique challenges relative to the deer-dominated wildlife profile of North American AV development.

29.7 Chapter Summary

Environmental extremes—heavy dust, flooding, potholes, wildlife—are specific operational challenges that standard perception handles poorly. Each requires engineering responses: sensor protection, modality redundancy, region-specific training data, and behavioral envelopes that adjust under adverse conditions. Regional variation is large; an AV program targeting global deployment must invest in region-specific training and validation. Safe graceful degradation is the unifying design pattern: perform confidently in nominal conditions, but slow down and widen margins when extremes are detected.

29.8 Exercises

1. For the flooded-road example, propose a third independent cue that could serve as a tiebreaker.
2. For the wildlife-zone example, compute the speed-reduction impact on a 200-km route with three such zones.
3. Propose a dust-robustness test for a perception system. What conditions must it pass?
4. Compare camera, LiDAR, and radar performance under heavy dust. Which is most robust?
5. Design a pothole-crowdsourcing protocol for a fleet. What data flows, what privacy considerations?
6. Propose three specific perception improvements tailored to African-rural deployment.
7. For the TDOA-flooded fusion, what false-positive rate is acceptable? Justify.
8. Compare production AV perception performance on European vs South African roads. What systematic gaps exist?
9. Design a wildlife-zone behavior envelope that balances safety with time-efficiency.
10. Identify three environmental conditions not addressed in this chapter that would affect AV deployment in a specific region.

Chapter 30

Climate Change and Autonomous Vehicles

30.1 Introduction

The relationship between autonomous vehicles and climate change cuts two ways. AVs can *reduce* transportation emissions through electrification, efficient routing, platooning, higher utilization of shared vehicles, and smoother driving styles that cut fuel burn. They can also *increase* emissions through induced demand (cheap rides generate more trips), empty deadheading (cruising between passengers), and the energy footprint of continuous sensor, compute, and cloud infrastructure. Which effect dominates depends on deployment choices: the grid that powers them, the patterns of use, and the policy environment.

Three observations motivate the chapter.

First, AVs are *not automatically green*. An electric robotaxi on a coal-fired grid, cruising empty between fares, emits more per passenger-mile than a well-utilized hybrid with a human driver. Engineering matters; deployment matters more.

Second, the *largest emission levers* are vehicle-level (electrification, efficient routing) not algorithm-level. The AV stack's direct emissions from compute and data are a fraction of the vehicle's operating emissions.

Third, *climate adaptation* matters in both directions: AVs must operate reliably in a climate-changed world with more extreme weather, and climate infrastructure (charging, grid stability) must keep pace with AV deployment.

30.2 Vehicle-Level Emissions

30.2.1 Life-Cycle Assessment

A vehicle's climate footprint spans manufacture (embodied emissions), operation (tailpipe or grid), and end-of-life (recycling, battery recovery).

Phase	ICE sedan (t CO ₂ e)	EV sedan (t CO ₂ e)
Manufacture	7	12
Operation (200k km)	40	12–28 ¹
End-of-life	1	1
Total	48	25–41

EV manufacture emits more (battery production is energy-intensive) but is compensated by lower operational emissions on most grids. The crossover point depends on grid carbon intensity.

Example 30.1 (EV vs ICE break-even mileage). EV manufacture excess: 5 tCO₂e. Operational emissions differential:

- ICE: 200 gCO₂e/km
- EV on EU average grid: 60 gCO₂e/km

Break-even mileage.

$$\begin{aligned}
 d_{\text{BE}} &= \frac{5000 \text{ kg}}{(200 - 60) \text{ g/km}} \\
 &= \frac{5000}{0.140 \text{ kg/km}} \\
 &\approx 35,700 \text{ km.}
 \end{aligned}$$

After 36,000 km, the EV has paid back its manufacturing emissions premium. Fleet AV robotaxis cover this in 3–4 months of operation.

30.3 AV Operational Effects

30.3.1 Routing Efficiency

AVs can route optimally and share traffic state. Fleet-scale optimization reduces congestion-related emissions by 5–20 % in modeled urban scenarios.

¹Grid-dependent: 12 on very clean grid, 28 on coal-heavy.

30.3.2 Eco-Driving

Smoother acceleration and regenerative braking reduce energy consumption by roughly 5–15 % compared to typical human driving.

30.3.3 Platooning

Vehicles traveling in close-spaced convoys reduce aerodynamic drag; 10–20 % fuel savings for trucks in production platoons. Relevant mostly to freight, less so to passenger AVs.

30.3.4 Deadhead Miles

Robotaxis must reposition between fares: to pickup, to charge, to depot. Deadhead fraction in production operations runs 20–40 %. Each deadhead mile emits without revenue and without serving a passenger.

Example 30.2 (Deadhead impact on per-passenger-mile emissions). Fleet EV robotaxi:

- Energy per km: 180 Wh/km
- Grid carbon: 350 gCO₂e/kWh (US average)
- Revenue km per day: 200
- Deadhead km per day: 80

Per-km emissions.

$$\begin{aligned} e_{\text{km}} &= 0.180 \text{ kWh/km} \times 350 \text{ g/kWh} \\ &= 63 \text{ gCO}_2\text{e/km.} \end{aligned}$$

Per-passenger-mile emissions (accounting for deadhead).

$$\begin{aligned} e_{\text{pax-km}} &= e_{\text{km}} \times \frac{\text{rev} + \text{dead}}{\text{rev}} \\ &= 63 \times \frac{280}{200} \\ &\approx 88 \text{ gCO}_2\text{e/pax-km.} \end{aligned}$$

For reference, a typical 5-seat car with single occupant on the same grid: ~200 gCO₂e/pax-km. The robotaxi is still favorable, but 40 % of its advantage is erased by deadhead. Minimizing deadhead is the single largest operational lever for fleet emissions.

30.4 Induced Demand

If AVs dramatically reduce the effective cost of travel, people travel more. Meta-analyses estimate induced-demand elasticities of 0.5–1.5 for mobility cost reductions—a halving of per-mile cost could increase total miles by 25–75 %.

30.4.1 Modal Shift

Cheap AV rides can pull passengers *from* transit, walking, or cycling—modes with far lower per-passenger emissions. This is a specific induced-demand concern: improvements in AV economics can be climate-negative if they cannibalize lower-carbon modes.

30.4.2 Policy Interventions

Interventions proposed to capture the efficiency gains without the demand rebound include: congestion pricing, deadhead taxation, minimum-occupancy rules for AV services in dense zones, and transit integration requirements in ride-hailing licenses.

30.5 Compute and Data Emissions

30.5.1 Per-Vehicle Compute

A typical AV compute stack consumes 50–100 W continuously, or roughly 0.4–0.8 kWh per hour of operation.

Example 30.3 (Compute contribution to vehicle energy use). Robotaxi operating 10 hours/day, consuming total 40 kWh/day (vehicle + compute):

- Compute: $0.6 \text{ kWh/h} \times 10 \text{ h} = 6 \text{ kWh/day}$
- Traction: 34 kWh/day

Compute share.

$$\text{share} = 6/40 = 15\%.$$

15 % is non-negligible but far smaller than traction. Compute efficiency still matters; it is not the dominant emission lever.

30.5.2 Cloud and Training Emissions

Training large AV models consumes substantial compute. Training runs on foundation-scale perception models can emit 100–500 tCO₂e per run, amortized across thousands of vehicles that use the resulting model.

30.6 Climate Adaptation for AVs

AVs must operate in a climate-changed world. Challenges:

Stressor	Consequence for AV
More frequent heat waves	Thermal management stressed; battery degradation accelerates
Extreme precipitation	Flooded roads become more common; perception degrades further
Unusual weather patterns	Training data distribution shifts; models degrade
Infrastructure damage	Roads, markings, signs require re-mapping more often
Wildfires and smoke	Reduced visibility; air intake filtering needed

Adaptation is primarily a robustness and re-training issue rather than a fundamental architectural problem: models trained on broader weather distributions degrade more gracefully.

30.7 Industrial Deployment

The climate impact of AVs interacts with broader transportation electrification and grid decarbonisation trends. The International Energy Agency publishes annual EV outlooks tracking global EV adoption. The U.S. Inflation Reduction Act of 2022, the EU Fit for 55 package, and China's New Energy Vehicle policies are the major industrial-policy drivers of EV/AV adoption. Carbon-intensity data for electricity grids is published by national grid operators including the U.S. EPA's eGRID and the European Environment Agency. Bloomberg New Energy Finance publishes sector-specific carbon-intensity analyses. The full life-cycle assessment of EVs versus ICE vehicles depends sensitively on manufacturing-grid carbon intensity (heavily influenced by where batteries are produced) and operational-grid carbon intensity (where the vehicle is charged), as widely documented in peer-reviewed LCA studies.

30.8 Chapter Summary

AVs' climate impact depends on deployment choices more than on the technology itself. Electrification is the largest lever; routing efficiency, eco-driving, and platooning add meaningful but smaller gains. Deadhead miles and induced demand risk erasing the advantages. Compute emissions are 10–15 % of vehicle-level emissions on typical robotaxis—non-negligible but not dominant. Adaptation to climate-changed operating conditions becomes an ongoing robustness discipline.

30.9 Exercises

1. For the break-even example, compute the mileage if EV excess is 3 t and the grid is 80 gCO₂e/km.
2. Estimate per-pax-km emissions for a 2-passenger average occupancy instead of single.
3. Design a deadhead-minimization policy for a 500-vehicle fleet. What constraints bind?

4. Compare induced-demand and mode-shift effects; propose a metric that captures both.
5. Compute the climate impact of a 300 tCO₂e model-training run amortized across a 50,000-vehicle fleet over 3 years.
6. Propose three climate-adaptation features a production AV stack should include.
7. Estimate the climate benefit of 15 % eco-driving savings applied to the current US light-duty fleet.
8. Compare fleet robotaxi emissions to personal-EV ownership under different utilization assumptions.
9. Discuss the circular-economy implications of AV battery second-life use in stationary storage.
10. Identify a policy intervention that specifically addresses the deadhead problem without distorting normal operations.

Part IX

Frontier Technologies and Society

Chapter 31

Quantum Technologies for Autonomous Vehicles

31.1 Introduction

Quantum technologies—quantum sensing, quantum computing, and quantum communication—are moving from laboratory demonstrations toward niche commercial applications. For autonomous vehicles, three threads are emerging: quantum inertial sensors that could provide GNSS-independent navigation with unprecedented accuracy, quantum-assisted optimization for fleet-level scheduling, and quantum-safe cryptography for V2X communications. None is yet a production reality, but research programs at major AV developers and defense agencies are actively investigating each.

Three observations motivate the chapter.

First, quantum sensing is the *nearest-term* thread. Quantum gyroscopes and accelerometers have demonstrated drift rates orders of magnitude below conventional IMUs; miniaturization is the remaining barrier to vehicle deployment.

Second, quantum computing for AV applications is *speculative but increasingly explored*. Specific problems—large-scale fleet scheduling, combinatorial route optimization—map naturally to quantum-algorithmic templates. Practical quantum advantage is not yet demonstrated but the theoretical fit is strong.

Third, quantum-safe cryptography is a *defensive imperative*. Current V2X protocols rely on public-key schemes that large-scale quantum computers could break. Post-quantum cryptography is a real near-term engineering concern even as quantum computers themselves remain years away.

31.2 Quantum Sensing for Navigation

31.2.1 Cold-Atom Interferometry

A cloud of laser-cooled atoms is split into two wavepackets that traverse different paths and are recombined. Interference fringes depend on the acceleration or rotation of the apparatus during the interferometry—yielding extraordinarily sensitive inertial measurements.

Example 31.1 (Drift comparison: classical vs quantum IMU). A navigation-grade classical fiber-optic gyroscope has a bias stability of roughly 0.01°h^{-1} .

Integrated heading error after 10 hours GNSS-denied:

$$\begin{aligned}\theta_{\text{err,classical}} &= 0.01 \times 10 \\ &= 0.10 \text{ deg.}\end{aligned}$$

Cold-atom gyroscope demonstration: bias stability $\sim 1 \times 10^{-5}^\circ \text{h}^{-1}$.

Integrated heading error:

$$\begin{aligned}\theta_{\text{err,quantum}} &= 10^{-5} \times 10 \\ &= 10^{-4} \text{ deg.}\end{aligned}$$

$\sim 1000\times$ improvement. At 100 ms^{-1} speed, 1000 seconds of motion projects heading error into lateral position error of $\sim 0.17 \text{ m}$ (classical) vs $1.7 \mu\text{m}$ (quantum)—a difference between degrading navigation and preserving it.

31.2.2 Practical Deployment Barriers

Quantum inertial sensors have been demonstrated in laboratory and ship-scale configurations. For AV deployment:

- **Size:** current systems are suitcase-scale; automotive packaging requires $10\text{--}100\times$ miniaturization.
- **Power:** laser cooling is power-intensive; compatible with EV budgets but nontrivial.
- **Cost:** laboratory systems cost $100\text{k--}1\text{M USD}$; automotive economics need $1000\times$ reduction.
- **Vibration and thermal environment:** the automotive environment is more hostile than laboratory.

Commercial deployment in specialized applications (military autonomous ships, high-value mining, arctic-logistics fleets) is plausible within 5–10 years; mass-market automotive is further.

31.2.3 Quantum Magnetometry and Map-Based Localization

Earth's magnetic field has spatial variation that can be mapped and used for localization without GNSS. Quantum magnetometers (atomic, NV-diamond) measure the local field at sensitivities orders

of magnitude beyond classical magnetometers, enabling sub-meter localization against a reference map.

31.3 Quantum Computing for AV Problems

31.3.1 Fleet Routing as QUBO

Large-scale fleet dispatch is a combinatorial problem: assign vehicles to passengers, route through traffic, charge opportunistically. It maps to a Quadratic Unconstrained Binary Optimization (QUBO) problem:

$$\min_{\mathbf{x} \in \{0,1\}^n} \mathbf{x}^\top \mathbf{Q} \mathbf{x}, \quad (31.1)$$

which is the canonical form that quantum annealers (D-Wave) and QAOA-style gate-based algorithms target.

Example 31.2 (Quantum-vs-classical crossover). Classical mixed-integer-programming solver runtime for a $n = 100$ -vehicle fleet assignment: ~ 30 s.

For $n = 1000$: runtime scales poorly. Estimated ~ 1 h with aggressive heuristics; impractical for real-time dispatch.

Quantum-annealer runtime (D-Wave Advantage): claimed scaling favors specific problem structures. In benchmark tests on sparse-structure problems, crossover with classical heuristics is observed at $n \sim 10^4$ – 10^5 .

Current state: Practical quantum advantage for production fleet routing is not yet demonstrated; classical heuristics remain dominant. Research is ongoing.

31.3.2 Quantum Machine Learning

Quantum machine learning (QML) has been proposed for training perception models or reinforcement-learning policies more efficiently. Current evidence for quantum advantage on classical-data ML tasks is limited; the field is active but not production-relevant.

31.4 Quantum-Safe Cryptography

31.4.1 The Threat

Large-scale quantum computers can break RSA, ECDSA, and ECDH (the public-key schemes underlying most V2X security) using Shor’s algorithm. Estimates for when sufficiently-large quantum computers will exist range from 10 to 30+ years; the threat is real but temporally distant.

31.4.2 Harvest-Now, Decrypt-Later

Adversaries can record encrypted traffic today and decrypt it once quantum computers are available. For AV traffic—which contains safety-critical content, user location data, and potentially regulated

information—this is a genuine concern.

31.4.3 Post-Quantum Cryptography

NIST has standardized post-quantum algorithms (CRYSTALS-Kyber for key exchange, CRYSTALS-Dilithium for signatures). These are lattice-based and believed secure against quantum attack. Migration of V2X standards (IEEE 1609, C-V2X) to post-quantum cryptography is underway.

Example 31.3 (Post-quantum signature size and latency).

Algorithm	Signature size (bytes)	Sign latency
ECDSA-P256 (classical)	64	~0.1 ms
Dilithium-2 (PQ)	2420	~0.3 ms
Dilithium-3 (PQ)	3293	~0.5 ms

Signatures are 30–50× larger; latency is 3–5× longer. For V2X BSMs (basic safety messages) at 10 Hz, this is manageable but tightens the bandwidth budget. Aggregate bandwidth scales proportionally with PQ adoption.

31.5 Hybrid and Near-Term Realizable

Before full quantum deployment arrives, hybrid classical-quantum systems provide bridges:

- Hybrid crypto: use both classical and PQ algorithms simultaneously; security holds if either remains unbroken.
- Quantum-inspired classical algorithms: adaptations of quantum annealing techniques that run on classical hardware and sometimes match quantum-advantage claims.
- Classical-quantum optimization pipelines: classical preprocessing + quantum subroutine for the hard core.

31.6 Industrial Deployment

Quantum sensing and quantum computing have specific potential applications in AV systems. AOSense, Bosch, Honeywell, and others have published research on cold-atom interferometry for drift-free inertial navigation; DARPA has funded multiple programs on quantum inertial sensors. Quantum computing applications to logistics and routing have been explored by D-Wave Systems (Canada), IBM, IonQ, and others, with Volkswagen and Daimler having published pilot studies on quantum-routing problems on existing quantum hardware. Quantum-safe cryptography migration is being pursued in automotive following NIST’s standardisation of CRYSTALS-Kyber and CRYSTALS-Dilithium in 2024 as the post-quantum cryptographic standards. Practical quantum advantage for commercial automotive problems remains the subject of ongoing research rather than current production deployment.

31.7 Chapter Summary

Quantum technologies offer three threads relevant to AVs: quantum sensing (nearest-term, offering GNSS-independent navigation and magnetic-map localization); quantum computing (speculative, potentially useful for fleet optimization); and quantum-safe cryptography (already required for harvest-now-decrypt-later-safe V2X). None is production-ready today, but each is tracked by major AV programs and the trajectory is clear enough that engineering planning must account for them. The defensive case for post-quantum cryptography is the most immediate; offensive quantum advantages (quantum sensing, quantum optimization) remain on the horizon.

31.8 Exercises

1. For the drift-comparison example, compute the heading error after 100 hours GNSS-denied for both systems.
2. Discuss three specific AV scenarios where quantum inertial navigation would provide decisive value.
3. Estimate the bandwidth overhead for a 100-vehicle fleet upgrading from ECDSA to Dilithium signatures at 10 Hz BSMs.
4. Compare QUBO and mixed-integer programming for fleet dispatch. What problem structures favor each?
5. For the range example with quantum magnetometry, estimate localization accuracy using known magnetic-field spatial variation.
6. Design a hybrid-crypto V2X protocol that is secure against both classical and quantum attack.
7. Identify three quantum-inspired classical algorithms relevant to AV optimization.
8. Propose a procurement plan for quantum-ready AV systems: what technologies today, what next 5 years, what next 15?
9. Compute the signature overhead ratio for different PQ-security levels (Dilithium 2, 3, 5).
10. Discuss ethical concerns of harvest-now-decrypt-later attacks on AV traffic data. What mitigations exist?

Chapter 32

Geopolitics of AV Technology: The East-West Competition

32.1 Introduction

Autonomous-vehicle technology has become a flashpoint in the broader technology competition between the United States, its allies, and China. Export controls on AI chips, restrictions on cross-border vehicle data flows, reciprocal market-access limitations, and domestic-industrial-policy programs on both sides have turned AV development from a globally-integrated industry into an increasingly-bifurcated one. Engineers and program managers building AV systems now navigate a regulatory landscape shaped as much by geopolitics as by safety or commercial considerations.

Three observations motivate the chapter.

First, both sides view AV as strategic. Beijing has explicitly designated intelligent connected vehicles (ICV) a strategic industry; Washington has applied export controls to advanced chips citing national security. AV is not merely commerce; it is industrial-policy terrain.

Second, regulatory divergence is accelerating. Data-localization requirements, map-data restrictions, safety- certification regimes, and V2X spectrum policies differ between the two blocs. Vehicles designed for one market increasingly require significant re-engineering for the other.

Third, third countries and the global South will increasingly face choices about which technology stack to adopt, with long-term consequences for interoperability and sovereignty.

32.2 US and Allied Policies

32.2.1 Chip Export Controls

Beginning October 2022 and tightened repeatedly since, the US has restricted export to China of advanced AI accelerators (NVIDIA A100/H100 equivalents), advanced semiconductor manufacturing

equipment, and related know-how. These controls specifically target AI training compute relevant to AV model development.

Example 32.1 (Compute-capacity implications). A large AV-training run for a frontier model: roughly 5000 A100-equivalent GPU-months of compute.

Pre-restriction market access.

Chinese AV developers could source A100s at market prices; training infrastructure was effectively unconstrained.

Post-restriction.

Access to A100/H100 is restricted; domestic alternatives (Huawei Ascend 910B, others) have emerged but at lower per-chip performance.

Cost differential (approximate).

US access : baseline cost.

China access : $1.5\text{--}3 \times$ baseline,
higher chip count, lower utilization.

Effective compute cost for frontier AV training is meaningfully higher for Chinese programs, though the gap has closed as domestic chips have matured.

32.2.2 Data and Connected-Vehicle Rules

The US Commerce Department's January 2025 final rule on connected vehicles restricts integration of Chinese-origin vehicle connectivity systems (VCS) and automated-driving systems (ADS) in US-market vehicles from MY 2027 (VCS) and MY 2030 (ADS). The rule explicitly cites national-security concerns about vehicle data flows and remote-access potential.

32.2.3 Subsidies and Industrial Policy

The Inflation Reduction Act (IRA) and CHIPS Act direct substantial subsidies toward domestic vehicle and chip manufacturing. Domestic-content and domestic-processing requirements create effective barriers to foreign participation in key technology areas.

32.3 Chinese Policies

32.3.1 Intelligent Connected Vehicle Strategy

China's "Made in China 2025" and subsequent plans designated intelligent connected vehicles a strategic sector. Combined market access restrictions (foreign map data must be processed through licensed Chinese partners) and industrial-scale subsidies have produced rapid domestic AV-industry

growth.

32.3.2 Data Localization

Chinese data regulations (Cybersecurity Law, Data Security Law, Personal Information Protection Law) require vehicle- collected data to be stored and processed within China. High-resolution map data is especially sensitive and is subject to licensing regimes that exclude most foreign companies from direct participation.

32.3.3 V2X Spectrum

China has allocated V2X spectrum distinct from US allocations (C-V2X at 5.905–5.925 GHz in China), and has mandated C-V2X-first deployment. US policy has taken a longer and more contested path to C-V2X adoption. Vehicles designed for one market require re-engineering for the other at the V2X layer.

32.3.4 Market-Access Reciprocity

Chinese regulators have restricted Tesla's access to certain Chinese-market data and have imposed operational constraints citing national security at sensitive locations. These actions mirror, on a smaller scale, US restrictions on Chinese companies.

32.4 Technology Stacks Diverging

Layer	US-allied stack	Chinese stack
Compute chips	NVIDIA / Qualcomm / Mobileye / emerging domestic	Huawei / Horizon Robotics / emerging
Perception models	Proprietary, some open research	Baidu Apollo / domestic frameworks
Maps	TomTom / HERE / proprietary	AutoNavi / Baidu / Nav-Info
V2X	Extended debate over DSRC vs C-V2X	C-V2X mandated and deployed
Data storage	Flexible, subject to state privacy laws	Mandated domestic storage

Vehicles are increasingly stack-specific. A Waymo in China is conceptually possible but regulatorily implausible; a Baidu Apollo in the US faces growing restrictions.

32.5 Implications for Engineering

32.5.1 Design for Export-Control Compliance

AV architectures increasingly separate export-controlled from exportable components. Training compute (heavily controlled) runs on compliant infrastructure; inference compute and specific software stacks are engineered for the target market.

32.5.2 Dual-Stack Development

Large AV programs maintain duplicate technology paths for US and Chinese markets. Engineering effort is duplicated; organizational complexity increases; cross-pollination constrained.

32.5.3 Supply-Chain Resilience

Dependencies on single-source suppliers now carry geopolitical risk. Programs that used TSMC-produced Nvidia chips, for example, have been constrained by both export controls (outbound) and concerns about Taiwan-specific supply risk.

Example 32.2 (Supply-chain risk quantification). AV program sourcing:

- Primary AI compute: Nvidia chips from TSMC.
- Fallback: Qualcomm Snapdragon Ride.
- LiDAR: Chinese supplier (cost-advantaged).
- Radar: European supplier.

Scenario: export controls tighten restricting Nvidia to domestic only, and LiDAR sourcing restricted to US-allied suppliers.

chip re-qualification : 12–18 months engineering,

LiDAR sourcing : 24 months qualification at higher cost,

program delay : 18–24 months in worst case.

Mitigation requires early identification of single-source dependencies and parallel qualification of alternatives—at substantial ongoing cost.

32.6 Third Countries

Major third markets—India, Southeast Asia, the EU (partly), the Middle East, Africa, Latin America—face choices about which technology stack to adopt. Outcomes will vary:

- EU: tends toward US-allied but with distinct regulatory approach and its own industrial programs.
- Japan, South Korea: US-allied on core stack; own substantial domestic industry.

- India: policy preference for domestic production; technology sources mixed.
- Middle East, Africa: mixed adoption; Chinese firms have been more active in some markets.

The pattern is familiar from other technology-competition cases: bifurcation of the world into two overlapping technology spheres.

32.7 Industrial Deployment

Geopolitical dimensions of AV technology include the U.S. CHIPS and Science Act of 2022, EU export controls and the Inflation Reduction Act provisions affecting EV/AV supply chains, and China's Made in China 2025 initiative that explicitly prioritises intelligent and connected vehicles. The U.S. Commerce Department's January 2025 rule restricts the import and sale of connected vehicles incorporating Chinese or Russian software and hardware, with phased enforcement beginning in 2027 for software and 2030 for hardware. NIO, BYD, XPeng, and other Chinese AV/EV companies face restricted access to U.S. and increasingly EU markets due to tariffs and non-tariff barriers. Semiconductor supply chains for AV-grade chips, with TSMC (Taiwan), Samsung (Korea), Intel (United States), and SMIC (China) as critical players, are a focus of strategic-autonomy policies on both sides of the Pacific.

32.8 Chapter Summary

AV technology has become geopolitically strategic, and the US-China technology competition has shaped both the business landscape and the technical architecture of modern AV programs. Export controls on AI chips, data-localization requirements, market-access restrictions, and divergent spectrum and standards policies have produced two increasingly separate technology stacks. Engineering programs must accommodate the resulting complexity: compliance architecture, dual-stack development, and supply-chain resilience. Third markets face increasingly explicit choices between the two spheres. The trajectory toward further bifurcation is more likely than the previous era of globally integrated AV development.

32.9 Exercises

1. For the compute-capacity example, compute the relative cost of training a model at a $2\times$ chip-cost disadvantage.
2. Identify three specific AV technology areas where the US and Chinese stacks are most divergent, and three where they remain substantially similar.
3. Design an AV system architecture that supports deployment in both US and Chinese markets with minimum duplicate engineering.
4. Discuss the V2X spectrum divergence: what are the implications for cross-market operation?

5. Propose a supply-chain-resilience plan for an AV program with heavy dependence on Taiwan-produced chips.
6. Compare the impacts of the US 2025 connected-vehicle rule on global AV deployment patterns.
7. For a mid-sized third-country market (e.g., Brazil, Indonesia, Vietnam), analyze what factors favor one technology stack over the other.
8. Discuss the dual-use implications of AV technology. Are there specific AV capabilities that should be treated as military-sensitive?
9. Identify two specific technology transfer restrictions that would most affect a globally-integrated AV program.
10. Propose an industry-level standard for cross-market AV interoperability despite regulatory divergence. What would it need to address?

Chapter 33

Socio-Technical Perception and Human Factors

33.1 Introduction

An autonomous vehicle that is statistically safe but that passengers fear, that pedestrians mistrust, or that other drivers cannot predict is not a successful deployment. The social acceptance of AVs depends on more than their engineering correctness; it depends on how humans perceive, interpret, and interact with them. This chapter addresses the *socio-technical* layer: measuring public trust in AV systems, designing perception-transparency interfaces that show users what the car sees, and detecting human-override intent as the vehicle and driver share control.

Three observations motivate the structure.

First, trust is an adoption factor. Technological correctness does not guarantee adoption. A safer AV that people refuse to ride produces fewer real-world benefits than a slightly-less-safe AV that is widely used.

Second, transparency reduces fear. Passengers are more comfortable in AVs when they can see what the vehicle perceives—a visualization of detected objects, lane boundaries, and planned trajectory. Transparency is a passenger-experience feature that correlates with trust.

Third, mixed-autonomy systems are unavoidable for a long period. Level 2/3 systems with human handoff, remote operators assisting Level 4, and humans driving alongside AVs will dominate for decades. Detecting human override intent is essential to these configurations.

33.2 Public Trust Modeling

Trust in AVs varies by region, demographic, and individual experience. Understanding and influencing it is a distinct discipline from engineering the vehicle itself.

33.2.1 What To Measure

Dimension	Example indicator
Willingness to ride	Fraction that agree to first AV trip
Willingness to pay	Fare premium or discount relative to human-driven
Repeat usage	Fraction of first-time riders who use again within 30 days
Perceived safety	Survey-based scores on likert scales
Behavioral indicators	Pulse rate, posture, gaze patterns during ride

33.2.2 Regional and Demographic Variation

Trust varies substantially:

- Higher in markets with prior robotaxi exposure.
- Lower in markets with high-profile AV incidents.
- Varies with age (younger more accepting), education, and prior experience with automation.

Example 33.1 (Trust as a function of exposure). Longitudinal survey in a city rolling out robotaxi service. Willingness-to-ride measured at 6-month intervals.

Month	Willingness	Actual fleet incidents/million mi
0	24 %	N/A (pre-launch)
6	31 %	0.12
12	38 %	0.08
18	27 %	0.15 (one visible incident)
24	33 %	0.09

Observations.

- Baseline trust grows steadily with no-incident months.
- Single visible incident produces ~11-point drop, taking ~6 months to recover.
- Recovery trajectory depends on transparent communication about the incident and remediation.

Trust is asymmetric: it builds slowly and collapses quickly. Operating practice must be disproportionately cautious about incidents compared to their raw statistical impact.

33.3 Perception Transparency Interfaces

Passengers (and regulators) are more comfortable when they can see what the vehicle sees. Perception transparency interfaces display detected objects, lane boundaries, planned trajectory, and confidence indicators in the vehicle cabin or on an app.

33.3.1 Design Principles

Principle	Application
Show, don't tell	Graphical scene view rather than text alerts
Calibrated confidence	Visual encoding of detection certainty
Avoid over-disclosure	Don't display every noisy micro-detection
Consistent timing	Display and reality synchronized (lag kills trust)
Fallback clarity	Clearly mark when fallback / degraded mode active

33.3.2 Information Display Levels

- Minimal: driving status icon only (engaged, disengaged, manual).
- Moderate: bird's-eye view of ego + nearby vehicles.
- Rich: full scene with lanes, pedestrians, traffic lights, planned trajectory.
- Expert: confidence values, alternative trajectory hypotheses.

Tesla's Autopilot visualization and Waymo's passenger app are production examples at different points on this spectrum.

Example 33.2 (Transparency effect on passenger comfort). Study of passenger comfort during L4 robotaxi rides across two interface conditions.

Condition A: Minimal display.

- Displays engagement status only.
- Mean comfort score: 3.7/5.
- 22 % of rides show pulse elevation during events.

Condition B: Rich scene visualization.

- Real-time BEV with detected objects.
- Mean comfort score: 4.2/5.
- 11 % of rides show pulse elevation during events.

Comfort improvement.

$$\Delta\text{comfort} = 4.2 - 3.7 = 0.5/5,$$

$$\Delta\text{stress} = -50\% \text{ of rides with elevation.}$$

Transparency nearly halves the fraction of rides that produce detectable stress responses. This effect size is large; it is why all major robotaxi programs now include rich visualization.

33.3.3 Caveats

Transparency can also *undermine* trust if poorly executed:

- Displayed detections that visibly flicker or disappear reduce confidence.
- A trajectory that visibly deviates from the passenger’s expectation without explanation produces unease.
- Excessive detail overwhelms; minimal may be preferable for general passengers.

33.4 Human Override Awareness

In Level 2/3 systems, the human is in the loop. In Level 4 systems, remote operators can intervene. Detecting when a human is about to take over—or wants to—matters for smooth handoff.

33.4.1 Override-Intent Signals

Signal	What it indicates
Steering-wheel grip pressure	Driver preparing to intervene
Pedal hovering	Driver ready to brake
Eye gaze direction	Attention on road or away
Verbal alert in cabin	“watch that car”
Explicit request (button/voice)	Intent to take control or defer

33.4.2 Prediction-Based Handoff

The AV can predict impending override before it occurs: “the driver is gripping the wheel and gazing at an object we are not planning around; they will likely intervene soon.” Anticipatory handoff is smoother than reactive handoff.

Example 33.3 (Handoff latency with and without prediction). Scenario: Level 3 AV on highway; driver decides to take over for a complex merge.

Reactive handoff.

- Driver grabs wheel + turns: 400 ms.
- System detects override: 150 ms.
- System releases control: 50 ms.
- Total handoff latency: 600 ms.

Predictive handoff.

- System detects grip onset: 100 ms.
- System pre-flags override intent.
- Control released as driver actually turns wheel: 50 ms.
- Total handoff latency: 150 ms.

Benefit.

$$\Delta t_{\text{handoff}} = 600 - 150 = 450 \text{ ms.}$$

450 ms saved at highway speeds = 12 m saved before driver has full control. Difference between smooth handoff and a jerky one the driver blames on the automation.

33.4.3 Mode-Confusion Risks

A notorious risk class in mixed autonomy is *mode confusion*: the driver thinks the system is engaged when it is not, or vice versa. Several high-profile incidents trace to this. Mitigations:

- Unambiguous engagement cues (visual, audible, haptic).
- Regular re-engagement requirements.
- Explicit confirmation at mode transitions.
- Driver-monitoring systems that verify attention.

33.5 Predictable Behavior for External Observers

Beyond passengers, other road users—pedestrians, cyclists, other drivers—must predict the AV’s behavior. AVs that behave unpredictably are unsafe even when individually correct.

33.5.1 External Signaling

- Gentle approach patterns for crosswalks that signal intent to yield.
- External displays (emerging technology) showing intentions.
- Consistent maneuver timing that matches human expectations.

33.5.2 Conforming to Local Driving Culture

Driving cultures differ across regions (Chapter 27). An AV deployed globally must adapt to local conventions—aggressive where everyone is aggressive, cautious where everyone is cautious. A deviant AV is an unsafe AV.

33.6 Industrial Deployment

Public-trust dynamics around AVs have been the subject of extensive academic and industry research. The Pew Research Center, J.D. Power, AAA, and the MIT AgeLab publish recurring AV public-perception studies. Cruise's October 2023 incident produced a significant subsequent decline in public-trust measures for AV technology more broadly, illustrating the asymmetric dynamics of trust building and collapse. Perception-transparency interfaces have become standard features in production robotaxi services from Waymo, Cruise, and Apollo Go, and in ADAS interfaces from Tesla, Mercedes-Benz, and others. Driver-monitoring systems for Level 2/3 systems are increasingly required: Euro NCAP began requiring Driver Monitoring for top safety ratings from 2023, and U.S. NHTSA has explored mandates. Suppliers including Smart Eye, Seeing Machines, and Tobii ship production driver-monitoring camera systems.

33.7 Chapter Summary

The socio-technical layer of AV engineering addresses how humans perceive, trust, and interact with autonomous systems. Public trust varies by region and exposure; it builds slowly and collapses quickly. Perception transparency interfaces reduce passenger stress substantially and have become standard in production robotaxi deployments. Human override detection supports smooth handoff in Level 2/3 systems and remote-assistance in Level 4; anticipatory handoff outperforms reactive handoff. External predictability makes AVs safe for the humans around them. These dimensions are engineering concerns as much as PR concerns; they determine whether a technologically-correct AV is a deployable one.

33.8 Exercises

1. For the trust-exposure example, estimate recovery time if two incidents occur within 3 months.
2. Design a perception-transparency interface for a specific target user (elderly, child, expert).
3. Compare reactive and predictive handoff latency for a city-street scenario at 12 m s^{-1} .
4. For the transparency example, propose a design that achieves the comfort benefit without over-disclosure.
5. Design a driver-monitoring system that detects drowsiness within 1 minute of onset.
6. Identify three culture-specific driving conventions that an AV must adapt to when deploying in a new region.
7. Propose a metric for external-observer predictability that can be measured in deployment.
8. Discuss the trade-off between disclosing perception confidence and overwhelming the user.
9. Design a mode-confusion-prevention protocol for a Level 2 system.

10. Compare trust-building strategies for robotaxi services and private-ownership AVs; identify differences in approach.

Appendix A

Notation Reference

This appendix collects the conventions used throughout the book.

General

Symbol	Meaning
$\mathbb{R}, \mathbb{N}, \mathbb{Z}$	reals, naturals, integers
$a, b, c, \dots$	scalars (italic lowercase)
$\mathbf{x}, \mathbf{y}, \mathbf{u}$	column vectors (bold lowercase)
$\mathbf{A}, \mathbf{B}, \mathbf{K}$	matrices (bold uppercase)
$\mathbf{A}^\top, \mathbf{A}^{-1}$	transpose, inverse
$\det \mathbf{A}, \operatorname{tr} \mathbf{A}$	determinant, trace
$\mathbf{I}_n, \mathbf{0}_n$	identity matrix, zero vector of size n
$\ \mathbf{x}\ _p$	ℓ_p norm of $\mathbf{x}$
$\mathbf{x}^\top \mathbf{A} \mathbf{x}$	quadratic form
$\odot$	Hadamard (elementwise) product

Probability

Symbol	Meaning
$\mathbb{P}(A)$	probability of event A
$p(x)$	probability density function of X at x
$\mathbb{E}[\cdot], \operatorname{Var}[\cdot], \operatorname{Cov}[\cdot, \cdot]$	expectation, variance, covariance
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution
$H(p)$	differential entropy
$D_{\text{KL}}(p \ q)$	Kullback–Leibler divergence
$I(X; Y)$	mutual information

Geometry

Symbol	Meaning
$\text{SO}(n)$	special orthogonal group of dimension n
$\text{SE}(n)$	special Euclidean group of dimension n
$\mathfrak{so}(n), \mathfrak{se}(n)$	corresponding Lie algebras
$\mathbf{T}_{AB}$	pose of frame B expressed in frame A
$[\boldsymbol{\omega}]_{\times}$	skew-symmetric matrix for cross product
$\exp, \log$ (capitalized)	Lie group exp/log maps
$\text{Ad}_{\mathbf{T}}$	adjoint representation
$\pi(\mathbf{K}, \mathbf{T}, \mathbf{p})$	camera projection

Control and Estimation

Symbol	Meaning
$\mathbf{x}_t, \mathbf{u}_t, \mathbf{y}_t$	state, control, observation at time t
$f(\cdot), h(\cdot)$	dynamics, measurement functions
$\mathbf{F}, \mathbf{H}$	their Jacobians (linearized systems)
$\mathbf{Q}, \mathbf{R}$	process, measurement noise covariances
$\mathbf{P}, \boldsymbol{\Sigma}$	state covariance
$\mathbf{K}$	Kalman gain
$\pi(a s)$	policy probability
$V^{\pi}(s), Q^{\pi}(s, a)$	state-value, action-value functions
$\gamma \in [0, 1)$	discount factor
ρ	system response time (RSS), also occupancy

Machine Learning

Symbol	Meaning
$\boldsymbol{\theta}, \phi$	learnable parameters
$f_{\boldsymbol{\theta}}(\cdot)$	parameterized model
$\mathcal{D}$	dataset
$\ell(\cdot, \cdot)$	per-sample loss
$\mathcal{L}(\boldsymbol{\theta})$	training objective
$\nabla_{\boldsymbol{\theta}} \mathcal{L}$	gradient of loss w.r.t. parameters
η	learning rate
σ	activation function (usually ReLU or GELU)

Acronyms

Acronym	Expansion
ADS	Automated Driving System
ASIL	Automotive Safety Integrity Level
BEV	Bird's-Eye-View
CBF	Control Barrier Function
DDT	Dynamic Driving Task
EKF, UKF	Extended/Unscented Kalman Filter
FMCW	Frequency-Modulated Continuous-Wave (LiDAR, radar)
GNSS	Global Navigation Satellite System
HD map	High-Definition map
ICP	Iterative Closest Point
IMU	Inertial Measurement Unit
IRL	Inverse Reinforcement Learning
LiDAR	Light Detection and Ranging
LQR	Linear Quadratic Regulator
MDP, POMDP	(Partially Observable) Markov Decision Process
MOT	Multi-Object Tracking
MPC	Model Predictive Control
ODD	Operational Design Domain
RSS	Responsibility-Sensitive Safety
SAE	Society of Automotive Engineers
SLAM	Simultaneous Localization and Mapping
SOTIF	Safety Of The Intended Functionality (ISO 21448)
VIO	Visual-Inertial Odometry
V2X	Vehicle-to-Everything communication

Appendix B

Datasets and Benchmarks

This appendix summarizes the public datasets and benchmarks referenced throughout the book.

Perception and Detection

Table B.1: Major perception benchmarks.

Dataset	Modality	Tasks	Scale
KITTI [6]	stereo, LiDAR, GPS/IMU	2D/3D detection, tracking, odometry, stereo, scene flow	7k training frames
nuScenes [3]	6 cam, 5 radar, LiDAR, IMU	3D detection, tracking, prediction, BEV map, lidarseg	1k scenes, 20 s each
Waymo Open [11]	5 cam, 5 LiDAR	3D detection, tracking, prediction, segmentation	1150 scenes, 20 s each
Argoverse 2 [12]	ring cam, stereo, LiDAR	forecasting, 3D detection, map change	1000 scenes
Cityscapes [4]	stereo camera	semantic/instance/panoptic segmentation	5k fine, 20k coarse
BDD100K [13]	camera, GPS	multi-task (det, seg, drivable, lanes, weather)	100k videos
Mapillary Vistas	crowd-sourced cam	urban semantic segmentation	25k images
SemanticKITTI	LiDAR	LiDAR semantic/panoptic segmentation	22 sequences
ApolloScape	LiDAR, cam	lane, trajectory, stereo, 3D detection	multiple tracks
ONCE	LiDAR, 7 cam	detection, self-supervised pre-training	1M frames (unlabeled)

Motion Forecasting and Planning

Table B.2: Prediction and planning benchmarks.

Dataset	Scope	Horizon
Argoverse Motion Forecasting	vehicle trajectories with map	3 s
Waymo Open Motion	multi-agent, map, many classes	8 s future
nuScenes prediction	interactions in urban scenes	6 s
INTERACTION [14]	multi-country interactive sets	variable
nuPlan	closed-loop planning benchmark	simulation
Waymax	closed-loop simulation + datasets	simulation

Major Datasets in Detail

The summary tables compress what are individually substantial research artefacts. This section expands the three datasets that together account for the majority of citations in production AV perception research.

KITTI

KITTI [6] is the founding AV-perception benchmark. Released in 2012 as a joint effort between the Karlsruhe Institute of Technology and the Toyota Technological Institute at Chicago, by Andreas Geiger, Philip Lenz, Christoph Stiller, and Raquel Urtasun, KITTI defined the public-benchmark template that every subsequent AV dataset has followed: a calibrated multi-sensor data-collection vehicle, hand-annotated ground truth, public train/test splits, and a hosted online leaderboard for held-out test scoring.

The KITTI sensor suite consisted of two pairs of cameras (one colour and one greyscale, all 1392×512 pixels at 10 Hz), a Velodyne HDL-64E LiDAR (64-channel mechanical-spinning, 360° , 10 Hz), and a high-precision OXTS RT 3003 GPS/IMU. The sensor rig was mounted atop a Volkswagen Passat station wagon driven through Karlsruhe and its surroundings during 2011. The total recorded driving was modest by contemporary standards—several hours of data—but the released benchmark splits remain the reference for reproducibility comparisons.

KITTI’s task taxonomy was unusually broad for its time and remains influential. The benchmarks include stereo matching, optical flow, visual odometry, monocular and stereo 3D object detection, 2D/3D multi-object tracking, road and lane detection, and (added in SemanticKITTI [1]) point-wise semantic and panoptic segmentation of LiDAR. The KITTI-360 follow-up dataset extended the original with denser annotations, 360° camera coverage, and full semantic segmentation.

KITTI’s principal limitation today is scale: the labelled training set is roughly 7 500 frames, two to three orders of magnitude smaller than modern equivalents. Its weather, lighting, and geographic diversity are also constrained: Karlsruhe in summer daylight. As a consequence, KITTI numbers are no longer competitive evidence of state-of-the-art performance. Its enduring value is pedagogical and

reproducibility- oriented: implementations and ablations are routinely reported on KITTI to allow direct comparison against the hundreds of papers that preceded.

Waymo Open Dataset

The Waymo Open Dataset [11], released by Waymo LLC (an Alphabet subsidiary) starting in 2019 and expanded in successive releases through 2024, is the dominant large-scale benchmark for current AV perception and prediction research. The original release contained 1 150 scenes of 20 seconds each, totaling roughly 6.4 hours of driving. The dataset uses Waymo’s own production-class sensor suite: five LiDARs (one mid-range top-mounted plus four short-range units mounted around the vehicle perimeter) and five cameras (four side-facing plus one forward-facing), all time-synchronised. The recorded driving spans multiple cities including Phoenix, San Francisco, and Mountain View, and includes both daytime and night-time conditions and a range of urban contexts.

Annotations include 3D bounding boxes for vehicles, pedestrians, cyclists, and signs, with track-level identity for tracking evaluation. The labelling rate is high: hundreds of millions of 3D box labels across the dataset.

Beyond the perception dataset, Waymo released the **Waymo Open Motion Dataset (WOMD)** for trajectory prediction. WOMD contains over 100 000 segments of 20 seconds each with associated map data, totaling hundreds of thousands of unique agent trajectories. The released challenges have included motion prediction (predicting up to 8 seconds of future trajectory for surrounding agents), interaction prediction, sim-agents (controlling background agents in a closed-loop simulation), and most recently end-to-end driving from sensor input. The WOMD challenges held annually at the CVPR Workshop on Autonomous Driving have produced much of the published state of the art in recent years. The Waymax [8] closed-loop simulator complements WOMD with reactive simulation environments.

The dataset is distributed under a non-commercial research license; researchers must agree to terms before download. The combination of production-grade sensor calibration, large scale, geographic and temporal diversity, and continued investment from Waymo has made it the de facto reference dataset for current AV-perception publication.

nuScenes

nuScenes [3] was released in 2019 by nuTonomy (now Motional, the joint venture between Aptiv and Hyundai Motor Group) and was the first large-scale public dataset to combine multi-camera, multi-radar, and LiDAR sensing, making it the principal benchmark for multi-modal-fusion 3D detection research.

The dataset contains 1 000 scenes of 20 seconds each, totaling roughly 5.5 hours of driving across two cities: Boston (Seaport area) and Singapore (One-North area). The choice of these two cities gave the dataset deliberate left-hand-and-right-hand traffic exposure and distinctly different urban morphologies. The sensor suite consisted of six cameras providing full 360° coverage at 12 Hz, five 77 GHz radars (one front, two side-front, two side-rear), and one roof-mounted Velodyne HDL-32E

LiDAR at 20 Hz. A high-precision GNSS+IMU provided ground-truth pose. The dataset’s roughly 1.4 million annotated 3D bounding boxes cover 23 classes (vehicles, pedestrians, cyclists, animals, etc.) with per-frame visibility and pose attributes, plus an additional lidarseg release adding point-wise semantic-segmentation labels.

nuScenes has been particularly important for radar research, which is under-represented in KITTI and the early Waymo releases. The dataset launched the nuScenes Detection Score (NDS), a composite metric that combines mean Average Precision with translation, scale, orientation, velocity, and attribute errors; NDS has become a standard summary metric for 3D detection. The follow-up **nuPlan** dataset [2] provides a planning benchmark with 1 300 hours of expert driving, designed for closed-loop planning evaluation rather than open-loop trajectory regression.

Comparison and Selection Guidance

Dataset	Strength	Use it for
KITTI	reproducibility against historical literature	teaching, ablations, baseline replication
Waymo Open	scale, sensor diversity, current SOTA leaderboards	competitive benchmarking, motion prediction, end-to-end
nuScenes	multi-modal radar+camera+LiDAR fusion	3D detection with radar, cross-region generalisation

A research paper today routinely reports on at least two of the three to demonstrate generalisation across sensor configurations and geographies. Production stacks supplement public datasets with much larger internal fleet collections that public researchers do not have access to.

Simulators

- **CARLA**: Unreal-engine based urban simulator; widely used for end-to-end and perception research.
- **LGSVL / SVL Simulator**: Unity-based; strong vehicle dynamics.
- **SUMO**: microscopic traffic simulation at city scale.
- **highway-env**: light-weight Python environments for RL research.
- **MetaDrive**: procedurally-generated driving scenarios for RL.
- **CarMaker, dSPACE**: commercial simulators with certification-grade vehicle dynamics.

Open-Source Software

- **ROS / ROS 2**: messaging and middleware.
- **Autoware, Apollo**: open-source full-stack ADS platforms.
- **GTSAM, g2o, Ceres**: factor-graph and nonlinear least-squares solvers.
- **Open3D, PCL**: point-cloud processing.
- **MMDetection3D, OpenPCDet, Det3D**: 3D detection codebases.
- **trajectron++, scenarionet**: forecasting libraries.

Evaluation and Ethics Resources

- NHTSA Automated Driving Systems: Voluntary Safety Self-Assessment disclosures.
- UNECE WP.29 regulatory documents.
- California DMV autonomous vehicle disengagement and collision reports.
- Incident and crash databases (e.g., CA DMV, Waymo / Cruise transparency releases).

Accessing and citing these resources correctly—and honestly reporting stratified metrics over them—is itself a part of responsible ADS research practice.

Appendix C

Linear Algebra and Numerical Methods Refresher

This appendix collects the linear algebra identities and numerical methods used in the main chapters.

C.1 Vector Spaces and Norms

For $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{x}\|_1 = \sum_i |x_i|, \quad \|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^\top \mathbf{x}}, \quad \|\mathbf{x}\|_\infty = \max_i |x_i|. \quad (\text{C.1})$$

The induced matrix norm is $\|\mathbf{A}\|_2 = \sigma_{\max}(\mathbf{A})$, the largest singular value.

C.2 Eigen- and Singular-Value Decompositions

EVD. For diagonalizable $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$ with $\mathbf{\Lambda}$ diagonal. Symmetric $\mathbf{A}$ admits orthogonal $\mathbf{Q}$ and real $\mathbf{\Lambda}$.

SVD. Any $\mathbf{A} \in \mathbb{R}^{m \times n}$ admits $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with $\mathbf{U} \in \mathbb{R}^{m \times m}$, $\mathbf{V} \in \mathbb{R}^{n \times n}$ orthogonal and $\mathbf{\Sigma}$ non-negative diagonal. Key consequences:

- $\text{rank}(\mathbf{A}) = \#\{\sigma_i > 0\}$.
- $\mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$ is the Moore–Penrose pseudoinverse.
- $\min_{\text{rank}(\mathbf{B}) \leq k} \|\mathbf{A} - \mathbf{B}\|_F^2$ is achieved by truncating to the top- k singular components.

C.3 Matrix Identities

Woodbury (“matrix inversion lemma”).

$$(\mathbf{A} + \mathbf{UCV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}. \quad (\text{C.2})$$

Essential in Kalman filter reformulations.

Schur complement. For $\mathbf{M} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$, the Schur complement of $\mathbf{A}$ (assumed invertible) is $\mathbf{S} = \mathbf{D} - \mathbf{C}\mathbf{A}^{-1}\mathbf{B}$. Then $\det \mathbf{M} = \det \mathbf{A} \cdot \det \mathbf{S}$, and $\mathbf{M} \succ 0 \iff \mathbf{A} \succ 0, \mathbf{S} \succ 0$.

Matrix calculus (a few).

$$\frac{\partial}{\partial \mathbf{A}} \text{tr}(\mathbf{A}\mathbf{B}) = \mathbf{B}^\top, \quad \frac{\partial}{\partial \mathbf{A}} \log \det \mathbf{A} = \mathbf{A}^{-\top}, \quad \frac{\partial}{\partial \mathbf{x}} \mathbf{x}^\top \mathbf{A} \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}. \quad (\text{C.3})$$

C.4 Positive (Semi)definite Matrices

$\mathbf{A} \succeq 0$ iff $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0 \forall \mathbf{x}$. Equivalent conditions include: all eigenvalues non-negative; a Cholesky factor $\mathbf{L}$ exists with $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$; all leading principal minors non-negative.

Covariance matrices are symmetric PSD. Many estimators assume strict positive definiteness; numerical conditioning (e.g., Joseph-form Kalman updates) prevents accumulation of non-PD error.

C.5 Solving Linear Systems

Direct methods.

- Cholesky for symmetric positive-definite $\mathbf{A}$: fastest, most numerically stable.
- LU with partial pivoting for general square $\mathbf{A}$.
- QR for least squares and ill-conditioned systems.

Iterative methods. Conjugate gradient (CG) for SPD systems, GMRES for general; preconditioning is essential for convergence on large systems.

Sparse solvers. SLAM and bundle adjustment rely on sparsity-exploiting sparse Cholesky (CHOLMOD), sparse QR (SPQR), and block-sparse formulations from domain-specific libraries (Ceres, GTSAM).

C.6 Numerical Stability

- Condition number $\kappa(\mathbf{A}) = \sigma_{\max}/\sigma_{\min}$ bounds the amplification of input error in $\mathbf{A}\mathbf{x} = \mathbf{b}$.
- Never explicitly form $\mathbf{A}^\top \mathbf{A}$ if you can avoid it: the condition number squares.
- Use the Joseph form of the Kalman covariance update $\mathbf{P}_{t+1} = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{t+1}^- (\mathbf{I} - \mathbf{K}\mathbf{H})^\top + \mathbf{K}\mathbf{R}\mathbf{K}^\top$ to preserve symmetry and positive-definiteness.

C.7 Differentiation on Lie Groups

The composition rule for small perturbations on $\text{SE}(3)$ is $\mathbf{T} \leftarrow \mathbf{T} \exp(\hat{\boldsymbol{\xi}})$ for $\boldsymbol{\xi} \in \mathbb{R}^6$. Jacobians of residuals with respect to $\boldsymbol{\xi}$ replace classical multivariate Jacobians; their structure underpins the sparse SLAM Jacobians of Chapter 13.

C.8 Random Matrix and Covariance Propagation

Linear propagation. $\mathbf{y} = \mathbf{Ax} + \mathbf{b}$ implies $\boldsymbol{\mu}_y = \mathbf{A}\boldsymbol{\mu}_x + \mathbf{b}$, $\boldsymbol{\Sigma}_y = \mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A}^\top$.

Unscented transform. For nonlinear $\mathbf{y} = g(\mathbf{x})$, sample sigma points around $\boldsymbol{\mu}_x$ with covariance $\boldsymbol{\Sigma}_x$, evaluate g , reweight mean and covariance. Avoids Jacobians and is accurate to second order.

C.9 Graph Laplacians and Spectral Tools

Graph Laplacians appear in GNN-based prediction, pose-graph optimization, and consistent tracker initialization. $\mathbf{L} = \mathbf{D} - \mathbf{A}$ for adjacency $\mathbf{A}$ and degree $\mathbf{D}$; $\mathbf{L} \succeq 0$; the multiplicity of the zero eigenvalue is the number of connected components.

Bibliography

- [1] Jens Behley, Martin Garbade, Andres Milioto, Jan Quenzel, Sven Behnke, Cyrill Stachniss, and Juergen Gall. SemanticKITTI: A dataset for semantic scene understanding of LiDAR sequences. In *ICCV*, 2019.
- [2] Holger Caesar, Juraj Kabzan, Kok Seang Tan, Whye Kit Fong, Eric Wolff, Alex Lang, Luke Fletcher, Oscar Beijbom, and Sammy Omari. nuPlan: A closed-loop ML-based planning benchmark for autonomous vehicles. *arXiv preprint arXiv:2106.11810*, 2021.
- [3] Holger Caesar et al. nuScenes: A multimodal dataset for autonomous driving. In *CVPR*, 2020.
- [4] Marius Cordts et al. The Cityscapes dataset for semantic urban scene understanding. In *CVPR*, 2016.
- [5] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio López, and Vladlen Koltun. CARLA: An open urban driving simulator. In *CoRL*, 2017.
- [6] Andreas Geiger, Philip Lenz, and Raquel Urtasun. Are we ready for autonomous driving? the KITTI vision benchmark suite. In *CVPR*, 2012.
- [7] Cole Gulino, Justin Fu, Wenjie Luo, George Tucker, Eli Bronstein, Yiren Lu, Jean Harb, Xinlei Pan, Yan Wang, Xiangyu Chen, John D. Co-Reyes, Rishabh Agarwal, Rebecca Roelofs, Yao Lu, Nico Montali, Paul Mougín, Zoey Yang, Brandyn White, Aleksandra Faust, Rowan McAllister, Dragomir Anguelov, and Benjamin Sapp. Waymax: An accelerated, data-driven simulator for large-scale autonomous driving research. *NeurIPS Datasets and Benchmarks Track*, 2023.
- [8] Cole Gulino, Justin Fu, Wenjie Luo, George Tucker, Eli Bronstein, Yiren Lu, Jean Harb, Xinlei Pan, Yan Wang, Xiangyu Chen, John D. Co-Reyes, Rishabh Agarwal, Rebecca Roelofs, Yao Lu, Nico Montali, Paul Mougín, Zoey Yang, Brandyn White, Aleksandra Faust, Rowan McAllister, Dragomir Anguelov, and Benjamin Sapp. Waymax: An accelerated, data-driven simulator for large-scale autonomous driving research. In *NeurIPS*, 2023.
- [9] Quanyi Li, Zhenghao Peng, Lan Feng, Qihang Zhang, Zhenghai Xue, and Bolei Zhou. MetaDrive: Composing diverse driving scenarios for generalizable reinforcement learning. In *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.

- [10] Shai Shalev-Shwartz, Shaked Shammah, and Amnon Shashua. On a formal model of safe and scalable self-driving cars, 2017. *arXiv:1708.06374*.
- [11] Pei Sun et al. Scalability in perception for autonomous driving: Waymo open dataset. In *CVPR*, 2020.
- [12] Benjamin Wilson et al. Argoverse 2: Next generation datasets for self-driving perception and forecasting. In *NeurIPS Datasets*, 2023.
- [13] Fisher Yu et al. BDD100K: A diverse driving dataset for heterogeneous multitask learning. In *CVPR*, 2020.
- [14] Wei Zhan et al. INTERACTION dataset. In *arXiv:1910.03088*, 2019.